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Heinemann *Interactive* *Science* 2

Christine Watson
Wendy Aylmer

Heinemann

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22 Salmon Street, Port Melbourne, Victoria 3207
World Wide Web hi.com.au
Email info@hi.com.au

Offices in Sydney, Brisbane, Adelaide and Perth.
Associated companies, branches and representatives throughout the world.

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First published 2000

2003 2002 2001 2000

8 7 6 5 4 3 2 1 0

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Publisher: Malcolm Parsons

Editor: Catherine Greenwood

Text designer: Liz Nicholson

Cover designer: Giulia de Vincentis

Illustrations: Ophelia Leviny, Ross Gray, Boris Silvestri, Andrew Plant, Alan Laver

Photograph researcher: Janet Pheasant

Typeset in 9/12 Centennial by Palmer Higgs

Film supplied by Type Scan, Adelaide

Printed in Hong Kong by Wing King Tong

National Library of Australia
cataloguing-in-publication data:

Watson, C. (Christine).

Heinemann interactive science 2.

ISBN 0 86462 374 7

1. Science - Juvenile literature. 2. Science - Problems, exercises, etc. I. Aylmer, Wendy. II. Title. III. Title:

Interactive Science 2.

500

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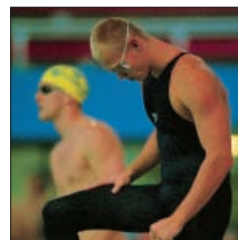
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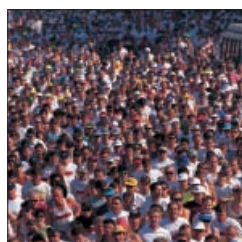
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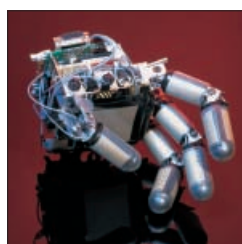
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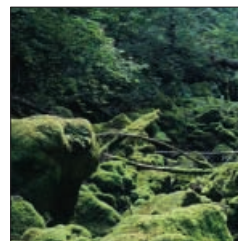


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Introduction

The Heinemann Interactive Science series has been specifically designed to assist schools in the implementation of the New South Wales Science Syllabus Stages 4 and 5. This resource will provide students with a wide range of learning experiences to achieve the knowledge and understanding, skills and values and attitudes outcomes of the syllabus.

Features of Heinemann Interactive Science

Chapters begin with a short **Introduction** which places the content in a **context** which will engage students and stimulate their interest. The chapter context gives coherence to the knowledge and understandings, skills and values and attitudes being developed. The chapter **outcomes** are descriptions of the intended student outcomes which could reasonably be expected to be demonstrated by students who successfully complete the chapter. They address both core and options outcomes from the syllabus.

Each chapter contains two **Science in Focus** double-page spreads which specifically address the mandatory **Prescribed Focus Areas** of the syllabus. These spreads are:

- History of science: 1.2, 5.9, 10.1
- The nature and practice of science: 2.7, 3.9, 4.1
- Implications for society and the environment: 2.9, 4.10, 6.4, 8.9, 8.10
- Applications and uses of science: 3.3, 5.4, 7.9, 9.6, 10.9
- Current issues, research and development: 1.9, 6.7, 7.7, 9.3

The double-page spread format of Heinemann Interactive Science provides all the content, experiments, discussion topics, activities and questions for a lesson. Each spread gives teachers the flexibility to treat a topic in whatever way is most appropriate for their students.

Experiments are detailed guided investigations relevant to each topic. Suggested extensions encourage students to investigate further, often using a more open-ended approach.



This icon is used to identify experiments for which safety glasses should be worn.



This icon is used to identify experiments that are regarded as having significant and identifiable risks associated with them. The Teacher's Resource and Assessment Disk includes extra advice on minimising these risks.

The **Using Science** activities include a wide variety of learning experiences such as model making, the manipulation and interpretation of given data and diagrams, group discussions and debates.

Talk about involves students sharing and developing their personal understandings. These may be used as a lesson introduction, to place the topic in context and establish the students' prior knowledge.

Each double-page spread contains a range of question types to reinforce students' understanding of the concepts. **Think about** questions ask students to go further with the concept they are learning about, to consider the application of the concept to different situations. **Find out** questions encourage students to research from a range of sources information related to the topic. They are aimed at extending students to obtain a greater depth of knowledge and understanding. Additional **Activities** are provided on many spreads. These additional learning experiences could be used as homework or assignment tasks. Many of the Find out and Activities involve students in writing tasks such as descriptive reporting, recounting, experimental reporting and narration.

Fact files are snippets of information designed to heighten students' interest in the concepts covered in each spread.

Each chapter concludes with a **Chapter Review**. These include exercises to reinforce the development of vocabulary, questions to check knowledge, and higher order questions to check application of knowledge. Some 'challenge' material is also included as additional or extension work.

Christine Watson

Teacher support

The **Teacher's Resource and Assessment Pack** consists of a teacher's resource disk written by **Monika Khun** and a student's book. The disk contains assessment and course advice. The safety guidelines are of particular interest to less experienced teachers. A range of worksheets for each chapter is also provided. One worksheet per chapter addresses a functional approach to literacy. An end-of-chapter test with answers provided and answers to textbook questions will also save teacher preparation time.

About the Authors

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Stage 4 Correlation Grid

The following grid references the **Stage 4 knowledge and understanding outcomes** to the relevant chapters of **Books 1 and 2**. The Teacher's Resource and Assessment Disk contains detailed information on mapping core content and assessment. A sample Work Program is available on hi.com.au.

	Book One Chapter	Book Two Chapter
	bold = principal source	
	<i>italics</i> = secondary source	
Outcomes		
4.1 identifies historical examples of how scientific knowledge has changed people's understanding of the world.	5 6 11	1 5 10
4.2 uses examples to illustrate how models, theories and laws contribute to an understanding of phenomena.	1 2 7 8 9 10	2 3 4
4.3 identifies areas of everyday life that have been affected by scientific developments.	2 3 4 5	3 5 7 9 10
4.4 identifies choices made by people with regard to scientific developments.	1 4 8 9	2 4 6 8
4.5 describes areas of current scientific research.	6 7 11	1 6 7 9
4.6 identifies and describes energy changes and the action of forces in common situations.	4 5	3 4 5
4.7 describes observed properties of substances using scientific models and theories.	2 3	1 2
4.8 describes features of living things.	6 7 8	6 7
4.9 describes the dynamic structure of Earth and its relationships to other parts of our solar system and universe.	9 11 10	10 9
4.10 identifies the factors affecting survival of organisms in an ecosystem.	8	8
4.11 identifies resources used by humans and where they are found and describes ways in which they are exploited.	4	9
4.12 identifies, using examples, common simple devices and explains why they are used.	4 5 6 8 9 11	10 5

CHAPTER

1

SUBSTANCES

We live in a 'chemical world'. Everything from the air we breathe and the water we drink, to the ground we stand on, is made of chemicals. Even our bodies are made of chemical substances. If we could reduce a human body down to the simplest substances possible we would find about 22 different elements. Each of these elements is composed of its own type of atom. The element oxygen makes up 65% of the mass of the human body. Carbon contributes 18%, hydrogen 10%, nitrogen 3% and calcium 2% of the total mass. The remaining 2% consists of phosphorus, potassium, sulfur, chlorine, sodium, magnesium and trace amounts of 11 other elements.

In a human body, the atoms are joined together in thousands of different ways to form molecules. When molecules contain different types of atoms, they belong to a group of substances called compounds. Water molecules, consisting of hydrogen and oxygen atoms, are the most abundant in the human body. While other simple molecules such as carbon dioxide are present, the human body consists of many complex molecules. These can contain hundreds of atoms arranged in long chains and rings. Many of these complex, large molecules are classified by chemists as carbohydrates, proteins, lipids and nucleic acids.

Complex and simple molecules form a mixture within the cells and tissues making up the skin, muscle, bone and blood of our bodies.

In this chapter you will learn about the elements, compounds and mixtures with which we come in contact in our daily lives.



Web links

CHAPTER OUTCOMES

Successful completion of this chapter will allow you to:

- Recognise that matter is made of atoms and molecules that are continuously moving and interacting.
- Distinguish between elements, compounds and mixtures.
- Describe how models and theories about matter have changed over time.
- Classify elements and compounds based on their properties.
- Identify symbols of common elements and use formulae to name simple compounds.
- Use models to explain the properties and behaviour of substances.
- Research some recent developments in creating new materials.

Sorting substances



FIGURE 1.1

How would you classify these everyday substances?

Talk about

You are given a collection of everyday substances and are asked to sort them into groups: milk, copper, salt, water, flour, sugar and detergent. Discuss how you would classify these into two or three different groups. For the substances placed in the same group, say what property or feature they have in common.

Everything around us, every material or substance is made of what scientists call **matter**. All matter is made up of tiny particles called **atoms**. Substances are rarely composed of single atoms. Most substances consist of groups of atoms joined together to form **molecules**. There are millions of different types of substances. In order to make their work easier, chemists classify this enormous range of substances into groups. Figure 1.3 shows one way chemists classify matter.

Mixtures

Most of the matter around us is in the form of **mixtures**. Milk, flour and detergent are all mixtures. For example, milk consists of water, proteins, fats, carbohydrates and minerals. The composition of milk, like all other mixtures, can vary. Full-cream milk, skim milk and high-calcium milk all have slightly different compositions. The taste, colour, texture and boiling temperature of different types of milk vary because of different compositions. Also, a mixture can usually be separated easily into its parts. In

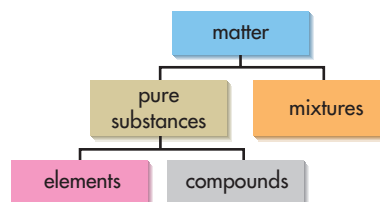


FIGURE 1.3

Classification of matter.

Heinemann Interactive Science 1 you learnt about some of the physical methods used to separate mixtures.

Pure substances

A **pure substance** is a single substance with nothing else mixed in. Copper, salt, water and sugar are pure substances. The composition of a pure substance is fixed. For example, a sample of pure water in Australia is identical in every way to a sample of pure water in Japan or India. The properties such as taste, colour, texture and boiling temperature will also be identical. Also, pure substances cannot be separated into simpler parts by any physical separation method.

Chemists classify pure substances into two groups, **elements** and **compounds**. Elements are made up of only one type of atom. Copper is an element. A pure sample of copper will contain only copper atoms. Compounds consist of two or more different types of atoms chemically joined together. Salt, water and sugar are compounds. A



FIGURE 1.2

Tap water is a mixture of water, dissolved gases and salts.

USING SCIENCE 1

MIXTURE IN A MUDDLE

**WHAT YOU NEED**

- mixture consisting of polystyrofoam balls, sand, iron filings and salt
- filter funnel and paper
- Bunsen burner, gauze and tripod
- evaporating basin • magnet
- 5 beakers • water
- matches

WHAT TO DO

You will be provided with a mixture of four substances and a collection of equipment. Your task

is to plan and carry out a series of separations so that each individual substance is recovered and placed in its own container.

DISCUSSION

- 1 Write down the separation methods you used to obtain each substance.
- 2 Examine the recovered substances and predict whether any could be separated further.
- 3 For the substances recovered which do you think may be elements and which compounds?

pure sample of salt contains sodium atoms chemically joined to chlorine atoms. A pure sample of sugar consists of carbon, hydrogen and oxygen atoms chemically joined together. In practice it is very difficult to obtain pure samples of elements and compounds. Most samples contain impurities (other substances) mixed with them. It is very difficult to remove the impurities completely.

Questions

- 1 **What is the difference between:**
 - a an atom and a molecule?
 - b an element and a compound?
- 2 **There are about 100 different elements but there are millions of compounds. How is this possible?**
- 3 **Copy Figure 1.3 into your book and add three examples of elements, compounds and mixtures to it.**
- 4 **The word 'water' is used to describe both a compound and a mixture. Explain this statement.**

Think about

- 1 **During an experiment Peter took too much compound from a bottle. At the end of the lesson he was about to put the unused compound back into its original container when**

**FIGURE 1.4**

Many products labelled as pure are in fact mixtures.

his teacher stopped him. Why did his teacher tell him to put it in a separate container?

- 2 **Is soft drink an element, a compound or a mixture? Explain your answer.**

Find out

Why can't compounds like water and salt be separated into their component elements by separation methods such as filtration and distillation?

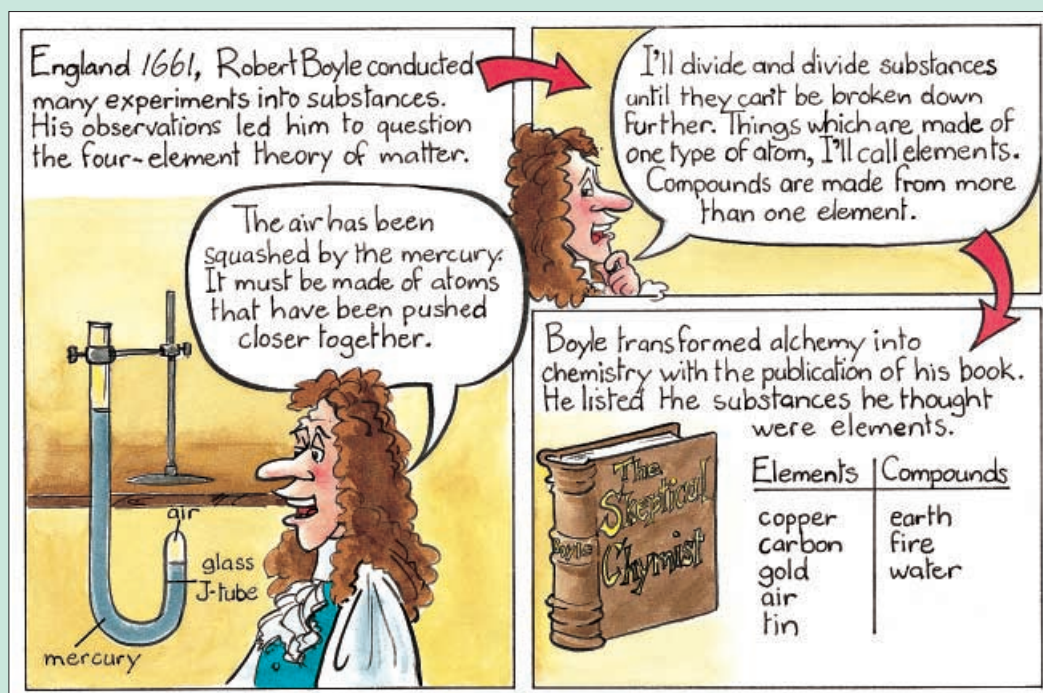
When chemistry began

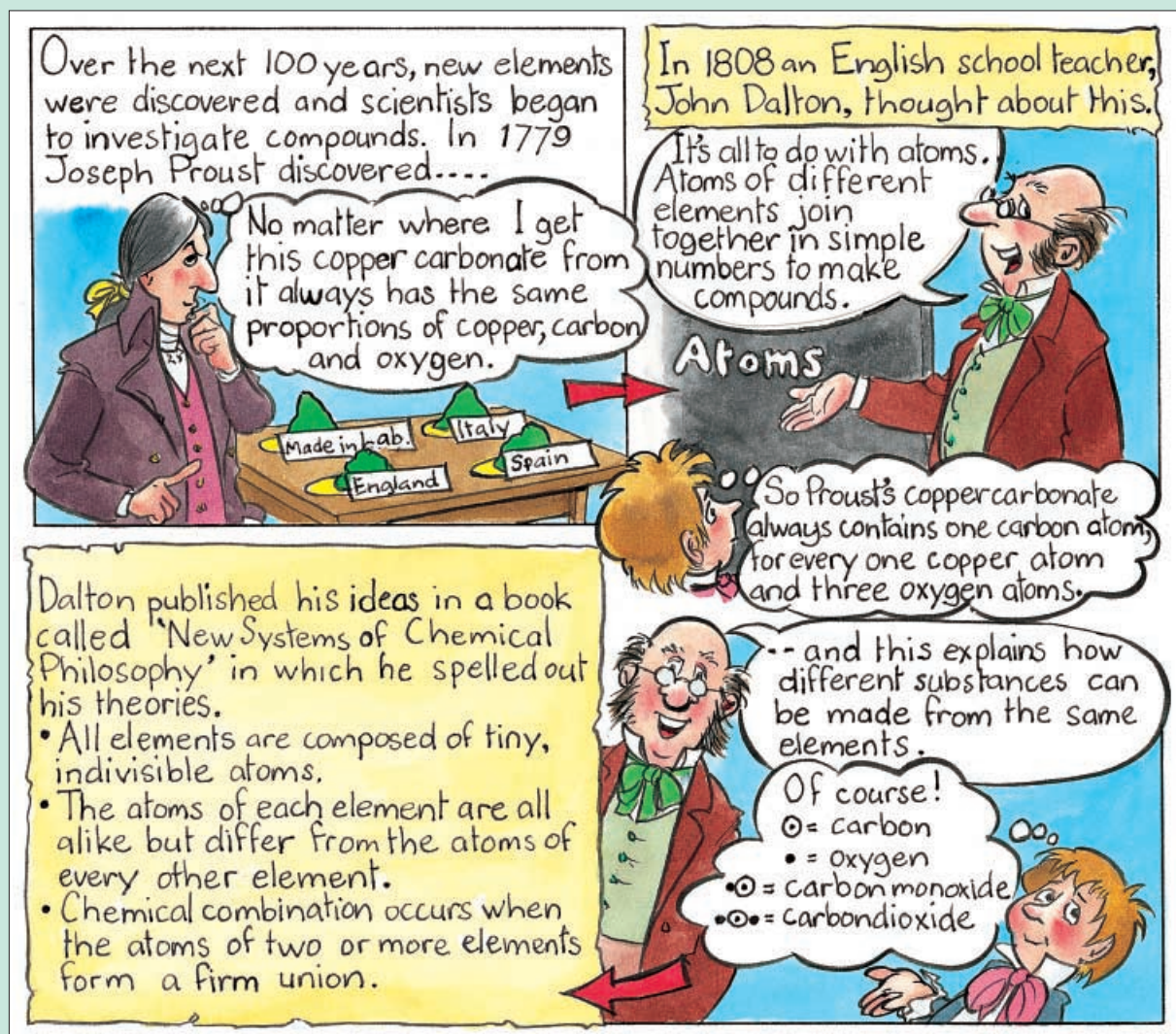
Before the beginning of modern chemistry, people were very influenced by the ideas of the ancient Greek philosophers Empedocles and Aristotle. Empedocles proposed that all matter was composed of four elemental substances—earth, air, fire and water—mixed in various proportions. Empedocles gave his four elements certain properties. Earth was dry and cold, water was wet and cold, air was hot and wet and fire was hot and dry. He said that when substances changed, the four elements could be separated and re-formed. For example, when a log of wood burned, the flames and sparks were the element fire. The smoke given off was the element air and the sap from the log was the element

water. The ashes and cinders which remained were the element earth. This seemed like a very good explanation at the time and these ideas spread among many ancient peoples including the Egyptians, Arabs, Indians and Chinese.

Many cultures observed one substance being transformed into another as they produced metals and medicines to use in their daily lives. Over the centuries, the idea developed that ordinary metals could be changed into gold. This led to what we call **alchemy**, an early form of chemistry. Alchemists discovered some new elements, such as arsenic, phosphorus and antimony. Modern chemistry grew out of alchemy in the 17th century.

FIGURE 1.5





Questions

- 1 Name and describe the four elements proposed by Empedocles by drawing and labelling a diagram of a burning log.
- 2 How did Robert Boyle change the idea of elements?
- 3 Boyle made some errors in his list of elements and compounds. What are they?
- 4 Robert Boyle said 'He that hath seen it hath more reason to believe it, than he that hath not'. What do you think he meant by this?

Find out

- 1 Robert Boyle defined the terms element and compound. However, many other scientists worked to identify substances. Select one of the following early chemists and find out what contribution he made to our understanding of elements and compounds: Antoine Lavoisier, Humphry Davy, Joseph Black, Henry Cavendish, Joseph Priestley.
- 2 Even though the work of Robert Boyle can be thought of as the beginning of modern chemistry, he still believed metals like lead could be turned into gold. Could this be possible, using modern technology?

FIGURE 1.6

FACT

Some elements have been known for thousands of years. Carbon, copper, gold, lead, mercury, silver and tin were used by ancient civilisations.

FILE

The building blocks

FIGURE 1.7

Helium is an example of a gaseous element. Helium is colourless and odourless at room temperature. Because helium is a light gas, it is used to inflate balloons and blimps.

Elements are the basic building blocks of all matter in the Universe. There are 112 known elements. Ninety-two of these occur in nature. The rest have been prepared by artificial means. Each element is unique; no two elements have the same atoms. Because of this, the properties of the

elements vary. Most elements are **solids** but some are **gases** and a few **liquids**. Elements also differ in how they react with other elements.

Symbols for elements

Each element has a short-hand symbol to represent it. John Dalton developed symbols for his elements in 1803, but these symbols were not easy to write or remember. A Swedish chemist, Jons Berzelius, produced a system of chemical symbols based on letters in 1811. This system is still in use today—it's like an international language. Symbols consist of one or two letters from the English or Latin names of the elements. The first symbol is always written as a capital letter.

FIGURE 1.8

Gold, platinum and diamond, used in making jewellery, are examples of solid elements.



FIGURE 1.9

The mercury used in clinical thermometers is a liquid element. Mercury is the only metal that is a liquid at room temperature. Because the metal and its vapour are poisonous, mercury must be handled carefully. It is used in some thermometers because it expands and contracts readily with changes in temperature.



Questions



- 1 How many elements occur naturally and how many have been artificially manufactured to date?
- 2 Which element is the most abundant in the Universe?
- 3 Explain two ways how the use of symbols could aid communication.

TABLE 1.1 Some common elements.

Element	Symbol	Melting point (°C)	Boiling point (°C)	Date of discovery
Aluminium	Al	660	2450	1825
Bromine	Br	-7	58	1826
Calcium	Ca	838	1490	1808
Carbon	C	3730	4830	before 1 AD
Copper	Cu	1083	2600	before 1 AD
Gold	Au	1063	2970	before 1 AD
Hydrogen	H	-259	-253	1766
Iodine	I	114	183	1811
Iron	Fe	1540	3000	before 1 AD
Lead	Pb	327	1740	before 1 AD
Magnesium	Mg	650	1110	1808
Mercury	Hg	-39	357	before 1 AD
Nitrogen	N	-210	-196	1772
Oxygen	O	-219	-183	1774
Phosphorus	P	44	280	1669
Potassium	K	64	760	1807
Silver	Ag	961	2210	before 1 AD
Sodium	Na	98	892	1807
Sulfur	S	119	445	before 1 AD
Tin	Sn	232	2270	before 1 AD

4 Kalium, natrium and argentum are the Latin names for three elements in Table 1.1. Work out the English names for these elements.

5 Use the information in Table 1.1 to answer the following questions.

- Which element has the lowest melting point temperature?
- Which element has the highest boiling point temperature?
- Which element is the most recently discovered?

d How many elements were discovered between 1800 and 1900?

6 Use the melting point and boiling point temperatures to determine which elements are solids, liquids and gases at 20°C. Draw up a table with three columns for solids, liquids and gases. Enter the name and symbol of the elements in the correct column. Give your table a title.

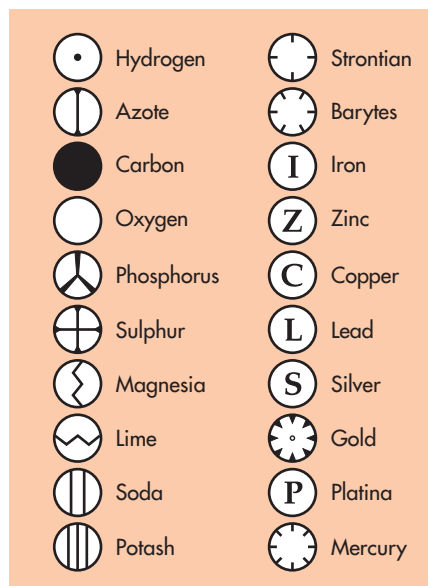


FIGURE 1.10
Dalton's elements and symbols.

FACT

FILE

Hydrogen is the simplest element. Over 90% of the Universe is made up of hydrogen—the same hydrogen that was created during the Big Bang, the explosion that created our Universe. A small amount of a few other simple elements was formed in the Big Bang. All the other elements have been formed from these in nuclear reactions in the depths of stars. The rarest naturally occurring element is astatine. There is only about one-third of 1 gram of this element in the Earth's crust.

Metals and non-metals



FIGURE 1.11

Copper wire is used in electrical cables; helium gas is used in air ships; aluminium is used in ships' hulls although the keel contains lead; match heads contain phosphorus. Identify the metals and non-metals present.

There are many ways of classifying elements. For example, you could divide them into solids, liquids and gases or into **metals** and **non-metals**. Elements are classified on the basis of their physical features or properties. Properties such as

lustre (shine), **malleability** (ability to be bent), **ductility** (ability to be drawn into a wire) and **conductivity** (ability to conduct heat and electrical energy) are used to separate metals from non-metals.

EXPERIMENT 1

COMPARING ELEMENTS

AIM

To compare the properties of elements and classify them as metals and non-metals.

MATERIALS

- samples of the elements sulfur, copper, iron, carbon, zinc and tin
- emery paper
- power supply
- lamp
- wires with alligator leads

METHOD

1 Connect the lamp to the power supply in a simple circuit. Place each sample between two alligator clips in the circuit. If the lamp glows when the switch is on, the element conducts electricity.

- 2** Use the emery paper to rub each sample and decide whether it is shiny or dull.
- 3** Try to bend each sample. If it breaks it is brittle.
- 4** Record your results in a table.

Element	Does it conduct electricity?	Is it shiny or dull?	Is it malleable or brittle?

DISCUSSION

- 1** Metals conduct electricity and are shiny and malleable. Which elements would be classified as metals?
- 2** Non-metals usually do not conduct electricity and are dull and brittle. Which elements would be classified as non-metals?

Period	Group																		III	IV	V	VI	VII	VIII			
	I	II																									
1	1 H																										2 He
2	3 Li	4 Be																			5 B	6 C	7 N	8 O	9 F	10 Ne	
3	11 Na	12 Mg	Transition elements										13 Al	14 Si	15 P	16 S	17 Cl	18 Ar									
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr									
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe									
6	55 Cs	56 Ba	57 La	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn									
7	87 Fr	88 Ra	89 Ac	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Uun	111 Uuu	112 Uub															
Lanthanides			58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu											
Actinides			90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr											

The metalloids

The eight **metalloids** have properties between those of metals and non-metals. They are also called **semiconductors**. Chemists often use a 'staircase' line to divide metals and non-metals on the periodic table. The metalloids are either side of the 'staircase'. They are the 'elements in between'.

Germanium and silicon have similar properties: part metal, part non-metal. Shiny to look at, grey or silver in colour, they are easy to break (brittle). They are typical metalloids. They are also semi-conductors used in transistors and solar cells. They have revolutionised the electronics industry through their use in computer chips and communication equipment.

Questions

- 1 List five properties that sodium, iron and aluminium have in common.
- 2 List five properties that nitrogen, sulfur and iodine have in common.
- 3 What is a metalloid? Name two examples.

FIGURE 1.12

The periodic table, showing all the elements. Apart from hydrogen, all elements on the extreme right are non-metals. Most elements are metals. The metalloids are found in between the metals and non-metals.



TABLE 1.2 Properties of metals and non-metals.

Property	Metals	Non-metals
Conductivity of heat	good conductors	usually poor
Conductivity of electricity	good conductors	usually poor
State	usually solid except for mercury	solids, liquids or gases
Lustre	shiny, metallic surface when scratched	dull when scratched
Malleability	malleable	brittle
Ductility	can be drawn into wire	cannot be drawn into wire
Colour	usually grey or silver except for gold and copper	coloured or colourless

Find out

Carbon can be found in two main forms, graphite and diamond. Graphite is an excellent conductor of electricity and diamond is an excellent conductor of heat, yet carbon is a non-metal. Does this fit the trend for non-metals? Try to account for these observations.

Making and breaking compounds

Talk about

Which of the following substances are elements and which are compounds: water (H_2O), iron (Fe), gold (Au), salt (NaCl)?

Although there are over 100 elements there are over a million different compounds. Compounds contain more than one type of atom chemically joined together in a definite ratio.

For example, water is a compound. It contains two different types of atoms—hydrogen and oxygen. When these atoms combine a molecule is formed. One molecule of water contains two atoms of hydrogen and one atom of oxygen, hence its **chemical formula** is H_2O . Hydrogen and oxygen in any other ratio is not water.

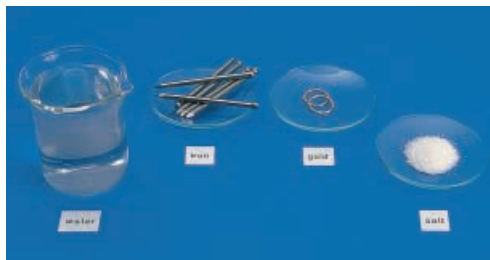


FIGURE 1.13
Water, iron, gold and salt.

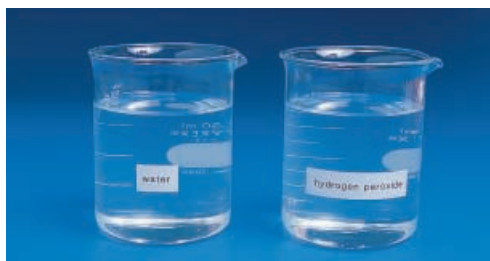


FIGURE 1.14
Hydrogen peroxide and water may look alike, but they have very different properties.

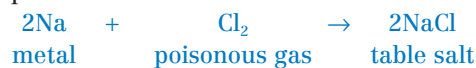


FIGURE 1.15
Salt, chlorine and sodium. Compounds often have different properties from the original elements.

Another compound which contains only hydrogen and oxygen is hydrogen peroxide. It is useful for cleaning wounds and as a bleach. Hydrogen peroxide has the chemical formula H_2O_2 , which means that it contains hydrogen and oxygen in the ratio of 2:2.

A chemical formula (such as H_2O or H_2O_2) is often used to represent the number and kinds of atoms present in one molecule of the compound.

Compounds usually have different properties from the original elements. For example, table salt, sodium chloride (NaCl), is made from sodium (Na), a metal, and chlorine (Cl), a gas, according to the equation:



In this reaction, the sodium and the chlorine are said to be **reactants** (because they react to produce a new substance) and the sodium chloride is said to be the **product** (because it is produced by the reaction). The reaction is shown as a **chemical equation** with reactants on the left-hand side and products on the right-hand side.

Questions



1 Are the following substances elements or compounds?

- a oxygen gas, O_2
- b bicarbonate of soda, NaHCO_3
- c hydrogen gas, H_2
- d chlorine gas, Cl_2
- e nitric acid, HNO_3

2 What is a chemical formula? Give an example.

3 Name the reactant(s) and the product(s) in these reactions.



EXPERIMENT 2**MAKING A COMPOUND****AIM**

To make a compound by causing two elements to combine.

MATERIALS

- tongs
- Bunsen burner
- magnesium ribbon
- heatproof mat

METHOD

- Examine the magnesium ribbon.
- Hold the magnesium by using the tongs.
- Place the end of the magnesium ribbon in the Bunsen flame and allow it to burn.

4 Remove from the flame. Do not look directly at the light from the burning magnesium.

5 Place the tongs on the heatproof mat after it has stopped burning and examine the substance formed.

RESULTS

- Write a description of the reactant magnesium.
- Another reactant from the air has combined with the magnesium. Suggest what this substance is.
- The product is a compound called magnesium oxide. Write a description of its appearance.
- Complete the word equation:

$$\begin{array}{l} \text{magnesium} + \dots \rightarrow \dots \\ \text{(element)} + \text{(element)} \rightarrow \text{(compound)} \end{array}$$

EXPERIMENT 3**BREAKING A COMPOUND****AIM**

To break or decompose a compound.

MATERIALS

- hot plate
- sugar
- metal pan/lid
- beaker

METHOD

- Examine the sugar.
- Place 1 teaspoon of sugar on the metal pan and sit it on the hot plate.
- Observe the sugar; as it starts to change hold the inverted beaker over the top of it to collect some vapour on the inside of the glass beaker.

RESULTS

Write a description of the sugar before, during and after heating. Also describe the substance that collected on the inside of the beaker.

DISCUSSION

- The black residue on the metal pan is an element. Suggest a name for it.
- The droplets of colourless liquid inside the beaker are a compound. Suggest a name for it.
- Complete a word equation for the decomposition of sugar by heat:

$$\begin{array}{l} \text{sugar} \rightarrow \dots + \dots \\ \text{(compound)} \rightarrow \text{(element)} + \text{(compound)} \end{array}$$

b carbon + oxygen → carbon dioxide



c magnesium + oxygen → magnesium oxide

$$2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$$
Activities

- 1** Make a model. Using a molecular model kit or toothpicks and small styrofoam balls, make a model

to represent H_2O and H_2O_2 .

- 2** In your class, what is the ratio of:

a girls to boys?

b teachers to students?

c tennis players to basketball players?

- 3** In a group, determine the ratio of nitrogen (N) to oxygen (O) for the following compounds: NO_2 , N_2O_5 , NO .

FACT

FILE

Energy in the form of heat, light or electricity is often required to make atoms form or break bonds.

Modelling atoms and molecules



FIGURE 1.16

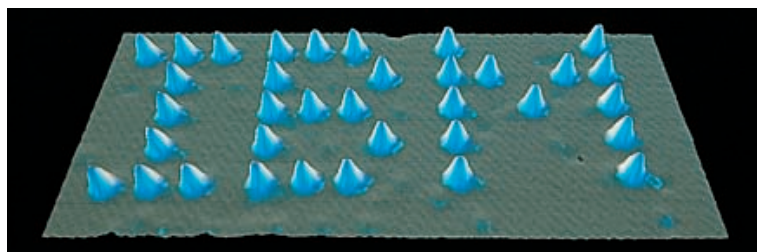
The letters IBM spelled out in xenon atoms. The scanning tunnelling microscope reads the position of atoms like a blind person reads Braille.

Talk about

How do we know air exists? What evidence could you use to convince someone that air was real?

How do we know atoms and molecules exist?

Atoms and **molecules** are incredibly tiny particles. They are too small to be seen with anything except the most powerful electron microscopes (see Figure 1.16). Evidence that atoms and molecules exist can be found by observing their effects. In the same way, we can see evidence that air exists, when the wind moves leaves on a tree.



EXPERIMENT 4

MOVING ATOMS

AIM

To see the effects of atoms and **molecules** moving, called Brownian motion.

MATERIALS

- microscope
- power supply
- Brownian motion box
- pen and paper

If a Brownian motion box is not available, use a teat pipette to inject smoke from a burning piece of string into a small glass cell and cover with a cover slip. Light from the side will enhance viewing.

METHOD

- 1 Connect the wires of the Brownian motion box to the power supply.
- 2 Place the box on the microscope stage.
- 3 Turn on the power supply and focus the microscope on the smoke particles in the box.

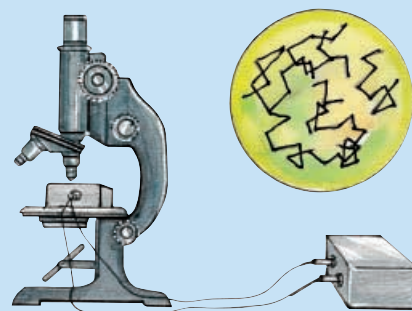


FIGURE 1.17

Watching Brownian motion at work.

RESULTS

Draw a picture of what you see. Use lines with arrows to show the movement of the smoke particles. Smoke is made up of billions of minute particles floating in the air. They are tiny, but visible under a microscope. When you watch them you will see them zigzag chaotically. This is believed to be caused by the atoms and molecules of the air moving incredibly fast and crashing into the smoke particles.

Using models

Many of our observations of the physical and chemical behaviour of substances can be explained by the kinetic theory of matter. This gives us a **model** which we used to explain the behaviour of solids, liquids and gases in *Heinemann Interactive Science 1*. Using this model we imagine atoms are like tiny spheres which can join together to form molecules and crystals and break apart again in chemical reactions.

Our model can also be used to compare mixtures and compounds. Mixtures and compounds are similar in that they are both made up of different types of atoms. Mixtures and compounds are different in several ways. For example, a mixture of iron and sulfur atoms can be easily

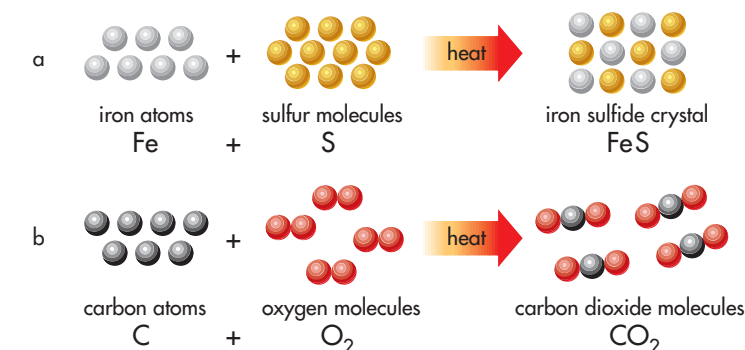


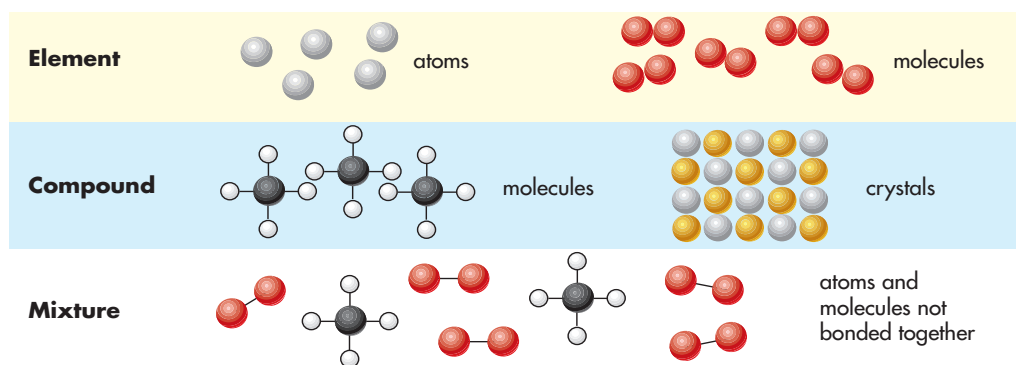
FIGURE 1.18

Using the model to show elements combining to form compounds.

separated but a compound consisting of iron atoms bonded to sulfur atoms cannot. Also, the substances in a mixture keep their properties, whereas in a compound the elements lose their individual properties. A compound can be completely different from the elements it contains.

FIGURE 1.19

Using the model to show atoms, molecules and crystals in elements, compounds and mixtures.



Questions

- 1 Explain what causes Brownian motion.
- 2 Use examples to compare and contrast mixtures and compounds.
- 3 Explain why it is not possible to have an atom of a compound.
- 4 Look at Figure 1.20. For each molecule say whether it represents an element or a compound and give its chemical formula.

Activity

Examine samples of iron, sulfur and iron sulfide. Describe their physical appearances. What tests could you carry out to compare their physical and chemical properties?

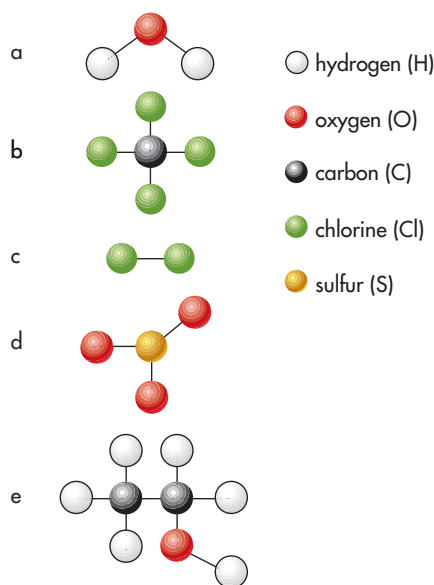


FIGURE 1.20



A Scottish scientist, Robert Brown, first saw evidence for moving particles in 1827. He looked at pollen grains in water under a microscope. He saw the pollen move in a jerky way as if pushed around by the water. This is called Brownian motion. However, it was Albert Einstein who said in 1905 that it provided evidence for the kinetic theory of matter.

What's in a name?

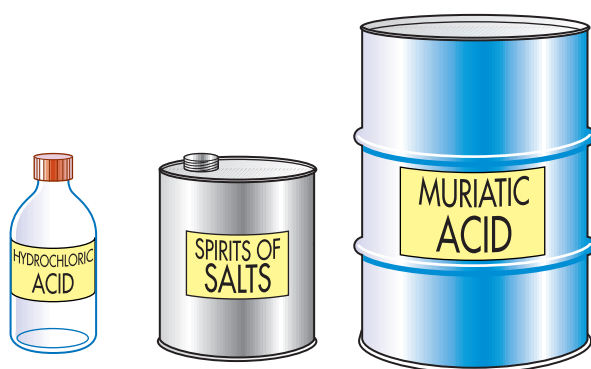


FIGURE 1.21
One compound but several names.

FACT

Mercury is the only metal that is liquid at room temperature. Its chemical symbol Hg is from the Latin *hydrargyrum* (liquid silver). It is also known as quicksilver.

FILE

Over the years substances have been given different names. For example, the compound with the formula HCl(aq) has been known as spirits of salts, muriatic acid and hydrochloric acid. Although these common names have served their purpose, chemists need to be able to communicate clearly with other chemists. They need to know they are talking about the same substance. So a standard system of naming compounds has been developed based on simple rules that everyone can use.

The standard or systematic name of a compound gives us more information than a common name. For example, HCl has the systematic name hydrogen chloride. This tells us that this compound consists of two elements, hydrogen and chlorine. However, some compounds are still known by their common names. For example, we still call H_2O water, NH_3 ammonia and NaCl salt.

Being able to read chemical formulae and name compounds is a useful skill to help understand chemical substances.

TABLE 1.3 Names and formulae of radical groups.

Name	Formula
Carbonate	CO_3
Hydroxide	OH
Nitrate	NO_3
Sulfate	SO_4
Phosphate	PO_4

Chemical formulae

The formula of a compound gives us two pieces of information. The symbols tell us what elements are in the compound and the numbers tell us the ratio of each type of atom present. For example, MgCO_3 tells us that this compound contains magnesium, carbon and oxygen atoms in the ratio 1:1:3. Or another way of saying the same thing is that for every one magnesium atom, there is one carbon atom and three oxygen atoms. The reason why the atoms combine in this way can be explained by knowing their valency. You will learn more about this in *Heinemann Interactive Science 3*.

Multiple compounds

When some elements form compounds they can combine in more than one ratio. Different compounds with different formulae and properties are produced. The naming of these compounds must show how they vary. For example, the elements nitrogen and oxygen can combine to form several different compounds. N_2O is called dinitrogen oxide, NO is called nitrogen monoxide and NO_2 is known as nitrogen dioxide.



RULES FOR NAMING COMPOUNDS

We will concentrate on naming simple compounds which contain only two or three different elements. These compound have two parts to their names.

1 Compound containing two elements, one metal, one non-metal.

- Write the name of the metal element first.
- Shorten the name of the non-metal and add 'ide' as a suffix.
- Write the name of the non-metal second.
For example, MgO = magnesium oxide.

2 Compounds containing two non-metal elements.

- If hydrogen is present, write its name first.
- If no hydrogen is present, the solid non-metal or the one closer to the bottom left-hand side of the periodic table is written first, e.g. carbon, sulfur, phosphorus.
- Prefixes are used on the names of the non-metals to indicate the number of atoms present: 'mon' = 1, 'di' = 2, 'tri' = 3, 'tetra' = 4, 'penta' = 5, 'hexa' = 6. (Note, 'mon' is not used in front of the first non-metal.)
- Shorten the name of the second non-metal and add 'ide' as a suffix and write its name.
For example, SO_3 = sulfur trioxide.

3 Compounds containing three elements.

Many compounds contain a group of atoms called a radical. Look at Table 1.3 for examples.

- If a metal is present it is named first.
- If no metal is present, hydrogen is named before other non-metals.
- The radical is named second.
For example, H_2SO_4 = hydrogen sulfate, ZnCO_3 = zinc carbonate.

Questions



1 Explain why chemists mostly use systematic chemical names instead of common names.

2 Copy and complete Table 1.4.

3 Draw ball and stick diagrams to represent the following molecules.

- SO_3
- CCl_4
- NH_3
- HBr
- H_2O

4 Use the rules to name the following compounds. (Symbols of elements are shown in section 1.3 and the periodic table in section 1.4 shows metal and non-metals.)

- KI
- AgCl
- PbO
- Na_2S

TABLE 1.4

Formula	Name of elements present	Number of each type of atom in formula	Total number of atoms in formula
KNO_3			
Na_2S			
FeCO_3			
H_3PO_4			
$\text{Mg}(\text{OH})_2$			

- CaH_2
- CO
- CO_2
- PCl_3
- PCl_5

5 Use Table 1.3 to name the following compounds:

- SnCO_3
- HNO_3
- AgOH
- AuNO_3
- MgSO_4
- AlPO_4

Classifying compounds

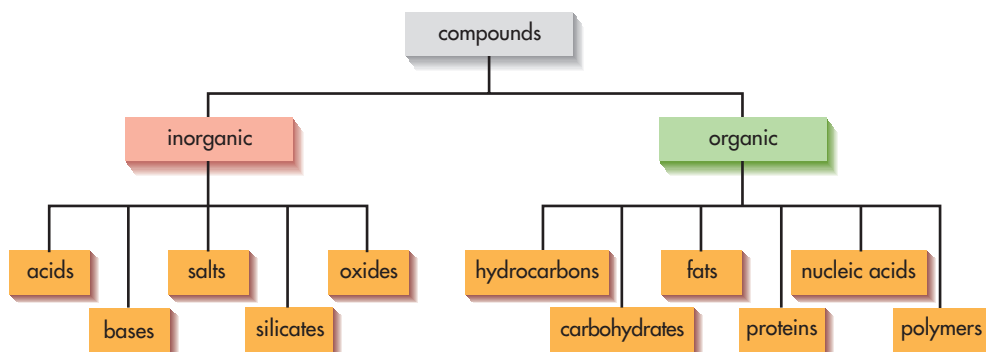


FIGURE 1.22



FIGURE 1.23

In some parts of northern Australia, swimming in the ocean during summer can be dangerous due to the presence of the **sea wasp**, also called the **box jellyfish**. When they sting, sea wasps inject a **basic** substance into the victim which can affect the heart and the ability to breathe. In order to **neutralise** the sting, vinegar—a weak **acid**—is poured over the wound. Medical advice is sought. Often, bottles of vinegar are provided in first-aid boxes on beaches. Mouth-to-mouth resuscitation may be required and/or antivenene. In severe cases, sea wasps have been known to cause death within a few minutes.

There are millions of compounds, and new ones are constantly being discovered or made. Chemists try to organise compounds into groups to make communication easier. They use two broad groupings, inorganic and organic. **Inorganic compounds** contain elements other than carbon. These compounds make up the non-living structures of the Earth. **Organic compounds** contain carbon and are found in or are made by living things. Each of these two broad groups is further divided into smaller groups based on their properties. Some are shown in Figure 1.22. We will focus on two groups of compounds, acids and bases, to learn more about their properties.

Talk about

You may have heard the words acid, base and neutralise. In groups of three or four:

- give at least three examples where you have heard each of these words used;
- compare your examples with other groups in the class; and
- try to decide what each word means.

What are acids?

Acids are very common substances which are used in everyday life. Acids may be classified as strong or weak. Strong acids are potentially dangerous because they are **corrosive**—they can burn through solid objects. Car batteries, for example, contain sulfuric acid. Hydrofluoric acid is used to etch glass, and hydrochloric acid is used to



FIGURE 1.24

Acids are all around you!

clean mortar from bricks. Weak acids, however, can be safe enough to eat and drink. Citrus fruits and tomatoes contain *citric acid* and grapes contain *tartaric acid*. The bubbles in soft drinks are created by *carbonic acid*.

All acids have the following properties.

- They have a sour taste.
- They turn blue litmus paper to red.
- They react with metals to form hydrogen gas.
- They react with baking soda and limestone to form carbon dioxide gas.
- They react with bases in a process called **neutralisation**.

What are bases?

Bases are commonly used in the home as cleaning agents. Strong bases, like sodium hydroxide and potassium hydroxide, are found in oven and drain cleaners. The

Bases which are soluble in water are called **alkalis**.



CURRENT ISSUES, RESEARCH AND DEVELOPMENT

New materials

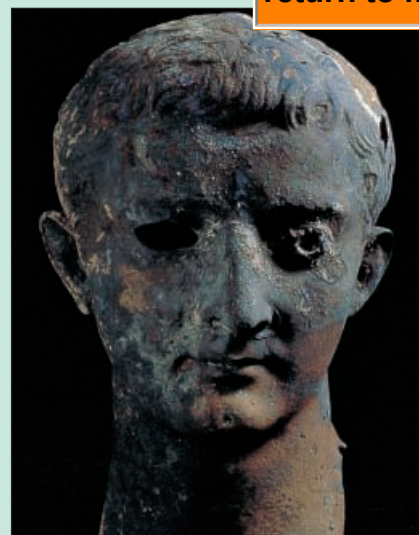


FIGURE 1.26
Materials over time: bronze, steel, aluminium, plastics, alloys and synthetic materials.

FACT

An alloy is produced by melting a metal and combining it with some other metal or non-metal to give a mixture with improved properties.

FILE

Talk about

The rise in technology through the ages has been related to the materials available and their properties. We often describe the progress of civilisation in terms of the stone age, bronze age, iron age and so on. The 20th century is sometimes called the aluminium age; however, synthetic and composite materials have increasingly been used. Make a list of the materials you have used today. Compare your list with others and write a summary of the most used chemicals or materials. What do you think is a good name for the age that we live in?

New materials are becoming highly sophisticated. Cars now contain about 10% by weight of plastics, and this value is expected to increase further. Optical fibres are replacing metal wires in communications technology. Super-strong alloys and super-magnets are reducing the mass of metals used in cars and electrical appliances.

Are metals advanced materials?

Pure aluminium was virtually unknown about 200 years ago. Now it is alloyed

with tiny amounts of other metals, like magnesium, to make it one of the most versatile metals of today. **Alloys** have major advantages over their individual metal components.

Rare earth metals (lanthanides) are used to make very powerful super-magnets. Ferrite is used in ordinary magnets, cobalt makes a strong magnet, but neodymium (a rare earth) can make a supermagnet. In medicine supermagnets are used in extremely sensitive probes that provide scans inside the human body.



FIGURE 1.27
In Japan, high-speed levitation trains operate on powerful supermagnets.

USING SCIENCE 2

NEW MATERIALS

WHAT YOU NEED

Sources of information—magazines, reference books, computer databases, the Internet, newspaper articles, pamphlets, video or TV programs, poster paper and library resources.

WHAT YOU DO

Work together in small groups of two or three. Each group is allocated a new material or chemical to investigate.

Research and determine the key uses for the selected new material. Include a summary of the components or chemicals from which it is made.

Plan, produce and present a poster showing the components, the properties and uses of the new material.

Examine the presentation of each group poster and list the most important uses of each new material. Discuss how small variations in the composition of materials change their properties and affect their use. An alternative is to write a story about a new material, explaining its advantages over conventional products. Try a marketing approach.

Examples could be drawn from the following:

- 1** The new body: many parts of the human body can be replaced by artificially made components. These need to be chemically inert and resist corrosion. Examples include the artificial heart; ceramics for knee, hip and bone replacement; stainless steel or titanium alloy pins; and plastics and silicones for implants and prosthetics.
- 2** Modern ceramics can have the strength of steel yet resist cracking, corrosion and extremely high temperatures. They are used in knives, scissors, car engines (experimental at this stage), and bulletproof tiles.
- 3** Supermagnets. These contain rare earths and are much smaller, lighter and stronger than conventional magnets. Many new and lighter products are being developed.
- 4 Superconductors.** In 1986 a ceramic superconductor made of oxides of copper, yttrium and barium was discovered which below a certain temperature could conduct an electric current without loss of energy.

**FIGURE 1.28**

Ceramics, stainless steel or titanium alloys are used for joint or bone replacement parts.

**FIGURE 1.29**

Ceramic engines are lighter than metal engines and withstand temperatures up to 800°C.

**FIGURE 1.30**

Over 33 000 ceramic tiles are needed to cover the space shuttle to protect it from extremes in temperature during launch and re-entry.

5 Synthetic fibres and composites can be made from polymers, which are carbon-based molecules formed into long chains. Kevlar and Nomex are light-weight, durable and fire-resistant, and are stronger than steel.

6 Metal alloys: alloys made of titanium and nickel have a built-in 'shape memory' (they can return to their original shape after distortion) and are used in robotics, micro-switches, micro-motors and computer technology. When the composition of metal alloys is changed, new properties are created.



Using scientific language

- Define the following words or terms.

matter	atom	molecule
element	compound	mixture
bond	metal	non-metal
lustre	malleability	ductility
conductivity	organic	inorganic
acid	base	
- Select at least four words from the list above and create a limerick to help remember their meanings.



Check your knowledge

- Draw a diagram to represent:
 - a molecule of an element containing three atoms
 - a molecule of a compound containing three atoms
 - a molecule of a compound made of three elements and five atoms
 - a mixture of an element and a compound
 - a mixture of two elements.
- Beryllium is a metal. Write down five properties of this substance.
- Iodine is a non-metal. Write down five properties of this substance.
- Choose five elements that you are familiar with. For each element, describe as many uses for it as you can.
- Choose five compounds that you are familiar with. For each compound:
 - describe as many uses for it as you can
 - identify the elements from which it is made.
- Choose five mixtures that you are familiar with. For each mixture, try to identify the components from which it is made.
- What do we call early 'chemists'?
 - What did they hope to achieve?
- What contribution did Robert Boyle and John Dalton make to the science of chemistry?
- Name the following compounds:
 - K_2O
 - HF
 - $MgCl_2$
 - HNO_3
 - Na_3PO_4
- Explain neutralisation and give an everyday example.



Web page

Apply your skills



- Tap water contains dissolved substances such as chlorine which is added during water treatment.
 - Why is chlorine added to the water?
 - Why can't you purify water by filtering it?
 - Explain how you would get pure water from tap water.
- What is wrong with a label which says 'pure natural mineral water'?
- What evidence suggests that atoms and molecules:
 - are small?
 - move around all the time?
- Is air an element, a compound or a mixture? Explain your answer.
- When calcium carbonate is heated, it breaks down to form calcium oxide and carbon dioxide.
 - Write down a word equation for this reaction.
 - Explain why this reaction is an example of decomposition.
- Some elements such as gold have been known to people for thousands of years. Others have only recently been discovered. How is gold different from more recently discovered elements?

Challenge yourself



The elements which make the Earth's crust differ in abundance from those in the oceans and atmosphere. Represent the information below by either divided bar graphs or sector graphs.

Elements in the Earth's crust: oxygen 46.6%; silicon 27.7%; aluminium 8.1%; iron 5.0%; calcium 3.6%; others 9%.

Elements in the Earth's oceans: oxygen 86%; hydrogen 10.8%; chlorine 1.9%; sodium 1.1%; others 0.2%.

Elements in the Earth's atmosphere: nitrogen 78.1%; oxygen 20.9%; argon 0.9%; others 0.1%.



Tester

CHAPTER

2

CHANGING SUBSTANCES

Modern chemistry has created an amazing array of substances to improve our quality of life. Raw materials from the environment, such as minerals, crude oil, water and gases, are used by chemists. They begin by separating the useful components out of the raw materials. These materials are then changed to create new substances which provide us with food, medicine, fibres for paper, clothing and furniture, building materials, cleaning products for ourselves and our homes and fuels.

Occasionally substances are created which don't have an immediate use but may have some interesting properties. Two such chemical curiosities which are sold in toy stores are 'slime' and 'silly putty'. Slime is a semisolid liquid or gel which can be moulded into a shape but slowly flattens into a puddle. It is made when a polymer such as polyvinyl acetate is mixed with borax and water. The slime can be stretched into a thin film if pulled slowly. However, if pulled sharply it snaps. Also if it is poured from a container and the container is tipped upwards slightly it will continue to flow out by self-siphoning.

Silly putty was created by scientists who were trying to make synthetic rubber. They made a silicon polymer called polyborosiloxane which has some of the properties of rubber. Silly putty is elastic—like rubber it can bounce. But like slime, it cannot keep its shape. When pounded by a hammer silly putty shatters into pieces. The pieces can be stuck back together again. If dropped into liquid nitrogen, rubber will become brittle and shatter but not silly putty—it can still bounce.

In this chapter you will investigate the physical and chemical changes which substances can undergo, and discover how this can be used by chemists to produce useful substances.

CHAPTER OUTCOMES

Successful completion of this chapter will allow you to:

- Explain the difference between physical change and chemical change and give examples of both.
- Detect evidence of a chemical change.
- Use the particle theory to explain physical changes.
- Determine experimentally whether a gas is hydrogen, oxygen or carbon dioxide.
- Give examples of common reactions and describe them by using word equations.
- Give examples of chance observations which led to discoveries through the understanding of the observer.
- Describe the manufacture of some household products.
- Demonstrate a commitment to safe personal and community practices in the home and workplace.

Everything changes

Everywhere we look, we see substances undergoing changes. Some changes such as rain falling, seeds sprouting, rocks wearing away and snow melting occur naturally. Many changes are brought about by people. For example, wood pulp is changed into paper, crude oil is changed into petrol, iron ore is changed into steel and cotton fibre is changed into clothing.

Some changes are slow. They take place over weeks, months or years. Changes such as trees growing, cars rusting and tyres wearing away occur so slowly we hardly notice them. Other changes occur quickly. Burning is an example of a fast change. Explosions are even faster.

The changes to materials which we see around us can be classified as either **physical changes** or **chemical changes**.

During a physical change no new substances are formed. Some of the properties (characteristics) of the substances change, but the substance is still the same. A change in the size, shape or state (solid, liquid or gas) of a substance is a physical change.

Examples of physical changes include chocolate melting, glass bottles breaking and the filament in a light globe glowing white hot. One feature of physical changes is that they can usually be easily reversed. Melted chocolate can be changed back into solid chocolate. Broken glass can be recycled into a new bottle and when the electricity is turned off the filament in the light globe returns to its usual appearance.

In a chemical change, one or more new substances are formed. For example, when wood burns, smoke and colourless gases form, leaving behind ash (carbon). These are new substances different from those originally present in the wood. When iron tools are left outside, a crumbly brown layer forms over them. The iron combines with oxygen in the air to form a completely new substance, iron oxide (rust). Cooking food is an example of a chemical change. When you boil an egg, bake a cake or barbecue sausages, the carbohydrates, fats and proteins in the foods are changed into new substances.

Chemical changes are also called **chemical reactions**. In a chemical reaction the substances you begin with are called the **reactants**. The atoms which make up the reactants become rearranged in the reaction to form new substances called the

FIGURE 2.1

Changes to materials occur around us all the time. For example, bread goes mouldy if left for a day or two, and wet washing dries when put on a line.



FIGURE 2.2

The freezing of water and the melting of ice are examples of physical changes. When water freezes to become ice, it is still the same substance. Ice looks different from water, but we can always melt the ice to produce water again.



FIGURE 2.3

Fireworks are a good example of chemical changes. The coloured sparks and flames are created by the burning of various metallic compounds. These different compounds are responsible for the many spectacular colours associated with fireworks. The ash, smoke and gases formed are the new substances.



products. It is important to understand that no atoms are created or destroyed in a chemical reaction. What do change are the bonds which hold the atoms together. The bonds inside the reactants are broken then new bonds form to hold the atoms together in the products. This breaking of bonds and rearranging of atoms means it is often much harder to reverse chemical changes. Sometimes it is impossible. When a stick of dynamite explodes, releasing heat energy and gases in all directions, it is impossible to reverse the reaction.

Questions



- 1 What is the difference between a physical change and a chemical change?**
- 2 State whether the following changes are physical or chemical.**
 - clothes drying on a line
 - iron rusting
 - dynamite exploding
 - ice-cream melting
 - a toaster element getting hot
 - milk going sour
 - toasting bread
 - concrete setting
 - writing with a pen
- 3 Give an example of a slow change and a faster change other than the ones already mentioned.**

Think about



Elizabeth, Ben and Matthew watched an Alka-seltzer tablet fizz and bubble as it disappeared in a beaker of water. Their teacher told them the bubbles were carbon dioxide gas. She asked them to explain where they thought the gas came from. Matthew said that the gas was built into the tablet when it was made. Ben said the gas was dissolved in the water and the tablet released it.



FIGURE 2.4

Wood-fired barbecues are fun, but it is amazing how much wood you can go through! Even though some ashes remain, most of the mass of the wood leaves as smoke and gases. This is because a chemical change has occurred.

Elizabeth said the gas was formed from atoms in the Alka-seltzer tablet and the water. Whose explanation is most like what actually happens? Pick one of the explanations and design a way of testing it.

Activity



Using invisible ink is a fun way of observing changes to substances. You will try two different methods to compare the reversibility of the changes.

Method 1. Dip a paint brush in some phenolphthalein and write on a sheet of paper. Allow the paper to dry then spray it with dilute ammonia solution. Observe what happens when the ammonia evaporates.

Method 2. Dip the paint brush in lemon juice and write on a sheet of paper. Allow the paper to dry, then hold the side of the paper with the writing on it close to a candle flame. The writing will gradually appear, does it disappear when removed from the flame?

FACT

FILE

You are a walking chemical factory in which many different chemical changes are continually taking place. Raw materials such as food and air enter your body. These are physically and chemically changed into compounds which can be used in other chemical reactions. Chemical reactions are needed to provide energy for warmth, movement and life processes. Building new cells for growth and repair also requires chemical reactions.

Identifying physical and chemical changes

FACT

Some chemical reactions do not change colour, give off a gas or form a precipitate. In these cases it may be possible to use an indicator to show that a reaction has taken place. An indicator is a compound which changes its colour when a specific type of substance is present. Litmus, iodine and potassium dichromate can all be used as indicators.

FILE

It is sometimes difficult to tell the difference between a physical change and a chemical change. It is not possible to see atoms being rearranged into new substances. Therefore, it is necessary to look for evidence of a new substance with different properties appearing. The following can be used as evidence of a chemical change.

- **The production of a gas**
When a cake is cooked, the presence of bicarbonate of soda allows carbon dioxide gas to be released, which causes the cake to rise. The gas that is produced tells us that a reaction has taken place.
- **A permanent change in colour**
The colours displayed when fireworks explode are due to chemical reactions.
- **The formation of a precipitate**
A precipitate is a solid that forms when two solutions are mixed. The chemical change shown in Figure 2.5 can be written as an equation as follows:

$$\begin{array}{lcl} \text{lead nitrate} & + & \text{potassium iodide} \rightarrow \\ (\text{colourless}) & & (\text{colourless}) \\ \\ \text{lead iodide} & + & \text{potassium nitrate} \\ (\text{yellow}) & & (\text{colourless}) \end{array}$$

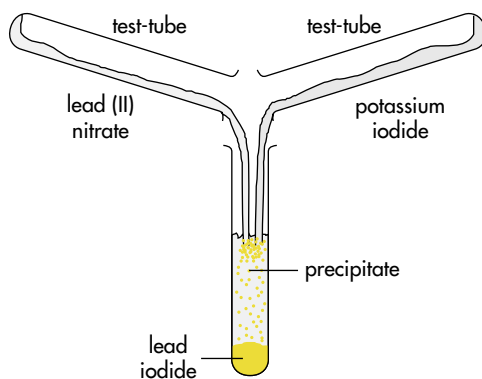


FIGURE 2.5

The formation of a precipitate is evidence of a chemical change. When solutions of lead nitrate and potassium iodide are mixed, a yellow precipitate of lead iodide is formed.

Also in a chemical change energy may be given out in the form of heat, light, sound or electricity. Some physical changes absorb or give out heat energy so this alone cannot be taken as proof of a chemical change. Even if none of the evidence described can be found, a reaction may still have produced a new substance. It may be necessary to carry out chemical tests to identify it.

Questions



- 1 What three pieces of evidence can indicate that a chemical reaction has taken place?
- 2 When orange juice is added to milk, a precipitate is formed.
 - a What is a precipitate?
 - b Is this a physical reaction or a chemical reaction?
- 3 If a silver spoon is used to eat an egg and is not washed afterwards, sulfur in the egg reacts with the silver and forms a black coating on the spoon according to the following equation:



What indicates that this is a chemical reaction rather than a physical reaction?

- 4 Give an example of a chemical reaction which produces:
 - a heat
 - b light
 - c electricity.
- 5 Design an advertising brochure to promote the 'slime' you made in Experiment 1.
What are its best features? Who do you think would buy it?

EXPERIMENT 1

LOOKING FOR EVIDENCE

**AIM**

To observe changes and find evidence for chemical reactions.

METHOD

- 1 For each of the following experiments, record the title in your book.
- 2 State whether you think a reaction has taken place. If you think a reaction has taken place, what evidence is there to support this? (There may be more than one piece of evidence.) Record the evidence.

EXTENSION

Write word equations for the chemical reactions you have observed.

A MAKING SILVER**MATERIALS**

- test-tube containing 5 mL silver nitrate solution
- copper wire
- pencil

METHOD

Warning: Be careful not to spill the silver nitrate solution as it stains skin and clothing.

- 1 Wind the copper wire around a pencil.
- 2 Hook the wire over the edge of the test-tube.
- 3 Record your observations.

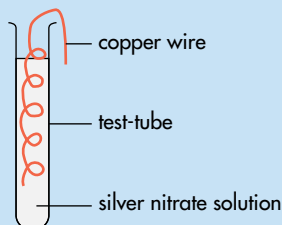


FIGURE 2.6
Experimental set-up.

B SLIME**MATERIALS**

- beaker
- measuring cylinder
- 20 mL borax solution
- paper towel
- stirring rod
- 25 mL PVA glue
- food dye

METHOD

- 1 Place about 25 mL of PVA glue into the beaker.

- 2 Add two or three drops of dye.

- 3 Measure 20 mL of borax and slowly add it to the glue, stirring constantly.

- 4 When the slime has solidified, wash it thoroughly in running water and dry it with the paper towel.

C HEATING COPPER CARBONATE**MATERIALS**

- test-tube
- Bunsen burner
- 1 teaspoon copper carbonate
- safety glasses
- test-tube holder

METHOD

- 1 Place the copper carbonate in the test-tube.
- 2 Use the test-tube holder to heat it with the Bunsen burner.
- 3 Keep the test-tube moving to heat it evenly until no further changes occur.

D MIXING ACID AND ALKALI**MATERIALS**

- thermometer
- 5 mL hydrochloric acid in test-tube
- 5 mL sodium hydroxide in test-tube

METHOD

- 1 Measure the temperature of both solutions.
- 2 Pour the contents of one test-tube into the other.
- 3 Measure the temperature of the solutions combined.

E MAKING FIZZ**MATERIALS**

- test-tube
- 5 mL vinegar
- 1 teaspoon of sodium bicarbonate
- test-tube stand

METHOD

- 1 Pour vinegar into the test-tube.
- 2 Add sodium bicarbonate and observe.



Explaining physical changes

FACT

FILE

Diamond crystals are so strong that when you heat them enough to make the carbon atoms break loose, they are moving so fast that the diamond bypasses the liquid phase and turns into a gas instantly. We call this process 'sublimation'.

FIGURE 2.7

If the molecules can move freely then we have water or steam, depending on how close the molecules are.

Talk about

Draw up a table with three columns and label them solids, liquids and gases. Under each heading write down as many properties of each as you can think of.

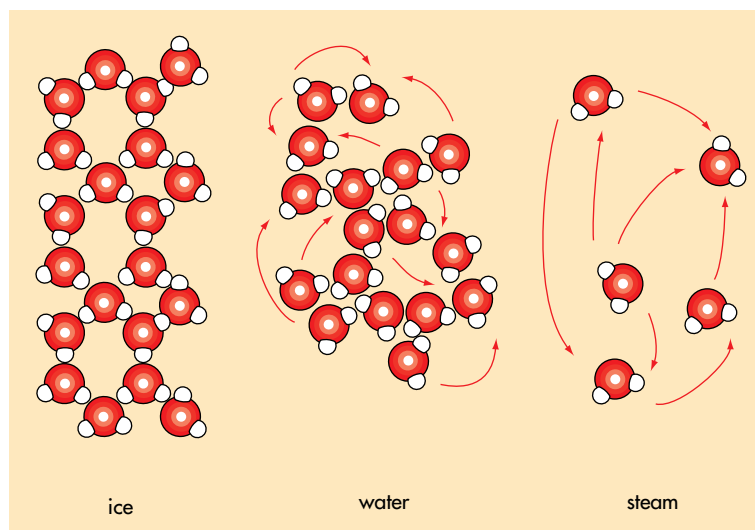
Just about everyone knows the difference between a solid, a liquid and a gas. But if something like water is made up of molecules which are all the same, then how can it be a liquid one minute, a solid (ice) at another time, and a gas (steam) at yet another time? The properties of ice, water and steam are very different.

In terms of molecules, the main difference between solids, liquids and gases is the distance between the molecules and the speed at which they are moving. If the molecules are all very closely bonded together we have a solid. If they are close, but not tightly bonded together, then we have a liquid. And if they aren't bonded together and aren't close, then we have a gas. Gas molecules can be pushed closer together, so gases can be compressed, but liquids and solids are usually incompressible because their molecules are already very close.

We know we can turn ice into water by melting it, and we can turn that water into steam by boiling it. The thing both of these processes have in common is heating. So what exactly is heat? **Heat** is a kind of energy, and in terms of molecules and atoms, temperature is determined by how fast the atoms or molecules vibrate or move. Atoms and molecules are always moving—even carbon atoms in a diamond are vibrating despite being locked very tightly into place.

The bonds which hold the oxygen and hydrogen atoms together in a water molecule are strong and very difficult to break. There are much weaker forces of attraction between the water molecules themselves. When water is cold enough, the molecules pack together and form a solid, but as we add heat, we force the molecules to vibrate and eventually move apart from each other and form a liquid. When we heat liquid water we are making the molecules move faster and harder. Near the surface of the water there are some molecules which will be moving fast enough to break away from the other molecules and fly off into the air. This is what steam is, water molecules moving so fast that they don't bond together any more.

If temperature measures how fast molecules are moving, then as fast molecules escape from the surface of the water there are fewer fast molecules left. The fewer fast molecules there are, the lower the temperature. This means that as the water molecules escape they are actually carrying off heat with them, and so the rest of the water is cooler. If the steam is trapped, the fast molecules can bounce back into the water, which heats it up again.



EXPERIMENT 2

MOLECULES AND HEAT

**AIM**

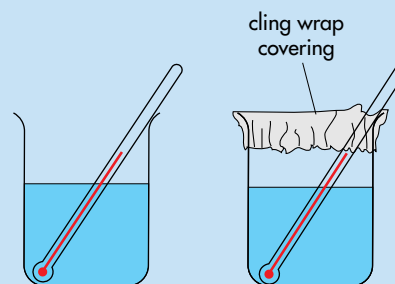
To investigate heat and molecules.

MATERIALS

- Bunsen burner, tripod and gauze mat
- 1 large beaker
- 2 medium beakers
- oven mitt
- 2 thermometers
- cling wrap
- stop-watch
- pencil and paper

METHOD

- 1 Three-quarters fill the large beaker with water.
- 2 Set up your Bunsen burner, tripod and gauze mat, and boil the water.
- 3 Using the oven mitt, carefully pour equal amounts of the boiling water into the other two beakers.
- 4 Place thermometers into each of the two beakers and cover one with cling wrap.

**FIGURE 2.8**

Which one will cook the fastest? Why?

5 Record the temperatures of the two beakers and start the stop-watch.

6 Continue to record the temperatures every 30 seconds for the next 10 minutes.

RESULTS AND DISCUSSION

Graph your results and try to explain any differences you notice between the two beakers.

This is the same way water **evaporates**. Just above the surface of water there is always a layer of steam (or water vapour). Since molecules are always smashing into each other, sometimes some will be moving fast enough and at the right angle to escape from the liquid. Many of these molecules will crash back into the surface, but not all. If there is a breeze blowing them away before they get a chance to crash back into the water then the water eventually 'dries up'. This is the same as why blowing on a hot drink helps to cool it down faster.

Questions

- 1 What things do steam and ice have in common? In what ways are they different?
- 2 What is heat? What can it do to substances?
- 3 Why can't you have liquid diamond?

**FIGURE 2.9**

Getting rid of the fastest molecules.

Think about

- 1 What are some ways to keep a hot drink hot? Explain how you think each works.
- 2 Do puddles get colder or warmer as they evaporate? Explain your answer.
- 3 Draw a labelled diagram showing water evaporating from a beaker. Show molecules in the liquid, just above its surface and leaving the beaker.

Reactions with hydrogen

FIGURE 2.10
Hydrogen–oxygen welding
torches can produce flames with
temperatures as high as
4000°C.

Hydrogen is the most abundant element in the Universe. It is a colourless, odourless gas. Pure hydrogen makes up less than 0.000 001% of the atmosphere. It is the lightest or least dense of all elements. Hydrogen is much less dense than air and so it quickly floats upwards through it. This means it can be used in weather balloons and airships.

Mixtures of hydrogen and air will explode if a flame is brought near. However, if the mixing is controlled hydrogen

can be made to burn with a hot, clean flame. When mixed with pure oxygen it can be used in welding torches hot enough for use under water. Hydrogen can also be used as a fuel for transportation. Experimental cars and buses have been built to run on hydrogen. The space shuttle uses hydrogen as a fuel in its engines.

Hydrogen is also used in industry to manufacture a range of useful substances such as methanol and ammonia. It is also used to turn vegetable oil into margarine.

EXPERIMENT 3

THE BURNING BOAT



NOTE: Your teacher will do this as a demonstration in the fume cupboard or behind a safety screen. Sodium is a very reactive and dangerous metal.

AIM

To observe the reaction of the element sodium with water.

MATERIALS

- pea-sized amount of sodium
- empty pneumatic trough
- water
- 10 cm × 10 cm piece of paper
- phenolphthalein
- safety screen

METHOD

- 1 Half fill the pneumatic trough with water, and add several drops of phenolphthalein.

- 2 Your teacher will now float the piece of paper carefully on the water, and place the sodium onto the paper.
- 3 Students should remain behind the safety screen. The teacher should wear safety goggles.

DISCUSSION

- 1 List your observations from this demonstration.
- 2 What do you think made the sodium, and so the boat, move around the trough?
- 3 Phenolphthalein caused the pink colouring. This is because phenolphthalein turns pink when a basic compound is present. When sodium reacts with water, sodium hydroxide, a base, is formed, according to the equation

$$\text{sodium} + \text{water} \rightarrow \text{sodium hydroxide} + \text{hydrogen}$$
 Which of the substances in this reaction are elements and which are compounds?
- 4 Is sodium a metal or a non-metal? How did you decide?

Hydrogen + oxygen → water

Most of the hydrogen on Earth is found in water. Water is a compound formed from hydrogen and oxygen. A sample of water is made up of many molecules. Each water molecule is made up of one oxygen atom and two hydrogen atoms. Water can be formed by reacting hydrogen gas and oxygen gas:



FIGURE 2.11

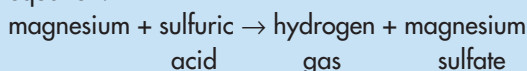
The *Hindenburg* disaster. Hydrogen gas was often used to fill airships until the *Hindenburg* exploded into flames in 1937, killing 35 passengers.

EXPERIMENT 4

MAKING AND TESTING HYDROGEN GAS



When a flame or spark is put into hydrogen gas, a 'pop' sound can be heard. This is how you can determine whether the gas is hydrogen. Hydrogen gas can be made by reacting magnesium metal with sulfuric acid as shown in the following word equation:



AIM

To produce and test for hydrogen gas.

MATERIALS

- 5 mL dilute sulfuric acid
- 2 cm strip of magnesium ribbon
- test-tube and rubber stopper
- test-tube rack • matches

METHOD

- 1 Place 5 mL of sulfuric acid into the test-tube.

- 2 Add the magnesium strip to the acid and immediately put the rubber stopper in the test-tube.

- 3 When the reaction is complete, confirm that the gas produced is hydrogen by removing the rubber stopper and *immediately* placing a lighted match in the mouth of the test-tube.

DISCUSSION

- 1 Draw a diagram of the experiment to show what happened to the magnesium when it was placed in the acid.
- 2 Describe the appearance of the gas produced.
- 3 How did you know when the reaction was completed?
- 4 When testing for hydrogen gas, why does the match need to be placed in the mouth of the test-tube as soon as the rubber stopper is removed?
- 5 Describe in your own words how hydrogen can be produced and tested in the laboratory.

Questions



- 1 Write a description of hydrogen.
- 2 Why is hydrogen suitable for weather balloons but is no longer used in airships?
- 3 What makes hydrogen a good fuel?
- 4 Explain, in your own words, why the test for the presence of hydrogen gas is often called the 'pop test'.
- 5 Calcium reacts with water to

produce hydrogen gas in the same way as sodium does. Write a word equation for this reaction.

- 6 Zinc reacts with sulfuric acid in the same way as magnesium does. Write a word equation for this reaction.

Find out

How are methanol and ammonia manufactured and what are they used for?



FIGURE 2.12

The space shuttle's external tank carries liquid hydrogen and oxygen. The two are mixed in the engines.

Reactions with carbon dioxide

FIGURE 2.13

Carbon dioxide is dissolved in soft drinks such as cola and lemonade to make them 'fizzy'. It is forced into the liquid under pressure so that more will dissolve. When a soft drink bottle is opened, some of the carbon dioxide gas escapes, causing the 'fizz'.

Carbon dioxide is a colourless, odourless gas. It makes up only 0.03% of the atmosphere. Carbon dioxide is more dense than air and so it will sink to the ground. It also prevents substances from burning. This property means that carbon dioxide is used in fire extinguishers.

Many reactions that occur in the kitchen involve carbon dioxide gas. The bubbles in soft drinks and champagne, the rising of bread, the fluffiness of sponge cakes and the holes in honeycomb are all due to the effects of carbon dioxide.



EXPERIMENT 5

MAKING AND TESTING FOR CARBON DIOXIDE



Carbon dioxide can be produced by dissolving marble chips (which contain calcium carbonate) in hydrochloric acid as shown by the following equation.

calcium + hydrochloric → carbon + water + calcium
carbonate acid dioxide gas chloride

The following tests can be used for carbon dioxide.

- Place a lighted taper in a test-tube containing gas. If the taper goes out, the gas is probably carbon dioxide.
- If carbon dioxide is bubbled into limewater, the limewater turns milky.

AIM

To produce and test for carbon dioxide gas.

MATERIALS

- small marble chips
- test-tube rack
- 2 test-tubes
- one-holed stopper
- delivery tube
- limewater
- 5 mL dilute hydrochloric acid

METHOD

- Put a few marble chips into the bottom of a test-tube.
- Half fill the other test-tube with limewater.

- Pour the acid into the test-tube containing the marble chips. Stopper the test-tube with the rubber stopper connected to the delivery tube. Place the delivery tube into the test-tube containing the limewater.

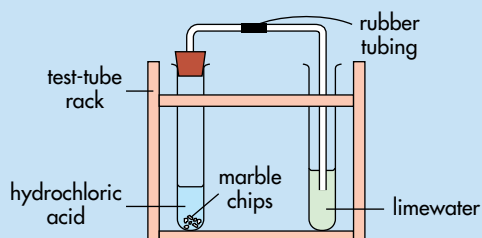


FIGURE 2.14

If carbon dioxide is bubbled through limewater, the limewater will turn milky.

- Record your observations in your workbook.

DISCUSSION

- What would have happened to the limewater if oxygen, rather than carbon dioxide, had been produced in the experiment?
- Describe, in your own words, how carbon dioxide can be produced and tested in the laboratory.

Fermentation

Yeast is a fungus that can be found on the skins of many fruit, including grapes and plums. Yeasts are living things that need energy to grow. They obtain their energy by converting sugar into alcohol and carbon dioxide, as shown in the following equation:



This process is called **fermentation**. Bread-making and brewing both involve fermentation.

In the presence of yeast, the sugar in the bread dough is converted into alcohol and carbon dioxide. As the dough is warmed and the reaction begins, carbon dioxide bubbles are formed. The bubbles are trapped in the dough and expand as the dough is warmed, causing it to rise. Once the temperature reaches a certain level, the yeast is killed and the alcohol evaporates.

Fermentation is used to produce the alcohol in wine, beer and some foods. In the case of wine and beer, glucose (a type of sugar) is acted upon by the yeast to produce alcohol.



FIGURE 2.15 (far left)
What makes bread rise?

FIGURE 2.16 (left)
Breathalysers rely on a chemical reaction to detect the presence of alcohol. The crystals in the breathalyser are initially orange, but change to green if the user has a blood alcohol level of more than 0.05%.

Questions



- 1 Explain, in your own words, how to produce and test for carbon dioxide gas.
- 2 The graph opposite represents the size of bread dough as it undergoes each stage of cooking. Copy the graph into your book and mark on it the points at which:
 - a the dough was left in a warm place to rise
 - b the dough was put in the oven.
- 3a What are the advantages and disadvantages of roadside blood-alcohol testing?
- b What is the legal blood-alcohol limit for probationary drivers?
- c Do you think that breathalysers in public bars are a good idea? Explain your answer.
- 4 Suggest why champagne is 'bubbly' but red and white wines are 'still'.

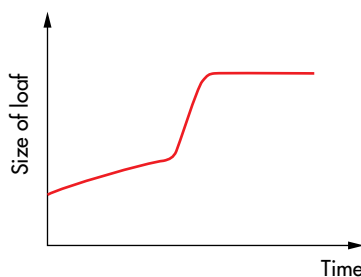


FIGURE 2.17
The size of bread dough at each stage of cooking.

Think about



The Andrios family make their own 'home brew' each year by fermenting specially cultivated wine grapes. This year, however, something went wrong. Three weeks after they had bottled their wine, the bottles exploded. Explain how this could have occurred.

Find out



Wine that has 'gone off' tastes like vinegar. What causes wine to 'go off', and why does 'off' wine taste like vinegar?

Reactions with oxygen

FIGURE 2.18

All living things require oxygen to live. Fish use the oxygen dissolved in water. They take this in through their gills.

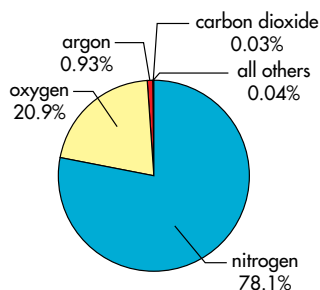


FIGURE 2.19

Nitrogen and oxygen are the main gases in air.

FACT

Pure oxygen is used in oxy-acetylene welding and cutting. When acetylene burns in oxygen, the temperature reaches 3200°C. This is hot enough to melt most metals, which can then be cut or welded together.

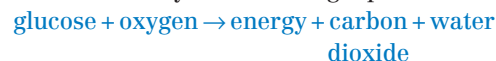
Talk about

What factors could prevent fish from obtaining all the oxygen they need?

Oxygen is a clear, colourless, odourless gas. Although it only makes up about 20% of the air we breathe, it is the most **reactive** gas in the air. Oxygen is involved in many of the reactions we see around us, from the discolouration of fruit to the rusting of metals. It is needed by living things to survive, and is necessary for the burning of fuels.

Living things cannot exist without oxygen. An important reaction for living things is **respiration**. During respiration, oxygen reacts with the glucose in our bodies to produce energy, as well as carbon dioxide and water. The glucose comes from

the food we eat and the oxygen comes from the air we breathe. Respiration can be summarised by the following equation:



As well as being used by living things, oxygen is necessary for burning, or **combustion**. Without oxygen, burning would not be possible.

Fuels are substances that we burn to obtain energy, usually in the form of heat. Natural gas is a fuel that is composed mainly of methane. Methane can be burnt in gas fires, ovens and central heating systems. The energy from the reaction is used to warm houses, heat water and cook food. The reaction is shown in the following equation:

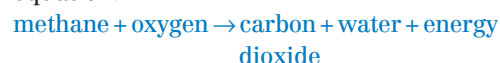


FIGURE 2.20

The energy from respiration allows us to carry out our daily activities, including exercise, and also helps keep us warm.



FIGURE 2.21

For thousands of years, humans have used fire for keeping warm and for cooking food.

Petrol, oil, coal and wood are also used as fuels. Each is a mixture of compounds that contain carbon and hydrogen. When these fuels burn in plenty of oxygen, they all produce carbon dioxide, water and energy.

Iron reacts with the oxygen in the air to form iron oxide, or rust, according to the following equation:



FIGURE 2.22

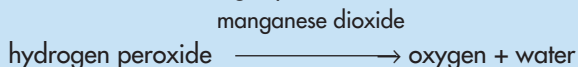
Pure iron is a silvery-white metal. In a moist environment, iron combines easily with the oxygen in the air to form the substance called iron oxide, or rust.

EXPERIMENT 6

MAKING AND TESTING FOR OXYGEN



Oxygen gas can be produced by the decomposition of hydrogen peroxide. A **catalyst**, manganese dioxide, can be used to speed up the reaction as shown in the following equation:



To test for oxygen, place a glowing splint in the container of gas. If the splint glows brighter or ignites, then the gas is oxygen.

AIM

To produce and test for oxygen gas.

MATERIALS

- small amount of manganese dioxide
- 5 mL hydrogen peroxide
- test-tube
- glowing splint
- rubber stopper

METHOD

- 1 Place a quarter of a spatula of manganese dioxide in a test-tube.
- 2 Pour the hydrogen peroxide into the test-tube and quickly stopper the flask.
- 3 While the reaction is bubbling, remove the rubber stopper and quickly place a glowing splint into the test-tube just above the bubbles. Observe and record what happens. (Oxygen gas is heavier than air so it does not immediately escape from the test-tube.)

DISCUSSION

- 1 Draw a diagram of the experiment that shows what happened to the hydrogen peroxide when the manganese dioxide was added to it.
- 2 Describe the appearance of the gas produced.
- 3 Describe in your own words how oxygen can be produced and tested in the laboratory.

Questions



- 1 Draw a flow diagram that shows how to produce and test for oxygen gas.
- 2 Air contains about 20% oxygen. Would a fuel burn faster in air or in pure oxygen? Explain your answer.
- 3 Write the equation for respiration and combustion.
 - a In what ways are they similar?
 - b In what ways do they differ?
- 4 Is the formation of rust a physical change or a chemical change? Why do you think so?

- 5 Lauren was interrupted by her younger brother, Bryce, who was holding a half-eaten apple. 'Look at this apple, Lauren,' Bryce said excitedly, 'it's rusting!'
 - a Explain why Bryce described the apple as 'rusting'.
 - b How could the rate at which the apple turned brown be slowed down?

Find out

What can be done to prevent or slow down the rusting of iron?



Foundation worksheet 2



Multimedia

THE NATURE AND PRACTICE OF SCIENCE

Chance discoveries in chemistry

FIGURE 2.23

Tyres made from synthetic vulcanised rubber. These tyres are built up of alternating layers of rubber and textile often incorporating steel wire.



What do rubber, safety glass and Teflon have in common? They are materials that were developed by chance, accidental discoveries which have made improvements to everyday living. In each case, the person making the accidental discovery was not just lucky, they were observant. They also realised that what they had observed was new and important.

Talk about

Have you ever made a chance discovery by being observant? Have you ever found an unusual rock, fossil or shell? Louis Pasteur was a famous scientist who made many important discoveries. He said 'In the fields of observation chance favours only the prepared mind'. What do you think he meant by this? How can you 'prepare your mind'?

Rubber

Natural rubber (latex) comes from certain trees found in tropical regions. It is elastic and can be used to waterproof materials. However, it has the tendency to become soft and sticky on hot days and hard and stiff on cold days. An American inventor, Charles Goodyear, became interested in rubber. He wanted to discover a way of improving the properties of rubber. He wasn't a trained chemist and so he worked by trial and error, mixing many substances with the rubber without success. Goodyear heard

that other people had been mixing sulfur with rubber, so he tried it too.

In 1839, by chance, some of the rubber-sulfur mixture came in contact with a hot stove. To Goodyear's astonishment the rubber had become dry and flexible. He immediately tested this new form of rubber and found that it stayed dry and flexible when heated and cooled. Goodyear then set about experimenting to find the best way of making this new rubber. He called his process **vulcanisation**.

Vulcanised rubber has many uses such as in shoe soles and hoses. It was used to make car tyres but so much is needed that a synthetic form of rubber had to be developed.

Safety glass

Natural glass forms from the intense heat of volcanic activity on sand (silica). How people discovered a method of making glass is not known. Perhaps this was an accidental discovery by ancient people who built fires on sandy beaches. However, we do know that the Egyptians were moulding glass as early as 1400 BC.

Glass is a strong material but very brittle. It shatters into sharp fragments. This property led to an important discovery by a French chemist named Edouard Benedictus in 1903. Benedictus had been investigating a plastic called cellulose nitrate. He accidentally dropped a flask on

a hard floor. He observed that the flask shattered but did not fall apart. When Benedictus examined the flask he found a thin film of the plastic inside it which had held the broken pieces together. He kept the flask but thought no more about it until he read about several automobile accidents where the passengers were badly cut by the shattered glass. Benedictus realised his earlier discovery could be put to good use and he immediately set to work to make the first safety glass. He did this by using the cellulose nitrate as a glue to hold two sheets of glass together. Today safety glass has many uses in buildings and vehicles.

Teflon

This amazing material was discovered by a young chemist named Roy Plunkett in 1938. Plunkett worked in a research laboratory in New Jersey, USA. One morning he opened a new cylinder of the gas

tetrafluoroethylene. He was puzzled about why the cylinder which should have contained 1000 g of gas only contained 990 g. After opening up the tank he found a curious white powder which he realised must have formed from the gas. He investigated the white waxy powder and found that it had some remarkable properties.

Plunkett's discovery was named polytetrafluoroethylene. It later became known as Teflon. This new plastic was not affected by hot corrosive acids, did not dissolve in solvents and could be heated to 500°C without burning. These properties made it suitable for use as a protective covering on the first atomic bomb, space vehicles and on the moon suits worn by the Apollo astronauts. It is so unreactive that it is used to coat pacemakers and other artificial parts because the body doesn't reject it.

It is also used as a coating on cookware, dental floss and as a stain repellent for clothing and furnishings.



FIGURE 2.24
Safety glass can be used in many situations.

Questions



- 1 Describe three properties of natural rubber.
- 2 What happens to natural rubber when it is vulcanised?
- 3 How did Edouard Benedictus's discovery benefit people?
- 4 Which property of teflon makes it suitable for use:
 - a on dental floss?
 - b as a coating on glass eyes?
 - c as a lining for chemical containers?
 - d as a stain-repellent?
- 5 What personality traits did Charles Goodyear, Edouard Benedictus and Roy Plunkett have in common?

Find out



- 1 Charles Goodyear's discovery may have depended on an element of luck, but he did not lead a lucky life. Research Charles Goodyear and write a recount of his life.
- 2 How is glass manufactured?



FIGURE 2.25
Teflon is extremely slippery. When used as a coating on cookware it provides a non-stick surface. However, many technical problems had to be solved to get the Teflon to bond to the metal surface.



FIGURE 2.26
Teflon-coated frying pans were on sale 10 years before the first man walked on the Moon. Teflon was used for the outer skin of the space suits worn on the Moon. It is also used on nose cones, heat shield and fuel tanks of space vehicles.

Reactions in industry

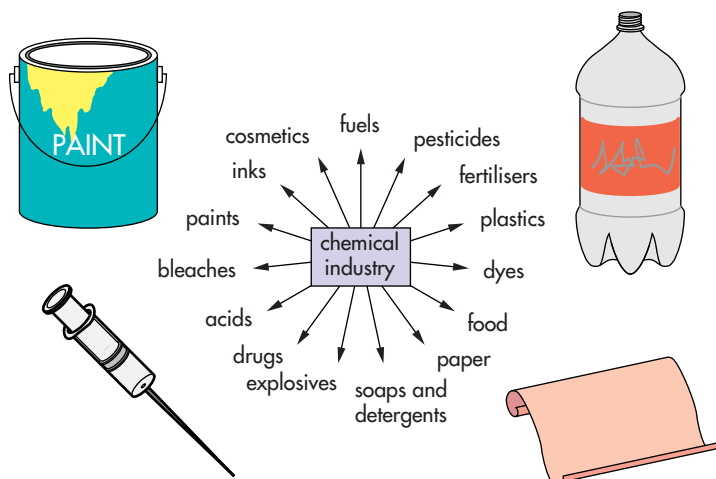


FIGURE 2.27

We rely on the chemical industry to make many of the things we use every day. Imagine life without some of these products! What did people use before each of these products was developed?

Many of the things we use in our daily lives, such as soap, plastics and cosmetics, have been made in large quantities by chemical industries. Chemical reactions are involved in producing these useful materials. In some cases these reactions may be quite simple. In other cases the reactions involved are more complicated.

Detergents and soap

Anything that works with water to get things clean is called a **detergent**. There are two common types of detergents:

- **soaps**, which are made from animal fats or vegetable oils
- **soapless (synthetic) detergents**, which are made from oil.

EXPERIMENT 7

CLEANING UP YOUR ACT

Caution: 20% sodium hydroxide is very corrosive. Handle with care. Plastic gloves should be worn.

AIM

To make soap.

MATERIALS

- 10 mL olive oil
- 10 mL methylated spirits
- evaporating dish
- distilled water
- beakers
- hotplate
- 10 mL 20% sodium hydroxide solution
- 30 mL concentrated sodium chloride solution
- safety glasses

METHOD

- 1 Heat 20 mL of water in a beaker on a hotplate.
- 2 Place the olive oil, sodium hydroxide and methylated spirits in the evaporating dish. Heat and stir until thick.
- 3 Sit the evaporating dish on top of a beaker containing cold water.

- 4 Add hot distilled water and stir.
- 5 Add sodium chloride solution and stir.
- 6 Scoop off the soap that forms on the top.

DISCUSSION

- 1 How pure is your sample of soap?
- 2 How useful is your product as a soap?
- 3 Is your soap safe to wash with?
- 4 There are many different types of soap, each useful for different purposes. Name as many different types of soap as you can.

EXTENSION

- 1 Compare the effectiveness and quality of soaps made from different types of oil (such as castor, sesame, safflower or peanut oil, or mixtures of these).
- 2 Compare the effectiveness of different brands of commercially produced soap.

Soap has been used for more than 4500 years. In recent years chemists have made a wide range of synthetic (processed) detergents that can be more effective than soap.

Plastics

You probably own many things made of plastic. Walkmans, CD players, cassettes, pens, toothbrushes and even nylon socks

are all made of plastic! Plastics are compounds of carbon that can be moulded into different shapes. The starting materials for making plastics are usually obtained from oil (petroleum).

Plastics are all **polymers**, which means that they consist of very long molecules made by joining many small molecules together, just like a string of beads. The small molecules are called **monomers**.

EXPERIMENT 8

MAKING PLASTIC FROM MILK



Most plastics are made in laboratories or chemical plants from oil and natural gas. However, a simple plastic can be made from **casein**, the curds that form in milk.

AIM

To make plastic from milk.

MATERIALS

- 100 mL milk
- beaker
- sieve
- paper towel
- safety pin
- varnish
- shape to use as a mould
- hotplate
- tempera paint
- 5 mL vinegar
- stirring rod

METHOD

- 1 Pour the milk into a beaker and warm it on a hotplate. Do not boil.
- 2 Add the vinegar gradually. Stir until the milk separates into curds (solid) and whey (liquid).
- 3 Strain the curds (chemically known as the protein casein) and discard the whey.
- 4 Squeeze out any remaining liquid from the curds.
- 5 Dry the curds with the paper towel.
- 6 Pack the curds into the mould and leave for a few days to dry.

EXTENSION

- 1 If you want to make a brooch, push a safety pin into the curds while they are still soft.
- 2 When your plastic has set, it can be painted and varnished.

Questions



- 1 What is the difference between a soap and a detergent?
- 2 Conduct a survey to find out which cosmetics are used by the members of your class. Present your findings in a column graph.
- 3 Bacteria can grow in most cosmetics, so preservatives must be used. For products that are mostly water-based, a pinch of boric acid should be added. For acidic products a few crystals of salicylic acid will prevent bacterial growth, while for non-acidic products a few drops of

tincture of iodine should be added. Explain why bacteria can grow quickly in cosmetics.

Think about



Apart from improving appearance, list three other functions of cosmetics.

Find out



Use your library resources to research the chemical reactions involved in:

- a the production of paint
- b disposable batteries.

IMPLICATIONS FOR SCIENCE AND THE ENVIRONMENT

Living with chemicals



FIGURE 2.28
Common household products can pose risks if not used, stored and disposed of correctly.



FIGURE 2.29
Chemicals may carry a hazard symbol.

Many new products have been developed to improve our standard of living. They help to clean our bodies and homes, destroy pests, provide fuel for motors and treat ill health. However, some chemicals can become hazardous to the environment and the health of people if not used correctly. It is essential that people understand the risks in using these chemicals and know how to reduce those risks.

Talk about

Look at Figure 2.28. Make a list of the types of products in this cupboard. Suggest what hazards these chemicals could pose to the people who use them. Is this a safe way to store these chemicals?

Chemical hazards

Many chemicals have the potential to harm us if not used correctly. Some chemicals are poisonous or toxic, causing vomiting, unconsciousness and even death. Others are slow acting and may cause cancer or damage to unborn babies. Some chemicals are irritants, causing inflammation to the skin or airways. Others only affect a few sensitive people. Chemicals can be corrosive, producing a chemical burn. Combinations of chemicals can also be explosive or flammable.

Even if chemicals are not a direct health risk, they may harm the environment. For example, pesticides can kill useful animals as well as the pests.

Chemicals can enter the body in three ways: by inhalation, skin absorption and direct ingestion. Once inside, chemicals are treated by the body in three ways: they can be excreted, stored in organs and tissues, or chemically changed into another substance.

Safe use of household chemicals

When working in the science laboratory or the workplace we are required to follow safety rules for handling chemicals to reduce the risk of accidents. There is just as much potential for accidents with chemicals to occur in homes. Every year thousands of calls are made to Poisons Information Centres. Each week many children are admitted to hospitals with chemical poisoning, and some die. People need safety rules for using chemicals in the home.

The risk of accidents with chemicals can be greatly reduced if the following simple rules are followed.

- 1 Read the label on the container.
- 2 Follow the directions and safety instructions fully.
- 3 Store chemicals correctly, this may mean locking medicines up and using childproof containers. Do not store chemicals in drink containers.
- 4 Dispose of chemicals correctly. Do not pour dangerous chemicals down the drain. Find out from your local council what the correct disposal method is for each chemical.



Questions



1 Use a dictionary to find the meanings of the following terms: risk, hazard, toxic, inflammation, inhalation, ingestion, excrete.

2 Look at Figure 2.30 and answer the following questions.

a Which group of products were most responsible for children being poisoned in the home?

b Name five examples of chemicals which could be included in household personal products.

c In which group would you place woodworking glue?

3 When using chemicals in the home, what could you do to prevent:

a inhaling chemicals?

b absorbing chemicals through the skin?

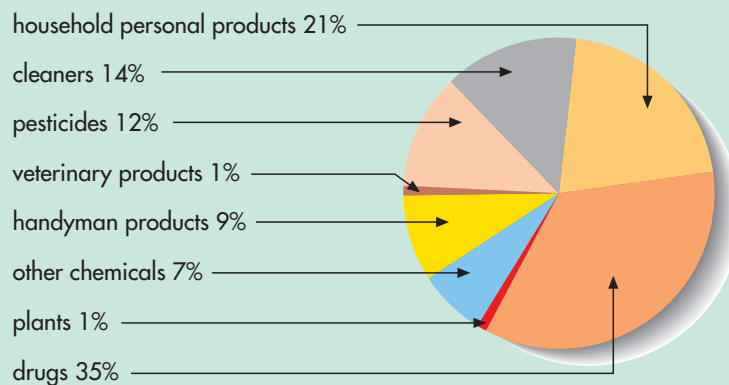
c ingesting chemicals?

Find out



1 Use reference books or the Internet to find out what the hazards are for common products such as detergent, disinfectant, bleach and paracetamol.

2 How does your local council dispose of toxic chemical wastes?



Activities



1 Select a household product and examine the label on the container carefully. Write down six pieces of information found on the label. Look for the following information: product name; directions for use; safety instructions for preventing an accident or what to do in the event of poisoning; ingredients; safe storage details; and safe disposal information. Is there any information missing from the label?

2 Organise a class debate on the topic 'Chemicals cause more harm than good'.

FIGURE 2.30

Products involved in child poisoning (based on calls to New South Wales Poison Information Service 1991).





Using scientific language

Design a crossword with clues which use the following words: polymer, evidence, chemical, precipitate, state, physical, evaporate, hydrogen, oxygen, carbon dioxide, detergent, toxic.



Check your knowledge

- Decide whether each of the following changes is physical or chemical:
 - gas burning in a heater
 - a plant growing
 - bread going mouldy
 - an egg cooking in a frying pan
 - oil boiling
 - ice melting
 - sandstone cliffs being worn away by wind
 - sugar dissolving
 - breathing.
- 'When a cake is cooked, a chemical reaction occurs.' What evidence supports this statement?
- Luke adds a piece of magnesium metal to lemon juice. A gas is produced. Luke suspects that the gas is hydrogen. How could he test his hypothesis?
- A colourless odourless gas extinguishes a flame. How could you prove it was carbon dioxide?
- Describe three methods for making carbon dioxide gas.
- How is soap different from detergent?
- What four simple rules should be followed to prevent accidents with chemicals in the home?



Apply your skills

- Make a list of 10 reactions that occur around the home, other than those described in this chapter.
- Draw a labelled diagram to show the laboratory equipment and the set-up you could use to:
 - fill a balloon with hydrogen gas
 - test to see whether carbon dioxide is breathed out by humans
 - identify the gas that causes the bubbles in fizzy soft drinks.

- For each experiment in question 2, list four possible dangers associated with the experiment and suggest how each danger could be reduced.
- Copy and complete the following word equations.
 - glucose + oxygen $\rightarrow$ _____ + _____
 - iron + _____ $\rightarrow$ iron oxide
 - hydrogen peroxide $\rightarrow$ _____ + water
 - potassium + water $\rightarrow$ potassium hydroxide + _____
 - magnesium carbonate + _____ $\rightarrow$ magnesium chloride + water + carbon dioxide
- Three gases are mixed together in a sealed container—oxygen, carbon dioxide and hydrogen. Draw a diagram of how you would expect the gases to be found inside the container.
- Collect three labels from household chemicals, spread them out and paste each of them in the centre of a separate sheet of paper. Label each piece of information shown on the label according to the information it provides. Include an explanation of the reasons for the information.

Challenge yourself

The table below shows the colour changes of some indicators.

Indicator	Spex	Yom
Spex	in the presence of X	yellow
Spex	in the presence of X and Y	pink
Yom	in the presence of X	green
Yom	in the presence of X and Y	green

Spex and Yom were added in turn to separate solutions, A, B, C and D. The results can be seen below.

Indicator	Spex	Yom
A	pink	green
B	yellow	green
C	no change	no change
D	no change	green

Which solution contains X but not Y?



CHAPTER

3

FORCE AND MOTION

What do sharks and dolphins have in common? Both are helping scientists to overcome drag. Drag is the resistance to motion an object experiences when passing through a liquid or a gas.

The drag exerted by water is considerable. If you have ever tried to run knee deep in water you would appreciate the strength of this resistance force. Also the faster an object travels in water, the greater is the drag. The drag is produced in a thin layer of water in contact with the surface of the moving object. At low speeds this layer can be smooth but as speed increases small eddies begin to form in the water. At high speeds, the eddies grow into large-scale turbulence which greatly reduces motion. However, dolphins have been observed to swim at speeds which can outpace ships. Scientists realised that their secret must lie in their skins. Dolphin skin wrinkles when they swim at high speed, whereas sharks have a covering of scales with microscopic ridges. Both surfaces appear to reduce turbulence and hence drag.

This discovery has been put to good use by designers of jet aircraft and racing yachts. Recently this knowledge has been applied to racing swimsuits. Designers have developed a polyester/polyurethane fabric which has strips of Teflon forming riblets adhered to it. When used in swimwear this fabric produces less drag than bare skin. In a full length suit it can reduce drag by 23% compared to conventional nylon/Lycra suits.

In this chapter you will investigate a number of forces which change the motion of everyday objects.

CHAPTER OUTCOMES

Successful completion of this chapter will allow you to:

- Identify changes that take place when forces are acting.
- Give examples of contact and field forces.
- Use an equation to calculate the speed of an object.
- Describe technological changes in engines and their impact on people.
- State that all objects exert a force of gravity on all other objects.
- Describe friction and resistance forces and identify everyday situations where they act.
- Describe how scientists construct hypotheses and models to explain observations.

Motion is everywhere



FIGURE 3.1 (above)
The skateboard rider pushes backwards with his foot. The ground pushes the rider forward.

FIGURE 3.2
(above right)
Nowhere to go but down!

FIGURE 3.3 (right)
A stationary skateboard—a case of balanced forces or forces?

Talk about

Look at the rider in Figure 3.1. If she stops pushing, the skateboard will stop. What causes the skateboard to stop? The rider in Figure 3.2 has just made a jump. Are there any forces acting on the skateboard and rider while airborne? The skateboard in Figure 3.3 is not moving. Are there any forces acting on it?

Everything is moving in our world, from the smallest atom to the largest galaxy in space. Blood flows through our bodies, water flows along rivers, micro-organisms swim over surfaces and even the atoms in rocks are vibrating.

All movement is affected by **forces**. Forces are pushes and pulls. Forces can start objects moving or stop already moving objects. They can also speed up, slow down or change the direction of motion.

Forces can be exerted when two objects actually touch one another. These forces are called **contact forces**. For example, riding a skateboard involves pushing on the ground to get it moving. **Friction** acts when the wheels of the skateboard rub against the ground. This works to slow the skateboard down and eventually stop it. Other forces can act even when objects are not touching. These are called non-contact or **field forces**. **Gravity** is an example of a force which can be exerted on an object



from a distance. For example, the airborne skateboard rider experiences the force of gravity pulling him towards the ground.

Forces often go unnoticed because they act against each other. The stationary skateboard will remain at rest even though forces are still acting on it. The force of gravity pulls the skateboard downward towards the Earth. However, it does not move, because at the same time, the Earth is pushing the skateboard upwards. The two forces are equal but opposite in direction. The forces are balanced. The motion of an object will only change when one force is greater than the other. The forces are then described as unbalanced.

USING SCIENCE 1

MARBLE MOTION

WHAT YOU NEED

- marbles
- cardboard
- cardboard tubes
- plastic tubing
- Plasticine
- any other materials you like

WHAT TO DO

1 Working in groups of three or four, construct a display that produces as many pushes and pulls as possible from the movement of marbles.

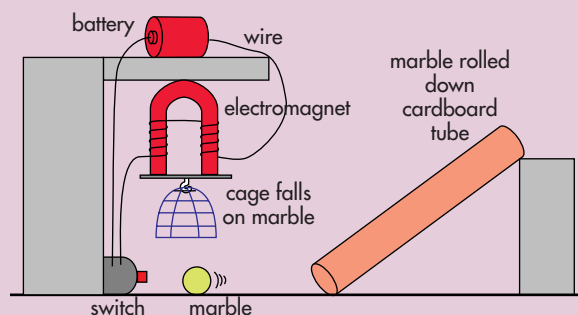


FIGURE 3.4
A possible marble machine.

2 To help you think about your construction, here are two examples:

Example 1: A marble rolls down a slope and operates an electric switch that controls a series of lights, or a buzzer, or releases a metal ball held by an electromagnet.

Example 2: A marble rolls onto one end of a see-saw, the other end of which is attached to a rubber band or a weak spring. The stretching of the band or spring just touches a row of dominoes ...

EXTENSION

Arrange to slow the marble down when it reaches a certain point in the display. This could be achieved in many ways, for example by using sandpaper, water (or another liquid), or something sticky; or by making the marble operate some other part of the machine.

DISCUSSION

1 What causes the motion when a marble falls from your hand or rolls down a slope? Is the cause a push or a pull?

2 Describe what causes the motion at each point of your marble construction.

Questions



- 1** What is motion?
- 2** Can you think of some pushes and pulls that you can encounter in your everyday life? List three of each and their effects.
- 3** You throw a ball straight into the air. Explain why the ball rises so high before it falls to the ground. Include in your explanation any pulls or pushes that you think might be affecting the flight of the ball.
- 4** Once the ball has landed it might roll for a while before coming to a stop. Explain why you think the ball slows down and stops. Why doesn't it keep on going forever? Are there any pushes or pulls acting on it?

5 Looking at the picture below showing a car dashboard and controls. Identify the controls that would need a pull or push to make them operate.



FIGURE 3.5
A car dashboard.

FACT

FILE

Gases, like everything else, are made up of particles. These particles are very small—about one ten-millionth of a millimetre in diameter. The particles that make up a gas are constantly moving. The average speed of a gas particle is 100 m/s.

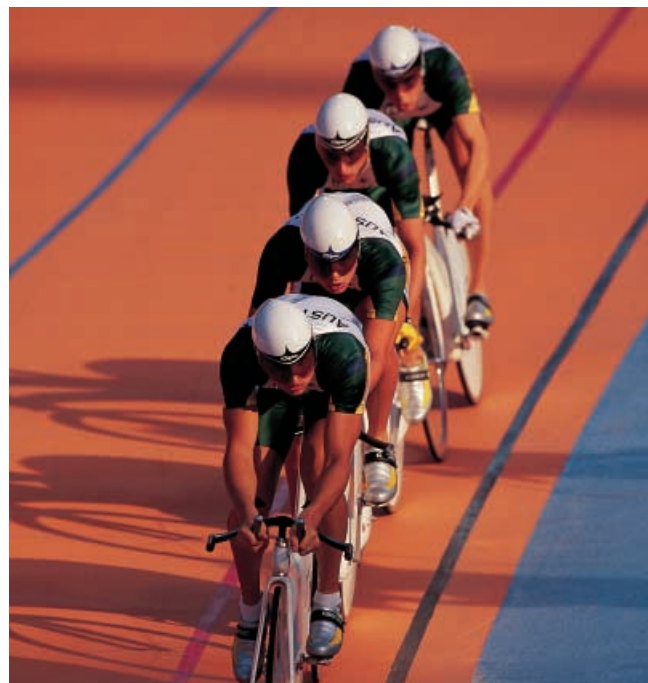
Comparing motion

Motion, distance and time

FIGURE 3.6 (right)
Who is the fastest hurdler in this race? Since all the hurdlers will travel the same distance, the fastest one will be the one who records the shortest time.



FIGURE 3.7 (far right)
Who is the fastest cyclist in this race? In cycling time trials, cyclists see how far they travel in a certain time.



EXPERIMENT 1

30-METRE MOTION



A professional athlete needs to know how fast he or she is running in order to achieve a better performance in future. In the case of a 100-metre hurdler, all that is needed to determine the athlete's speed is to measure the time, since the distance is always the same.

AIM

To measure the time taken for students to cover a certain distance.

MATERIALS

- stopwatch (accurate to 0.1 second)
- long tape measure or trundle wheel

METHOD

- 1 Measure and mark out a distance of 30 metres on the school oval.
- 2 Measure the times taken for various people to cover the 30 metres. They can walk, run, hop or even crawl.

- 3 Record the results in a table like the one below.

Person	Time (measured to a tenth of a second)

DISCUSSION

- 1 Plot your results on a column graph to give a quicker way of comparing results.
- 2 What conclusions can you draw about the motion of various members of your class?
- 3 How accurate was your measurement of 30 metres? Could it have been more accurate? Does it matter, as long as the distance was the same for everyone?

EXPERIMENT 2

10-SECOND MOTION

**AIM**

To find out how far a toy can travel in a fixed time.

MATERIALS

- stopwatch
- long tape measure
- three or four moving toys (such as model cars) with or without motors

METHOD

- 1 Measure the distance travelled by each toy in 10 seconds. Use the stopwatch to set the starting and finishing times.
- 2 Record the results for all toys tested by members of your class in a table like the one below.

Toy	Time (measured to a tenth of a second)

DISCUSSION

- 1 Plot your results on a column graph to give a quicker way of comparing results.
- 2 What conclusions can you draw about the motion of the various toys that were tested by members of your class?

Motion and speed

If you wanted to compare the motion of your class members with the motion of the toys you tested in the two previous experiments, how would you do it? Who or what was the fastest? Although it was easy to see who was the fastest mover in your class, and which was the fastest toy, it is more difficult to compare the students with toys.

The methods used to compare motion in the two experiments relied on either the distance or the time being fixed. This is rarely the case in real life.

We need a way to compare motion in situations where the distance and/or the time taken varies. The way to do it is to use a measure that combines the distance travelled and the time taken. This measure is called **speed**.

Average speed is the distance an object travels in a certain amount of time. To calculate the average speed of an object, you divide the total distance the object travels by the time it travels. We can therefore describe average speed by the following formula:

$$\text{average speed} = \frac{\text{total distance}}{\text{time taken}}$$

Example

An insect travels 20 centimetres in 4 seconds. How fast does it travel?

Since

$$\text{average speed} = \frac{\text{total distance}}{\text{time taken}}$$

then the insect's

$$\begin{aligned} \text{average speed} &= \frac{20 \text{ centimetres}}{4 \text{ seconds}} \\ &= 5 \text{ centimetres per second} \\ &= (5 \text{ cm/s}) \end{aligned}$$

Dividing distance by time has given us a way of comparing one motion with any other motion. Now we can compare the motion of a toy crossing the floor of a room with that of a car speeding down a highway. However, we can make these comparisons only if we use the same *unit* to measure speed in each case.

Questions



- 1 Use your results from Experiment 1 to calculate the average speed of each student.
- 2 Use your results from Experiment 2 to calculate the average speed of each toy tested.
- 3 Why is it more accurate to say that you measured the average speed rather than the speed of students and toys in question 1 and 2?
- 4 Calculate the class average of all the students and compare it with the average speed of all the toys tested in your class. What can you conclude?
- 5 Convert the two speeds calculated in question 4 into kilometres per hour (km/h) and metres per second (m/s).

FACT

FILE

Moving fast in the animal world is an advantage. The fastest animal on land is the cheetah. It has a record speed of 101 km/h, compared to 43.3 km/h for humans. The fastest sea animal is the sail fish with speeds up to 110 km/h compared to around 8 km/h for humans. In the air, the spine-tailed swift holds the record for level flight at 171 km/h. However, the peregrine falcon can reach speeds of 350 km/h in a dive.

APPLICATIONS AND USES OF SCIENCE

Travelling faster

FIGURE 3.8

The latest designs in motor cars are put through their paces. Are they advances in technology, or a waste of money?

Speeding up the human race

In most parts of the world today people are able to travel at a relatively high average speed compared to the average speed of even just a hundred years ago. In fact, our modern society depends on it. Every day people travel distances in a relatively short time. For example, while it might have taken your great grandparents an hour to walk to school, a bus or car can now get you there in 10 minutes or so. The distances many people travel to work might have taken days to walk. Business people may even fly thousands of kilometres for a meeting and return the next day.

Talk about

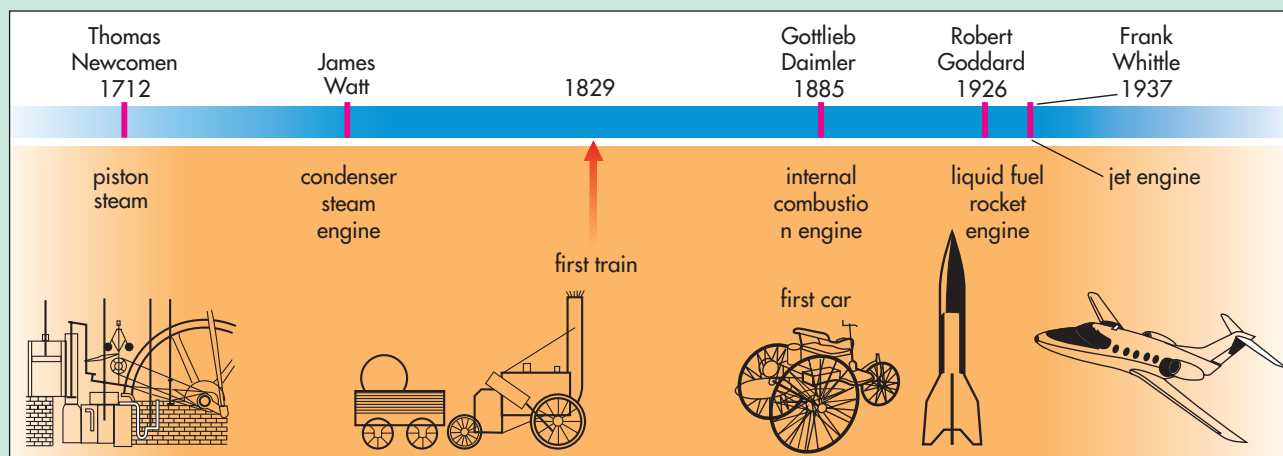
Organise a class discussion based on the following statements:

- 'Faster is better. Motor car racing has produced better and safer car technology.'
- 'The basic design of cars has not changed since they were invented. New cars are developed simply to make money for their manufacturers.'

The history of speed is directly linked to the development of the **engine**. Scientific and technological advances in engine design over the last three centuries have allowed people to travel faster. Engines freed people from their dependency on horses and wind for transportation.

FIGURE 3.9

Timeline for engine development.



USING SCIENCE 2

SPEEDING UP

Carry out the following activities to demonstrate the spectacular rise in the average speed of humans over just the last 200 or so years.

1 Estimate the average speed of the cars in the world.

Hint: Consider that most car journeys are probably in large cities, involving a lot of starting, stopping and crawling in traffic jams. The next time you travel some distance by car, work out your average speed for the whole trip. Your estimate does not have to be very accurate.

2 Estimate the average speeds of bicycles, trains and passengers aeroplanes.

You could simply make an informed guess or you could measure the speed of students who ride bicycles to school and use train and airline timetables to make your calculations.

3 Average human walking speed is between 4 and 6 km/h. Calculate by how much our average speed has increased as a result of mechanical transport such as bicycles, cars, trains and aeroplanes. For example, it is twice, five times, ten times?

EXTENSION

Copy the following table and, using your answers to the previous questions, complete the second column.

Compare the average speeds with the fastest recorded speeds shown in the third column.

Vehicle	Estimate average speed	Fastest recorded controlled speed (km/h)	Year recorded
Car		over 1000 (jet-powered vehicle)	1983
Bicycle		about 245	1985
Train		515	1990
Aircraft		3530 (jet aircraft)	1976
Ship		556 (jet-powered)	1977

You could produce a large class poster showing the increase in the average speed of humans over the last few hundred years.

DISCUSSION

Have we benefited from this increase in speed? While we can travel at greater speeds than ever on our freeways, car accidents cause many deaths and injuries and cost the community millions of dollars in medical bills. Wouldn't it be better if we had an efficient public transport system, where the opportunity for accidents is much reduced?

Questions



1 Estimate the average speed (in km/h) at which you would travel to school:

- a** if you travelled by your normal means (including car, train, bus, tram)
- b** if you walked the entire distance.

2 Convert your answer to question 1 into metres per second (m/s).

3 In 1988 Kerry Ann Saxby broke the women's record for road walking when she covered 10 kilometres at an average speed of about 14.5 km/h. How does this compare to your average speed in going to school? Again, convert your answers into metres per second.

Activities



1 Collect three newspaper articles about car accidents in your area. Write a letter to your local newspaper giving suggestions as to how such accidents might be avoided.

Find out



- 1** Select one type of engine to research. Find out what scientific and/or technological developments were made to improve the engine. What land, sea or air speed records were set for your chosen engine? Present your findings as a poster.
- 2** What percentage of road accidents in your state are caused by drivers speeding? Collect information that will allow you to work this out, then represent your findings graphically.

FACT

FILE

Astronauts are the ultimate speedsters. To get into orbit they must reach speeds of around 28 000 km/h. However, the Apollo astronauts reached much greater speeds on their missions to and from the Moon. Apollo 10 exceeded 39 897 km/h during re-entry in 1969.

Bend, squash and stretch

FIGURE 3.10

(above left)
Elastic golf ball, distorted as it collides with the club.

FIGURE 3.11

(above centre)
When the bow is drawn, ready to fire, its shape is distorted.

FIGURE 3.12

(above right)
Compressed foam rubber in the lining of sports shoes protects the feet from damage.

When a golf ball is hit with a club, both the ball and the club temporarily change shape. To shoot an arrow from a bow, the bow is first bent out of shape. The bow then returns to its original shape when the arrow is released. Each step a jogger takes squashes or compresses the foam rubber lining inside their sport shoes. The foam rubber returns to its original shape in between steps.

In each of these examples forces have changed the shape of materials. However, each material has returned to its original shape and in doing so has exerted a force itself. Materials which return to their original shape after being bent, squashed or stretched are said to be **elastic**. Materials that do not return to their original shape after a force is applied are described as **inelastic** (or plastic). Clay, putty and lead are examples of inelastic materials.

Which is the most elastic, a rubber band or a steel spring? It might surprise you to

find out that steel is in fact more elastic than rubber! This is because steel can regain its original size and shape better than rubber can after a distorting force is removed.

The difference between steel and rubber can be explained by considering the forces which hold atoms and molecules together. When materials are stretched or compressed the atoms and molecules in them are pulled apart or pushed closer together. These particles exert both forces of attraction and repulsion which are responsible for the material returning to its original size and shape.

Elastic limit

All elastic materials have a limit to how much they can be stretched or compressed. Large forces can stretch a rubber band or steel spring so much that they will not return to their original size or shape. They will remain permanently distorted.

FIGURE 3.13



USING SCIENCE 3

DESIGN AND MAKE A MOVING TOY

TASK

You have been asked to design and develop a moving toy for a toy company. The toy should be re-useable. You will need to provide all the materials, and the cost of production should not exceed \$10.

STEP 1: WRITING YOUR BRIEF

Before you begin, there are a number of questions you need to consider:

- 1** Who is the moving toy for? It is important that you decide who the toy is for before you do anything else. What age group is the toy for?
- 2** What sort of moving toy is required? You need to decide what sort of moving toy to make. Will it be a flying toy? Or perhaps a toy that moves a ball around, like a pinball machine? Finding the answers to these questions will help you write a brief for your toy design. An example of a brief would be: 'To design a flying toy for five year olds'. Whatever type of moving toy you decide to make, be sure it is simple to make and can be made in the time available. Don't make the action of the toy too complicated. You will have succeeded with the task if your toy works when you have finished.

STEP 2: DESIGNING YOUR TOY

- 1** First you need to decide what is going to make the toy move. For example, will you use a rubber

band, a spring, a wind-up motor or an inflated balloon? What other types of mechanisms can produce motion? To get some ideas, look at how other moving toys work.

- 2** Next, draw a sketch that clearly shows how the toy works and what materials and work are required to make it. Make sure that all the materials are readily available. You may be able to re-use parts of old toys that no longer work. If you have to buy any materials, make sure they are cheap.

- 3** Show the drawing to your teacher who can check it to make sure the design is feasible. Are any changes needed to the design? Do you need to try out a few ideas before you decide on a final design?

STEP 3: TESTING YOUR DESIGN

Before you start building your toy, test out different ways of using the source of energy that you have chosen. For example, if you have chosen a rubber band or a wind-up motor, try different ways of controlling the motion.

STEP 4: BUILDING YOUR TOY

Good luck!

STEP 5: TOYS ON PARADE

Set up a display of all the toys made by the class to illustrate the different forces involved. Determine whether each toy is operated by pushes or pulls.

Questions

- 1** Look at Figure 3.13 and answer the following.
 - a** Name the object that the force or forces act(s) on in each situation.
 - b** Is the force pulling or pushing?
 - c** Select the statement which best describes what the force is doing to each object: stretching it; compressing it; starting it moving; stopping it when already in motion; changing its direction of motion; balancing another force and preventing motion.
- 2** How would each of the following materials behave after being gently stretched? Predict which would be elastic and which inelastic: chewing



gum, copper wire, nylon thread, Plasticine, cling wrap.

Think about

Ben inflates a rubber balloon.
Which material is compressed?
Which material is stretched?

Activities

- 1** Design a fair experiment to test your predictions in question 2. Remember to control variables such as starting length of material, amount of stretching force applied and method of measuring final length after the force is removed.
- 2** Design an experiment to find the elastic limit of a rubber band.



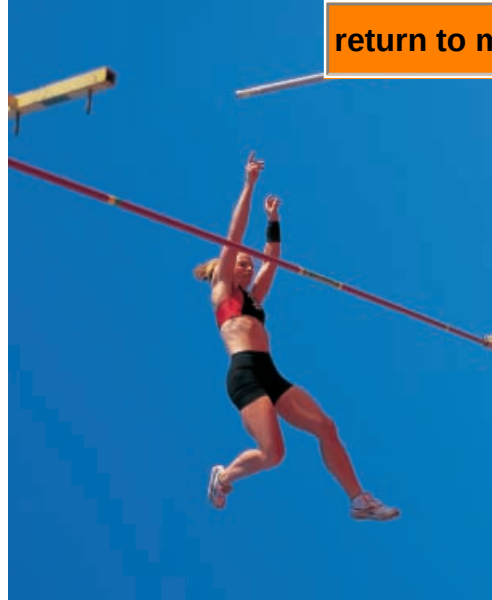
FACT

FILE

Centrepont Tower sways back and forth when a strong wind blows. However, it remains perfectly safe because of its 'elastic properties'.

What goes up must come down!

FIGURE 3.14
Falling over the pole vault bar.



[return to main menu](#)

In our everyday experience we expect that objects thrown into the air will fall back to the ground. In fact our very understanding of what is up and what is down is directly linked to this occurrence. The force which causes objects to fall is called gravity.

Talk about

What do you already know about gravity? Form a small group, discuss the following statements and decide if they are true or false. Only large objects have gravity. There is no gravity on the Moon. Gravity is a pulling force. The further you move away from the Earth the weaker the force of gravity becomes.

A universal force

Gravity is sometimes described as a universal force because wherever there is matter, there is gravity. It acts between everything in the Universe from the largest galaxy to the smallest atom. Everything

FIGURE 3.15
Galaxy NGC 2997, gravity on a grand scale!



that has mass exerts a force of gravity. Gravity is an extremely weak force, so weak in fact that one of the objects must be massive before the effect of this pulling force can be felt. Massive objects like stars and planets can exert a large gravitational force of attraction on each other.

Force field

The Earth's gravitational pull extends out into space, in all directions. The further you move away from the centre of the Earth the weaker the force becomes. Wherever the gravitational pull of the Earth can be felt a gravitational field is said to exist.

The law of gravity

English scientist Sir Isaac Newton thought about why objects fall to the Earth and what keeps the Moon circling the Earth. He realised that the same force which kept the Moon in orbit around the Earth and the Earth in orbit around the Sun caused objects to fall to the ground. Newton worked out mathematically how the force of gravity could be calculated. He stated his ideas in his law of universal gravitation. This law says that all objects are attracted to one another by the force of gravity. How big the force is depends on the mass of the objects and the distance between them. The force is greatest when the masses are large and the distance is small.

Weight and mass

You may recall from *Heinemann Interactive Science 1* that **weight** and **mass** are not the same. In science, these words have special meanings. Mass is a measure of the amount of matter in an object, whereas weight is a measure of the force of gravity acting on a mass. Mass is measured in grams (g) or kilograms (kg). Weight is

measured in **newtons** (symbol N) (named in honour of Sir Isaac Newton).

Because the force of gravity is not the same all over the Earth, the weight of an object can vary. Gravity is stronger at places which are nearer the centre of the Earth. For example, it is stronger at the poles than at the Equator. It is also stronger at sea level than on top of mountain peaks.



FIGURE 3.16
Sir Isaac Newton.

USING SCIENCE 4

INTERPRETING A DIAGRAM

WHAT TO DO

Look at the information given in Figure 3.17 and answer the discussion questions below.

DISCUSSION

- 1 What is the diagram showing?
- 2 At which position is gravity the strongest? How do you know?
- 3 At which position is gravity the weakest? How do you know?
- 4 What happens to the mass of the astronaut, the greater the distance from the Earth?
- 5 What happens to the weight of the astronaut, the greater the distance from the Earth?

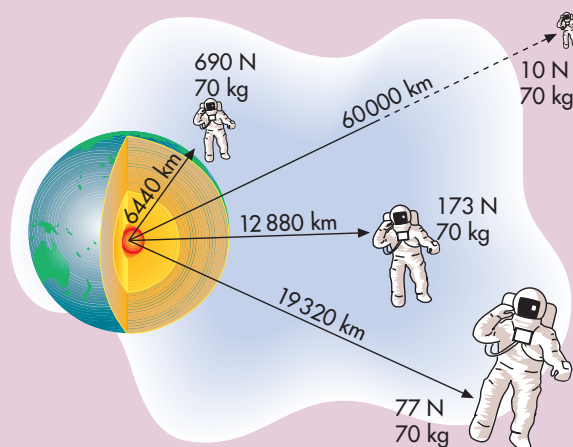


FIGURE 3.17

Questions

- 1 In your own words write a definition of gravity.
- 2 On Earth 1 N is roughly the weight of an object with a mass of 0.1 kg or 100 g. Use this information to calculate the approximate weight of a:
 - a 2.5 kg bag of oranges
 - b 0.4 kg book
 - c 375 g can of soft drink
 - d 50 kg person



Think about

- 1 If the Earth's gravity is pulling on the Moon, why doesn't it crash into the Earth?
- 2 The Moon contains about one-sixth the mass of Earth. What force would the Apollo astronauts who walked on the Moon's surface have experienced compared to:
 - a on Earth?
 - b in their space vehicle between Earth and the Moon?

Find out

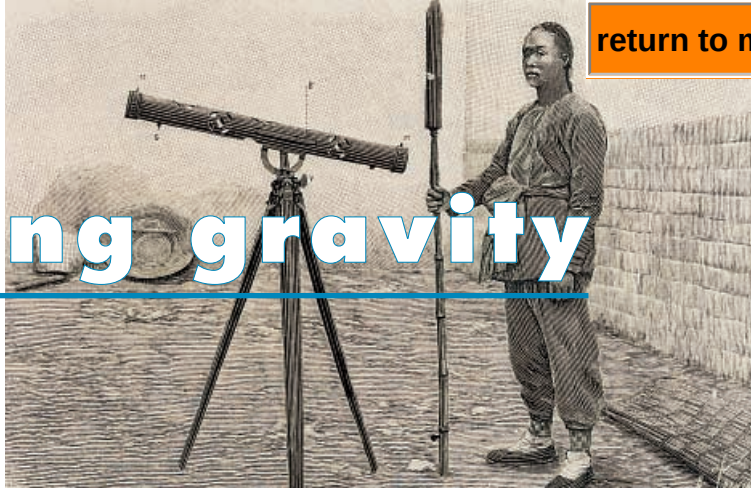
Gravity is a field force. What other forces act on objects from a distance?



Overcoming gravity

FIGURE 3.18

Rockets were invented by the Chinese. This device was used to launch rockets at an enemy.



Imagine you are playing cricket on top of the tallest mountain. When you hit the ball, it travels through the air a distance before gravity pulls it back to the ground. The harder you hit the ball, the faster it goes. A fast moving ball will travel further before falling back to the ground. So one answer to overcoming the force of gravity lies in speed. If you could hit the ball so hard that it travelled at 28 000 km/h then it would go into orbit around the Earth. If the ball could reach a speed of around 40 000 km/h it could escape the pull of Earth's gravity and travel away from the Earth, without ever falling back.

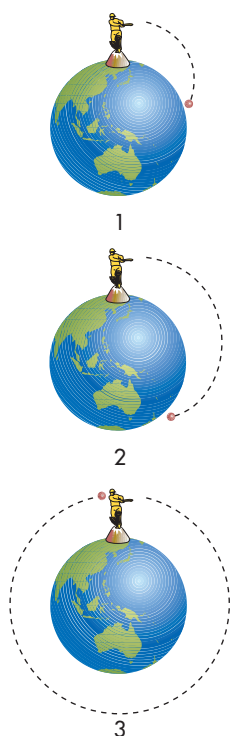


FIGURE 3.19

The answer to overcoming gravity lies in obtaining great speed. Enormous forces are required.

Rocketry

The only engine powerful enough to reach the speeds needed to go into orbit or escape the Earth is the rocket.

Talk about

How do you think the rocket launcher in Figure 3.18 worked? Would it have been accurate?

Although rockets seem to be symbols of the space age, they were first invented in China thousands of years ago. However, it is less than 50 years since the first rocket went into orbit around the Earth, and less than 40 years since the first rocket went to the Moon. But what exactly *are* rockets?

One thing that all rockets have in common is the way they move.

What you have just done in Experiment 3 is to behave like a simple rocket. Rockets work in exactly the same way; that is, they 'throw' matter behind them which pushes them forward. This type of motion was first

EXPERIMENT 3

INVESTIGATING ROCKET MOTION

AIM

To find out how rockets work.

MATERIALS

- smooth floor
- chair on castors
- six heavy telephone books

METHOD

- 1 Sit on the chair with your feet off the ground and the telephone books on your lap.
- 2 Throw three telephone books away from yourself as hard as you can. Observe what happens.

- 3 Try this again but this time, once you have thrown the first three books, throw the other three *in the same direction as you are moving*. What happens now?

- 4 Experiment to see whether how hard you throw the books affects the results.

DISCUSSION

Summarise what you have found out in this experiment. Compare your summary with those of others in the class.

**FIGURE 3.20**

Every action has an equal and opposite reaction.

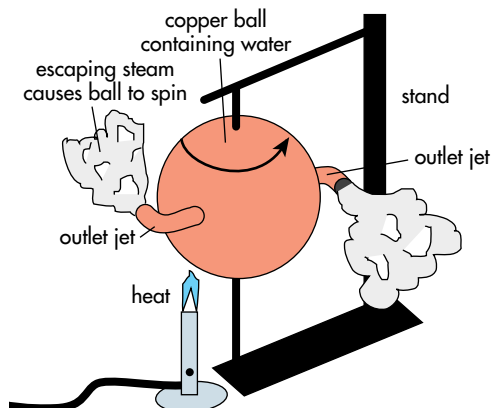
described by Isaac Newton over 200 years ago and is now known as Newton's third law of motion. This law is usually expressed as:

'Every action has an equal and opposite reaction'.

The action is the force which you exerted when throwing the books; the reaction is the force which the books exerted on your hands. These forces are equal and opposite. The harder you threw them, the faster you moved backwards. The same thing happens when you turn on a hose without holding the end. If the water comes out fast, the hose is pushed in the opposite direction as a reaction. Because the end of the hose is attached to the rest of the hose, it whips around as long as the water keeps coming out. If you hold the hose while you turn the water on fast, you will feel the reaction force push against your hand.

Questions

- 1 Jet engines work in the same way as rockets, except that they use air from the atmosphere as part of their fuel. Why can't space rockets do this?
- 2 If you had your back to the edge of a cliff, what could happen if you threw a brick as hard as you could in front of you? Explain your answer.
- 3 In what way was Hero's engine 'rocket-powered'?
- 4 What is Newton's third law of motion? What does it mean?

**FIGURE 3.21**

A model of the steam engine built thousands of years ago in ancient Greece by a mathematician called Hero.

- 5 How would a machine-gun-powered skateboard work?
- 6 Draw and label the forces on a rocket.
- 7 Complete the following sentences. Rockets move by shooting some substance in the _____ direction to their direction of travel. They were first used by people in _____ to launch _____. Modern rockets have _____ and _____ to stabilise them as they move through the air.

Activity

Design and draw a picture of your own rocket-powered transport.

Find out

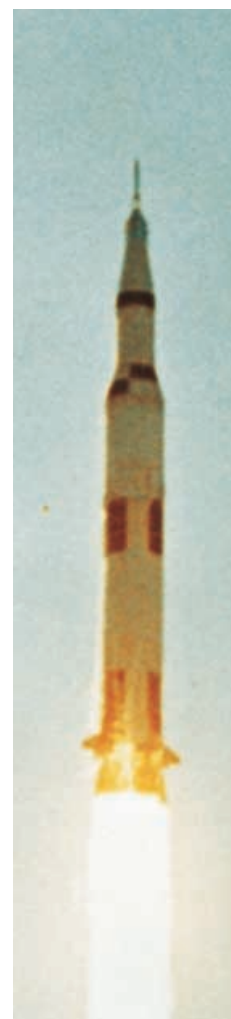
- 1 What do you think 'thrust' means? Look up its meaning in a dictionary and write a sentence about rockets using the word 'thrust'.
- 2 What did Sir Isaac Newton say in his first and second laws of motion?

**FIGURE 3.22**

Modern rockets have fins and nozzles which allow them to be 'steered'.



Although the Chinese invented gunpowder and were able to use it to make fireworks (skyrockets), the first known practical use of rocket power was a steam engine that spun around as it shot out jets of steam.



The force of friction

Friction is a contact force. It exists whenever two surfaces rub against each other. Friction works in a direction opposite to the direction of motion. It opposes motion.

Friction exists between surfaces even before they are moving. When you push against a heavy object such as a refrigerator, which doesn't move, you can feel the effect of **static friction**. Static friction is the force responsible for preventing motion from starting.

With someone helping you, together you may be able to overcome the static friction and get the refrigerator moving.

However, friction still exists, because if you stop pushing, the refrigerator will stop moving. The friction which exists between moving surfaces is called **sliding friction**.

FIGURE 3.23

Arrows can be used to show both the size and the direction of the forces acting on an object. In this case, static friction prevents motion.

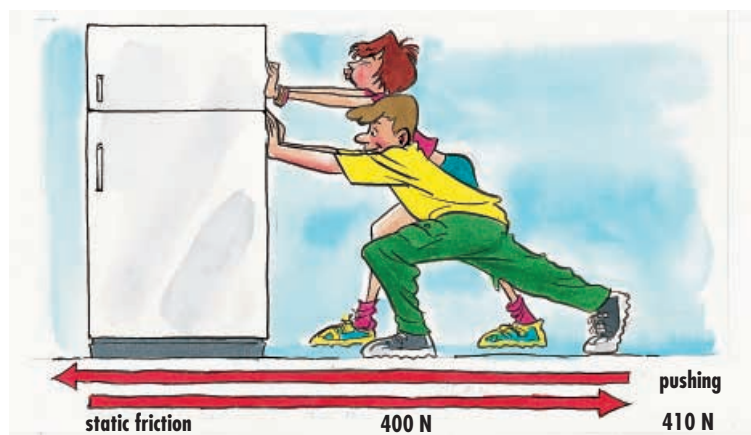
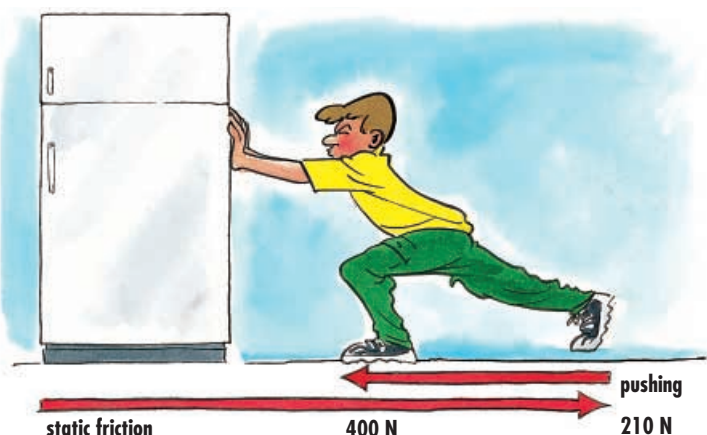


FIGURE 3.24

The pushing force now overcomes the static friction.

Understanding friction

No surface is completely smooth. Even this page which appears smooth is in fact made up of a rough mat of fibres. You just need a microscope to see them.

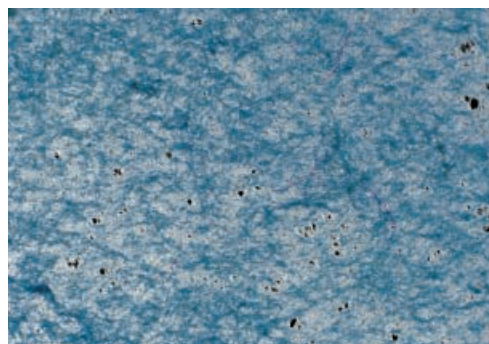


FIGURE 3.25

It may appear smooth, but looking through a microscope reveals paper's true roughness.

However, friction does not depend on the roughness of the surfaces. It depends on the actual surface area in contact. So two smooth surfaces may experience more friction because they have more area in contact than two rough surfaces. Friction depends on the amount of force which pushes two surfaces together. The greater the force pushing two surfaces together, the greater the area of contact between them.

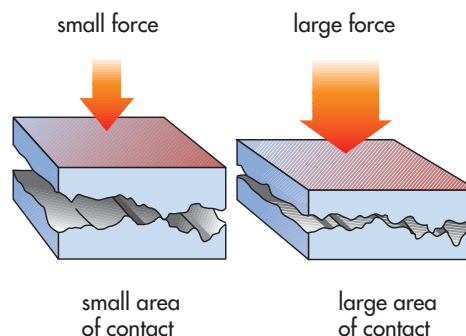


FIGURE 3.26

The more force pushing the surfaces together the greater the friction.

Friction pluses and minuses

Friction, like gravity, is a force we cannot escape from in our everyday lives. It is a force which can be useful but can also cause problems.

TABLE 3.1 Benefits and problems of friction.

Benefits	Problems
Prevents slipping	Produces heat
Allows us to control the direction and speed of motion	Causes surfaces to wear away
Allows us to stop moving	Causes energy to be wasted in moving machines

EXPERIMENT 4

SLIP, SLIDING AWAY



AIM

To measure the slip angle needed to overcome static friction between wood and other materials.

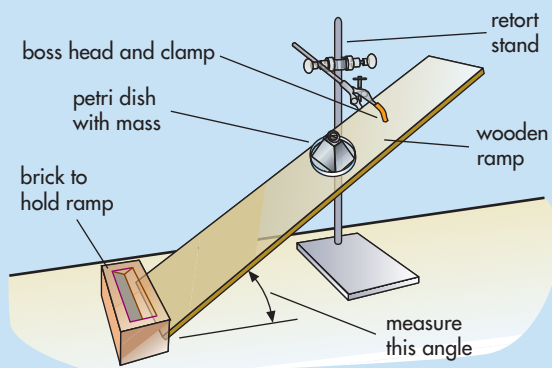


FIGURE 3.27

MATERIALS

- wooden ramp
- retort stand, boss head and clamp
- materials to test such as cardboard, sandpaper, carpet, rubber
- protractor
- brick
- 4 plastic petri dishes
- glue
- scissors
- 500 g mass to hold petri dish on ramp

METHOD

- 1 Cut a circle from each material to be tested large enough to cover the base of a petri dish.
- 2 Glue the materials to the petri dishes.

- 3 Set up the ramp as shown, at an angle of 25° .
- 4 Place a petri dish with a mass sitting in it at the top of the ramp. (If it moves, reduce the starting angle.)
- 5 Gradually increase the angle of the ramp until the dish first slips.
- 6 Record the 'slip angle' for the material being tested in a results table.
- 7 Repeat the test to obtain three measurements and then average them.
- 8 Repeat steps 4–7 for each material.

RESULTS

Material	Test 1	Test 2	Test 3	Average slip angle

DISCUSSION

- 1 What variables have been controlled?
- 2 Why was each material tested three times and the results averaged?
- 3 Draw a diagram of the petri dish on the slope. Use arrows to show the direction of the forces acting on it. Label each arrow.

EXTENSION

- 1 Try changing the surface of the ramp and compare the static friction between different surfaces.
- 2 Investigate varying the size of the mass in the petri dish, keeping all other variables the same.

Questions

- 1 What is friction?
- 2 What is the difference between static and sliding friction?
- 3 Give an example of a situation where friction is an advantage and a disadvantage.



- 4 Sliding down a rope may burn your hands. Explain why.

Find out

How can we reduce the effect of friction in machines?



Join the resistance!

FIGURE 3.28

Model T Fords and a modern car. Which would experience the most air resistance? Why?



FACT

FILE

General Motors have recently developed a car called the Impact. It experiences as little air resistance as an F-16 jet, 30% less than any other car. The Impact has a smooth skin all over, including underneath, where most cars are anything but smooth!

Talk about

What do you think is happening in Figure 3.29? What are the tapes for?

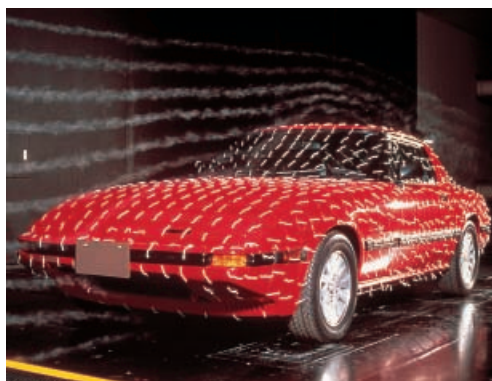


FIGURE 3.29

Air resistance is one of the many frictional forces that surround us in the real world. Many scientific ideas about forces appear to be untrue if air resistance isn't taken into

account. Air resistance acts to slow everything down. Engineers in many different fields spend much of their time battling to reduce its effects.

For a car travelling at normal speeds, most of the force produced by the engine goes towards overcoming air resistance. Car designers know that one of the best ways to increase fuel efficiency in their cars is to build them in a more **aerodynamic** shape. One of the most obvious trends in car design over time has been a movement from boxy, tall, straight-fronted bodies to the much more fluid shapes of today.

The amount of air resistance experienced by an object is affected by many factors, the most important of which are the speed at which it is moving; the size of the cross-section it presents to the air; its overall shape; and the surface texture.

For a long time scientists assumed that the smoother a surface, the less air resistance there would be. This is why cyclists and other athletes shave their bodies

USING SCIENCE 5

BUILD A WIND TUNNEL

WHAT YOU NEED

An electric fan, a large cardboard box, a range of objects to test, paper tapes, sticky tape.

WHAT YOU DO

Build the wind tunnel by removing two ends of the cardboard box. You may need to reinforce the

remaining sides. Set the fan up firmly at one open end. Attach tapes to each object with sticky tape, running away from the end which will face the fan. What kind of shape do you think will be most aerodynamic? Can you design a shape with less 'flutter'?

improve performance. Lately, however, scientists have discovered that there are some special cases of roughness which out-perform the smoothest surfaces.

By examining the skins of sharks scientists found that by running many shallow grooves around a body in the direction of motion, the churning effect known as turbulence can be reduced. Turbulence is one of the many parts of air resistance. By using the grooves, overall air resistance can be reduced significantly.

Although it is most commonly thought of as a nuisance, in some situations air resistance is absolutely essential. Skydivers know that adapting their air resistance will fine-tune their speed in flight. The way they are able to do this is by changing the size of their cross-section. To reduce air resistance, and therefore accelerate, skydivers pull their arms and legs in tightly together. Because air resistance is dependent on velocity, the upward force of air resistance increases as the diver speeds up, until it balances with their weight force again, and the acceleration ceases. At this point a skydiver is said to have reached **terminal velocity** (velocity is a scientific term which means speed in a given direction).

If skydivers wish to slow down, they spread out their arms and legs. Most skydivers choose to fall this way because it lengthens the time spent in free fall. If skydivers, who have been falling wrapped up tightly like balls, suddenly spread themselves out flat, the increase in air resistance would hit them as though they'd just run into something.

Opening a parachute works in the same way. By increasing the size of the cross-section presented, air resistance is increased. The increased air resistance is now bigger than the weight force, which produces an acceleration in the opposite direction to the motion. The skydivers slow down, until their drop in speed reduces the air resistance to the point that it again balances the weight. This new speed is also called terminal velocity, because it is the final velocity reached. If the parachutes have opened fully, it shouldn't be too 'terminal' when the skydivers land!



FIGURE 3.30
A skydiver in free fall.



FIGURE 3.31
Increasing resistance creates negative acceleration.

Questions

- 1 List three factors that increase air resistance.
- 2 Explain what is meant by 'terminal velocity'.
- 3 If I open out my arms during free fall, why do I slow down?

Think about

Modern golf balls are designed especially to reduce air resistance so they can travel further. Starting with cars and golf balls, make a list of items which should be designed to reduce air resistance.

Activity

Design a parachute for an egg. Can you build a parachute so that an egg can fall from a second-storey window and not break.



Foundation
worksheet 2



Internet assignment

FACT

FILE

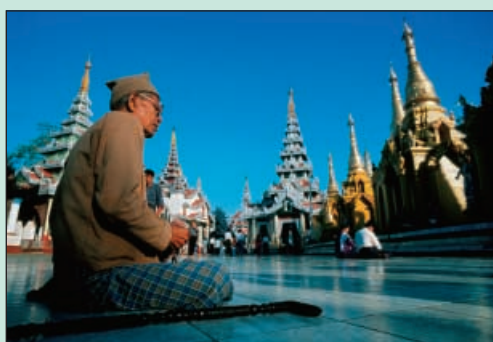
Skydivers leap out of aeroplanes flying anything up to about 3.7 km above the ground, and free-fall without parachutes at about 190 km/h. Parachutes must be opened 600 m above the ground for a safe landing.

THE NATURE AND PRACTICE OF SCIENCE

Can we ever be still?

FIGURE 3.32

Meditation is often used as a relaxation technique to 'still the mind'. Look carefully at the photograph. List the things you can see that would be moving and the things that would not be moving. Are there more moving things or still things? Are there any moving things that you cannot see?



If you were to sit alone in a room and keep as still as possible, it might appear that nothing was moving, especially if the curtains were drawn, the windows closed and there was no noise outside. Despite this apparent stillness, you would actually be moving in a most amazing way.

If standing at the Equator, in 24 hours we would travel a distance of about 40 000 km as the Earth spins on its axis. Meanwhile the Earth is orbiting the Sun at about 30 km/s. The solar system is moving at about 20 km/s towards Vega, one of the stars nearest to the Sun in our own galaxy. The Sun and its nearest stars like Vega are orbiting the centre of our galaxy at about 200 km/s. And of course our galaxy is also moving through space ...

FIGURE 3.33 (right)
The pendulum is set swinging along a line on the floor.

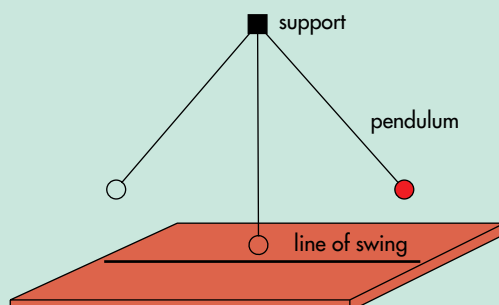


FIGURE 3.34

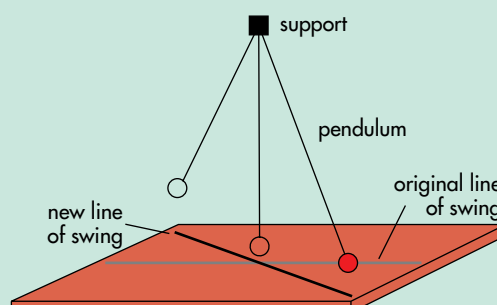
(far right)
After some time the pendulum is found to be swinging along a different line. It appears to have twisted around.

Why don't we feel the motion of the Earth? One reason is that we often only know we are moving when the motion changes. For example, you wouldn't realise the car you were in was travelling at 100 km/h if it weren't for the bumps and jolts and twists. You would certainly know you were moving, however, if the car suddenly accelerated or slowed down, or turned a corner sharply.

How do we know the Earth is spinning?

From a spaceship or from another planet it is possible to observe that the Earth is spinning. But how did people work it out before space flight?

In 1851, long before the first space flight, Jean Bernard Léon Foucault carried out experiments that showed that the Earth was spinning. He used a pendulum about 67 metres long with a 28 kg ball on the end. After the pendulum had been swinging for some time, Foucault found that the pendulum was swinging in a different direction.



Foucault realised that it was not the pendulum that was changing direction, rather that the Earth was spinning beneath it. Because we are spinning with the Earth, it appears to us that the swing of the pendulum is changing direction. However, there are no forces acting on the pendulum to make it do this.

If the pendulum were at the North or South Pole, you could imagine the Earth rotating beneath it once every 24 hours. If you were watching the pendulum, you would think it was changing its direction of swing.

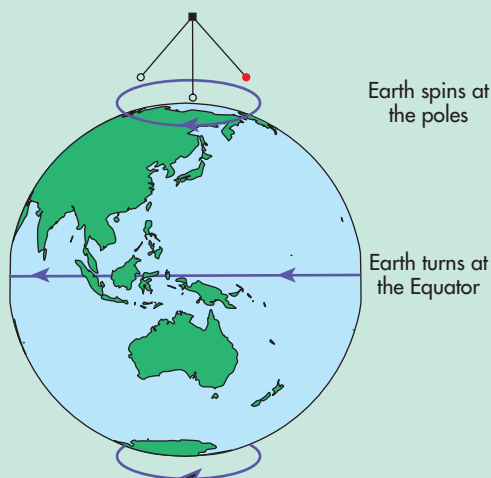


FIGURE 3.35

The pendulum appears to turn clockwise at the North Pole and counterclockwise at the South Pole. There is no effect at the Equator. Between the poles and the Equator, the effect lessens as you approach the Equator.

EXPERIMENT 5

A FOUCAULT PENDULUM

AIM

To make a model of Foucault's pendulum experiment.

MATERIALS

- 4 retort stands and clamps
- strong thread
- pendulum bob or heavy metal ball (about 20 mm in diameter)

METHOD

- 1 Unscrew two of the retort stands from their bases.
- 2 Set up the pendulum framework as shown in the diagram.
- 3 Hang the pendulum bob from the centre of the top rod.
- 4 Set the pendulum swinging in line with the frame.
- 5 Grasp firmly the lower parts of the two vertical rods. Twist the frame quickly while the pendulum is still swinging.

DISCUSSION

- 1 What did you notice about the direction of the pendulum's swing after you gave the frame a sharp twist?

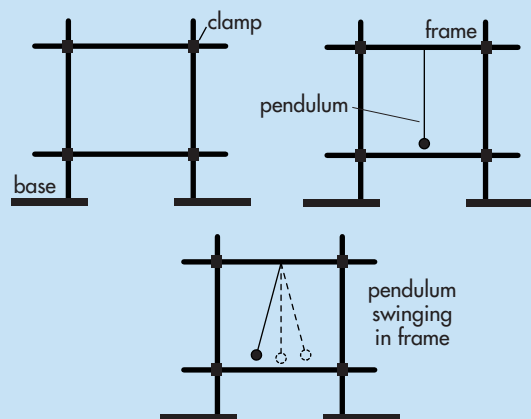


FIGURE 3.36

Experimental set-up: the frame for a Foucault pendulum.

- 2 Compare this activity to the actual experiment that Foucault carried out. What is the frame equivalent to? What does the twist represent?

EXTENSION

If there is a suitable place in your school, set up a large Foucault pendulum. It doesn't have to be 67 metres long, but the longer you can make it, the better the results. (You must suspend the pendulum from a hook which can pivot freely on a point.)

Questions

- 1 List two ways we can tell that the Earth is moving.
- 2 What is a pendulum? What must be done to make it move?
- 3 Draw a picture of the pendulum used by Foucault, showing its motion.



Think about

When the pendulum reaches the top of its swing, it stops and then moves back in the opposite direction. What do you think causes this? What causes the pendulum to slow down and eventually stop?





Using scientific language

Give a brief explanation of each of the following terms:

motion	newton	force
friction	speed	gravity
mass	weight	air resistance



Check your knowledge

- For each type of motion listed below, discuss how it might have been produced and whether the cause would have been a push or pull.
 - The movement of the hands of a mechanical clock.
 - The flight of an arrow.
 - The circular motion of an egg-beater.
 - A bicycle freewheeling down a slope.
 - The upward movement of an elevator.
 - An artificial satellite orbiting the Earth.
- Explain why you don't have to know the difference between mass and weight in order to run a supermarket on Earth. Would you need to know the difference if you were in charge of providing supplies to a colony on the Moon? Why?
- 'Friction opposes motion, but you can't have motion without friction.' Discuss this statement.
- In each of the situations below say whether friction is a benefit or a problem.
 - Starting a fire by rubbing two sticks together.
 - Jumping from a plane with a parachute.
 - A vacuum cleaner motor getting hot during use.
 - Tread wearing off bike tyres



Apply your skills

- A car travels 25 kilometres in 0.5 hour, then travels a further 35 kilometres in the next 1.5 hours. Calculate:
 - the average speed (km/h) for each section of the journey
 - the average speed (km/h) for the whole journey.
- As car tyres wear, the tread disappears and they become dangerous because they no longer grip the road surface. However, tyres on Formula One racing cars are smooth with no tread, but they have very good grip on the road surface. Why do you think this is?

Distance moved	Time taken	Average speed	Example
10 cm	20 min		snail
100 m		50 m/s	
	0.1 s	50 m/s	
	4.5 h	100 km/h	
3000 m	10 s		the sound of thunder
40 000 km (the circumference of the Earth)		300 000 km/s	

- Copy the following table. Fill in the missing entries by calculating the distance, time or average speed in each case. Don't forget to include the unit with your answers.
In each case, suggest an example from real life that might perform at the average speed given. Some have been done for you.
- Look at the photograph of the skydiver in a free-fall descent on page 57. He/she is falling because the force of gravity is pulling them down.
 - What other force is acting on them and in which direction is it acting?
 - How do skydivers control the speed at which they are falling?

Challenge yourself



- Take an empty plastic soft drink (PET) bottle and think of ways of crushing it. One way is to place increasing loads on the bottle until it gives way. Try this in two ways: applying the load on the side and applying it on the top. Which requires the least force? Can you think of some way of measuring the force you used?
 - When you have some idea of the force required to crush the bottle, carry out the following activity. Put the bottle into very hot water for a few minutes, leaving the lid off. Screw the lid on tightly and plunge the bottle into a large basin of iced water. What force crushed the bottle? Did the force come from inside or outside the bottle? Was the force greater or less than the force you applied to crush the bottle? Was it a push or pull?
- Fill a glass with water until it is overflowing. Slide a postcard over the top, making sure no air is trapped. Hold the glass in one hand and place the palm of the other on the card. Turn the glass upside down and carefully remove the hand placed against the card. What force is holding the water in the glass?

CHAPTER

4

ELECTRIC CIRCUITS

Lightning is one of nature's most awesome displays of electricity. It occurs when conditions are just right during a thunderstorm. Inside storm clouds violent draughts push water droplets and ice crystals up and down, causing static electricity to build up. Positively charged particles gather in the upper regions of the cloud while negatively charged particles collect at the cloud's base. This separation of charge effectively turns the cloud into a giant battery capable of discharging a sudden enormous spark. The ground beneath the storm cloud becomes positively charged. A lightning flash begins when a burst of negatively charged electrons is attracted down to the positive ground. This discharge is called a lead stroke and it can travel between 160 and 1600 km/s. Following the lead stroke, a return stroke is directed back along the same path at much greater speeds, up to 140 000 km/s. This burst of electrical discharge reduces the charge differences temporarily. However, conditions inside the storm cloud soon allow the charge differences to build up, and repeated flashes of lightning follow. The flash we see in a strike of lightning occurs because the discharges change the surrounding air molecules, causing them to emit light as they are heated to a temperature of around 30 000°C.

The current in a stroke of lightning can reach 200 000 amperes, more than enough to reduce human flesh to bone and ashes.

In this chapter you will investigate electricity and discuss how to use it safely.

CHAPTER OUTCOMES

Successful completion of this chapter will allow you to:

- Describe how scientists construct theories to explain observations.
- Explain what is meant by electricity and what is necessary for a current to flow.
- Recognise that electrical energy can be converted into other forms of energy in a circuit.
- Construct and draw simple circuits.
- Distinguish between conductors and insulators and explain the behaviour of each.
- Compare series and parallel circuits.
- Identify everyday situations in which electromagnets are used.
- Demonstrate a commitment to safe practices when using electricity.
- Identify choices which need to be made when considering a method of power generation.

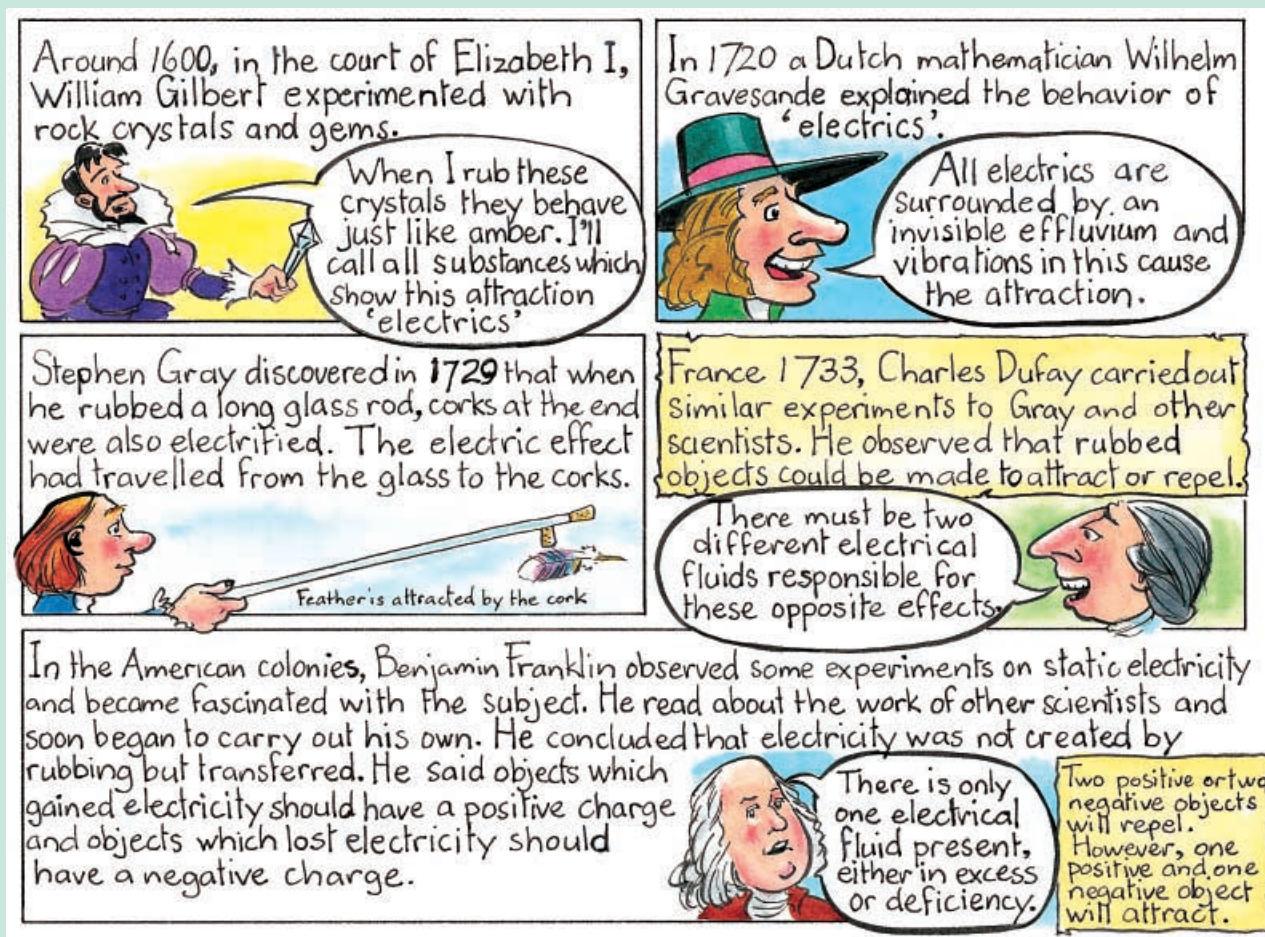
THE NATURE AND PRACTICE OF SCIENCE

Moving charges

Throughout history people have been fascinated by **electricity**. Over 2500 years ago the ancient Greeks observed the effects of **static electricity**. They discovered that a natural substance they called *elektron* could be rubbed and made to attract light objects. We call this substance *amber*. From around 1600, early experimenters began to investigate the effects of static electricity. They also attempted to

explain their observations. Many theories about electricity were put forward, each based on new experiments. Theories were altered or expanded to explain new observations. Scientists often quarrelled about whose **theory** was the best. Over time human understanding of electricity grew as new devices were invented that allowed scientists to explore this phenomenon further.

FIGURE 4.1



Many scientists argued about the single and double Fluid theories of electricity. However, all ideas were based on static electricity. Devices which produced a flowing current were not invented until an accidental discovery by Luigi Galvani in 1780.

The frog's leg twitched when I touched it with the scalpel. There must be some type of animal electricity.

Galvani set out to find the cause of the twitching frog's leg. He discovered that the leg would twitch when hung from a brass hook against an iron rail.

News of Galvani's discovery reached another Italian scientist, Alessandro Volta, who set to work repeating Galvani's experiment.

These contractions in the frog's legs are not caused by animal electricity but by the connection of two different metals through a moist body.

Volta conducted many experiments and found that if he connected a circuit of conductors electricity would flow through it.

If you touch both the top and bottom discs you will experience a real shock.

In 1800 Volta connected a pile of zinc and copper discs in pairs separated by a piece of cloth soaked in acid. He had invented the first battery. His invention allowed the study of electricity to make great strides over the next 200 years.

FIGURE 4.2

Questions

- 1 Explain in your own words the meaning of 'theory'.
- 2 Why do theories change over time?
- 3 Describe three early theories about electricity.
- 4 How do we explain, using modern ideas about atoms and charges, what happens when an object is charged by rubbing?

Find out

A number of inventions allowed scientists to investigate the behaviour of electricity. Find out how the following devices worked and who invented them: Leyden jar, electroscope and electrophorus.

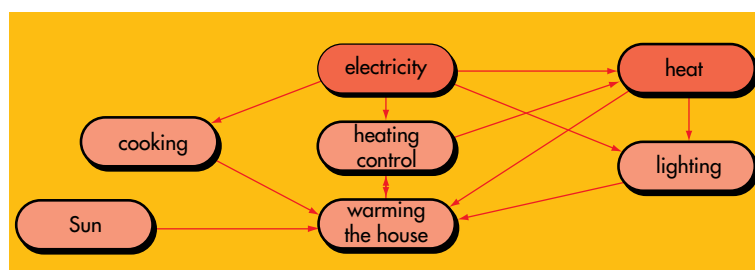
Activity

Build a voltaic pile by using two different metals, filter paper and two alligator leads. You may use coins, metal sheet or foil to make each cell.

Throwing some light onto electricity

FIGURE 4.3

Concept map showing the connections between electricity and heat in the home.



Talk about

Write down ten words that you think are related to electricity.

The whole class should list their words on the blackboard and agree on the meanings.

Now draw a diagram showing the connections between the words. This type of diagram is called a concept map or 'mind map' because it shows how you think the various words are related. It is a useful way of thinking out your ideas. Figure 4.3 is an example of a concept map.

EXPERIMENT 1

MAKE YOUR OWN TORCH

AIM

To investigate, then design and make, a torch.

PART A

MATERIALS

- torch (students could bring one from home)
- screwdriver

METHOD

- 1 Work in pairs.
- 2 Make sure the torch works.
- 3 Take the torch apart carefully so that it can be put together again.
- 4 As you examine it, think of an explanation of how the electricity flows and how the energy is brought to the globe.

DISCUSSION

- 1 Record your explanation by using a large labelled diagram to help make it clear.
- 2 Now explain how you think the torch works to another pair of students.

PART B

MATERIALS

- globe
- 6 V dry cell
- 3 insulated wires with the ends bared
- rubber bands or Blu-tac to hold the connections onto the globe and cell
- 2 paper clips to make a switch

METHOD

Working in pairs, connect the globe, the cell and the paper clips so that by using the clips as a switch, they work like a torch.

DISCUSSION

- 1 When you have made your 'torch', join with another group to check your previous explanation of how torches work. Make any changes to your explanation that you think are necessary.
- 2 The whole class could discuss the various explanations. What are the strengths and weaknesses of each? You should think up challenging questions. You might agree that only one explanation is the best or that two or even three explanations have good points.

How good is your explanation of how a torch works?

Does the explanation tell us:

- why heat and light are produced in the globe and not in the wires?
- how the brightness of the globe can be increased?
- why the cell goes flat eventually?

If it does you are probably on the right track!

The scientific view

The torch consists of a **circuit** or complete pathway for an **electric current** to flow through. This circuit consists of a number of **components**: the **cell**, the globe and a switch. Each component has at least two connection points called **terminals**. For an electric current to flow, the components must be connected from terminal to terminal by conducting material such as metal strips or wire. The electric current consists of a continuous flow of charged particles called **electrons**. The cell provides the energy to move the electrons in the circuit. Electric current flows all the time through all parts of a circuit. The current is the same all around the closed loop. If a break occurs in the loop, the electrons stop moving and no current flows. The switch is used to break the circuit and control the flow of current.

Electrical energy is different from electric current. Current transfers the electrical energy from the cell to the components of the circuit, where it is converted into other forms of energy such as heat, light or **kinetic energy**. The globe converts some of the electrical energy carried by the current into light and heat energy. Cells go flat when all the stored **chemical energy** they possess has been converted into other forms of energy. A flat cell can no longer provide the 'push' to move the electrons in the circuit. The energy the cell gives the electrons is called the '**voltage**'. Different cells produce different voltages. The bigger the voltage supplied by the cell the more energy is available for conversion into other forms of energy. Cells can be connected together to produce a bigger voltage. The bigger the voltage the brighter a globe will glow. The voltage can be measured with a voltmeter.



FIGURE 4.4

Electricity in our everyday life.

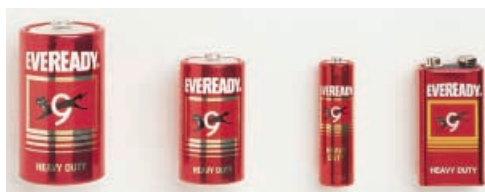


FIGURE 4.5

The scientific name for each of these familiar objects is a cell, although we call them batteries in everyday language. A battery is actually two or more cells connected together.



FIGURE 4.6

Torches are very familiar electrical devices, but do we know how they work?

Questions

- 1 What is a component? Give three examples of components.
- 2 Does a circuit work if only one terminal of a component is connected?
- 3 Do you think the current gets used up as it goes round a circuit? Explain your answer.

Think about

- 1 How does your explanation in question 3 above compare to the scientific view? Explain why you agree or disagree with the scientific view, or why you still need to be convinced.
- 2 Working in pairs, draw one or more concept maps showing how you think the following terms are related: current, cell, positive, negative, energy, metals, heat energy, resistance.

FACT

FILE

Some fish have specialised muscle cells that can act like tiny batteries. The electric catfish can generate up to 300 V, which they use to stun their prey. The electric eel has over 10 000 tiny electric organs allowing it to generate an incredible 550 V, which can even kill a person.

Conductors and insulators

FIGURE 4.7

The pole carrying electricity to your home consists of conductors and insulators. Which materials carry the current and which prevent the current from escaping from the circuit?

Electric current consists of a flow of charged particles, usually electrons, through a complete path or circuit. Some substances allow a current to flow through easily. These are called **conductors**. Copper wire is a good conductor of electric current. Other substances do not allow electric current to flow through so easily. These are called **insulators**. Both conductors and insulators play important roles in electric circuits.



EXPERIMENT 2

CONDUCTORS AND INSULATORS



AIM

To test substances to determine whether they are conductors or insulators.

MATERIALS

- 1.5 V cell and holder
- connecting wires
- 1.5 V globe in holder
- 2 alligator clips to hold test materials
- test materials—a range of metals and plastics, wood, tap water, salty water, the insides ('leads') of several different pencils, and any other materials you can collect

METHOD

1 Set up a circuit with the cell, globe, wires and alligator clips as shown in the diagram.

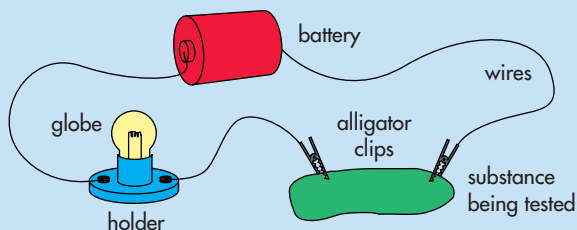


FIGURE 4.8

A circuit to test materials for conductivity.

2 Touch the alligator clips together to complete the circuit. Check that the globe lights up. (If it doesn't, trace around the circuit and check all the connections.)

3 Copy the following table into your book. Leave enough room to include all the materials you are going to test.

Material	Conductor (conducts electricity)	Insulator (doesn't conduct electricity)

4 Clip the alligator clips onto one of the test materials.

5 Record whether the material is a conductor or an insulator.

6 Repeat the experiment for all the materials you wish to test.

DISCUSSION

What kinds of materials make good conductors?
What kinds of materials make good insulators?

Explaining electrical conductivity

Why is it that some substances will allow electrons to move through them and others will not? To explain the difference between conductors and insulators we need to know something about the atoms which make up all substances. All atoms consist of a positive **nucleus** surrounded by negative **electrons**. Conductors are made of atoms with some electrons which are free to move around. Most metals, especially gold, silver, copper, aluminium and mercury are good

conductors. These metals have large numbers of free-moving electrons.

On the other hand, insulators have electrons which are held very tightly to their atoms. This makes electron flow difficult, almost impossible. Wood, rubber, plastic, glass, ceramics and most other non-metals are good insulators. Substances which hold onto their electrons tightly are said to resist the flow of electricity. All substances have a certain amount of **resistance**. It is possible to compare the amount of resistance by measuring the electric current in a circuit. An **ammeter** is used to measure the current. It measures current in the unit **amperes** (or amps for short).

FACT

FILE

If the voltage is large enough even an insulator can be made to conduct electricity. Tap water is normally a poor conductor but at 240 V a current large enough to kill a person can flow through it.

EXPERIMENT 3

COMPARING CONDUCTORS

AIM

To measure the size of the current flowing through several different wires.

MATERIALS

- ammeter
- 12 V power supply
- 1.5 V globe in holder
- 4 leads, 2 with alligator clips to hold wire
- 3 × 20 cm lengths of copper, iron and nichrome wire

METHOD

- 1 Set up the circuit as shown, making sure to connect the positive terminal of the power supply to the positive terminal of the ammeter.
- 2 Touch the alligator clips together to complete the circuit. Check that the globe lights up and read the size of the current on the ammeter.
- 3 Use the alligator clips to connect one of the wires into the circuit and observe the globe and read the ammeter.
- 4 Repeat step 3 for the other wires, record your observations and ammeter readings.

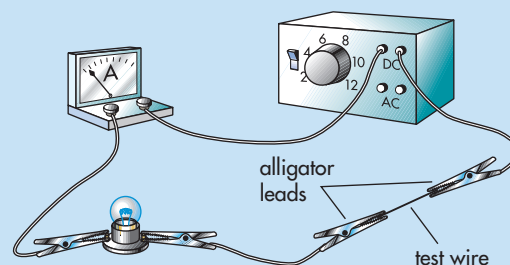


FIGURE 4.9

DISCUSSION

- 1 Which wire was the best conductor? How do you know?
- 2 Which wire had the highest resistance to the flow of electric current? How do you know?
- 3 Did any of the wires give a reading on the ammeter but not cause the globe to glow? If yes, try to explain why.
- 4 Was this a fair test? Were all the variables kept the same?

Semiconductors and superconductors

All substances can conduct electricity to some extent, but conductors do so 1 million million million times better than insulators. In between conductors and insulators are **semiconductors**, which conduct electricity better than glass but not as well as metals. Substances such as selenium, silicon and germanium are very useful semiconductors. They are widely used in microchips found in all electronic devices.

Superconductors are substances which are normally insulators. However, when cooled to extremely low temperatures (around -270°C) they can conduct very large currents because resistance reduces to zero.

Questions

- 1 What is a conductor? Give three examples of conductors.



- 2 Do some conductors conduct better than others? Can you suggest why?
- 3 Which materials are good insulators? Give three examples.
- 4 Why do electrical wires or leads have a plastic coating?
- 5 Draw a diagram of a globe and label all the parts as either conductor or insulator.

Think about



Carbon in the form of graphite (found in pencils) is a good conductor of electricity but it is a non-metal. Make a prediction about the way carbon atoms hold onto their electrons in graphite.

Find out



What voltage is needed to turn air into a conductor?

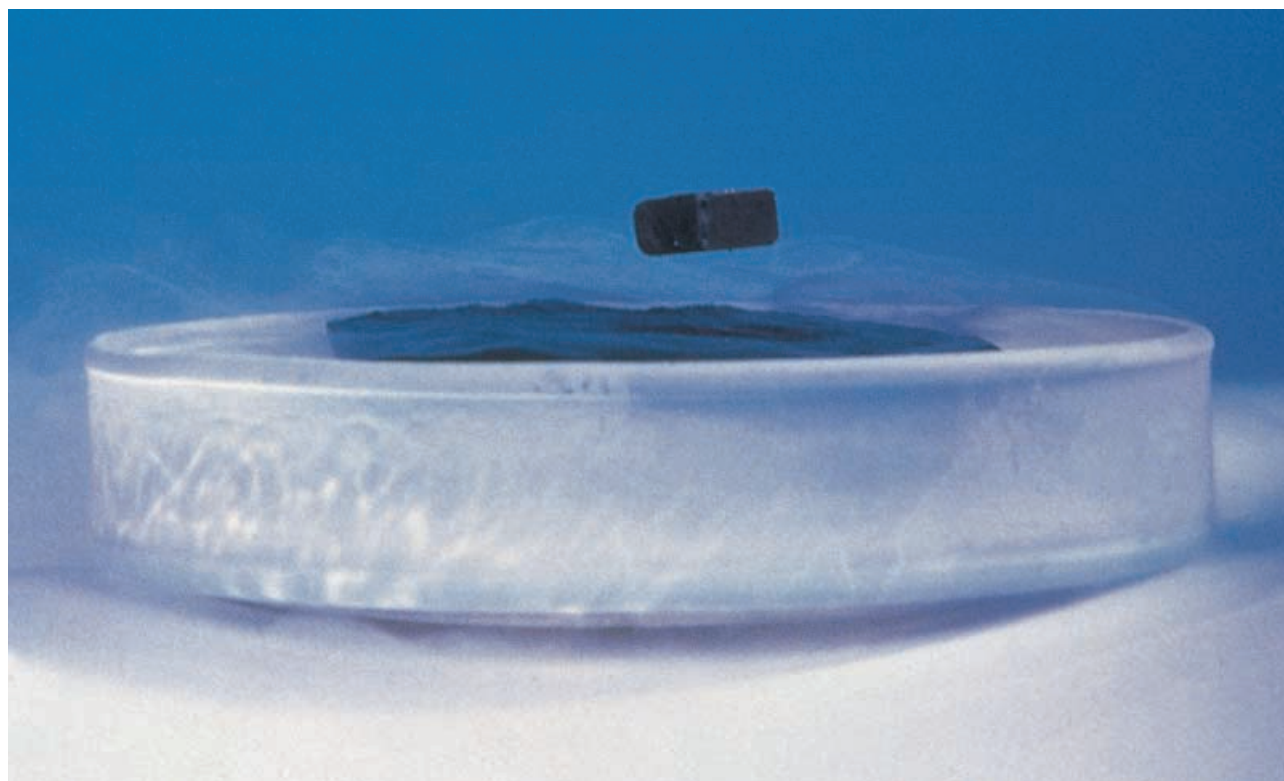


FIGURE 4.10

One feature of superconductors is that they can float above magnets.

Circuit diagrams

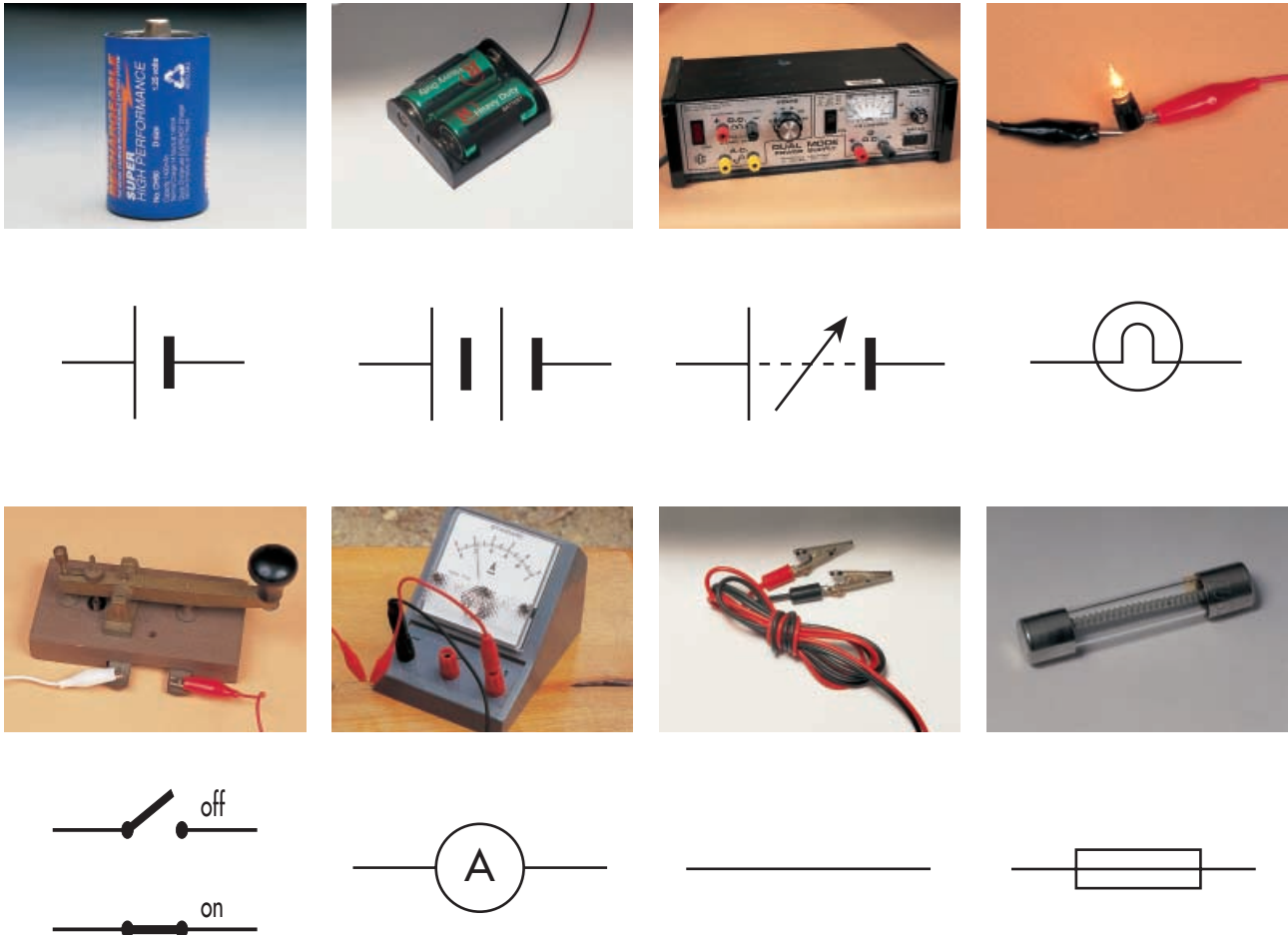
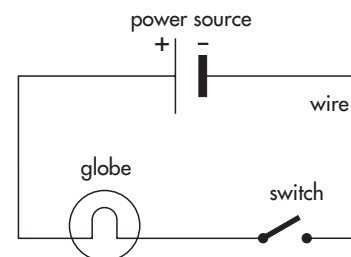


FIGURE 4.11
Common circuit components and their symbols.

When following directions it helps to have a plan or a map. Electricians use plans to show how the components in a circuit are connected. These plans are called **circuit diagrams**. Each component is given a

symbol to make the circuit diagram easier to draw. All the connections in a circuit diagram are usually shown as straight lines joined at right angles.

FIGURE 4.12
A simple circuit with its matching circuit diagram. Note the straight lines and right-angled corners.



USING SCIENCE 1

DRAWING CIRCUIT DIAGRAMS

Draw circuit diagrams for each of the circuits shown below.

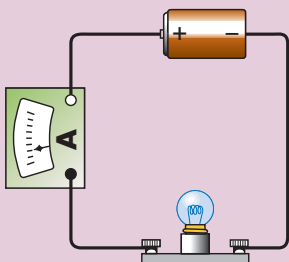


FIGURE 4.13

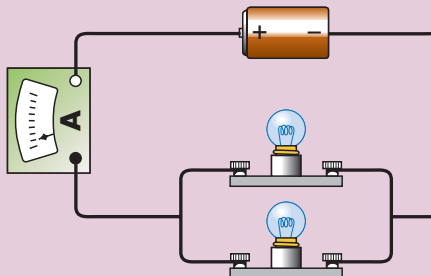


FIGURE 4.15

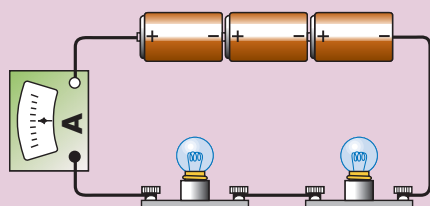


FIGURE 4.14

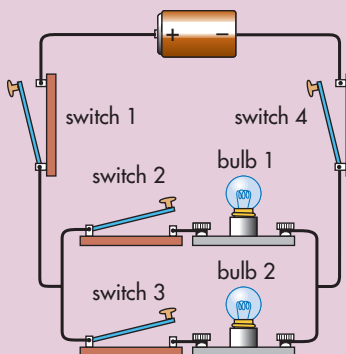


FIGURE 4.16

FACT

FILE

In an electric cell, chemical reactions take place that produce an excess of electrons at the negative terminal. At the positive terminal a deficiency of electrons builds up. The build-up of these equal but opposite charges prevents the chemical reactions from continuing unless the cell is connected into a circuit.

Which direction?

Scientists thought about electricity before they knew about electrons. Benjamin Franklin proposed that an object with an excess of electric charge should be called positive and an object with a deficiency of electric charge should be called negative. He said that the charge would flow from the positive to the negative. Later research showed that in metals electrons actually travel from places with a negative charge to places with a positive charge.

Today, electricians still use Franklin's convention and say that current flows from the (+) to (-). Benjamin Franklin was a great scientist; he had a 50/50 chance of picking the right direction but missed.

Questions

- 1 Where in your home could you find a circuit diagram?
- 2 What advantages would people gain by using circuit diagrams?
- 3 Use the symbols in Figure 4.11 to draw a circuit diagram for a torch.
- 4 Draw a circuit diagram that contains two globes, connecting wires and a cell. Could you have drawn it any differently? Explain your answer.

Find out

There are many other components which can be connected in a circuit. Find out the symbol for three other components and what each one does in a circuit.



FIGURE 4.17
Benjamin Franklin.

Series and parallel circuits

FACT

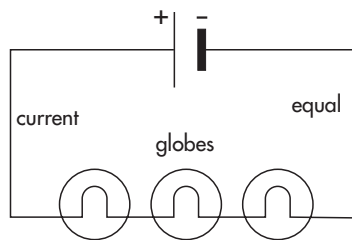
FILE

The most common type of cell is the zinc-carbon dry cell. It consists of a carbon rod which acts as the positive 'electrode' and a zinc outer casing which acts as the negative 'electrode'. In between the two electrodes is a paste containing ammonium chloride, zinc chloride and manganese dioxide. This type of cell supplies 1.5 V. Combining six 1.5 V cells in series produces a 9 V battery.

Two different circuit arrangements can be used to connect a number of light globes to a single power supply.

Series circuits

In one type of electric circuit, the electric current is made to pass *through every globe or component one after the other*. This is called a **series circuit**. Christmas tree lights are sometimes wired in this way. If one globe 'blows', however, the circuit is



Series circuit

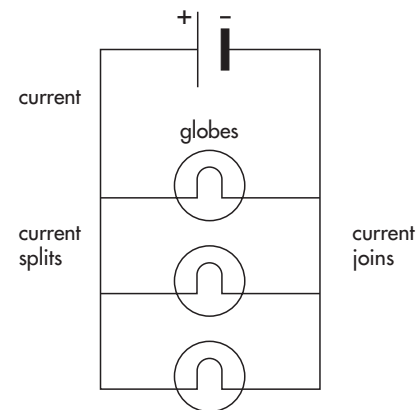


FIGURE 4.18
A series circuit and the corresponding circuit diagram.

broken and all the globes go out. Can you explain why?

Parallel circuits

In a **parallel circuit**, the current splits up and *portions go through each globe or component*. In a parallel circuit there is more than one way for the current to go around the complete circuit. If one globe blows, only the current in that part of the circuit will stop flowing.



Parallel circuit

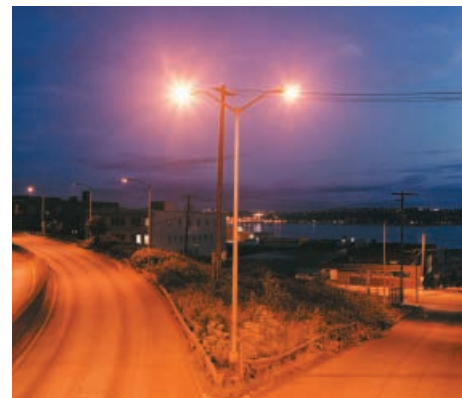


FIGURE 4.19
A parallel circuit and the corresponding circuit diagram.

EXPERIMENT 4**OPERATION OF SIMPLE CIRCUITS****AIM**

To examine the operation of series and parallel circuits.

MATERIALS

- 2 cells
- 1 switch
- 2 globes with holders
- connecting wires

METHOD

- 1 Set up each of the circuits shown in the diagram.
- 2 Copy the circuit diagrams into your book. Next to each circuit diagram, note the brightness of each globe (for example, dim, normal or bright) and record the effect of unscrewing and tightening each globe in turn.

DISCUSSION

- 1 In which circuits are the globes
 - a in series?
 - b in parallel?
- 2 Which type of circuit would be most convenient for wiring the lights in a house?
- 3 Summarise your findings on the operation of series and parallel circuits.

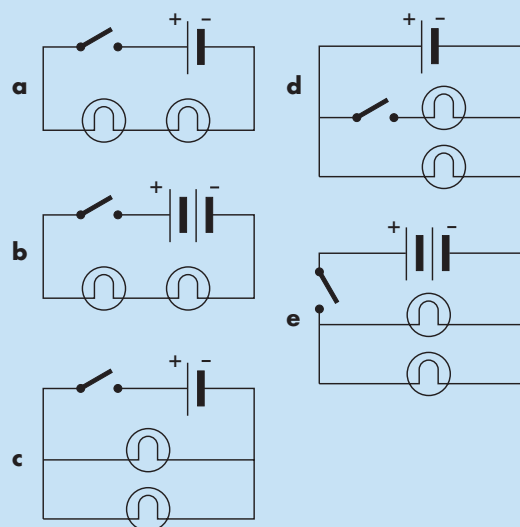


FIGURE 4.20
Experimental set-up for five electric circuits.

Questions

- 1 Complete the following sentences.

The _____ drives the current around the circuit. A circuit in which all of the current flows through one globe/component after another is called a _____ circuit. When the current splits in a _____ circuit, it eventually joins up again before reaching the battery.

2a Draw a circuit diagram for a circuit that contains two globes, a battery and an open switch connected in series by conductors.

b Would the globes in this circuit be alight or not? Explain your answer.

Activities

- 1 Design and build an electric circuit that demonstrates how an interior light in a car comes on when one or more doors are opened.
- 2 Design and construct a simple burglar alarm for your bedroom that rings a bell when anyone steps into the room.
- 3 Try connecting electric cells in series with a switch and a globe. Connect a voltmeter in parallel with the cells by connecting the positive terminal of a cell with the positive terminal of the voltmeter. Measure the voltage for one, two and three cells in series.

FACT

FILE

Thomas Edison, an American inventor, developed the first electric light globe in October 1879. It burned for 40 hours before failing. Only 2 months later, on New Year's Eve 1879, he floodlit the main street of Menlo Park, New Jersey, in a public demonstration.

Play it safe

Talk about

What causes electrical accidents? What happens when a person receives an electric shock? How can we prevent these types of accidents from happening at home and at school?

Electricity is a quick, clean and reliable form of energy. However, care must be taken when using it. When people ignore safety rules electrical accidents can occur. Each year hundreds of people in New South Wales experience electric shocks from faulty circuits or unsafe practices. About 25 people die each year as a result of electrical accidents. For this reason it is important that everyone makes a commitment to play it safe when using electricity.

What can go wrong

When all components in an electric circuit are in good working order, with all wires covered in insulation and all connections secure, then electricity is safe to use. However, when insulation wears away and exposes wires, and connections work loose, the circuit become dangerous.

If an exposed wire which is 'live' (carrying electric current) comes in contact with a conductor such as the metal frame of an appliance, then that appliance also becomes 'live'. The electric current flows into the frame instead of taking its usual path back along the neutral wire. This alternative path is called a **short circuit**. Short circuits are very dangerous because the current increases as it takes the easier path with less resistance through metal. The high current may cause a spark or a fire as wires overheat.

Fuses and safety switches

Fuses, circuit breakers and safety switches protect us from dangerous short circuits. Fuses work when a large current melts the fuse wire and breaks the circuit. Circuit breakers and safety switches work automatically to switch off the electricity before a fire can start. Safety switches (also called earth leakage devices) cut off the current in less than 30/1000 of a second, preventing a fatal accident.

FIGURE 4.21

Scene of fire caused by an electrical short circuit.

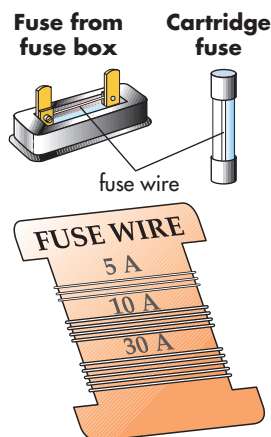
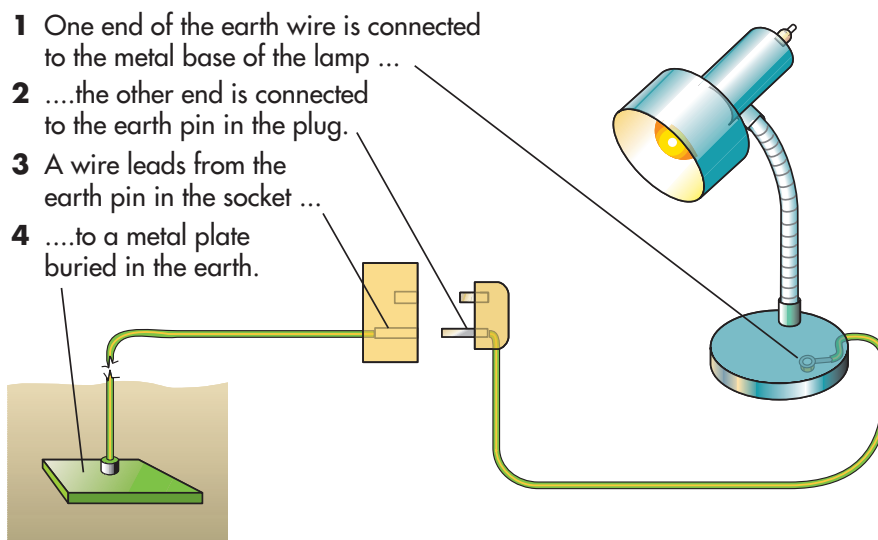


FIGURE 4.22

A range of safety switches and fuses.

**FIGURE 4.23**

Trace the path the earth wire takes.

EXPERIMENT 5**SHORT CIRCUITS AND FUSES****AIM**

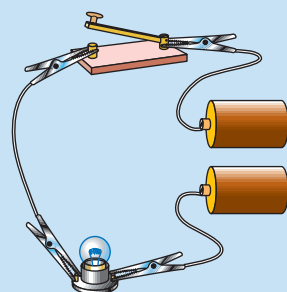
To observe a short circuit and find out how a fuse works.

MATERIALS

- 2 × 1.5 V cells
- 1.5 V globe in holder
- switch
- screwdriver
- large cork
- 2 long pins
- 5 A fuse wire
- 4 leads with alligator clips

METHOD**PART A**

- 1 Connect the cells then make a simple circuit as shown in Figure 4.24 and close the switch.

**FIGURE 4.24**

- 2 Place the metal end of the screwdriver so that it contacts the alligator clips between the globe and the switch as in Figure 4.25.
- 3 As soon as you observe what happens, remove the screwdriver.

EXPERIMENT 5

CONTINUED

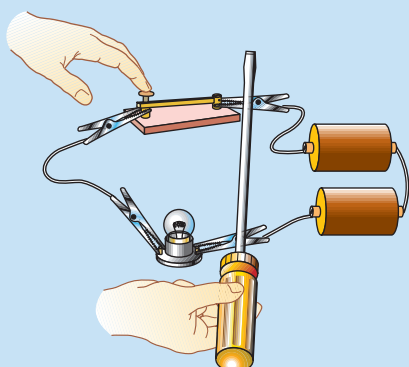


FIGURE 4.25

PART B

- 1 Construct a simple fuse from corks, pins and 5 A fuse wire.
- 2 Connect the fuse into the circuit from Part A.
- 3 Again use the screwdriver to create a short circuit and observe what happens.

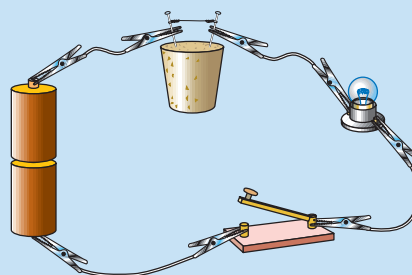


FIGURE 4.26

DISCUSSION

- 1 In Part A what effect did the short circuit have on the globe?
- 2 In Part B what effect did the short circuit have on the fuse wire?
- 3 How might fuses be used to protect circuits and prevent accidents?

Down to earth

Many electrical plugs have three pins. Each pin is connected to a different wire. There is a live wire, a neutral wire and an earth wire. The live and neutral wires carry the current when the switch is turned on. The earth wire is for safety. The earth wire connects appliances with metal surfaces to the ground. If the metal surfaces become live the current flows safely down to the earth wire.

Questions

- 1 Why are loose connections and broken insulation dangerous?
- 2 Describe what happens in a short circuit.
- 3 Explain how a fuse works.

Find out

- 1 What colour insulation is used on the three wires in electrical cable?

- 2 Why do some appliances' plugs only have two pins?
- 3 What should you do if you are outside during a thunderstorm?

Activity

Design a safety poster on one of the following electricity safety rules.

- Don't overload power sockets with too many double adaptors.
- Never use electrical appliances near water in a bath, sink or pool.
- Never use appliances with frayed or damaged cords.
- Never remove a plug without turning off the switch first.
- Never attempt to repair faulty switches or appliances yourself. Only use a qualified electrician.

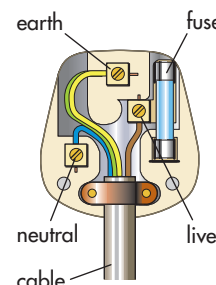


FIGURE 4.27
Inside a three-pin plug.

Electricity and magnetism

Whenever an electric current flows through a conductor it creates a **magnetic field** surrounding the conductor and this causes a **magnetic force**. Any magnets or magnetic materials nearby, such as iron or steel, will be affected by the magnetic force. This force can be put to good use.

An electric current can also **magnetise** close magnetic objects. This happens when you record music on a cassette tape. The tape is covered in a very thin layer of magnetic material. To record the music the tape recorder magnetises the tape, leaving an 'imprint' of the music. This is why video and cassette tapes should be kept well away from magnets.

EXPERIMENT 6

MAGNETIC FIELDS FROM ELECTRIC CURRENTS

AIM

To detect and use the magnetic field created when a current flows along a wire.

MATERIALS

- 2 m insulated copper wire
- small plotting compass
- DC power supply (2–12 V variable)
- 150 mm steel nail

METHOD

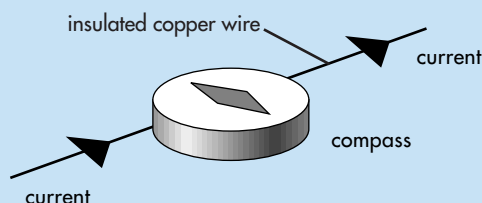


FIGURE 4.28

Experimental set-up to investigate magnetic fields caused by electric currents.

1 Connect the wire to the output of the low-voltage DC power supply. Make sure that the output is set to its lowest voltage.

- 2** Stretch the wire out so you have a straight length of at least 1 m. Hold the compass over the wire.
- 3** Switch the current on and watch what happens to the compass needle.
- 4** Change the orientation of the wire and repeat the experiment.
- 5** Swap the end of the wire that is connected to the power supply to change the direction of the current. Repeat the experiment.

DISCUSSION

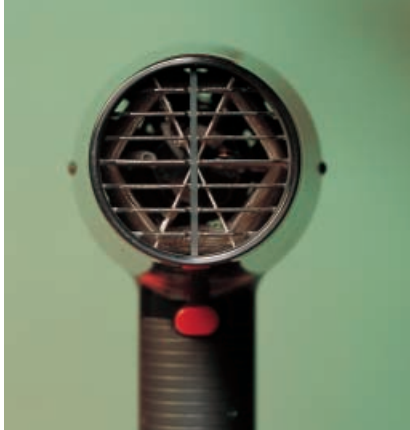
- 1** Does the compass needle always react when the current flows? Explain your answer.
- 2** Describe what happens when the direction of the current is reversed.

EXTENSION

- 1** Wind the wire around the 150 mm nail, leaving enough wire to make a connection to the power supply.
- 2** Switch the current on and bring the compass to each end of the nail in turn.

DISCUSSION

Is there any difference in the behaviour of the compass needle at the two ends of the nail?


FIGURE 4.29

Electric motors contain small magnets. In a hairdryer, the electric current flowing through the motor creates a magnetic force that pushes against the magnets. This force keeps the motor turning to blow air past the heater.


FIGURE 4.30

Small electric currents in the tape head of a cassette player magnetise the magnetic coating on the tape, leaving a magnetic 'imprint' of the music.


FIGURE 4.31

In a speaker electric currents flowing through a coil connected to the cone create forces that push against a permanent magnetic field. The movement of the cone in the air causes the sound we hear.

EXPERIMENT 7

ELECTRIC CURRENTS FROM MAGNETIC FIELDS

AIM

To detect the current created in a wire by a magnetic field.

MATERIALS

- 2 m insulated wire
- cardboard tube (about 50 mm in diameter)
- sensitive voltmeter or galvanometer
- bar magnet

METHOD

- 1 Wind the length of insulated wire around the cardboard tube.
- 2 Connect the ends of the wire to the voltmeter.
- 3 Quickly push the magnet into one end of the tube and pull it out again while watching the voltmeter.

DISCUSSION

- 1 What did you observe on the voltmeter?
- 2 Where is the electrical energy coming from?

Questions

- 1 List five everyday objects that contain an electromagnet.
- 2 Complete the following sentence.
A magnetic field always exists around a wire while it carries an ____.
- 3 The diagram opposite shows an electromagnet and a steel rod with a hook. Weights can be attached to the hook.
 - a Copy the diagram into your book. Complete the circuit to include a battery and a switch.
 - b With a small current flowing, the electromagnet will just pick up the steel rod with a 200 g weight attached. How could the electromagnet be made to pick up a heavier weight?

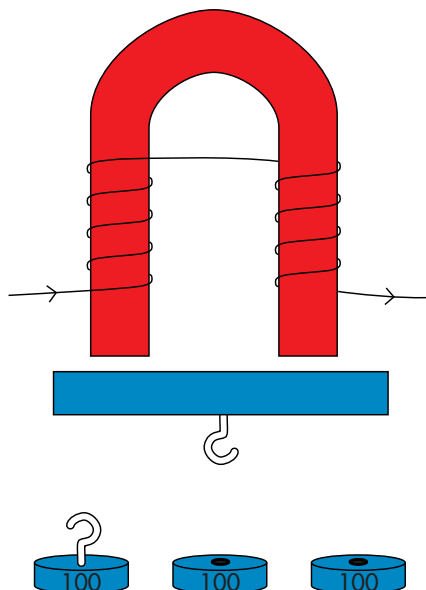


FIGURE 4.32
An electromagnet.

FACT

FILE

Samuel Morse, who gave his name to Morse code, first sent messages over long distances by using an electromagnet to move a pencil resting on a piece of paper. This was the birth of modern telecommunications.

Electricity: making new things

As well as producing a magnetic field, an electric current also produces heat energy whenever it flows through a conductor. In fact, the heat produced may be so great that it makes the conductor glow brightly. This is how normal household light globes work.

Conductors that heat up are called heating elements. They can be used to heat many different substances including air, water and even the concrete slabs on which houses are built.

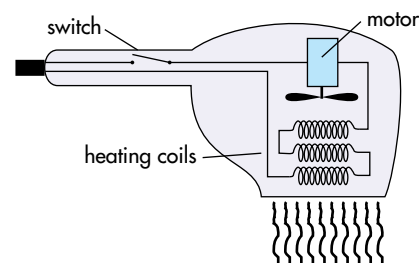


FIGURE 4.33

In an electric hairdryer, an electric motor drives a fan that forces cold air past a heating element.

EXPERIMENT 8

THE HEATING EFFECT OF AN ELECTRIC CURRENT

AIM

To produce heat energy from electric energy.

MATERIALS

- heating element made from nichrome wire
- polystyrene cup
- thermometer
- low-voltage DC power supply
- connecting wires

METHOD

- 1 Connect the heating element to the low-voltage DC supply.

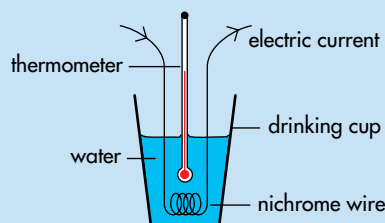


FIGURE 4.34

Experimental set-up to produce heat from electricity.

- 2 Completely cover the heating element with water. Take the temperature of the water and record it in your book.

- 3 Set the voltage to 2–4 V and switch on the power supply.

- 4 Stir the water gently with the thermometer. Switch off the power supply after 2 minutes. Record any change in temperature.

DISCUSSION

- 1 Did the temperature of the water fall, stay the same or rise?
- 2 Describe the energy changes that took place.

EXTENSION

Repeat the experiment by using similar amounts of different liquids instead of water. You could use different strengths of cordial or different household liquids such as cooking oil and detergent. Present your results in the form of a table and compare the temperature changes in the liquids over a 2-minute period.

Important: Get your teacher's approval before using any liquids in this experiment.

Electrolysis

An electric current can also cause chemical reactions to take place. One such process, called **electrolysis**, is used widely in industry.

Aluminium is obtained from aluminium oxide (from the ore bauxite) which is

dissolved in molten sodium aluminium fluoride. Current is passed through this liquid and aluminium is deposited. The molten pure aluminium can then be poured into moulds to set.

In **electroplating** a thin layer of metal is deposited on an object to improve its appearance or stop it rusting. Cutlery and teapots are often silver-plated in this way.

EXPERIMENT 9

THE CHEMICAL EFFECT OF AN ELECTRIC CURRENT

AIM

To produce a chemical reaction from an electric current.

MATERIALS

- 250 mL beaker
- 150 mL copper sulfate solution
- copper electrode
- low-voltage DC power supply
- connecting wires with alligator clips
- something metal to electroplate (such as a locker key)

METHOD

- 1 Pour the copper sulfate solution into the beaker.
- 2 Connect the copper electrode to the positive (+ or red) terminal of the power supply. Attach the alligator clip to the object to be electroplated and connect it to the negative (– or black) terminal.
- 3 Be careful not to allow the copper electrode to touch the object you are electroplating.
- 4 Set the power supply to 6 V and turn it on. Dip the object and the copper electrode into the copper sulfate solution. Leave them there for about a minute.

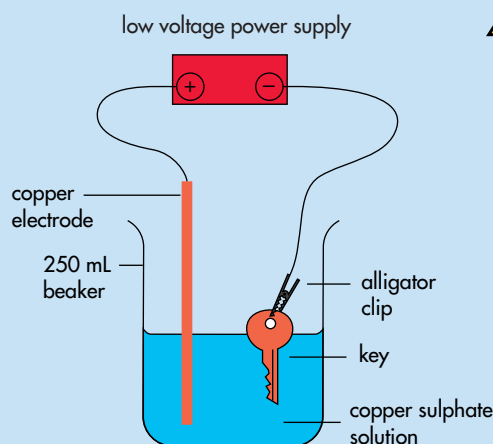


FIGURE 4.35

Experimental set-up to investigate chemical reactions produced by electricity.

DISCUSSION

What do you notice happening to the object? Describe any visible changes.

EXTENSION

Weigh the copper electrode as accurately as you can before and after the experiment. What do you notice?

Questions

- 1 Complete the following statement.

An electric jug heats water by converting ____ energy into ____ energy.

- 2 Name five things that use electricity to produce heat energy.
- 3 List some everyday items that are electroplated.

Find out

The invention of the battery in 1800 made it possible to use an electric current to break down compounds and discover many new elements. Find out which elements were discovered by Humphry Davy using electrolysis.



Web page



Internet assignment

Buying and using electricity

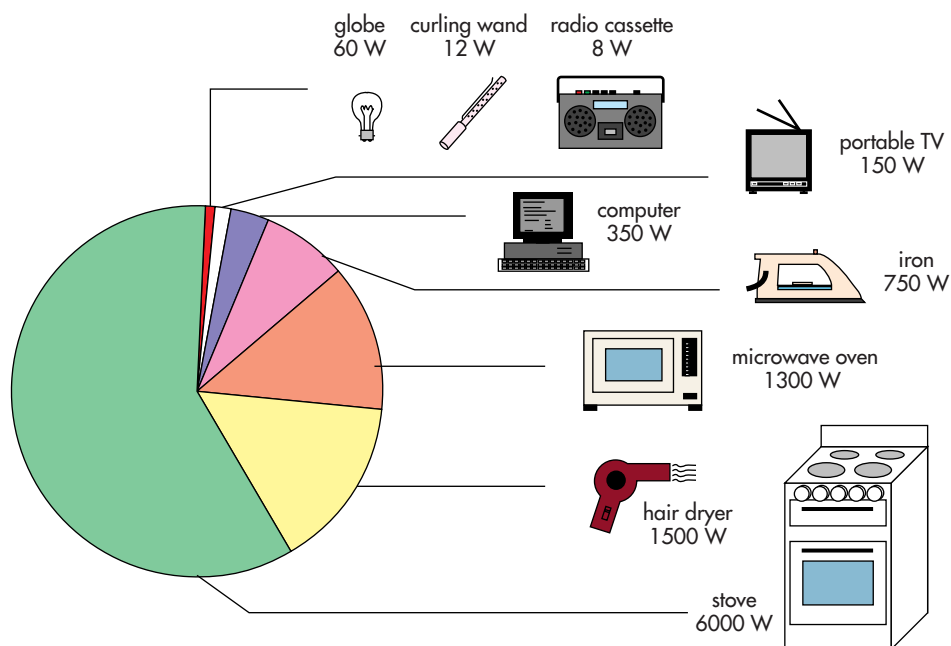


FIGURE 4.36

The sector graph shows the relative power consumption of some common household appliances.

When you buy an electrical appliance, the **power** it uses will be written on it somewhere. The power tells us how much electrical energy the appliance uses each second.

Power is measured in **watts (W)** or **kilowatts (kW)**:

$$1000 \text{ W} = 1 \text{ kW}$$

FIGURE 4.37

A power consumption label. How much power does this appliance use?



Every time you use an appliance, the electrical energy it uses is registered on an **electricity meter**. The electricity supply authority regularly sends somebody to read your meter and then sends you a bill for the electricity you have used. Electricity is sold in **kilowatt hours**, sometimes called **units**.

Number of kWh used =

power in kW $\times$ number of hours used

Examples

- A 1 kW (1000 W) heater operating for 1 hour would use 1 kWh of electrical energy.
- A 6 kW stove operating for 1 hour would use 6 kWh of energy.
- A 100 W (0.1 kW) light globe operating for 10 hours would use 1 kWh of energy.

Remember, the more kilowatt hours used, the bigger the electricity bill!

Questions



- What does the term 'power' mean in relation to electricity?
 - What unit is used to measure it?
- Electricity is sold in units of electrical energy. How are these units calculated?
 - Which would use more units of electrical energy:
 - a 5 kW microwave oven used for 1 hour?
 - a 3 kW microwave oven used for 1 hour?
 - If 1 kWh (1 unit) of electrical energy costs 12 cents, how much would it cost to run a 2 kW heater for 2 hours?

Activities



- Choose six electrical appliances that you use regularly.

For each appliance:

- find out its power rating
- estimate how many hours, on average, it would be used each day
- calculate the total number of



FIGURE 4.38

An electricity meter.

kilowatt hours it uses

- calculate the cost per day of using it (Hint: look at an electricity bill to find out cost per unit.)
- suggest how you could reduce the cost of using it.

Do not examine any electrical appliance unless it is switched off and unplugged.

2a Write a detailed 'energy diary' for yesterday, listing the things you did and the electrical appliances you used. Don't forget to include the lights, TV, radio and hot water service (if it's electric), as well as items that run on batteries.

- Now rewrite your diary assuming there was no electricity of any kind. In what ways would your lifestyle be different?

Find out



Interview a grandparent or other elderly person about life when they were young. What would their typical 'energy diary' have been like?



In all electric households the largest single electricity cost comes from heating water. Hot water accounts for 35% of the total energy bill each year. So any strategy which saves hot water also saves money. Fixing dripping taps, washing clothes in cold water, fitting a water-saving shower head and turning down the thermostat setting on the hot water system, will all help to reduce electricity bills. However, the best way to save money is to install a solar hot water system.



Extension
worksheet

IMPLICATIONS FOR SOCIETY AND THE ENVIRONMENT

Coal or nuclear generation?

At the start of the 21st century, fossil fuels such as natural gas, oil and coal are being used in greater quantities than at any time in the past. The world's accessible oil reserves are rapidly being used; however, geologists agree that there are still enormous reserves of coal around the

world. Some scientists estimate that there is enough coal to meet our needs for the next 200 years. Others argue that alternative means of large-scale electricity generation, such as nuclear reactors, should be established now in order to conserve fossil fuels for future generations.

The case for coal-fired generation



FIGURE 4.39
A coal-fired power station.

Coal has been used for hundreds of years as a source of energy. It was the fuel that powered the Industrial Revolution of the 18th and 19th centuries, and launched the 'electric era' of the 20th century.

Until the 1960s coal was the single most important source of the world's

primary energy. In the late 1960s it was overtaken by oil, but by the year 2010 it is expected that coal will once again be the main source of primary energy. Of all the coal consumed, 60% is used in electricity generation, 25% in making steel, and 15% for other purposes.

As the world's population grows, the demand for energy is increasing. Coal is the only abundant form of fuel that can meet this demand safely and economically. Other fuels will become scarcer and more expensive. Current known oil reserves, for example, will run out within 50 years. There are sufficient reserves of coal for it to remain the major fuel for the next 200 years.

New technology means that coal will be used more efficiently in the future. The proportion of greenhouse gases generated by coal-fired power stations will drop steadily.

Based on an article published by the
World Coal Institute.

The case for nuclear generation

Nuclear power generation is a fact of life in many industrialised countries. Australia, however, has so far limited its involvement to the mining of uranium, the raw material used to make the fuel for nuclear reactors.

There are three main reasons for increasing the use of nuclear power.

- It will help to conserve diminishing reserves of fossil fuels, especially oil.
- It has fewer adverse effects on public health and safety than coal. It also has less environmental impact.
- In many situations it is more economical than the alternatives.

Using uranium as the fuel

About a barrowful of uranium ore is needed to produce a spoonful of 'yellow-cake'. Overseas this is processed to yield 20 g of enriched uranium fuel which remains in a nuclear reactor for about 3 years.

When the fuel is removed and reprocessed, about 20 mL of high-level liquid waste remains. This can then be processed into about 1 cubic centimetre of Pyrex glass, although it is still highly radioactive. Other wastes are also produced, but they are much less significant.



Using coal as the fuel

About 2 tonnes of black coal (or 6 tonnes of brown coal) are needed to generate the same amount of electricity. The ash waste produced can vary from a couple of barrow-loads to half a tonne, depending on the type of coal. Burning the coal produces 6 tonnes of carbon dioxide, the gas that is the main cause of the greenhouse effect.

Based on an article published by the Australian Mining Industry Council.

FIGURE 4.40

A nuclear power station.

Activities



Imagine that you lived in a part of Australia that had a very bad unemployment problem. PHB, a major company, shows interest in establishing a large factory in the region. The factory would require a new power station to be built close by.

1 Set up a class 'court of inquiry' to make a recommendation on the type of power station to be built. Choose class members to become 'expert witnesses'. It will be their job to present the cases for and against the

various alternatives. The rest of the class should be divided equally to form the 'research team' for each presenter.

2 Compose a letter to the editor of your local newspaper expressing your concerns regarding the building of a power station or expressing your hopes for more local jobs.

3 Using a video camera or tape recorder, interview members of your class who support different types of power stations. Think carefully about the questions you ask. Edit the tape to provide a balanced view.





Using scientific language

- Define each of the following words or phrases:
 battery
 insulator
 electric current
 magnetic field
 parallel circuit
 static electricity
 conductor
 electric circuit
 electromagnet
 negative charge
 series circuit
 voltmeter
 volt
- Now construct a crossword puzzle using the definitions as the clues. Give your puzzle to a friend to complete.



Check your knowledge

- What is an electric current?
- How do electrons behave in
 a insulators?
 b conductors?
- Give the two uses of conductors of electricity and two uses of insulators of electricity.
- a Explain how lightning occurs.
 b What is a 'lightning strike'?
- Why do all the light globes stop working when a single globe is unscrewed from a set of Christmas tree lights?
- Label the following circuits to show whether they are in series or in parallel.

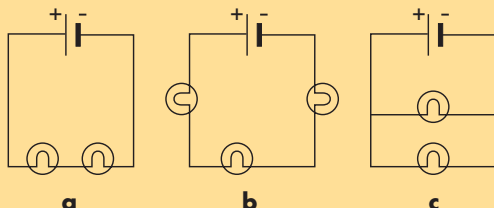


FIGURE 4.41
Three circuit diagrams.



Apply your skills

- You have two objects and you know that one of them is negatively charged. How can you find out whether the other object is negatively charged, positively charged or neutral?
- Draw a circuit in which:
 a a light is switched on when a cupboard door opens
 b an alarm sounds when a child falls out of its pram.
- A colour television has a power rating of 200 W.
 a Calculate how much energy the television uses in 3 hours and 45 minutes of use.
 b If electricity costs 12 cents per kilowatt hour, how much does it cost

to run the television for this amount of time?

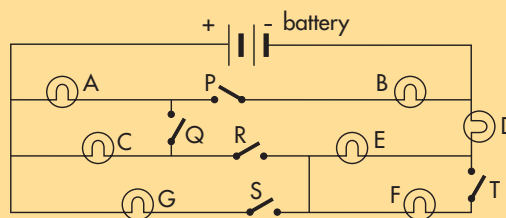


FIGURE 4.42
Circuit diagram.

- The diagram above shows an electric circuit consisting of globes and switches connected to a battery. Which switch or switches must be closed (allowing the current to flow) to make the following globes light?
 a A and B
 b D, E and G
 c A, C, D and F
- Name six ways that someone who heats water with electricity could use less energy.
- A fuse is needed to protect several appliances wired into the same circuit. Should the fuse be wired in series or in parallel with the rest of the circuit? Why?

Challenge yourself



- On a recent TV commercial a manufacturer of batteries claimed that their products last 'up to four times longer than other batteries'. Devise a way of testing their claim. Think about the different uses of a battery: low use, such as in a torch; and high use, such as in a cassette player. Present your findings clearly in the form of a poster.
- The following scientists each made great contributions to our present knowledge and use of electricity: Thomas Edison, André Marie Ampère, Charles Augustin de Coulomb, Michael Faraday, Luigi Galvani, Joseph Henry, Nikola Tesla, Alessandro Volta. Create a timeline that shows each scientist's contribution.
- Design and construct a rain-detecting device. It should consist of an electric circuit that causes an alarm to sound when rain starts to fall.
 a How does your rain-detector compare with those of other students of the class?
 b Suggest three uses for your rain-detector.

CHAPTER

5

LIGHT AND SOUND

Technologies which use either light or sound energy have had a great impact in medicine.

Surgeons use an instrument called an endoscope when performing operations. It allows them to see the body's internal structures through a very small opening or incision. A long probe fitted with a bright light and a tiny camera is inserted through the incision. This is connected to a viewing screen which magnifies the image. The light travels along an optical fibre which allows the surgeon to 'see around corners' because the light is reflected internally within the fibre.

Sound waves can also be used for diagnosis in medicine. Ultrasound uses sound waves reflected from body tissue to monitor organs such as the heart or the development of a foetus during pregnancy.

Both endoscopy and ultrasound rely on the reflection of wave energy.

In this chapter, you will investigate some properties of sound and light and find out more about technologies which use either light or sound energy.



Web links

CHAPTER OUTCOMES

Successful completion of this chapter will allow you to:

- Compare the speeds of sound and light.
- Describe the way sound and light are reflected.
- Discuss the uses of sonar and ultrasound.
- Describe how the frequency of light relates to its colour.
- Relate the pitch and loudness of a sound to its wave form.
- Compare sound and light waves.
- Explain how the eye works.
- Discuss how ideas about light and sound have changed.
- Identify causes of concern with noise pollution.

The speeds of sound and light

FACT

FILE

About 350 years ago an Italian scientist called Galileo became the first person to try to measure the speed of light. He took a lantern to the top of a hill at night while a helper took a lantern to the top of another hill a few kilometres away. Galileo shone his lantern towards his helper, and as soon as he saw Galileo's lantern the helper shone his own lantern back. By counting how many times his heart beat between shining his lantern and seeing his friend's lantern shine back, Galileo tried to estimate how long it had taken for the light to travel from one hill to the other and back again.

Light and sound travel at different speeds. If you have ever been to a sports game in a big arena you might have noticed a delay between seeing the umpire blow a whistle and hearing it. Or you might have seen someone hit a ball with a bat and heard the sound shortly afterwards. This happens because sound travels slower than light.

When you are close to the place where a noise is being made you don't notice the delay between seeing the cause of the noise and hearing the noise itself. When the light and sound have to travel much further (such as from one side of an oval to the other), the light gets to you before the sound does.



FIGURE 5.1

You see the smoke a fraction of a second after the pistol goes off, but you hear the sound nearly a whole second later.



FIGURE 5.2

Galileo's experiment to measure the speed of light.

The speed of sound

Sound travels about 330 metres in a second. For example, if you were 660 metres away from the source of a noise, the noise would take 2 seconds to get to you.

The speed of light

Light travels nearly 1 million times faster than sound: it travels at 300 000 000 m/s (300 000 km/s). In fact, scientists believe that light is the fastest thing in the Universe.

Questions



- 1** When you hear a plane flying overhead and look up to see it, why is it never exactly at the point where the sound seems to be coming from?
- 2a** If a bullet is shot towards a target at a speed of 300 m/s, would the sound of the gun reach the target before the bullet? Why?

EXPERIMENT 1**FLASHING MESSAGES****AIM**

To repeat Galileo's experiment (see Fact File) to estimate the speed of light.

MATERIALS

- long tape measure or trundle wheel
- stopwatch
- paper and pen
- 2 torches with flash buttons

METHOD

This experiment is best done at night or late on a winter's afternoon. You will need to work in pairs.

1 Find a suitable place to carry out the experiment (such as on the school oval). Mark the places where you and your partner will stand, and measure the distance between them. Try to make it a distance that will divide exactly into a kilometre (such as 100 m, 200 m, 500 m).

2 Work out a hand signal so that you and your partner know when to start each part of the experiment.

3 You will need to time the interval between you shining your torch at your partner and seeing your partner's torch shine back. Try this a few times to get used to using the stopwatch.

4 Carry out the experiment and record your results.

5 Repeat the experiment, but this time measure how long it takes the light to go from you to your partner, back to you, back to your partner, and back to you a second time.

6 Measure how long it takes for the signal to go between you both two, four, six, eight and ten times. Make sure you record the times.

DISCUSSION

1 Draw a picture of your experimental set-up and explain what you did.

2 Present your results in a table. For each trial, work out how long it took the signal to cross the oval once. How many times would the signal have to cross the oval to travel 1 km (1000 m)? How long would this take?

3 Light travels 1 kilometre in 0.000003 seconds. How does this compare with your results? Can you explain why your results might be different?

4 What can you conclude about measuring the speed of light?

EXTENSION

Repeat the experiment above by using a loud noise as the signal instead of light. How do the results compare with the actual speed of sound?

b Would the flash of light from the gun reach the target before the bullet? Why?

3 How fast would a plane have to be going to break the sound barrier (speed of sound)? How many kilometres would this be in 1 minute? How many kilometres would it travel in an hour?

4 How many kilometres per hour would you travel if you were going at the speed of light?

5 It is possible to estimate the distance to a thunderstorm from the time interval between when the lightning flash is seen and the thunder clap is heard. What would the time

interval be if the storm was 1 km away from you?

Think about

1 Distances in the Universe are so large that one of the units used to measure distance is the light-year. It is defined as the distance that light travels in a year. The closest star to our solar system is Alpha Centauri. It is 4.3 light-years away, i.e. light takes 4.3 years to reach us from Alpha Centauri. How far away is Alpha Centauri in kilometres?

2 Why is it true to say that when we look at the stars in the sky at night, we are looking into the past?

FACT**FILE**

The Sun is so far away that if it suddenly stopped shining, it would take 8 minutes for the last rays of light to reach the Earth—we wouldn't know until 8 minutes after it had happened.

Light reflects



FIGURE 5.3

For as long as people have been walking on the Earth they must have been looking at their reflections in water, polished metal or mirrors.



Over 2000 years ago, the famous Greek philosopher Euclid wrote 'light travels in straight lines called rays'.

Early thinkers such as Euclid created a scientific model to try and explain what they observed. They supposed light energy moved as **rays** in straight lines. When many rays were travelling together, they were called **beams of light**.

When rays or beams of light hit smooth shiny surfaces they bounce off them. This is called **reflection**. The **law of reflection** says that the reflected rays always leave the mirror at the same angle that they hit. If we want to measure these angles, by convention we measure them from a line at right angles to the mirror. This line is called the **normal**.

The two angles have special names—the **angle of incidence** is the angle at which the rays hit the mirror and the **angle of reflection** is the angle at which the rays leave the mirror.

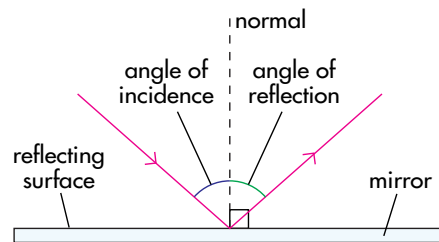


FIGURE 5.4

Reflection at a plane (flat) mirror.



FIGURE 5.5

The writing appears back to front, but does it appear upside down?

Mirrors come in many different shapes and sizes. Flat mirrors are called **plane mirrors** and are the easiest to study.

When you look into a mirror you see an **image**. Sometimes we are fooled by what we see. Look at some writing in a mirror and it looks the wrong way round. Hold your pen as if you are writing—do you appear right-handed or left-handed in the image?

The image in a plane mirror is exactly the same size as the object in front of it and is located exactly the same distance behind the mirror as the object is in front of it.

Arrange a few objects in front of a mirror and look carefully at the image to check this.

Questions

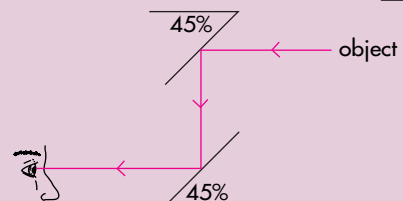


- 1 What is a plane mirror?
- 2 Copy and complete Figure 5.8 to show what happens when each of the rays or beams hits the mirrors.
- 3 What is the law of reflection?
- 4 Compile a list of everyday uses of mirrors.
- 5 Write the word **AMBULANCE** as it would appear on the front of an ambulance so a driver looking at it through a car mirror would read it 'correctly'.

USING SCIENCE 1 PERISCOPES

A periscope can be used to look over the heads of people in a crowd or to look above the water in a submarine. Arrange two plane mirrors at each end of a tube as shown in Figure 5.6. Experiment with the length of the tube and the size of the mirrors.

FIGURE 5.6
The periscope set-up.



EXPERIMENT 2

INVESTIGATING REFLECTION



AIM

To show that the angles of incidence and reflection are equal.

MATERIALS

- plane mirror
- sheet of paper
- ruler
- ray box—single slit/multiple slits
- power supply
- Plasticine
- protractor and pencil

METHOD

- 1 Connect the ray box to a power supply. Place a single slit in the ray box so that a single beam of light is produced.
- 2 By attaching small pieces of Plasticine at each end of the mirror, fix it so that it stands vertically on the sheet of paper as shown in the diagram. Draw a fine pencil line along the reflecting edge (back) of the mirror.
- 3 Move the ray box until the beam of light can be seen to move across the paper, hit the mirror and reflect.
- 4 Make small dots with a pencil along the centre of the beam before and after it strikes the mirror.
- 5 Remove the ray box and mirror. Join up the dots with a ruler. At the point the rays hit the mirror, use a

protractor to draw the normal. Measure the angles of incidence and reflection and record them in a table like the one shown.

6 Repeat for as many different angles of incidence as you can, giving each ray a number to distinguish between them.

7 Replace the single slit with the multiple one. On a separate sheet of paper trace three rays hitting the mirror. Make a similar table to record your results.

Ray number	Angle of incidence	Angle of reflection

What did you find about the angles of incidence and reflection?

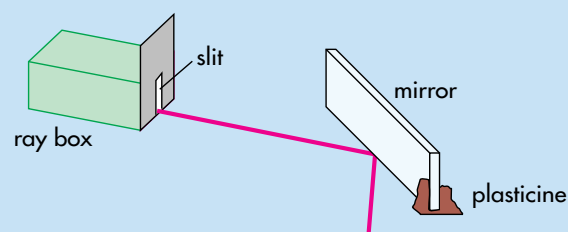


FIGURE 5.7
The experimental set-up.

Think about



I have a sticker on my car's rear window. It reads **SUPER-GLASS** when viewed from a car travelling behind. The writing also appears the correct way round when I look at it through my rear view mirror when driving. How can the writing appear correct both directly and from the view in the mirror?



Extension
worksheet

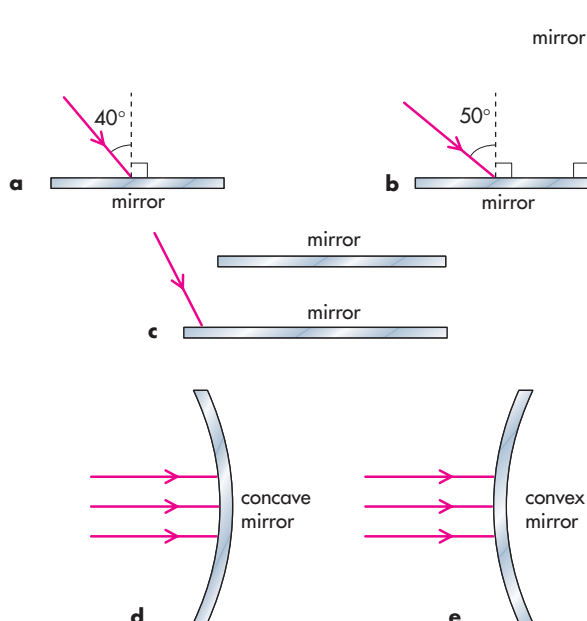


FIGURE 5.8
Rays of light hitting different mirror surfaces.

Sound reflects too!



FIGURE 5.9

The design of the Sydney Opera House Concert Hall takes acoustics into consideration. The dimensions of the hall and materials used were chosen to give a balance between absorption and reflection of sound.

FACT

The energy of sound is very small. It has been estimated that if the energy of sound was converted into heat energy it would take over 1 million people talking for over an hour to warm a cup of water to 50°C.

FILE

Sound energy, like light energy, can be made to reflect. Just as we need a shiny surface to reflect light, we need a smooth hard surface to reflect sound. A person standing close to a large brick wall may be confused as to where a particular noise is coming from because of reflections off the wall.

Echoes

An **echo** is hearing a sound for a second time because the sound is reflected. Because sound takes a second to travel about 300 m, shouting loudly some distance in front of a cliff or large building will produce an echo seconds later. The echoes produced by a clap of thunder can often be heard many seconds after the first sound is heard.

A small amount of echo can enrich the sound of someone singing or speaking. On

the other hand, if there are too many echoes, the sound becomes fuzzy and confusing. When engineers design large concert halls they need to think very carefully about echoes. The walls need to absorb most of the sound that hits them but reflect just enough to give a pleasant enrichment to the music.

This reflection of sound in the room is called **reverberation**. Look at the picture of the Sydney Opera House Concert Hall. The reverberation time for the Concert Hall is about 2.6 seconds, which is ideal for symphonic music. The acrylic rings suspended above the stage platform are designed to help the musicians on stage. The rings reflect sounds from the musical instruments back onto the stage so that the musicians can hear each other playing. The wall coverings tend to prevent echoes by absorbing the sound energy.

USING SCIENCE 2

ECHOES

Stand about 50 m away from a large building. Hit two pieces of wood together and wait for the echo. Move closer or further away and see if the time before hearing the echo increases or decreases. If you are very clever, you can clap the wood together in time with the echoes. A friend can then time how long each echo takes and if you know how far from the wall you are standing, you can estimate the speed that the sound is travelling!

EXPERIMENT 3**INVESTIGATING THE REFLECTION OF SOUND****AIM**

To find out how sound is reflected by a sheet of metal.

MATERIALS

- sheet of metal (aluminium or steel) about 200 cm × 100 cm
- 2 cardboard tubes approximately 100 cm long and 10 cm in diameter
- a loud ticking watch or alarm clock (not a digital one!)
- 2 retort stands and clamps
- masking tape
- sheet of paper, pencil, ruler, protractor etc.

METHOD

- 1 Use masking tape to attach the clock or watch to one end of the tube.
- 2 Clamp the sheet of metal between two retort stands so that it forms a vertical barrier.
- 3 Lay the tube with its open end pointing towards the sheet of metal.
- 4 Point the second tube towards the metal sheet.

Place your ear close to the end of the tube and turn the tube until you can hear the ticking of the clock.

5 Mark the positions of the tubes on the sheet of paper when the sound of ticking can be heard the loudest.

6 By continuing the lines on the paper, work out where and at what angles the sound reflects.

7 Repeat the experiment for several different angles.

8 Do your results agree with what you found for light?

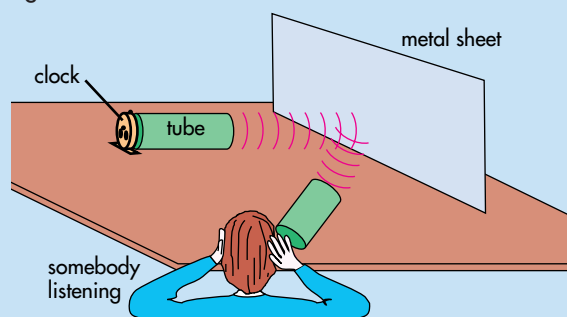


FIGURE 5.10
The experimental set-up.

Questions

- 1 Sound travels about 340 metres every second. Mario fires a starting pistol in front of a large brick wall. He hears the echo exactly 1 second later. How far is he from the wall? (Be careful . . . how far must the sound travel?)
- 2 What is an echo? Does sound reflect off flat surfaces in a similar manner to light?
- 3 Imagine you are designing a new hall for your school. It is to be used for assemblies and school concerts. Draw a plan of the hall and explain your choice of materials for the furnishings.
- 4 Steve Waugh hits a great six at the SCG. Charlotte first hears the

sound of the bat striking the ball a little after she sees the bat and ball make contact. She then hears the noise again, not quite as loud, a little time later. Try and explain what is happening to the sound energy. Use some or all of the following words: speed of sound, echo, absorb, reflect, speed of light.

Think about

Soundproofing materials are required to absorb sound, not reflect it. What type of materials are used in soundproofing houses? Design an experiment to test the effectiveness of some common soundproofing materials.

APPLICATIONS AND USES OF SCIENCE

Uses of sound

FIGURE 5.11

Sound can be fatal! Dolphins and many other marine creatures use sound to stun their prey before capturing and eating it.

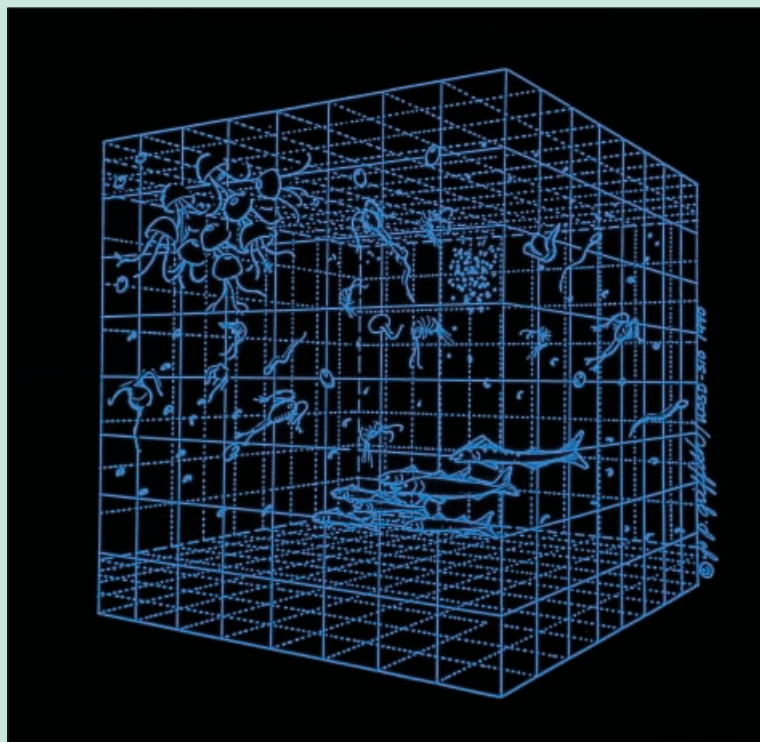


FIGURE 5.12

Sonar involves using echoes to estimate how far away things are. This illustration shows the sort of information that will be produced by a sonar system called 'Fish-TV'. It is being developed to study marine life under the sea.

Sound can be used for many purposes other than for talking and making music. Many animals have a sense of hearing that is much more sensitive than that of humans. Dogs are often used to guard people's homes because they can hear the slightest sound of an intruder. Bats use sound echoing off things around them to find their way while they are flying. Dolphins can use intense bursts of sound to stun fish they are hunting.

People have built machines that use sound in surprising ways. For example, it is common for fishing fleets to use **sonar** equipment to find schools of fish. This is similar to the way that bats use sound. The sonar unit sends out a loud sound from a speaker under the water. The sound travels through the water and is reflected back by whatever it hits, just as sound echoes off a building or a cliff when you shout. (Movies about submarines often have 'ping' noises in the background—this is the sound of the sonar.) Scientists also use sonar to find fossils or oil under the ground. Instead of a loudspeaker they use an explosion to cause a loud noise that travels through the earth.

Sonar is also used to allow doctors to 'look' at unborn babies. Sound of a frequency too high to be heard is made to travel through the mother's abdomen. The sound bounces back off the baby and is converted into a picture by a computer. This procedure is called an **ultrasound**.

In the future refrigerators may be made without any gases or compressors; instead, sound generators will cool the food. New engines are being developed that run on water, with sound being used to turn the water into a very fine mist. These are just some of the unusual uses that are being found for sound.

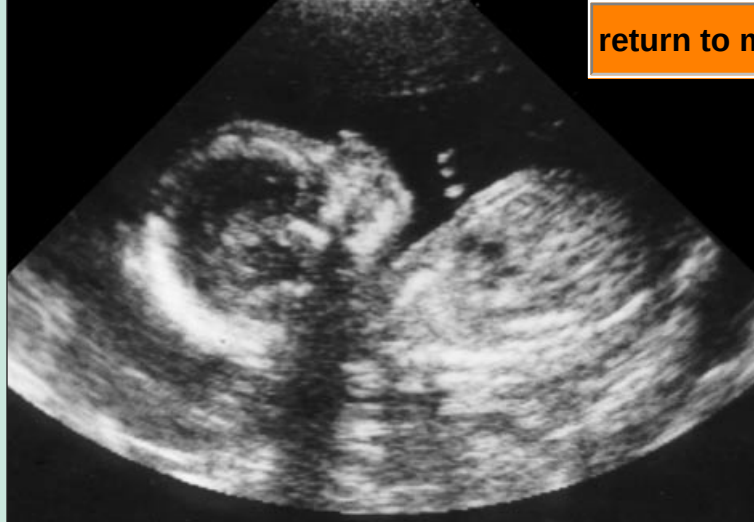


FIGURE 5.13

Ultrasounds are often used in early pregnancy to detect the sex of the baby and any physical deformities.

Questions



- 1 When blind people buy houses they often make clicking noises to find out how big a room is. What do you think they are doing?
- 2 How can you tell if a bottle is nearly full of water without looking at it?
- 3 List five ways you could use sounds that you've never tried before.
- 4 Why do you think some animals have better hearing than humans?
- 5 Why do you think dolphins would use sound to stun fish instead of catching the fish in their mouths?

Think about



Why do you think it could be extremely dangerous to drive at high speed with the car windows down?

Activity



Now is your chance to design a device that uses sound in a surprising way.

- a Draw up a plan of a mousetrap that is set off by a loud shout. You should be able to trigger the trap to close on the mouse by shouting when you see the mouse in the right place.
 - b Once you come up with a design, improve it so that the mouse will trigger the loud noise itself.
- Hint: remember that a loud sound can make things vibrate. If you set up your mousetrap carefully you can use these small vibrations to make other things move. For example, it doesn't take a very big vibration to make a domino fall over.

Extension: Build your mousetrap.

FACT

FILE

Some very low-frequency sounds (around 1 hertz) can have a strange effect on people—they can cause feelings of dizziness and sickness. One possible source of such low-frequency sound is driving very fast with the car windows open.



FIGURE 5.14

Would this be an effective mousetrap?

Intensities and frequencies of light

FACT

Albert Einstein first proposed how 'laser light' could be produced in 1917. The first true laser was made in 1960 by using a synthetic ruby crystal.

FILE

The frequency of light

Frequency means how often something happens. For things that happen very quickly, frequency is measured in **hertz**, which means 'number of times each second'. If you can blink your eyes very quickly, say 10 times a second, the frequency of your eye blinking is 10 hertz.

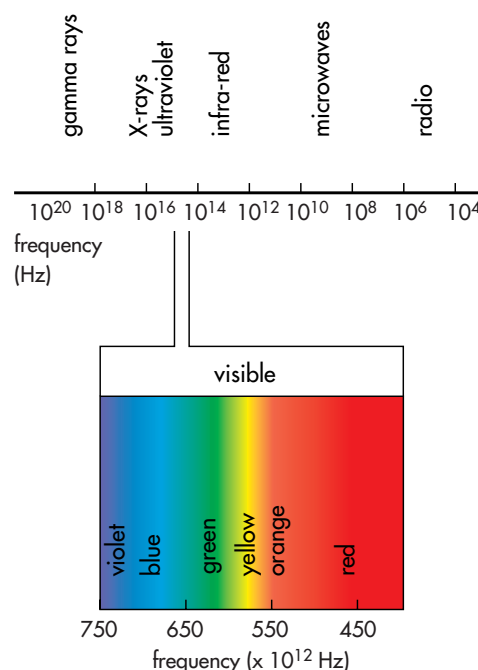
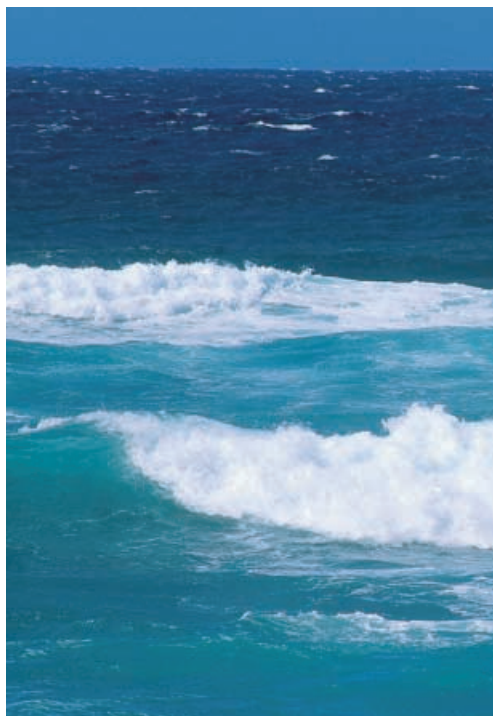
Scientists think that light is something like a wave, or small bits of waves. Unlike waves in the ocean, which travel through water, light doesn't need anything to travel in—it is just bits of pure energy that can travel through empty space.

The visible light that we see is only a very small part of all the radiation that comes from the Sun. The Sun also produces infrared and ultraviolet rays, X-rays, gamma rays, radio waves and microwaves. All these different types of radiation have their own specific frequencies and are grouped together to form the **electromagnetic spectrum**.

We see light of different frequencies as different colours. Violet, for example, has a higher frequency than red. The higher the frequency, the more energy the light has. For instance, ultraviolet (or UV) light, which we can't see, has much more energy than the light we can see—it can even burn our skin. A suntan or sunburn is caused by UV light from the Sun reaching our skin.

FIGURE 5.15 (right)
At the beach, the frequency of waves reaching the shore might be as high as 1 hertz in a storm. Visible light has frequencies of around 100 000 000 000 000 hertz, which means that there are about 100 million million waves of light hitting your eye every second.

FIGURE 5.16 (far right)
Visible light is only a small part of the electromagnetic spectrum.



USING SCIENCE 3

LASERS



Caution: Laser light should not be placed anywhere where it could shine into eyes. Never look directly into a laser beam.

Your teacher will set up a laser at the front of the classroom so that it is pointing towards the front wall.

How can the pathway of the laser beam be tracked through the air? What property of light is being used to see the beam through the air? How does a beam of laser light compare to a beam of light which comes from a film projector?

The intensity of light

The frequency of light determines its colour, but the **intensity** of light determines how bright it is. A light globe that has a high intensity is very bright: it gives off a lot of light. A light globe that has a low intensity is quite dull: it doesn't give off very much light.

The light from lasers

Talk about

Brainstorm the variety of ways that lasers can be used today. You should be able to think of examples of uses in communication, entertainment, medicine, industry and the military.

The word 'laser' is an acronym for Light Amplification by Stimulated Emission of Radiation. The light produced by a laser is special because the waves are lined up precisely, crest to crest and trough to trough, and they all have the same frequency (or colour). The beam can be so intense that it can cut through tissue or metal. Solids, liquids and gases can be used inside the laser as the medium that emits the light energy. The frequency of laser light produced depends on the nature of this medium.

Questions



1 How could you tell the difference between light of different frequencies?



FIGURE 5.17

Supermarket check-outs can process items very quickly if a laser is used to scan barcodes.

- 2** When you watch a video, the moving picture you see is actually a whole series of still pictures flashed onto the screen at the rate of 26 hertz. How many pictures are flashed every minute?
- 3** Which has a higher frequency: red light or yellow light?
- 4** Which is more intense, the Moon or the Sun?
- 5** What causes you to get a suntan?
- 6** Lasers are used to scan the barcodes on products at supermarket checkouts. Is it safe to use a laser where it could hit somebody's eyes? Explain your answer.



Intensities and frequencies of sound



Sound, like light, has different frequencies and intensities. With sound, changes in frequency change the **pitch** of the sound, ranging from low-frequency deep sounds (like trucks rumbling along a road, or earthquakes) to high-frequency sounds like

the whining of a jet engine or a dentist's drill.

We usually hear different intensities of sound as different degrees of loudness. High-intensity sounds are very loud, while low intensity sounds are often quiet.

EXPERIMENT 4

SEEING SOUND

AIM

To convert sound frequencies and intensities into something you can see by building a 'sonorscope'.

MATERIALS

- tin can with both ends cut off
- balloon that has been burst
- rubber band
- small pieces of mirror
- glue
- torch or sunlight
- tuning forks

METHOD

- 1 Stretch the broken balloon over one end of the tin can and hold it in place with a rubber band.

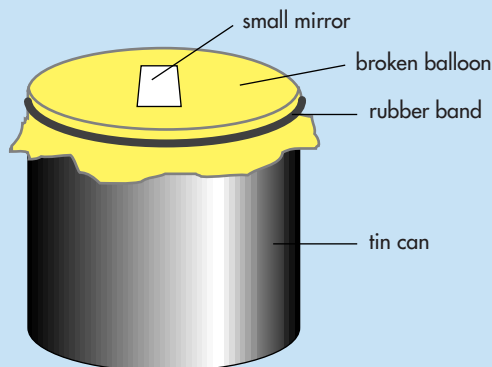


FIGURE 5.18

Your sonorscope should look like this.

- 2 Glue the small piece of mirror into the middle of the stretched balloon (on the outside). This is your sonorscope.

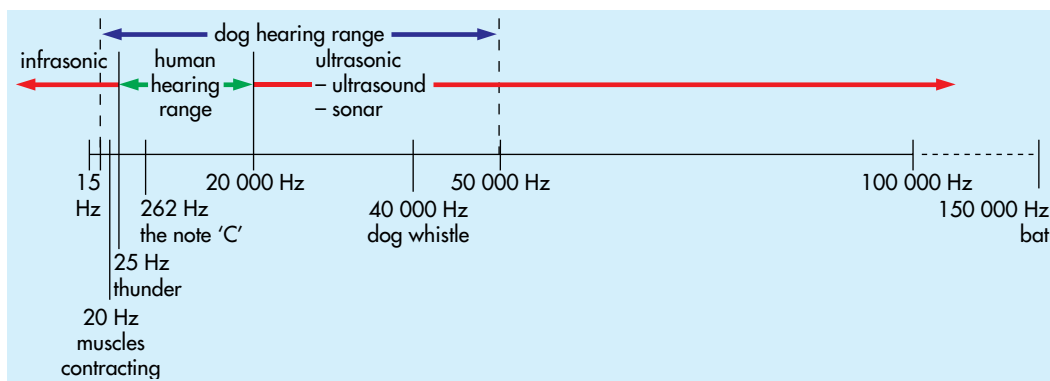
- 3 Hold the can with the open end near your mouth and use the torch or sunlight to reflect a spot of light from the piece of mirror onto a wall. Now speak into the can.

- 4 Observe what happens to the spot of light when you speak loudly and softly into the can. Try shouting.

- 5 Now use the tuning forks to see what effect they have. Holding the tuning fork by its base, hit one of the prongs against a table. Then press the base against the inside of the can.

DISCUSSION

- 1 What effect did the intensity of the sound have on the movement of the spot of light?
- 2 Tuning forks don't produce a high intensity of sound, but they have a constant frequency which is written on the stem in hertz. What difference do you notice between the pattern formed by your voice when talking quietly and the pattern made by a tuning fork? What difference in patterns do you notice between different tuning forks?
- 3 How do you think frequency affects the pattern?


FIGURE 5.19

Just as there is a light spectrum, there is also a sound spectrum.

Many things can make sounds. For instance, if you hit a wine glass it will ring at a particular frequency, and will always ring at that frequency no matter how hard you hit it. Hitting it harder, though, will produce a louder sound, or greater intensity of sound.

Questions



- 1 How does frequency influence sound?
- 2 How could you tell a high-intensity sound from a low one? Give four examples.
- 3 How does the sonoscope work?
- 4 How could you protect your wine glasses from opera singers on the loose?
- 5 How are voices and tuning forks different?
- 6 Look at a tuning fork. How many times do the prongs vibrate every second?

Think about



What would it be like to taste sounds? What would be your 'flavourite' music? What would it taste like? If you could hear your favourite food, what would it sound like?

Activity



Anything that vibrates can make sounds. Music is simply a mixture of different frequencies and intensities of sound. Experiment with sounds by using the following ideas.


FIGURE 5.20

If you hit a glass too hard it will break. You can imagine the sound it would have made if it hadn't broken: it would have been the same frequency, but just quite a bit louder than usual. So if you can play a loud sound of the right frequency, the glass will vibrate as if it were being hit. If you made the sound loud enough, the glass will vibrate as if it had been hit very hard, and could even shatter!

FACT

FILE

Some people have a rare condition that makes them experience sounds as tastes or feelings! For instance, they might like a certain song because it tastes good, or because it feels like a massage. Scientists think this condition is caused by confusion between certain nerves in the brain, but they don't totally understand it.

- Stretch strings across an ice-cream container to make a simple guitar.
- Fill bottles with different amounts of water so they make different sounds when you tap them with a piece of wood.
- Make a flute by blowing across the necks of bottles with different amounts of water in them.
- Make different kinds of drums from ice-cream containers filled with sand or pebbles.
- Work in groups to make an orchestra so that you can perform a song for your class.



How we see with our eyes

FIGURE 5.21
The eye of the beholder!

We carry in our heads two wonderfully precise optical instruments. They are capable of giving clear images both in the full sun and under a dark star-lit sky. They can focus automatically from mountains on the

horizon to grains of dirt on the palm of our hands.

The human eye is roughly spherical in shape with a diameter of about 2.5 cm. Each is set deeply within the skull to protect it from external damage.

USING SCIENCE 4

EYE DISSECTION

WHAT YOU NEED

Fresh eye—pigs' eyes are best but cows' eyes will do. Make sure they have not been frozen as this can destroy the inside structure. A sharp pair of scissors and a dissecting tray with plenty of paper towels is also necessary.

WHAT YOU DO

Carefully remove any fat or muscle from the outside of the eye. Observe its shape carefully, particularly the cornea. Locate the optic nerve coming out from the rear of the eye. It is white in colour and a few millimetres in diameter.

Now make a circular cut with the scissors around the outside edge of the cornea and carefully remove the front half of the eye. You will now be able to see the iris and be able to remove the lens. Check the shape of this lens. You will have seen that some liquid has spilled out. Between the cornea and the lens is a very watery substance and there is a thicker jelly-like substance between the lens and the retina.

Identify the retina and the blind spot—the point at which the optic nerve enters the eye.

QUESTIONS

- 1 What type of lens did you find in the eye?
- 2 Holding the lens carefully with tweezers, see how well it will magnify. Repeat, this time holding the lens in a beaker of water. Does the amount of magnification change, and if so, how?

- 3 Remembering what substances surround the lens in the eye, will the lens in an eye refract the rays a great deal or very little?



FIGURE 5.22
A cow's eye carefully dissected—identify all the parts!

EXTENSION

If you have access to another eye to dissect, carefully remove a small portion of the rear part. Without spilling any of the liquid, place a piece of tracing paper over the hole you made. If you take the eye to a window or point it at a bright light, you should be able to see an image formed on the tracing paper. Is the image the correct way up or upside down?

When we say someone has brown or blue eyes we are referring to the colour of the **iris**, a set of delicate muscles that controls the amount of light entering the eye. The 'black' **pupil** is just a hole.

You may notice that it is difficult to judge the depth of water in a pool because of a special property of light called **refraction**. Light rays bend (or **refract**) when they pass into a medium of different density. For this reason, water appears more shallow than it really is.

A similar situation exists with our eyes. Light is focused in our eyes because of **refraction**. As light moves from the air into our eye, it enters material of different density which causes it to bend.

Light enters the eye through a transparent curved window at the front, called the **cornea**. Light rays are refracted strongly at the surface of the cornea. As the rays continue to pass through the eye they are refracted a little more by a convex lens until they form an upside-down image on the light-sensitive back surface known as the **retina**. Here there are millions of tiny cells which respond to light energy by sending tiny electrical signals to the brain via the **optic nerve**.

Questions



- 1 Where does most of the refraction take place in the eye?
- 2 Explain why it is an important adaptation that the pupil of your eye grows very large under conditions of low light.
- 3 Explain why the dot disappears when you perform the blind-spot experiment (see Activity).

Think about



When you open your eyes under water, objects appear 'fuzzy' and indistinct. However, using goggles allows you to see clearly. Why might this be so? The explanation has

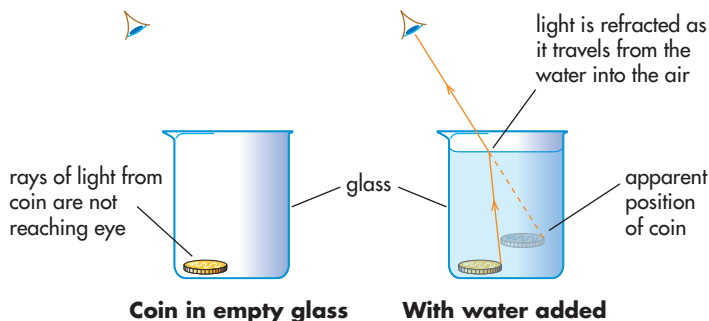
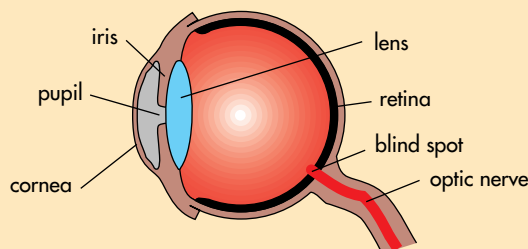


FIGURE 5.23

Try this experiment. When water is added to the glass, light rays travelling from the coin are refracted and can be seen by the observer. The coin comes into view and the water does not seem to be as deep.



cornea – the clear window on the front of the eye. Almost 90% of the bending takes place as light first enters the eye!

iris – this coloured muscular tissue contracts and relaxes according to how much light is present

pupil – the hole in the centre. Is large when there is little light and small when there is plenty

lens – a clear elastic piece of tissue – it puts the 'finishing touches' to the image by changing its shape slightly so that rays from all objects are brought to a clear image. This is called accommodation. When you look at something very close you can feel the lens being squashed!

retina – coats the back, inside surface of the eye. It is covered with millions of cells of two types, rods and cones, so called because of their shape

optic nerve – passes all the information from the light-sensitive cells in the retina to the brain. Where the optic nerve joins the retina there is a 'blind spot'.

something to do with where most of the refraction takes place!

FIGURE 5.24

The eye.

Activity



Find your blind spot: Hold this page at arm's length. Close your left eye and stare at the cross on the left of Figure 5.25. Move the page slowly towards your face and at some stage you will be aware that the dot has disappeared. The image of the dot is falling on your blind spot!

FIGURE 5.25



Fred Hollows—an Australian of the Year

FIGURE 5.26
Fred Hollows.



FACT

FILE

Blindness occurs ten times more frequently in Aboriginal and Torres Strait Islander people than in other Australians. Most of this is due to unoperated cataracts and an infectious disease called trachoma which is associated with a dry arid climate and lack of water.

Fred Hollows was a very special kind of Australian—a brilliant scientist, surgeon and doctor. He was also a person who used his talents for the benefit of under-privileged people not only in Australia but also in other parts of the world. He died of cancer while still relatively young and up to his death he was intent on passing on his knowledge to others around him so that they could continue with his work. After his funeral, attended by national dignitaries, he was buried in Bourke, with grief being shared by the Aboriginal community—a group of Australians with whom he had a special affinity.

What was his work? Giving back sight to thousands of blind people—and in doing so, developing a technique that cost only \$5 for each patient.

Many millions of people around the world suffer from **cataracts** in their eyes. Some part or all of the lens in their eyes becomes cloudy and light cannot pass through to form a clear image. What Fred Hollows achieved was a simple method of removing the damaged lens and replacing it with a mass-produced plastic lens.

The Fred Hollows Eye Centre in Katmandu is typical of the work taking place in Australia and Asia. It is estimated that there are presently over 1 000 000 cases of cataract in Nepal.

Nepalese doctor Dr Sanduk keeps Hollows' dream alive by operating on 35 to 40 patients per day, seven days a week—and all for no charge! Patients travel to the Centre by foot, lie on crude stretchers, and have their eyes operated on and covered. The next day they remove the pieces of cotton wool and they can see once more!

This is the Hollows philosophy. He once said any steady-handed plumber could do it. It has to be cheap and simple—and effective!

The plastic lenses required for the operations are now being constructed in small factories throughout Asia. To subsidise the cost of lenses the factories also manufacture surgical equipment or even CDs! Fred Hollows was outspoken but devoted to helping others less fortunate than himself. He believed in solving problems rather than patching them up.

Fred Hollows spent his all-too-short life teaching others how to save sight both by performing operations and by supplying the tools and equipment necessary to do it.

USING SCIENCE 5

DISCUSSION

Fred Hollows believed in solving problems rather than patching them up. Governments are often accused of patching things up rather than tackling problems. What do you think this means? Give an example if you can. Discuss your ideas with other members of the class.

What ways could other scientific and medical discoveries and procedures be used to help the sick

and underprivileged both here and in other countries? Are there any special problems associated with certain groups of the population in Australia? (Think about inner city areas, rural and outback areas, tropical areas.) How could affordable assistance be given?

Find out

What work did Fred Hollows do here in Australia, especially for the Aboriginal community?

What has the Fred Hollows Foundation achieved since Fred's death?

Prepare a short report on your findings.

Activity

Organise a social service activity such as a casual clothes day to raise money for the Fred Hollows Foundation or some other similar type of organisation.

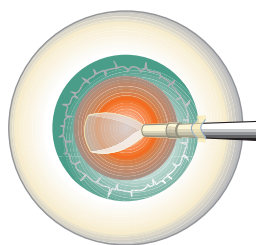
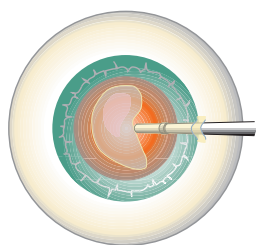


FIGURE 5.27

Cataract surgery. (top) A small incision is made near the front of the lens. Ultrasound is used to dissolve the damaged lens which is then vacuumed away. (bottom) The replacement lens is made of a soft material which can be folded to insert into the eye.



FIGURE 5.28

A patient at the Fred Hollows Eye Centre having an eye test.

HISTORY OF SCIENCE

Changing ideas

FIGURE 5.29

A microphone can change sound energy into electrical energy which can be seen as a wave pattern on a cathode ray oscilloscope.



Scientists are constantly making new discoveries, and scientific models are often modified over time. This is well illustrated by studying ideas about light and sound.

Talk about

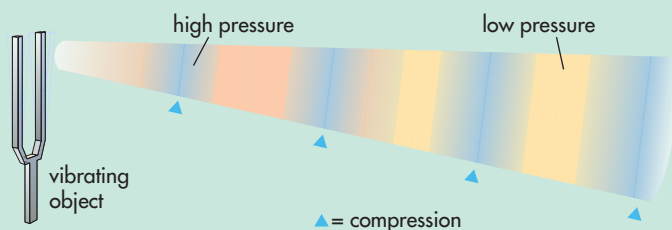
Does sound travel best through a solid, a liquid or a gas? Place your head down on a desk and tap the desk lightly with a pencil. Compare the intensity of that sound with the same tapping when you lift your head off the desk.



Scientific studies of sound

FIGURE 5.30

Vibrating ends of a tuning fork send out pulses of compressed air particles to form a compression wave.



scientist Robert Boyle demonstrated that sound needed either a solid, liquid or gas to carry it. He suspended a bell in a vacuum by a string and showed that there was no sound even though the clapper could be seen striking the bell.

Scientists began to accept the idea that sound travelled as **compression waves** with the energy being transferred from particle to particle in the medium that was transmitting it.

The invention of the microphone, telephone and phonograph in the 19th century allowed sound to be converted into electrical currents. Sound waves can now be analysed by a cathode ray oscilloscope.

Scientific studies of light

Light was far more difficult than sound to observe, measure and explain. Early Greek philosophers speculated about the nature of light but their understanding was limited because they did not know how the eye worked. Pythagoras thought that objects could be seen because they sent out particles in straight lines at high speed.

USING SCIENCE 6

SOUND WAVES AND THE CRO

Your teacher will set up a cathode ray oscilloscope with a microphone connected to it. Whistle into the microphone. Observe the wave form and compare how it changes for softer and louder sounds. Use the labels on Figure 5.31 to help describe the changes you see. Observe how the wave form changes for whistles of lower and higher pitch. Describe and summarise these changes.

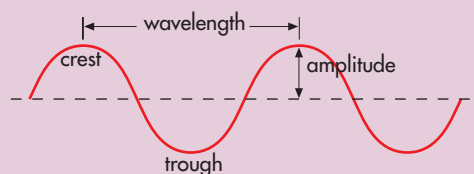


FIGURE 5.31
Parts of a wave.



Followers of Plato suggested that vision was the result of three types of particle fusing together—one stream from the object, one from the Sun and a third from the eye.

In 1665, Robert Hooke suggested that light travelled in a compression wave similar to sound. Physicists believed that light would have to travel through some medium so they suggested the existence of an invisible substance called **aether**. This would allow light to travel through space from the Sun. The Dutch astronomer and mathematician Huygens showed that reflection and refraction of light could be explained by assuming that it was a wave. Around the same time, Sir Isaac Newton believed that light consisted of a stream of particles and that both the reflection and refraction of light were explained by his model.

In the late 19th century, the work of James Maxwell, then Heinrich Hertz, established that light was an **electromagnetic wave**.

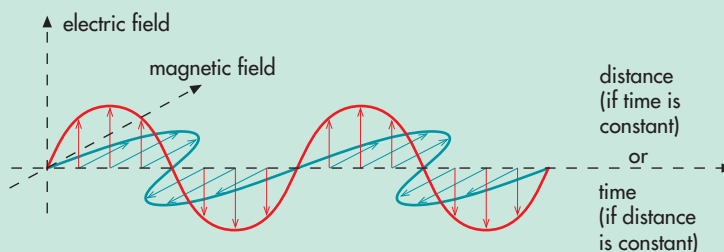
Particle nature of light

Even though light is a type of electromagnetic wave, it also has a 'particle' nature. When atoms absorb light, they absorb a fixed quantity of energy. This 'bundle of energy' can boost an electron inside the atom to a higher energy and be

Questions



- 1 Sound and light both travel in waves but they are of different type. Name each type of wave and give one difference between each type.
- 2 Draw the shape of a sound wave for a whistle as you would see it on the cathode ray oscilloscope. Underneath it draw:
 - a a louder sound of the same pitch
 - b a sound with the same loudness but higher pitch.
- 3 Until some time last century, scientists believed that light needed a special medium to allow it to travel



released when the electron drops back to a lower energy. A laser is designed to harness these **photons** (bundles) of light, resulting in an intense beam of amplified light with a uniform frequency.

Particle nature of sound

There is always something new to discover. Just as light exists in bundles of energy called photons, sound exists in bundles of energy called **phonons**. Scientists in France are currently developing a sound laser (a 'saser') which will amplify sounds. It has to work at temperatures close to absolute zero so that vibrations due to heat do not interfere. Astrophysicists have been looking for dark matter which many believe makes up most of the Universe. They are hoping that they can use the saser to make a very sensitive listening device which can pick up vibrations from atoms that make up this dark matter.

through space.

- a What was this medium called?
- b What properties would you expect it to have?
- 4 Which properties can be explained by assuming light has a wave nature?
- 5 Sound and light can both be considered to occur as packets of energy which have a particle nature. Name each type of particle.

Think about



Use the particle theory for the three states of matter to explain why sound travels best through a solid.

FIGURE 5.32
Representation of
electromagnetic wave.



Holograms give a three-dimensional photograph of an object by relying on a special property of light called interference. This is another property which can best be explained by assuming light travels in waves.

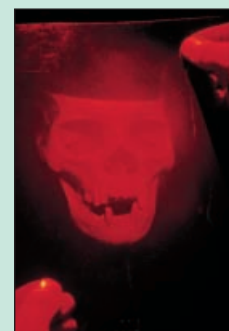


FIGURE 5.33
A hologram gives an all-round view of an object.

Noise pollution

FIGURE 5.34

Loud music may be damaging your hearing.



FIGURE 5.35

Alexander Graham Bell.

FACT

Alexander Graham Bell (1847–1922) is most famous for his invention of the telephone. The unit decibel is named after him.

FILE

Turn the radio down!

Many health experts are worried that young people are listening to music which is too loud. They are concerned that the loud music will cause either a temporary or a permanent loss of hearing. Although headphones don't produce a great deal of sound energy, the energy is close to the ear drum and is concentrated by the ear.

But listening to loud music is not the only problem faced today. Sometimes we have no choice in what we hear. We are surrounded by noise as we work and play. The noise of jet airliners, especially when they are taking off, can be more than just a nuisance. For those whose jobs are at the airport, there is a real risk of becoming

deaf. The families who live beneath the flight path may not suffer damage to their ears but the disruption to their sleep and other family activities can make their lives far from pleasant.

It is really important to be able to measure accurately how loud sounds are. Scientists measure sound levels with a special unit called a **decibel**—pronounced dess-i-bell and written as dB. The softest sound than an average ear can hear is called 0 dB. This is called the **threshold of hearing**. Various other noise levels can be measured on the decibel scale.

TABLE 5.1

Decibel level (dB)	Sound
0	quietest sound that can just be heard
20	whispering to a friend
40	a library or quiet classroom
60	ordinary talking in the laboratory or classroom
80	operating a vacuum cleaner/high street traffic
100	large orchestra playing loudly/jet plane directly overhead
120	rock band in concert
140	sound level at which the ears start hurting
160	standing close to a jet fighter before take off
180	sound level at which the eardrum splits or ruptures

USING SCIENCE 7

MEASURING SOUND



Ask your teacher for a sound level meter. This instrument will tell you how loud a particular sound is on the decibel scale. If your classroom is close to a noisy street, measure the sound levels at different times of the day. Make measurements at other places around the school. Make sure you ask permission first, as there may be other dangers present!

A safe working environment

In the workplace, it is the responsibility of the employer to create a safe and comfortable environment. Even around school you may see people wearing ear muffs for protection. What activities are likely to be happening? Wherever possible, noise levels have to be kept below 90 dB. Otherwise, ear protection has to be provided and worn. The table shows how the law limits the exposure to loud sounds.

TABLE 5.2

Sound level (dB) (1990)	Maximum daily exposure	Sound level (dB) (1997)
90	8 hours	85
93	4 hours	88
96	2 hours	91
99	1 hour	94
102	30 minutes	97
105	15 minutes	100
108	7 minutes	103
111	3 minutes	106

Breaking the sound barrier

When an aeroplane moves faster than the speed of sound, it pushes aside air particles with great force, creating a shock wave. This wave is like a bow wave which builds up at the front of a boat as it moves through water. It forms a cone of highly pressurised air particles. A **sonic boom** is heard when waves at the base of this pressurised cone, which travels along with the aeroplane, strike someone's ear. Sonic booms are so loud that they can shatter windows and burst people's eardrums.

The Concorde **supersonic** jet had its inaugural flight in 1970. It travels at 2600 km/h, which is twice the speed of sound. The jet carries only 100 passengers so high running costs, together with environmental concerns, have meant that it has not been as successful as expected. Its engines

produce nitric oxide gases which harm the ozone layer and it is only permitted to fly at supersonic speeds over the ocean because of the noise pollution caused by its sonic boom over land.

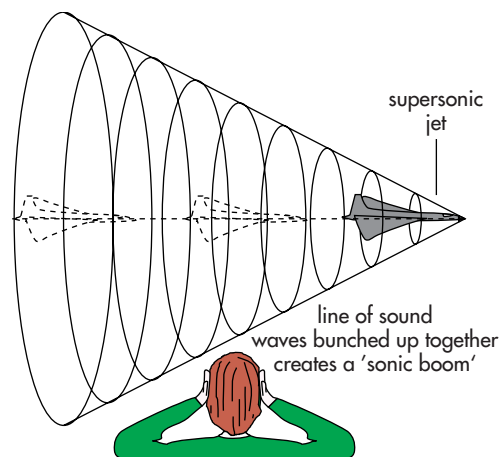


FIGURE 5.36

Shock waves moving in a cone from a supersonic jet cause huge pressure differences in air which are heard as a sonic boom.

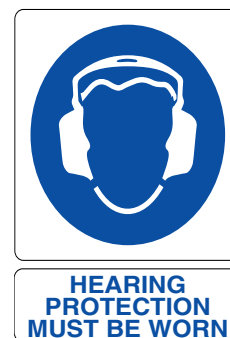


FIGURE 5.37

Hearing protection sign.

Questions

- Use the data on sound levels provided in the text to estimate:
 - the level of noise in the corridors at your school between lessons
 - the maximum daily exposure you should have to a jet plane directly overhead.
- The Concorde jet revolutionised air travel. Give reasons why this supersonic jet is only used on a very limited basis between Europe and North America.

Find out

NASA scientists are currently doing research into the design of supersonic jets. Name some of the changes they would like to make to the Concorde design.



FIGURE 5.38

Hearing protection being worn.





Using scientific language

Match each of the following words to the definitions below: laser, ultrasound, reflection, refraction, incidence, normal, echo, electromagnetic, compression, photon.

- Process needed to allow focusing of light in the eye.
- Angle at which a light ray hits a mirror.
- Produces an intense beam of light with definite frequency.
- Uses sound waves of a very high frequency.
- Name given to a packet or bundle of light energy which is absorbed by atoms.
- Used in sonar to estimate distance.
- A type of wave which needs particles to carry its energy.
- Process responsible for existence of echoes.
- X-rays and radio waves are examples of this wave type.
- Name for a line which is drawn at right angles to a mirror surface.



Check your knowledge

- List the similarities between sound and light.
 - List the differences between sound and light.
- How are the frequency and pitch of a sound related?
- Which has the higher frequency: red light or yellow light?
- What is the difference between UV rays, radio waves and a green light?
- How does sonar work? Draw a picture to help you explain your answer.



Apply your skills

- How far does light travel in 1 minute?
- How far does sound travel in 1 minute?
- If you were the third runner in a 4×100 metre relay, would you see the smoke from the starter's gun before you heard the sound? Why?
- Why do you think dolphins and whales use loud sounds to communicate with each other over long distances instead of using light to look at each other?
- A bushwalker standing at a lookout is directly opposite a steep cliff. If she calls out and hears the echo 4 seconds later, how far away is the cliff?
- Draw a scale diagram of a wave, with amplitude 1.0 cm and wavelength 4.0 cm.

- The speed, frequency and wavelength of a wave have the following relationship: $\text{speed} = \text{frequency} \times \text{wavelength}$. Use this relationship to find:

- the wavelength of the note middle C which has a frequency of 256 Hz
- the frequency of a beam of red light of wavelength 7.5×10^{-6} m.

Challenge yourself



- If the frequency of sound increased enough, would it become light? Why?
- What is a shadow? Why do they have blurry edges?
- Find out how people make sounds with their voices.
- Find out about some animals that rely on their hearing. Draw a table that summarises:
 - how each animal uses its sense of hearing
 - how important the other senses (sight, taste, touch and smell) are to each animal.
- Find out what things Galileo discovered. Report on one of his discoveries to the rest of the class.
- Describe the compound eye found in insects. What type of vision does it give?
- Find out how a laser works. Draw a simple diagram of its structure.
- In 1887, Michelson and Morley performed an experiment which suggested that there was no aether. Find out about this famous experiment.



Internet assignment



Tester

CHAPTER

6

ENERGY FOR LIVING THINGS

The first marathon was run in 490 BC. A Greek soldier, Pheidippides, ran approximately 40 km from the plains of Marathon to Athens to bring news of an overwhelming victory. The Greek soldiers had defeated the invading Persian army, which outnumbered them by three to one. Legend tells us that once Pheidippides had delivered this astounding news, he collapsed and died of exhaustion.

The first organised marathon was run in 1896 when the first modern Olympic Games were held. Today, a marathon is considered to be the most gruelling event in athletics. It is run over a distance of 42 km and it is always the last track and field event at the Summer Olympics. The competitors need a high level of fitness and endurance because they are depleting all their energy reserves in order to fuel their muscles and they are performing to the very limit of their capabilities.

Modern athletes have very structured training programs and follow diets recommended by sports nutritionists in order to perform at a top level. A good diet would usually be high in carbohydrates, have a moderate amount of protein and be low in fat. However, marathon runners usually follow a special diet before a competition to ensure that their bodies have the maximum energy reserves possible.

In this chapter, you will learn about the nutrients in food and how marathon runners can get the energy they need to run their race.

CHAPTER OUTCOMES

Successful completion of this chapter will allow you to:

- Identify components of food and their functions.
- Describe the process which allows the body to obtain energy from food.
- Test for carbohydrates, proteins and lipids.
- Describe the process of digestion.
- Identify the main parts of the digestive system.
- Discuss some of the issues associated with eating disorders.
- Identify the main parts of the respiratory system and the circulatory system.
- Explain how the digestive, circulatory and respiratory systems work together.
- Dissect a heart.
- Identify the way food is stored in plants and animals.

Food and energy

[return to main menu](#)


FIGURE 6.1

Spaghetti Napoli ... a thick, juicy steak cooked on the barbecue ... fried rice: all these are foods. Every living thing needs food in order to survive.

Talk about

It has been estimated that each Australian will consume an average of 17 beef cattle, 92 sheep, 4000 loaves of bread, 165 000 eggs, 15 pigs, 1170 chickens, 1 kangaroo, 500 kg cheese, 8 tonnes of fruit, 10 tonnes of vegetables and 10 000 litres of beer in a lifespan of 70 years. Do you agree with this estimate?

Food serves three major purposes in all living things. Food is needed to:

- supply the energy required for all activities.** When food is broken down in the body, energy is released which can be used for activities such as movement and heart beat.
- build new tissues.** Food provides the substances needed for making new cells and tissues.
- repair worn and damaged tissues.** The substances provided by food are needed to replace the cells which break down every day (such as red blood cells) and to repair damage such as cuts and broken bones.

The substances in food that your body needs in order to live and grow are called **nutrients**. **Carbohydrates**, **proteins** and **lipids** are three important types of nutrients in food. In addition, food contains three other useful substances: **vitamins**, **minerals** and **water**. We need all these nutrients to make up the structure of our bodies. In addition, carbohydrates and lipids, and sometimes proteins, may be broken down in the body and oxidised (burned) to release energy for our daily needs.

Energy needs

The total daily energy needs vary from person to person and depend on a number of factors including age, weight, sex, level of activity and environmental conditions. An average 13-year-old male, for example, may require approximately 10 500 kJ of energy per day, whereas an average 13-year-old female may need about 9000 kJ of energy per day.

A healthy diet

Evidence suggests that being overweight and eating a lot of fatty foods can increase the risk of heart disease, stroke and some types of cancers. The Australian Nutrition Foundation recommends choosing foods from the healthy diet pyramid to ensure a balanced diet and reduce the risk of these diseases.

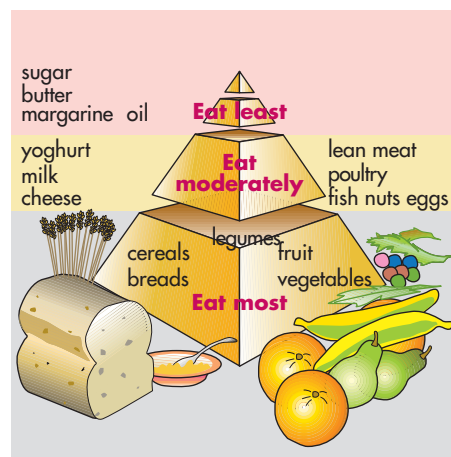


FIGURE 6.2

The Australian Nutrition Foundation recommends that foods should be eaten in the proportions shown in the healthy diet pyramid.

FACT

Your body is constantly making new cells to replace those that die. Humans make about 2 million new cells every second in order to replace damaged red blood cells. Skin cells need to be replaced every 19 days, whereas bone cells may not need to be replaced for up to 25 years.

FILE

EXPERIMENT 1

PEANUT POWER



The energy available from food can be measured by burning (oxidising) it. In the body, energy is released when food is oxidised during cellular respiration.

AIM

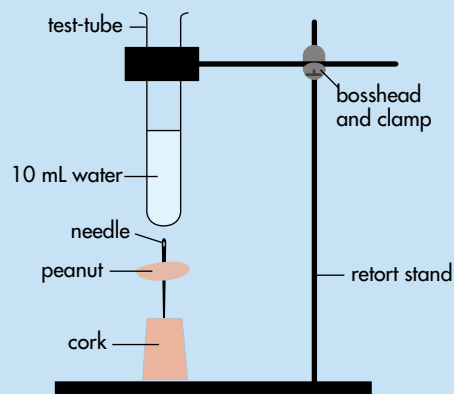
To determine the amount of energy which can be obtained from the burning of a peanut.

MATERIALS

- peanut
- 10 mL water
- test-tube
- balance
- thermometer
- cork
- long needle
- Bunsen burner
- retort stand, bosshead and clamp

METHOD

- 1 Pour 10 mL of water into a test-tube, then measure and record its temperature.
- 2 Weigh a peanut and record its mass.
- 3 Insert a long needle through the peanut and then into a cork so that the peanut sits above the cork like a 'flag' on the needle 'flagpole'.
- 4 Set the peanut alight, then allow the burning peanut to heat the water in the test-tube (clamp the test-tube as close as possible to the peanut).
- 5 When the peanut has finished burning, measure and record the water temperature again.

**FIGURE 6.3**

The experimental set-up.

DISCUSSION

- 1 Heat and other forms of energy are measured in **joules**. To heat 10 mL of water by 1°C takes 42 joules of energy. Calculate how much heat was released by your burning peanut.
- 2 How much heat would be released by burning 1 gram of your peanut?
- 3 Not all the energy released by your peanut could be recorded in this experiment. Explain why.

EXTENSION

Compare the energy available, per gram, of various types of nuts.

Questions

- 1 What is food?
- 2 What functions does food have in the body?
- 3 Draw a graph to show how your energy requirements change throughout the day.
- 4 Suggest why energy is needed by your body when you are asleep.
- 5 Name the types of food that we should eat most of.

Think about

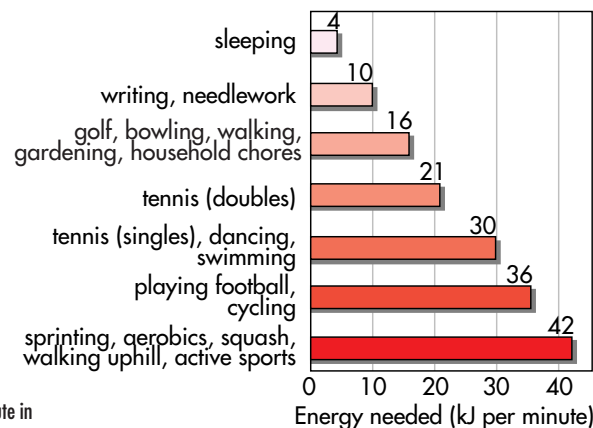
How would your energy needs be different from those of:

- a a baby?
- b an adult?

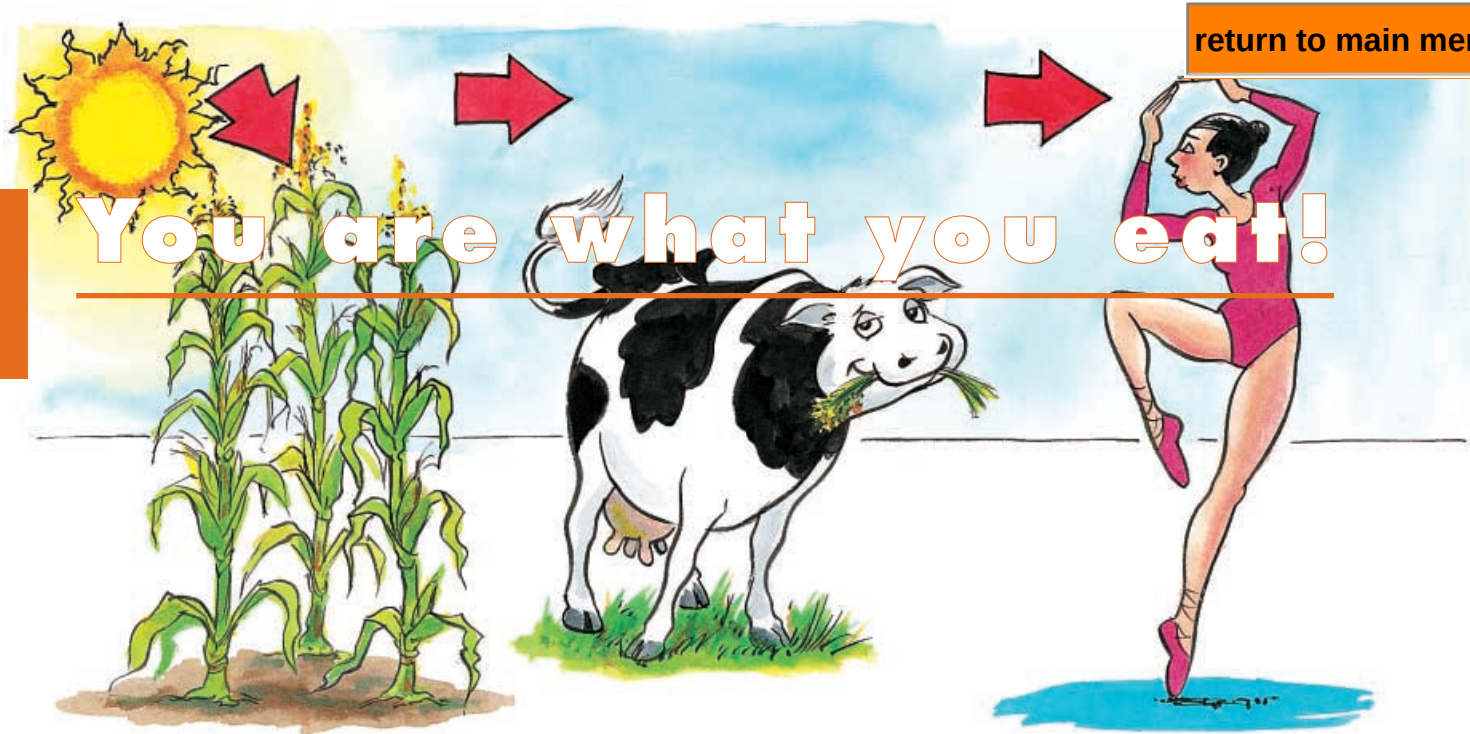
Activity

How much energy do you use?

Use Figure 6.4 to calculate the amount of energy you use, on average, every day.

**FIGURE 6.4**

Energy used per minute in various activities.

**FIGURE 6.5**

The Sun provides energy to crops. Eventually, that energy is passed onto us via the food we eat.

FACT**FILE**

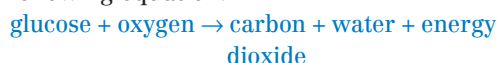
Constant dieting can change the body's metabolic rate, i.e. the body reduces the amount of energy it uses to keep functioning. When normal diet is resumed, the body responds by storing the extra kilojoules as fat. To lose weight and maintain a healthy weight, it is important to eat a balanced diet with a variety of foods in moderation and combine this with regular exercise.

Talk about

Have you ever wondered why some people can eat lots of food and remain slim? Or why others gain weight no matter what they eat? What exactly happens to food after we eat it? What is it used for?

The energy in food originally comes from the Sun as light energy. It is trapped by plants and used in photosynthesis to make the simple sugar glucose and the gas oxygen which contain energy stored in a chemical form. The plant cells can then use the glucose as a simple building block to assemble nutrients such as other carbohydrates (e.g. **starch**), protein and lipids. All these nutrients, with their stored chemical energy, can be passed along a food chain.

The energy stored in nutrients and oxygen can be released by a process called **respiration**. Respiration takes place in every living cell and the energy it releases is used for everyday cellular activities including growth and repair. The usual type of respiration is summarised in the following equation:



The glucose may be obtained directly from food eaten or be produced by breaking down other carbohydrates such as starch. If carbohydrate is not available, lipids and protein can be broken down to provide raw material for respiration.

The energy available from the food that we eat is measured in terms of the number of kilojoules (kJ) produced when the food is oxidised (burnt).

For a person's weight to remain constant, the energy intake should equal the energy used. If we take in more energy than we use, we gain weight. If we take in less energy than we use, we lose weight. When our body demands more energy than we are supplying, it either burns stored fat or breaks down protein, i.e. muscle mass. This is why when people suffer from starvation they experience severe muscle wasting.

Energy obtained from food may be converted to one of four forms of energy:

- 1 **Mechanical energy**—used for movement such as walking and blinking
- 2 **Chemical energy**—used for reactions in the body such as manufacture of complex molecules
- 3 **Heat energy**—required to maintain body temperature
- 4 **Electrical energy**—required for electrical impulses in nerves.

Questions

- 1 **What does your body require energy for?**
- 2 **Ozgur, a Year 8 student, claimed that cellular respiration was like burning. Explain why you agree or disagree with Ozgur's statement.**

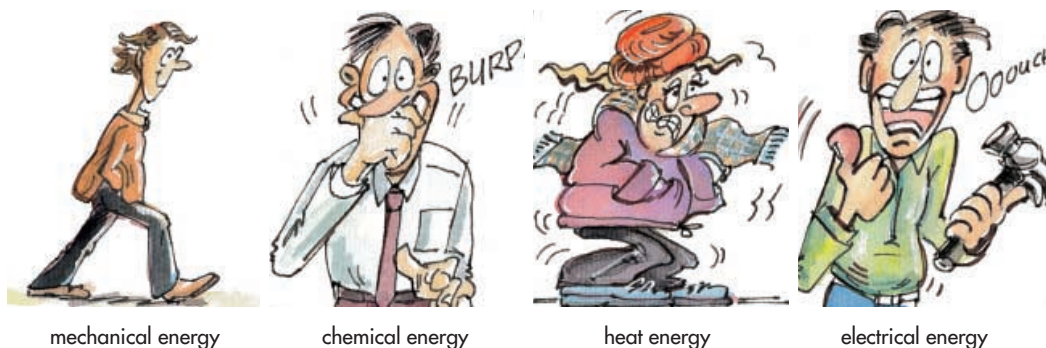


FIGURE 6.6
Food energy is required for many functions in the body.

USING SCIENCE 1

DO YOU EAT ENOUGH?

Make a list of all the food you would eat in an average day. Use a calorie counter book to determine the number of kilojoules that you take in per day. Use Table 6.1 to determine whether you eat too little, too much, or the right amount for a person your age and your height.

TABLE 6.1 Recommended energy intakes for adolescents.

Age (years)	Males	Females
10–11	8 100–9 100	7 300–8 200
11–12	8 700–9 800	7 700–8 700
12–13	9 200–10 300	8 100–9 100
13–14	9 800–11 000	8 400–9 500
14–15	10 500–11 800	8 600–9 800
15–16	11 100–12 500	8 700–9 900
16–17	11 700–13 200	8 800–10 000
17–18	12 000–13 500	8 800–10 000

Source: FAO-WHO-UNU (1985) Note: Level of energy expenditure: males 1600–1800 BMR, females 1500–1700 BMR.

- What happens when you don't get enough energy from food you eat?
- Sometimes, people may eat more food than their bodies require. What happens to the excess food?
- Suppose you have a breakfast consisting of one glass of orange juice (250 mL), one serve of porridge (30 g), half a cup of milk (125 mL) and one banana (150 g). Use the data in the table to calculate the energy provided by your breakfast.

Food	Amount	kJ
Orange juice	250 mL	480
Milk	250 mL	750
Porridge	100 g	1500
Banana	150 g	370

Activities

- You are a dietitian who has a male patient requiring a diet of 10 000 kJ per day. Use a calorie counter book and the healthy diet pyramid to plan his menu for one day.
- Make a collage, using magazines, of the different foods you eat.
- Conduct a survey of your class's favourite foods.
- State two ways in which the body reduces heat loss when you are cold.

Find out

Collect several different milk cartons from home. Compare the energy content of each type of milk.

FACT

Australians eat almost one-third of their meals outside the home. A survey done in 1992 by the Age newspaper found that each Australian household eats an average of five take-away meals per week.



Extension
worksheet

Testing foods



FIGURE 6.7

Almost every month a new food product can be seen on the supermarket shelves. Food scientists help create new foods and may analyse different diets to determine their carbohydrate, protein and lipid content.

FACT

FILE

Cellulose, or fibre, is a carbohydrate found in all plant material. It cannot be digested in the human body, but plays an important role in digestion because it helps the movement of food through the digestive system. Processed food contains little fibre.

The components of food

Carbohydrates are important energy-supplying substances. Energy is released when carbohydrates undergo respiration in the body's cells. If you eat more carbohydrates than your body can use, your body stores the excess energy produced as fat.



FIGURE 6.8

Sources of carbohydrates include potatoes, bread, rice, cereals and sugar.

Proteins are substances which your body uses to build up new cells and tissues. The body may also use some proteins as an energy source.



FIGURE 6.9

Sources of proteins include lean meat, fish, eggs, milk, cheese and lentils.



FIGURE 6.10

Sources of lipids include plant oils (e.g. safflower, palm, peanut, margarine) and animal fats (cheese, milk, butter, egg yolk).

Lipids are used by the body as an energy store. Solid lipids are called fats and liquid lipids are called oils. Lipids are also used to form some cell structures and as an insulating layer under the skin.

Questions



1 Give examples of foods which contain the largest proportion of:

- a carbohydrates**
- b proteins**
- c lipids.**

2 Foods are mixtures of carbohydrates, proteins and lipids. Consider a potato. Most people think that potatoes are made of carbohydrate. Although about 22% of a potato is carbohydrate, 2% is protein and the rest is water. Show this information in a sector graph.

Find out



Calcium, iron and iodine are all important minerals found in food. Find out why they are so important.

EXPERIMENT 2

WHAT IS IN FOOD?

**AIM**

To determine whether different food samples contain carbohydrates, proteins or lipids.

PART A TESTING FOR CARBOHYDRATES

Iodine solution can be used to detect *starch*, a type of carbohydrate. If starch is present, the colour of the iodine solution will turn from yellow to blue-black. Benedict's solution can be used to detect *glucose*, another type of carbohydrate. Substances containing glucose will change Benedict's solution from a blue colour to an orange-red colour when heated.

MATERIALS

- food samples
- mortar and pestle
- Bunsen burner
- Benedict's solution
- test-tubes
- water
- iodine solution

METHOD

- 1 Crush approximately 3 g of a food sample with 10 mL of water by using a mortar and pestle.
- 2 Divide the mixture into two test-tubes.
- 3 Add 3 or 4 drops of iodine solution to one test-tube. A dark blue colour indicates the presence of starch.
- 4 Add 5 mL of Benedict's solution to the other test-tube and heat gently. An orange-red colour indicates the presence of glucose.

PART B TESTING FOR PROTEIN

To find out if a food sample contains protein, it can be mixed with sodium hydroxide and copper(II) sulfate solutions. If the food contains protein, the colour of the copper(II) sulfate solution will turn from blue to purple.

MATERIALS

- food samples
- mortar and pestle
- dilute sodium hydroxide solution
- 1% copper(II) sulfate solution
- test-tubes
- water

METHOD

- 1 Crush approximately 3 g of a food sample with 5 mL of water by using a mortar and pestle.
- 2 Pour the mixture into a test-tube.
- 3 Add 5 mL of sodium hydroxide solution to the test-tube (caution: sodium hydroxide is caustic and can burn).
- 4 Slowly add 1 mL of copper(II) sulfate solution.
- 5 A pinkish-purple colour indicates the presence of protein.

PART C TESTING FOR LIPIDS

To find out if a food sample contains lipids, the sample should be dissolved in ethanol and filtered. The filtrate should be then added to water. If the food contains lipid, the water will become cloudy.

MATERIALS

- food samples
- mortar and pestle
- filter paper
- water
- test-tubes
- ethanol
- filter funnel
- rubber stoppers

METHOD

- 1 Crush approximately 3 g of a food sample with 10 mL of ethanol by using a mortar and pestle.
- 2 Filter the mixture into a test-tube.
- 3 Add 3 mL of water to the solution in the test-tube, stopper the tube and shake well.
- 4 A cloudy white **emulsion** indicates the presence of lipid.

DISCUSSION

- 1 Record the results of each experiment in a table.
- 2 Do these food tests enable you to determine which food sample has the greatest amount of carbohydrate/protein/lipid?

EXTENSION

Certain types of seaweed are used as food sources. If you have access to a beach, collect as many different types of seaweed as you can and analyse them for the presence of starch, protein and lipid. Present your findings as a poster.

IMPLICATIONS FOR SCIENCE AND THE ENVIRONMENT

Food preservatives and additives

FIGURE 6.11

Early immigrants to Australia took at least 100 days to get here from England. Many died on the journey because of inadequate food supplies. There was no refrigeration, so fresh food was scarce and meat was preserved in salt. Meals were simple because of the limited ingredients.

FACT

One form of radioactivity, gamma rays, can be used to preserve food. The gamma rays are very effective in killing any micro-organisms which might be present but the irradiated food should be of very high quality before treatment.

FILE

Talk about

On long journeys such as those shown in Figure 6.11, what precautions would be necessary regarding the selection and storage of food? In what ways is storing and transporting food over long distances easier today than it was 200 years ago?

Thirty years ago the shelves of an average supermarket may have contained 1500 different food items. Today there can be as many as 16 000 food items in your supermarket. This increase is partly due to the use of food additives. A food additive is a substance that:

- is not itself a food
- is not usually used as a typical ingredient in the preparation of food in the home
- is added to food to improve its quality, flavour, shelf-life or microbiological safety, or to reduce the food's preparation time in the kitchen.

Consumers want food to look appetising, retain its freshness and be economical and easy to prepare. For these reasons today's food manufacturers include additives in their products. The average shopper would find it difficult to buy a food product that contained no additives.

Additives can have one or more of the following purposes.

- **Maintaining freshness**
These additives protect against chemical changes in food, or at least slow these changes down. They also slow down the growth of micro-organisms, such as bacteria. They may be called **preservatives**, **anti-oxidants** or **enzyme inhibitors**.
- **Promoting sensory qualities**
These additives enhance food colour, flavour, smell, texture and appearance. They are known as **colouring agents**, and **taste, odour and texture modifiers**.
- **Making handling and processing easier**
Lubricants, **dough conditioners**, **emulsifiers**, **anti-caking agents** and **processing agents** are amongst the additives in this category.
- **Maintaining nutritional quality**
Additives such as **vitamins**, **minerals** and **essential nutrients** may be added to enhance or restore the nutritional value of food.

Fruit such as apples, pears and bananas usually turn brown quite quickly. This is an important consideration when preparing these foods. One way to keep the fruits looking fresher longer is to add a squeeze of lemon juice. The CSIRO has found that a


FIGURE 6.12

By law, all food additives used in a product must be listed on the label according to the type of additive it is (e.g. preservative, colour, emulsifier), followed in brackets by either the additive's actual name or its code number.

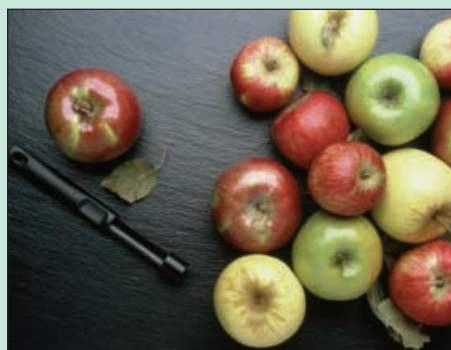
USING SCIENCE 2

WATCH WHAT YOU EAT

- 1 Collect five food labels and five drink labels. Read the information on each label carefully, then copy and complete the table opposite, listing all the additives used in each food and drink.
- 2 For each item, describe what it would be like if each of the additives was left out.

Name of food or drink	Name of additive or additive number	Type of additive	Purpose of additive

special gene (called the PPO gene for short) can be introduced into growing fruit trees as part of their cell structure. This gene controls the natural enzyme which causes the fruit to brown. It is now being developed by a company in Canada for use with a large variety of apples, pears and cherries. The fruit will not turn brown once it has been cut, and it will also be less likely to become bruised during handling and packaging.


FIGURE 6.13

Research is currently being carried out to produce fruit which does not brown when cut and which does not bruise easily.

USING SCIENCE 3

BROWNING OF FRUIT

Lemons, oranges and limes are all called citrus fruits because they contain citric acid. They also contain another acid, ascorbic acid (vitamin C). Design an experiment to test whether citric acid and ascorbic

acid are both important in stopping the browning of fruit. Take care to make your experiment a fair test. Perform the experiment if the materials are available and present your results as part of a scientific report.



Think about

Design an advertising campaign for the non-browning fruit which is being developed as a result of research by the CSIRO.



Find out

Some food additives cause health problems for some people including asthma, coughs, skin irritations, headaches and hyperactivity. What is MSG? Why is it used in some foods? What problems can it cause?



Digestive systems: breaking down food

FIGURE 6.14

By the time you eat your dinner, most of the food you ate at breakfast will be on its way to the tissues and cells of your body.

Talk about

In what way would you expect the food you ate today to be different from the food that has reached your cells? Why?

Before your body can use the food you eat, the food must be broken down into simpler and soluble substances. This breaking down of food is called **digestion**. Food must be broken down so that it can get into the bloodstream to be carried to all the cells in the body where it can be used for energy, growth and repair.

The organs of the body in which digestion occurs form the **digestive system**. As food moves through these organs, it undergoes both physical and chemical changes.

Mechanical and chemical digestion

There are two types of digestion: mechanical digestion and chemical digestion. **Mechanical digestion** is a physical process which breaks down food into smaller pieces. The teeth digest food mechanically by tearing and grinding it up. Muscles in the organs of the digestive system aid mechanical digestion by squeezing and pushing the food through the digestive system.

Chemical digestion changes the small pieces of food into simpler substances that dissolve in water. Chemical digestion occurs when **digestive juices** break food down from large molecules to smaller

ones. The digestive juices contain substances including acids, bases and **enzymes**. Enzymes are chemicals which speed up chemical reactions. Different glands throughout the digestive system produce these digestive juices.

Comparing animal digestive systems

Animals with different diets have different digestive systems.

Worms live on decaying material in the soil and store their food in their **crops**. They also have **gizzards** with strong, muscular walls to grind the food thoroughly. The soil and small stones that worms eat also help to grind the food. Digested food is then absorbed into the blood through the intestine. Undigested material such as soil is passed out through the anus.

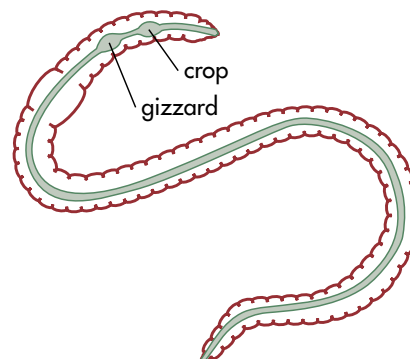


FIGURE 6.15

Digestive system of a worm.



Humans empty a total of 9 litres of digestive juices into their gut each day.

EXPERIMENT 3**WHY DO WE NEED A DIGESTIVE SYSTEM?****AIM**

To model some of the functions of a digestive system.

MATERIALS

- 6 test-tubes with rubber stoppers
- 1 teaspoon each of sugar, bread, ham, butter, carrot, orange
- water
- dilute hydrochloric acid

METHOD

1 Take six test-tubes and place a different sample of food in each.

2 Cover each sample with water, stopper each test-tube, and shake well. Record your observations.

3 Add 5 mL of hydrochloric acid to each tube. Stopper, and shake the tube gently. Record your results.

DISCUSSION

1 Describe what happens in each test-tube in step 2 of the Method.

2 Our digestive system contains chemicals which help break down food. Use your results in step 3 of the Method to explain what acid does.

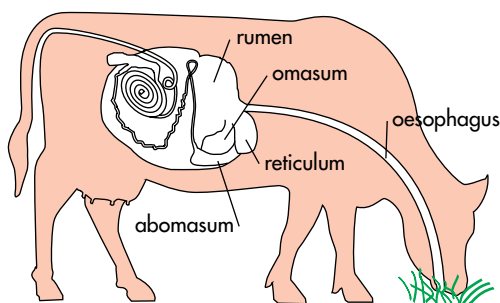


FIGURE 6.16
Digestive system of a cow.

Animals that eat only plant food like grasses and leaves need help from bacteria to digest the main part of their food, **cellulose**. Cows, sheep and kangaroos have a four-part stomach. After chewing and swallowing the food once, it is then brought back up and chewed again. This is why cows are generally seen chewing—it is called ‘chewing the cud’. After the second chewing, the food moves into another part of the stomach where bacteria do most of the chemical digestion.

Snakes swallow their food whole. Their jaws become unhinged so that they can swallow animals up to five times larger than themselves. The snake’s digestive juices are strong enough to digest bones, teeth and fur. Some snakes that swallow eggs whole have spines which bulge into their throats from their neck bones. The eggs are pierced by these spines and crushed by the jaw. The snake then spits out the shell and swallows the contents.



FIGURE 6.17

It may take over an hour for a snake to swallow a whole animal.

Questions

- 1 Why does food have to be digested?**
- 2 Explain the difference between mechanical and chemical digestion.**
- 3 Through which organ of the body is digested food absorbed into the blood?**
- 4 Why is it important for animals such as cows to chew their food, when animals like dogs may not need to chew food for a long time?**

Find out

Find out about the digestive system of an animal of your choice. Explain how the structure and function of the parts of its digestive system are suited to the animal’s diet. Present your findings in a poster then use it to present a talk to the rest of your class.

FACT

FILE

Young rabbits and koalas have to eat some of their mother’s droppings to get the bacteria their gut needs to break down the cellulose.



Multimedia

The human digestive system

2. Oesophagus (15 seconds)

A 25 cm long muscular tube which carries chewed food from the mouth to the stomach. Food is pushed down by the contraction and relaxation of the muscles in the oesophagus. This type of muscle action is called **peristalsis**.

5. Liver

The liver produces **bile**, which acts like a detergent to break down lipids into smaller droplets. This increases the surface area and so helps the enzymes break the lipids down.

6. Gall bladder

Bile is stored here until it is needed after meals. Bile moves to the small intestine through the bile duct.

8. Large intestine (12–24 hours)

Not all the food you eat can be digested. Undigested food continues through the digestive system into the large intestine where water is reabsorbed into the body. The remaining wastes are stored in the **rectum** until they are eliminated from the body through the **anus**.

1. Mouth (1 minute)

Digestion starts when you chew your food. The teeth grind food into small pieces and the tongue mixes the food with **saliva**, a watery liquid produced by the **salivary glands**. Saliva contains water, mucus and an enzyme. The enzyme breaks down starches into simpler sugar molecules. The water and mucus make food softer and easier to swallow.

3. Stomach (1–6 hours)

Muscles in the stomach churn and grind up the food, mixing it with **gastric juices**. These contain **hydrochloric acid** which breaks down proteins and destroys microorganisms, the enzyme **pepsin** which helps break down proteins and lipids, and **mucus** which protects the lining of the stomach from the acid. Food is stored here until the next stage of digestion.

4. Pancreas

Pancreatic juices contain **enzymes** to help digest carbohydrates, lipids and proteins. Pancreatic juices are delivered to the small intestine through the pancreatic duct.

7. Small intestine (5–6 hours)

The main digestive organ. Muscles in the intestinal walls continue to squeeze the food through the digestive system, mixing up food and digestive juices. The small intestine itself produces digestive juices which complete the digestion of carbohydrates, lipids and proteins. The walls of the small intestine are lined with finger-like projections called **villi**. The villi contain networks of **capillaries**. When food has been digested, it is small enough to pass through these villi to enter the **bloodstream**. Blood then carries the digested food to all body cells.

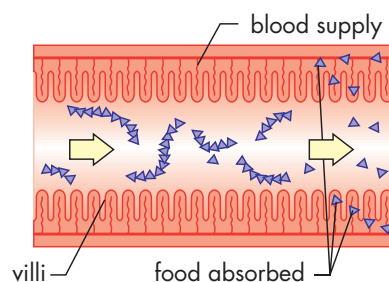
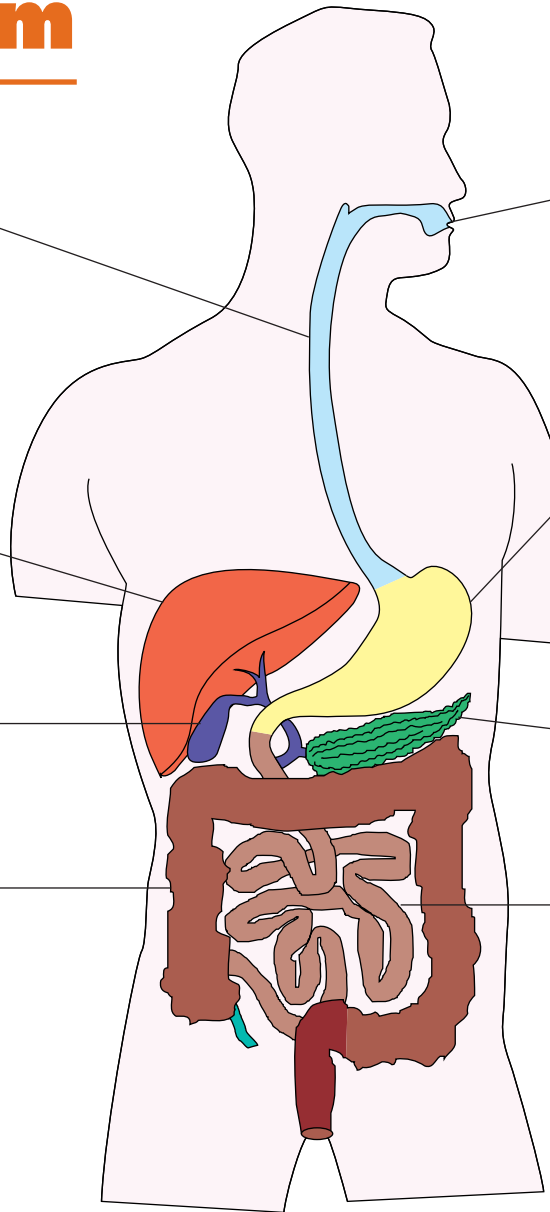


FIGURE 6.19

The inner surface of the small intestine is folded into villi. Many more cells are exposed to digested food than if it were a smooth tube.

FIGURE 6.18

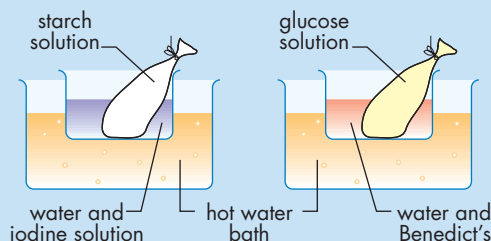
The human digestive system. Humans eat a variety of food and so we need a complex system and many different enzymes to digest our food.

EXPERIMENT 4**A MODEL OF ABSORPTION**

After food has been digested, the particles are so small that they can pass through the walls of the intestine into the bloodstream by **diffusion**.

AIM

To model how digested food passes through the intestinal wall into the bloodstream.

**FIGURE 6.20**

The experimental set-up.

MATERIALS (per group)

- 2 × 500 mL beakers
- 2 × 200 cm dialysis tubing
- starch solution
- iodine solution
- dropping pipette
- glucose solution
- Benedict's solution
- hot water bath

METHOD

- 1 Moisten both lengths of dialysis tubing under a tap and tie a firm knot at one end.
- 2 Fill one tube with starch solution, tie the top of the tubing and then rinse under a tap to remove any starch solution which may have spilt on the outside. Place the tubing in a beaker containing hot water and iodine solution. (Recall that iodine turns starch blue.)
- 3 Fill the other tube with glucose solution, tie the top of the tubing and then rinse under a tap to remove any glucose solution which may have spilt on the outside. Place the tubing in a beaker containing water and Benedict's solution, then set the beaker in a hot water bath. (Recall that glucose turns Benedict's solution orange-red when heated.)
- 4 Record the colour of all the solutions.
- 5 After 20 minutes, note any colour changes which have occurred in the solutions.

DISCUSSION

- 1 Explain any colour changes that have occurred.
- 2 How does this activity model what happens in the small intestine when food is absorbed into the bloodstream?
- 3 Iodine reacts with starch in cold water. Explain why a hot water bath was used in this experiment.

**Questions**

- 1 Imagine that you are a chicken and salad sandwich. Describe what happens to you after you enter a human mouth. Include descriptions of the changes to each part of you.
- 2 Construct a model of the human digestive system.

Activities

- 1 Place a piece of bread (a carbohydrate) in your mouth and chew it for 3 minutes. How does the taste change? Why?
- 2 Use a torch to look into a friend's mouth or look at your own in a mirror. Say 'ahhh'. What you see is

the entrance to the oesophagus.

Describe its appearance.

- 3 It is possible to sip water while you are upside down. In this case, the water is moving against gravity towards your stomach. Name the process responsible for moving water along your oesophagus to the stomach.
- 4 Place a tennis ball at the top of a stocking. Use your hands to squeeze the ball down towards the toe. This is similar to the action of peristalsis. Use this activity to explain how you think the muscles in the oesophagus could be arranged so that food can travel to the stomach.



Foundation
worksheet 1



Extra
worksheet

CURRENT ISSUES, RESEARCH AND DEVELOPMENT

Anorexia nervosa

FACT

Dehydration and starvation in anorexics reduce fluid and mineral levels in the body. Low levels of the minerals calcium and potassium can cause a disruption of the electric currents which flow through nerves. This 'electrolyte imbalance' can cause irregular heartbeat.

FILE



FIGURE 6.21
A victim of anorexia.

Anorexics feel they are 'too fat' and suffer pangs of guilt whenever they eat even 'normal' amounts of food. Eventually they become extremely thin (although they still think they are fat), their hair loses its lustre and may fall out, they suffer from the cold,

they are often constipated and the periods of female anorexics become irregular or stop altogether. Anorexics deny hunger although they are constantly racked by hunger pangs, and they are often 'exercise freaks'. Approximately 5% die from starvation or by suicide.

Anorexia mostly affects girls from middle- and upper-class families. It is 15 times more common in private schools than in government schools. Often, anorexia victims are well behaved, rarely cause trouble at home and do well at school.

There are many theories which try to explain why anorexia occurs. Some psychiatrists think that young people may become anorexic as a rejection of adulthood. Other psychiatrists suggest that anorexics feel powerless, and that by becoming anorexic they gain control, or power, over at least themselves.

Activities



1 Debate:

- a Society is responsible for promoting an individual's self-esteem.
 - b People should be taught not to judge other people or themselves by their weight—they should accept people without considering body size.
- 2 Conduct a survey to find the heights of students at your school and how much they would like to weigh. (Don't embarrass them by asking what they do weigh.) Compare these 'ideal' weights to the healthy range on a height/weight chart.

- a Is there a difference between males and females?

USING SCIENCE 4

CASE STUDY

THE PATIENT



FIGURE 6.22
'Before'.



FIGURE 6.23
'After'.

I always thought that if I could get my weight right, everything else would somehow fit into place. I spent so much time worrying about how much I weighed that I never worried about anything else. Even though I didn't realise it, all that year I was dieting. I just couldn't cope with Mum, Dad, school, my friends. It was easy with food because I could control it. I became better and better at losing weight. Eventually, all I had to eat everyday was an apple—peeled! We all knew at school about anorexia but I'd fool myself and convince myself I didn't really have it. When my mother finally dragged me to hospital, I just screamed and wanted to die. Even though I was 27 kilograms, I would look at myself in the mirror and just stand there and pinch myself where I thought I was fat. My body was red all over. I spent five months in hospital and nearly a year in therapy. I now know that you can eat the proper foods and exercise and be somebody.



FIGURE 6.24
The family.

THE MOTHER

Anne always told me that she'd eaten heaps at school or at a friend's house so she didn't feel like having tea with the rest of the family. Sometimes, she'd even cook the evening meal and refuse to eat at the table saying that she'd tasted the food while she was cooking it. But I noticed her getting thinner and thinner and it started to worry me. She became moodier and more introverted. I even asked her friends about her eating patterns, but they didn't say much. She always liked custard tarts so I tried buying her a custard tart as a treat three times a week. I tried to encourage her to eat—I even begged her to eat sometimes—but all that happened was that we'd end up fighting. I'd yell and she'd go up to her room and cry. One day, I just packed her bags and literally dragged her to the hospital.



FIGURE 6.25

THE DOCTOR

Admission notes:

Anne was admitted to hospital when she weighed 27 kilograms. Her hip and shoulder bones were sticking out. She was not allowed to leave hospital until she had reached and maintained a target weight of 48 kilograms. Because anorexia nervosa affects people physically and mentally, a range of health specialists were involved in Anne's treatment. She started group therapy, psychotherapy, family therapy, behaviour modification, hypnosis and reality imaging sessions with psychiatrists and psychologists to help her improve her self-image and relationships with others around her. Anne also had several sessions with a dietician to correct her eating habits.

- b** Is there a difference between different age groups?
- 3** Discuss: How skinny is 'too skinny'?
- 4** Collect two magazines or newspaper advertisements which rely

on the appearance of the models to sell a product. Re-design the advertisements so that the advantages of the product itself become the focus.

Gas exchange in living things

FIGURE 6.26

Insects have tiny holes in their exoskeletons for gas exchange.

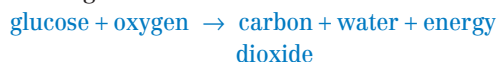
FACT

FILE

Sharks and rays do not have movable gill covers and so have to keep moving for water to pass over their gills.



All living things need oxygen to react with the food in their cells to produce the energy they need daily. **Cellular respiration**, the chemical reaction which releases energy in the cells of living things, also results in the production of carbon dioxide, which is a waste gas:



Living things need to obtain oxygen from the air in order to carry out respiration. In addition, living things need to rid themselves of the carbon dioxide produced during respiration because high carbon dioxide concentrations will poison their cells. **Gas exchange** is the process by which oxygen passes from the air into the body tissues and carbon dioxide leaves the body tissues and enters into the air.

Gas exchange in animals

Many small animals like worms and insects exchange gases through their outside covering. The oxygen goes straight into their body cells. Worms have a soft, moist

body covering that the gases can pass through—as long as they stay moist. Insects, however, have hard **exoskeletons**, through which gases cannot pass. To solve this problem, they have many tiny holes in their exoskeleton which lead into a branch of hollow tubes.

Water contains much less oxygen than air. Fish, therefore, need to work very hard to exchange gases. This is why most fish tanks have a bubbler. Fish carry out gas exchange through their **gills**. Gills are made up of rows of many tiny finger-like structures called **filaments**. Together these have a large surface area to exchange gases through. Instead of breathing water in and out, fish have to keep the water moving past their gills. Most do this by movements with their mouths and their gill covering. There is a rich blood supply to the gills—oxygen passes from the water into the blood, and carbon dioxide passes out into the water.



FIGURE 6.27

Energy is needed for everyday activities. This energy is obtained from the oxidation of food in the process of cellular respiration. Athletes often need large amounts of energy in their muscles. If their muscles don't get enough oxygen during exercise, athletes may experience painful cramps. Why do athletes need to breathe deeply after completing an event?

FIGURE 6.28

Gills enable fish to carry out gas exchange.

EXPERIMENT 5

GILL DISSECTION



AIM

To examine the features of gills in order to determine which features make them efficient.

MATERIALS

- a small, dead fish
- dissecting equipment
- beaker of water
- stereomicroscope

METHOD

- 1 Locate the gills and gill covering.
- 2 Poke a probe down through the fish's mouth to find its gullet, and then the path to the gills.
- 3 Carefully cut out the fish's gills and note their appearance in your workbook.
- 4 Place the gills into a beaker of water, noting the effect on the gill filaments.
- 5 Look at some gills under a stereomicroscope and draw a diagram of what you see.

RESULTS AND DISCUSSION

Draw a diagram and write a description of the gills:

- a that you dissected
 - b that you observed under a stereomicroscope, both in and out of water.
- 1 How could the gill covering help to pump water past the gills of a fish?
 - 2 What other purpose could the gill covering have?
 - 3 Do fish need to swallow water to breathe? Explain your answer.
 - 4 Why are gills red in colour?
 - 5 Why must gills be kept moist for the fish to 'breathe'?
 - 6 Why would reducing the gill surface in contact with the water kill the fish?

EXTENSION

Examine the gills of a yabbie. Explain how yabbies move water past their gills.

Large land animals like humans carry out gas exchange by **breathing** in and out of **lungs**. Lungs form part of the **respiratory system** of many large animals. Lungs are moist, and have a rich blood supply. Lungs are not like hollow balloons (except in frogs); instead they are made up of millions of tiny air sacs called **alveoli**. These have very thin walls and are each surrounded by a mesh of thin blood vessels called **capillaries**. Oxygen from the air you breathe passes through the thin walls and into the blood. At the same time, carbon dioxide passes from the blood out into the air spaces.

As you breathe, air moves in and out through your nose, down your windpipe and into your lungs so that some of the old air in your alveoli is continuously exchanged for new air.

Questions

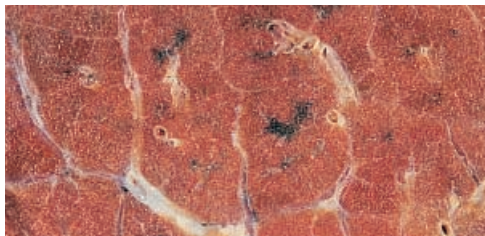


- 1 **Why do living things need oxygen?**
- 2 **All living things carry out gas exchange but not all animals breathe. Explain this statement.**

3a List four types of gas exchange systems.

b Which type of gas exchange system would each of the following animals have: bird, slug, spider, shark, oyster, wombat, ladybird, snake, koala? Justify your answer.

- 4 In what ways are gills and lungs:
 - a similar?
 - b different?
- 5 Why do you not find many worms in dry soil?
- 6 What features of gas exchange surfaces make them efficient?
- 7 If you have access to a spirometer, measure and compare the lung capacities of the students in your class. Graph your results.



FACT

FILE

There are about 700 million alveoli in an adult's lungs. Together the surface area is 100–150 square metres, or about half the size of a doubles tennis court.

FIGURE 6.29

Lungs allow many large animals to breathe in air which is high in oxygen and low in carbon dioxide and breathe out air which is low in oxygen and high in carbon dioxide.

The human respiratory system

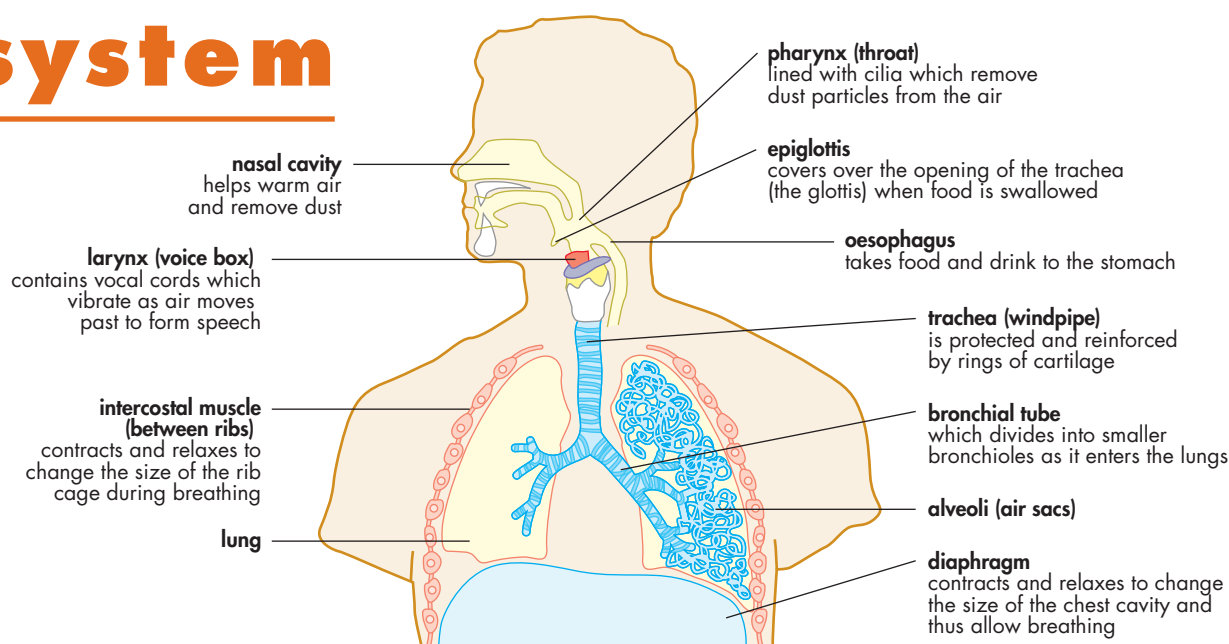
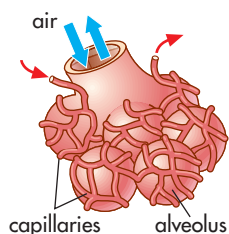


FIGURE 6.30
The human respiratory system.

Talk about

After a lot of exercise, you breathe faster and your heart beats faster. Why does this happen?

FIGURE 6.31
Exchange of gases in the lungs occurs at alveoli. Each alveolus is surrounded by capillaries which allow oxygen to diffuse into the blood from the air in the alveolus. At the same time, carbon dioxide diffuses from the blood back into the alveolus. Blood enters the lungs with a high carbon dioxide concentration and low oxygen concentration. It leaves the lungs with a high oxygen concentration and low carbon dioxide concentration.



The human respiratory system is designed to provide the human body with a plentiful supply of oxygen for use in respiration and to excrete the waste gas carbon dioxide. Breathing is an important part of this process as it allows air to enter the lungs where there is a very large surface area for gas exchange. This exchange of gases occurs between tiny air sacs called alveoli and the blood in the capillaries which surround them.

The term breathing refers to the inhaling and exhaling of air, while respiration refers to the chemical reaction which takes place in all body cells, using oxygen and simple sugars and releasing carbon dioxide, water and energy.

USING SCIENCE 5

TESTING EXHALED AIR

Test your own exhaled air in the following ways.

a Using a straw, blow your exhaled air into a test-tube containing 10 mL of clear limewater. Observe any changes and explain your observations. Be careful to blow into the limewater. Do not draw the limewater into the straw.

b Take two pieces of dried cobalt chloride paper. Add a drop of water to one piece and observe the change. Now breathe your exhaled air onto the second piece of dried cobalt chloride paper and observe. What substance have you detected in your exhaled air?

The mechanism of breathing

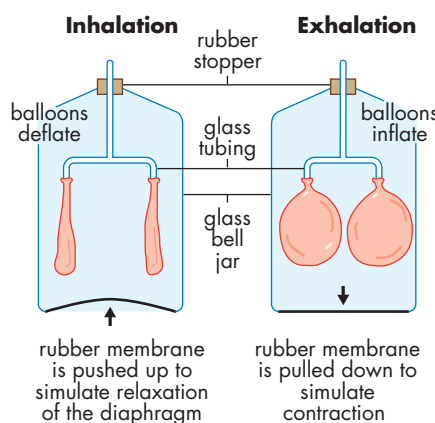
When you inhale, your **diaphragm** contracts and moves downwards. At the same time, the ribs are lifted upwards and outwards, making the chest cavity larger. The lungs, with the air they already contain, expand to fill this cavity, causing a reduction in air pressure. Outside air pressure is greater, so air moves through the nose down the trachea and bronchi into the lungs. Next the diaphragm relaxes and the muscles around the ribs also relax, making the chest cavity smaller and therefore increasing the air pressure within it. This forces air to be exhaled.

FIGURE 6.32

Model lungs. The diagrams show a model for how the diaphragm functions in breathing, but they do not give a full explanation of breathing. How could the model be improved?

TABLE 6.2 Composition of air.

Component	Inhaled air (%)	Exhaled air (%)
Oxygen	21	16
Carbon dioxide	0.03	4
Nitrogen	79	79
Water vapour	variable	saturated



FACT

FILE

Hiccups are caused by spontaneous contraction of the diaphragm when it is stimulated by the vagus nerve (a nerve associated with digestion). This leads to a rapid intake of breath, interrupted by a quick closing of the larynx. This causes the noise of the hiccup.

USING SCIENCE 6

EXAMINING A PAIR OF SHEEP LUNGS

Your teacher will organise this as a class demonstration.

- 1 Identify the trachea and feel the rings of cartilage surrounding it. What is their function?
- 2 Insert a plastic tube with a large syringe attached into the trachea.
- 3 Use the syringe to force air into the trachea. Do

the lungs inflate?

- 4 Trace the pathway of the trachea to find where it divides into the bronchi. Trace this pathway as far as you can.
- 5 Feel the lungs and describe the appearance.
- 6 Cut a section of lung and observe what is inside. Use a stereomicroscope to help your observations.

Questions

- 1 Explain the difference between breathing and respiration.
- 2 Why is it important that you breathe faster when exercising?
- 3 Draw a flow chart to summarise the steps in inhaling and exhaling.
- 4 Explain why lung tissue has a very small mass.
- 5 When learning first aid, you will often be shown how to perform expired air resuscitation on a patient who has stopped breathing. Predict the composition of gases in the air breathed out by the patient while this air resuscitation is being conducted.

Think about

Design an experiment to measure your lung capacity. Perform the experiment with other members of your class.

Find out

Tobacco smoke contains chemicals that can damage your lungs, even if you are a passive smoker (i.e. someone who breathes air which contains smoke from other people's cigarettes). The chemicals include tar, nicotine and carbon monoxide. What is the effect of each of these chemicals?

FACT

FILE

Sleep apnoea is a breathing disorder in which breathing stops for 10 seconds or more, perhaps 300 times a night. Eventually carbon dioxide in the blood rises and triggers a breathing reflex.

The human circulatory system

FIGURE 6.33

Heart disease accounts for the greatest number of deaths every year in Australia. Open heart surgery is a common operation following circulatory problems.



FACT

FILE

Your heart pumps blood through a network of more than 100 000 km of blood vessels. This is more than twice the distance around the Earth at the Equator.

FACT

FILE

Your capillaries are so narrow that red blood cells are flattened when passing through them.

The **circulatory system** consists of the **heart**, **blood** and **blood vessels**.

Blood is pumped around the body by the heart and carries nutrients, oxygen and wastes around the body. If the blood stops moving, then the animal will die.

Why do animals need a circulatory system? Each cell in the body needs oxygen—especially active cells in muscles and the brain. Oxygen only enters our body through the lungs. Here it is taken into the blood. The blood then takes it to all body cells. The distance between the lungs and other cells is so great that the oxygen could not make its own way there. You can probably work out the same type of pathway for nutrients, and for wastes.

The heart

Your heart is a muscular pump about the size of your fist. It is located in the centre of your chest with the bottom end tilted towards the left side of your body.

Blood vessels

The blood vessels involved in circulation are **arteries**, **veins** and **capillaries**. Blood is carried towards the heart in veins and away from the heart in arteries. Capillaries

are microscopic tubes, sometimes as small as one cell in diameter, which join arteries and veins. The exchange of gases, nutrients and wastes occurs between the capillaries and cells.

Blood

Blood is a mixture of **red blood cells**, **white blood cells**, **platelets** and **plasma**. The red blood cells carry oxygen, while the plasma carries digested foods, hormones and wastes such as carbon dioxide. White blood cells are involved in helping the body 'fight' against infection and disease and platelets are involved in the clotting of blood.

Questions



- 1 Name the major parts of the human circulatory system, and explain their functions.
- 2 Why do we need a beating heart?
- 3 Calculate how many times your heart beats every day.
- 4 Why do some animals not need a transport system?
- 5 Find out about the transport system of another animal and present it as a fully labelled diagram.
- 6 Trace a drop of blood through the body from the right side of the heart.

Think about



Imagine that your best friend needs a heart transplant. No human donor is available, but the heart of a pig (specially bred so that its heart won't be rejected by the human body) is available. What should be done? Is the decision easy? Do you agree

EXPERIMENT 6

DISSECTING A SHEEP'S HEART

**AIM**

To dissect a sheep's heart to identify its main features. Sheep have similar hearts to humans.

MATERIALS

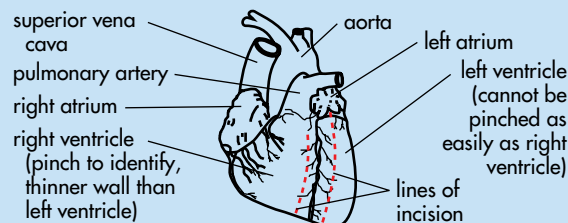
- sheep's heart
- dissection implements (dissecting board, forceps, tweezers, scalpel, scissors)
- antiseptic solution
- paper towelling

METHOD

Follow Figure 6.34 to dissect a sheep's heart.

RESULTS AND DISCUSSION

Identify the four chambers and the major blood vessels. Identify **valves** inside the heart.

**FIGURE 6.34**

Dissect the sheep's heart along the dotted lines.

Draw a diagram of your dissected sheep's heart and write a description of each structure you identify. Suggest why one side of the heart has thicker walls than the other side.

What is the function of the valves in the heart?

with breeding pigs solely for the use of humans? Are there any risks involved in transplant operations?

Activities

1 You can feel the pumping action of the heart in your arteries. This is a pulse. Find the pulse in your wrist. Count the beats in 1 minute:

- when you are sitting down
- when you lie down
- when you run on the spot for 1 minute. Graph the time it takes for your pulse to return to normal.

2 Look at slides of arteries and veins under a microscope. Note the thickness of the vessel walls and the shape of the vessels. Draw up a table of their similarities and differences. Consider Figure 6.35.

3 What happens to the contents of the blood as it goes:

- through the lungs?
- through the liver?
- through the intestines?
- past body cells?
- through the kidneys?

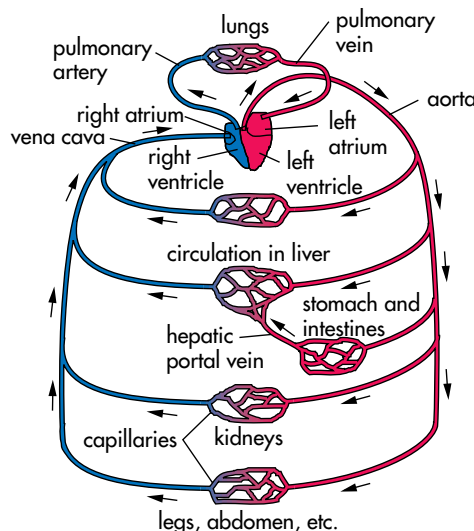
**FIGURE 6.35**

Diagram of the circulatory system of a human.



Foundation worksheet 2



Animation



Interactive

4 How many pumping chambers are there in a human heart?

5 From which side of the heart, and which chamber, is blood pumped:

- to the lungs?
- to the rest of the body?

6 Arteries take blood away from the heart, and veins take blood towards it. Capillaries join arteries and veins. Identify the arteries, veins and capillaries in the diagram.

Storage systems in living things

FIGURE 6.36

The stored food for a plant makes a good meal for many animals.

FIGURE 6.37

Animals that hibernate live on the food stores in their bodies for many months.



FIGURE 6.38

Animals such as whales and seals that live in cold climates store a lot of fat because it provides good insulation.

Why do plants and animals need storage systems?



Plants and animals use most of their food for energy and growth. Excess food that is not used is stored, and can be used when food is scarce.

Plant storage systems

Plants make sugar as their food. Much of the food made in a leaf is carried away, dissolved in water, to other parts of the plant. Some of this food is used by the plant for growth and repair and some of it is stored. Most plants store their food as starch. **Starch** is a large molecule formed when sugars join together.

Many plants have special storage organs for their starch. The best known example is the potato. Starch is stored in the bulbous, fleshy part of the potato plant. The sugar cane plant, on the other hand, stores sugar in its stem. Seeds and fruit contain stored nutrients for the new plant until it can make its own food. Leaves can also store starch.

Animal storage systems

Vertebrates, including humans, store most of their excess food as fat. Some animals have special storage areas for fat, for example a camel's hump and some lizard's tails.

Human storage systems

In humans, the liver performs a large number of roles, including the breakdown of harmful chemicals, the manufacture of proteins and the storage of minerals and nutrients.

One of the most important functions of the liver is to control the amount of glucose in the blood. If too much glucose is present in the blood as it passes through the liver it is extracted and converted to a substance called **glycogen**, which is stored in the liver

EXPERIMENT 7**TESTING LEAVES FOR STARCH**

During the daytime, photosynthesis may produce sugar at a faster rate than it can be used or carried away from a leaf. Many plants convert this extra sugar into starch which is temporarily stored in their leaves.

AIM

To test leaves for the presence of starch.

MATERIALS

- leaf (students could try different leaves)
- beaker of hot water
- hot plate
- test-tube rack
- tongs
- dropping pipette
- methylated spirits
- test-tube
- petri dish
- iodine solution

METHOD

Warning: Methylated spirits is flammable. Do not handle methylated spirits near a flame.

- 1** Boil a leaf in a beaker of hot water for about 30 seconds.
- 2** Remove the leaf from the water and place it in a test-tube. Keep the water boiling.
- 3** Keeping well away from a flame, add methylated spirits to the test-tube so that the leaf is covered.
- 4** Place the test-tube into the beaker of boiling water and heat until the green chlorophyll is removed from the leaf.
- 5** Carefully remove the test-tube from the beaker of boiling water and remove the leaf.
- 6** Place the leaf back into the beaker of boiling water for a few minutes to soften it.
- 7** When the leaf is soft, remove it from the water and place it in a petri dish.
- 8** Add 4 drops of iodine on the leaf. Areas of starch turn blue. Other areas turn brown.

RESULTS

Draw a diagram to show areas where starch has been stored in the leaf.

and muscles. When the liver cannot store any more glycogen, it converts the excess glucose to **fats**, which are stored at various sites around the body. Humans store fat beneath the skin and in fat cells around organs including the intestines. Between meals, the glucose levels in the blood drop and some of the glycogen in the liver is converted back to glucose. The liver can store enough glycogen to supply the body with glucose for about 20 hours.

Questions

- 1** Why do animals and plants store nutrients?
- 2** What is the main storage substance in:
 - a** plants?
 - b** animals?
- 3** Why may starch build up in a leaf during the daytime?
- 4** Explain why someone who eats three large meals a day and doesn't get much exercise would put on weight. What could such a person do?

**FIGURE 6.39**

Glycogen stored in these athletes' muscles provides energy which enables them to engage in strenuous exercise. Muscle glycogen is not used to maintain blood glucose levels.

- 5** What types of problems would a person with liver disease expect to face? Explain your answers.
- 6** What type of food would you expect a marathon runner to eat in the 3 days before a major competition?

FACT**FILE**

Athletes competing in a marathon follow a special carbohydrate diet to increase the reserves of glycogen in their muscles. The process is called carbohydrate loading. It usually includes eating plenty of food rich in complex carbohydrates for 3 days before the endurance event and only training lightly.



Using scientific language

- The word *photo* comes from a Greek word meaning light and *synthesis* comes from a Greek word meaning to make. Use this information to write a definition of photosynthesis.
- Match the definitions below with the following words: valve, right atrium, left ventricle, trachea, alveolus, liver, diffusion, diaphragm, oesophagus, enzyme.
 - Process which allows oxygen to pass from lungs into blood.
 - Substance which helps in the digestion of food.
 - The chamber of the heart which pumps blood to the whole body.
 - The chamber of the heart which receives blood from the whole body.
 - Contraction and relaxation of this muscle allows breathing to occur.
 - Another name for the windpipe.
 - Peristalsis allows food to move down this tube.
 - Another name for an air sac in the lungs.
 - Produces bile to help digest lipids.
 - Structure which allows blood to flow in one direction only.



Check your knowledge

- What is food?
 - Why do we need food?
 - What are the three major types of foods and how can they be identified?
- What is the difference between physical digestion and chemical digestion?
- What happens to:
 - food during digestion?
 - digested food?
 - undigested food?
- Where, in the digestive system, is water mainly reabsorbed?
- Explain the process which allows inhalation and exhalation of air in the lungs.
- Why is glycogen sometimes called 'animal starch'?
- Where do animals store food?
- Explain why humans have a more complex circulatory system than worms.

Apply your skills



- Teeth are important in the physical digestion of food. Look carefully at your teeth in a mirror.
 - Draw diagrams which show the different types of teeth in your mouth.
 - Describe how each tooth type is used when you eat a piece of steak.
- Why would a human diet based largely on grass be unsatisfactory?
- Explain why the chemical digestion of starch begins in the mouth whereas the chemical digestion of meat does not start in the mouth.
- Make a list of the digestive juices produced in the body and describe the function of each.
- Explain the difference between gills and lungs.
- Draw a diagram of the human heart showing clearly the direction of blood flow through it.
- The heart is made of muscle yet you can hear it beating. Why is this so?
- Design a menu for an athlete who is preparing to compete in a marathon in 3 days' time.

Challenge yourself



- Prepare a poster which illustrates a menu for one day which is a good example of a well-balanced and healthy diet.
- Write a newspaper article which discusses aspects of anorexia.
- Find out about the appendix:
 - Where is the appendix in the digestive system?
 - What is the function of the appendix in rabbits, koalas and rats?
 - Why do humans have a small appendix?
 - What happens in appendicitis?
- Using gamma rays to irradiate food is considered quite controversial by some people. Find out their concerns. Write a discussion about the use of gamma rays to preserve food.



Internet assignment



Tester

CHAPTER

7

SKIN AND BONES

NASA scientists have been designing a new type of robotic arm. It will be used in 2003 on a space mission to an asteroid between Mars and Jupiter. Equipment for space missions must be light and compact and use very little power. Existing robotic arms have a complicated system of gears and hydraulics which are quite heavy and use a lot of power.

The scientists have developed an artificial muscle made from special polymer ribbons of carbon, fluorine and oxygen. When stimulated by a very small electric current, the artificial muscle contracts. These muscles act very much like the muscles in our body. Although these new artificial muscles are very weak, similar types of muscles may one day be used in artificial limbs here on Earth.

Australian scientists have also been working with other researchers around the world to develop artificial muscles. They have been using long thin tubes made of carbon, called nanotubes or bucky tubes. These tubes bend if they are placed in a salt water solution and an electric current is run through them. These nanotubes work like living muscle but they are much tougher. In the future, they may be used to control the flow of fuel and air in jet engines. They may also be used in artificial hearts and to make robotic limbs.

In this chapter, you will learn about types of muscles and how they work. You will also learn about skin and bones and some of the medical breakthroughs that are helping to treat injuries and diseases in muscles, skin and bone.



Web links

CHAPTER OUTCOMES

Successful completion of this chapter will allow you to:

- Explain how plants and animals keep their shape.
- Describe the human skeleton and its main functions.
- Identify the differences between bone and cartilage.
- Describe similarities and differences between the three different types of muscles.
- Be aware of ways of avoiding back injury.
- Describe the application of new technologies to replace human body parts.
- Describe the main functions of skin.
- Classify types of burns.
- Identify issues associated with skin care.

Keeping in shape

FIGURE 7.1

Coral polyps rely on the buoyancy of water to support their tentacles when they are feeding.

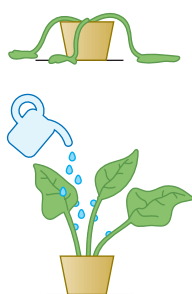


FIGURE 7.2

Water is important in helping provide support in plants. The cell wall supplies this support when the plant cells are turgid.

Talk about

Why do plants wilt? After a hot summer's day, plants which have wilted can sometimes be revived by watering them. How does water help a plant keep its shape?

Keeping plants in shape

Water is an important part of every living cell. It is a solvent which allows chemical reactions to take place within the cytoplasm of the cell and it is also used in some of these reactions. In addition, water in the cytoplasm and plant vacuoles pushes against the cell membrane and helps to

keep the cell in shape.

In plant cells, the cell membrane is surrounded by a rigid cell wall made of **cellulose**. When the cell membrane pushes up against the cell wall, the cell can be described as **turgid**. If the plant cell loses water, the volume of the cytoplasm and vacuoles is reduced and the cell membrane shrinks in from the cell wall. This causes a slight collapse in the cell wall and the cell is described as **flaccid**.

In many plants, the support needed to help a plant keep in shape is provided mostly by turgid plant cells. If the cells become flaccid, the plant wilts.

In most land plants, there is also special tissue, called **xylem**, which gives extra

EXPERIMENT 1

CHANGES IN POTATO TISSUE

AIM

To observe changes in potato tissue when it is in different surroundings.

MATERIALS

- 2 freshly cut potato cubes (similar in size)
- chemical balance
- labels
- tap water
- paper towel
- 5% salt solution
- 2 small beakers

METHOD

- 1 Label one beaker 'tap water' and the other beaker 'salt water'.
- 2 Weigh one beaker and record its weight.
- 3 Dry one potato cube with a paper towel.
- 4 Add the dried cube to the beaker and weigh the beaker and its contents.
- 5 Repeat steps 2–4 for the second beaker and potato cube.
- 6 Record the results in a table.
- 7 Add tap water to one of the beakers so that the

potato cube is completely covered by it. Add salt water to the second beaker to cover the potato cube. Leave for about 20 minutes.

8 Empty the tap water from the beaker. Remove the potato cube from the beaker and dry it and the beaker with paper towel.

9 Return the dried cube to its beaker and weigh the beaker and its contents again.

10 Record the results in a table.

11 Repeat the last three steps for the other potato cube in salt water.

12 Try bending both potato cubes.

DISCUSSION

- 1 Is there any change in weight for the potato cubes?
- 2 What substance could have moved in or out of the potato cube to cause the change in weight?
- 3 Compare the texture of each potato cube. What could account for any difference?
- 4 What happened to the cubes when you tried to bend them? Can you explain this?

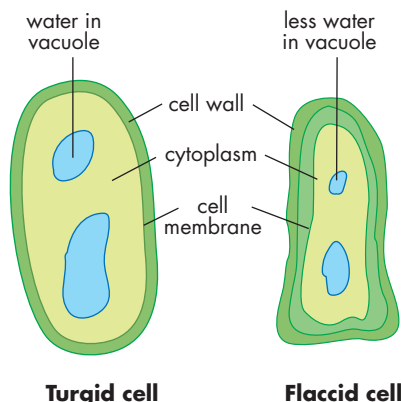


FIGURE 7.3

In a turgid cell, the cell membrane pushes up against the cell wall. In a flaccid cell, the cell membrane shrinks back, causing the cell wall to collapse slightly.

support. This tissue carries water through a plant from its roots to its leaves. The xylem consists of long cells with cellulose in the cell walls but they are given extra strength by another substance called **lignin**. The xylem cells are dead cells and they join end to end to make a tube. In woody plants, layers of these dead cells, reinforced by lignin, build up to form the wood in tree trunks.

Plants that live in water have no need for xylem and are supported by the water itself. Some water plants have air spaces or air bladders to help them float and make better use of the **buoyancy** of water to keep their shape.

Keeping animals in shape

Many simple animals live in water and rely on the buoyancy force of water for support. For example, a jellyfish loses its shape when it is washed up onto the beach. In more complex animals, the skeletons which provide support also help the animals to move.

Mammals, birds and reptiles are vertebrates. They have a skeleton inside their body, called an **endoskeleton**. Insects and spiders are arthropods, the largest group of animals. Every arthropod has an outer skeleton, called an **exoskeleton**.

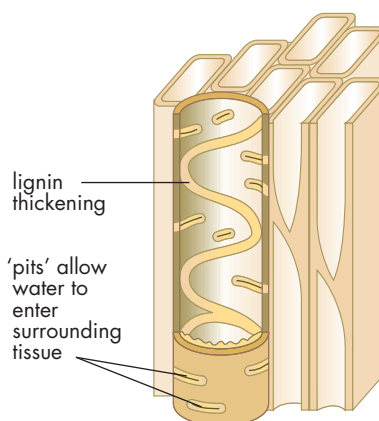


FIGURE 7.4

Xylem consists of long hollow cells which are dead. Lignin is often arranged in rings or spirals along the cell walls. It helps support the plant and makes a woody 'plant skeleton'.

FACT

FILE

The largest animal ever to have existed is the blue whale, *Balaenoptera musculus*, which can be up to 33 m long and weigh 180 tonnes. It can only reach this size because its body weight is supported by the buoyancy of water. All whales are crushed by their own body weight when they are beached. Their lungs cannot inflate properly to provide them with enough oxygen to survive.

Questions

- 1 Describe the difference between a flaccid and a turgid plant cell.
- 2 Name the substances which make the cell wall of xylem so rigid.
- 3 What is the function of xylem?
- 4 Why do animal cells provide less support than plant cells?
- 5 Why are the structures which support an animal generally much more complex than the structures which support a plant?
- 6 Imagine you were born with an exoskeleton. What difference would

there be in the way you grow and move?

Think about

Explain why the skeleton of a marine animal is much lighter than the skeleton of a similar-sized land animal.

Activity

Obtain a fish skeleton. Notice the spines along the backbone. What would their function be? What differences would you expect to see in the skeleton of a land vertebrate (e.g. a rabbit)?

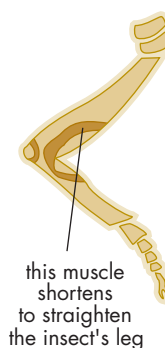


FIGURE 7.5

An important function of a skeleton is to allow movement. Muscles attach to the skeleton and cause movement about a joint when they get shorter (contract).

The human skeleton

FACT

FILE

The human skeleton consists of 206 bones. The largest bone is the femur or thigh bone and the smallest bone, the stirrup, is one of the tiny bones ('ossicles') in the middle ear.

Talk about

What would happen if all the bones in your body suddenly dissolved away? Brainstorm all the reasons why you need them.

The human skeleton has three main functions.

- **Support.** The skeleton helps give your body its shape and helps you stand upright. The bones in your skeleton have the strength of mild steel, although

they are much lighter. They can be compared with the framework of a building.

- **Protection.** Soft organs of your body are protected by your skeleton. For example, your skull protects your brain and your ribcage protects your heart and lungs.
- **Movement.** Muscles are attached to bones in the skeleton. Movement occurs at joints when these muscles apply forces to bones.

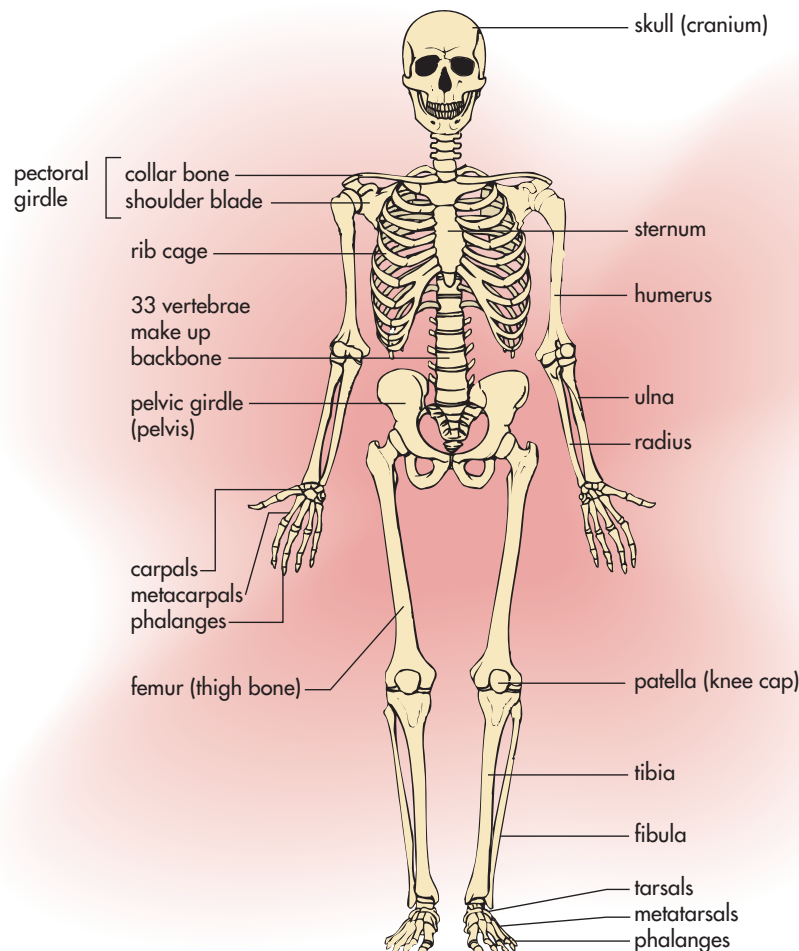


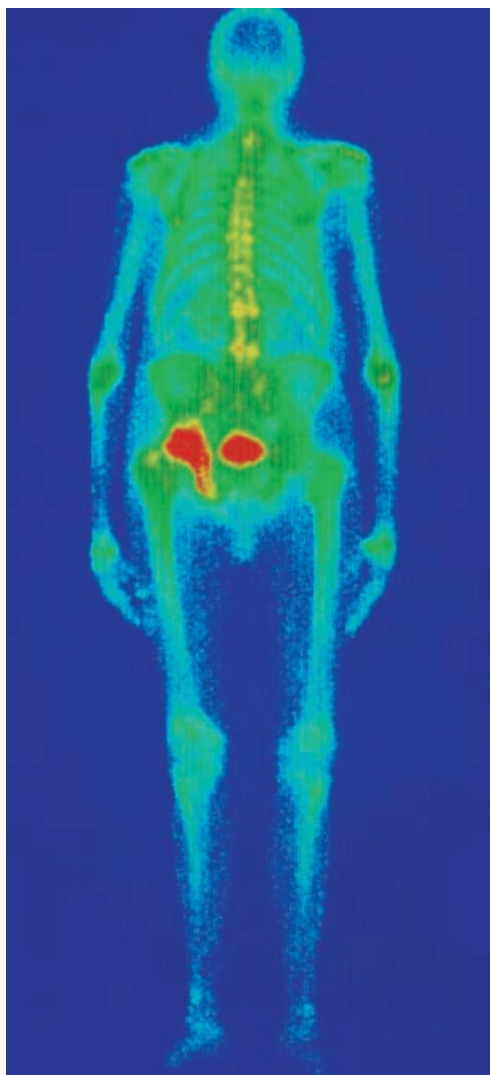
FIGURE 7.6
The human skeleton.

USING SCIENCE 1

THE HUMAN SKELETON

Study a model of the human skeleton and identify all the bones labelled in Figure 7.6. Identify which parts of the skeleton are used for protection and which for support. Count the bones in the hands and feet. Why do you think there are so many? What would be the

function of the pectoral and pelvic bones? Note the joints in the skeleton as well. You will be investigating these again in section 7.5. Which parts of the skeleton contain the softer and more flexible material called cartilage?



3 What is the advantage of having so many bones in the hands and feet?

4 Why is it important that the backbone consists of many vertebrae? What is the function of the projections (spines) of the vertebrae?

5 The front part of the rib cage, the sternum, consists of cartilage, which is softer and more flexible than bone. What would be the advantage of this compared with a sternum made of bone?

6 Bones in the body can be classified into two groups: bones which support and protect, and bones which allow movement. Draw up a table with examples of each type of bone.

7 Design a crossword, using names of bones in the skeleton, or compose a rap song which describes parts of the skeleton and their function.

Think about

How does the human skeleton compare with skeletons of vertebrates who walk on four legs?

Find out

X-rays and bone scans both involve exposure to small amounts of radiation. What special properties do radioactive substances like technetium have in order for them to be used safely in bone scans?

FIGURE 7.7

Bone scans detect a radioactive substance, technetium, which has been injected into the blood. 'Hot spots' can be detected as areas where blood builds up. This happens wherever there is an injury.

FACT

FILE

X-rays form an image of bones in the skeleton because they are absorbed more by bone than other tissue. Bone scans work on a different principle. A radioactive substance is injected into the bloodstream. It tends to collect in areas where there is an active blood flow, e.g. the site of an injury. The radioactive material gives out a form of radioactivity called gamma rays. Like X-rays, gamma rays can be detected by photographic film.

Questions

- What are the three main functions of the human skeleton?
- Name the soft body parts which are protected by the:

a skull	b rib cage
c pelvis	d vertebrae.



Word process

Bone and cartilage

FIGURE 7.8

Spongy bone is found in bones that need to support weight, like the pelvis. This type of bone provides strength but it is also light in weight.



FIGURE 7.9

As young children grow, the cartilage in their skeleton changes to bone. Compare the X-ray of a child's hand (top) with that of the adult hand (bottom).

Talk about

The skull of a newborn baby is very fragile. It is particularly important that the baby's forehead is handled carefully. Why is the skull so fragile? What other differences are there between the skeletons of a baby and an adult?

Bone and cartilage

A baby's bones are made up mostly of **cartilage**, a tough but flexible tissue. Gradually minerals, mostly calcium phosphate, are added and the cartilage hardens into

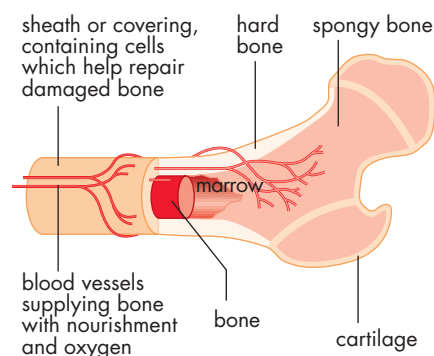


FIGURE 7.10

Long bones, like the femur, contain compact bone and spongy bone. They need a rich blood supply.

bone. However, there are still parts of the adult skeleton that contain cartilage: the lower part of the nose, the ears, the sternum, between the vertebrae and at the ends of bones in joints.

Bone is living tissue. The outer part of a bone has a good supply of blood vessels and contains special cells which can repair damaged bone cells. Long bones, like those in the legs and arms, need to be both strong and light. They are strongest at the ends, with compact bone and then spongy bone underneath. The shaft of the bone is hollow for lightness. A layer of cartilage at the end provides a cushion at the joint but it is also the region where growth occurs in children and young adults. The calcium phosphate minerals provide the hardness and special protein fibres (containing collagen and elastin) give bone toughness but flexibility.

The interior of the bone is richly supplied with blood vessels and contains **bone marrow**. Within the bone marrow are special cells called **stem cells** which are unspecialised. They can develop into red blood cells, white blood cells or platelets.

Bone marrow transplants are an important medical technique for patients suffering from cancer. Cancer treatments such

USING SCIENCE 2

OBSERVATIONS OF BONE

PART A

From your butcher obtain a thigh bone which has been sawn in half lengthwise. Identify the compact bone, spongy bone and bone marrow. Also identify the parts of the bone where muscles would attach.

PART B

Place a small chop bone in a beaker of 2 M nitric acid or acetic acid and leave it for a few days. Remove the bone carefully with tongs and wash it thoroughly to remove traces of acid. Try bending the bone and note any changes which have occurred. What material has been dissolved from the bone by the acid? What material remains?

as chemotherapy and radiation destroy rapidly dividing cells like cancer cells. However, bone marrow cells are also rapidly dividing and may be destroyed too. Patients may have some of their own bone marrow removed before treatment and then have it injected back afterwards. Patients suffering from leukaemia, a form of cancer where too many white blood cells are made in bone marrow, may receive donations of healthy bone marrow to replace their own.

Injuries to bones and joints

A break in a bone is called a **simple fracture**. A **compound fracture** occurs when

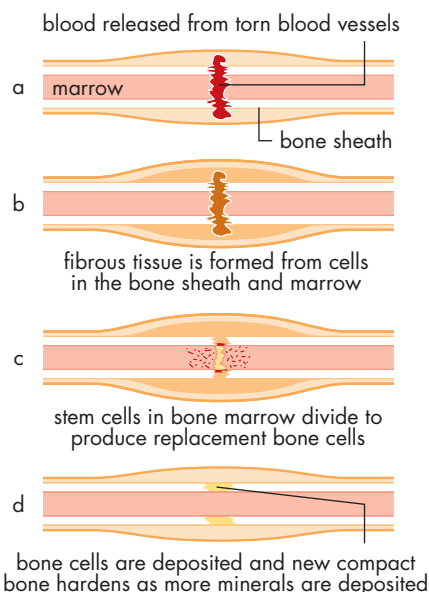


FIGURE 7.11

Repair of a broken bone may occur in stages over about 6 weeks if the fracture is immobilised in a plaster cast.

Questions

- 1 Name the parts of the skeleton where cartilage is found.
- 2 What material in bone is responsible for giving it hardness? What material is responsible for bone being flexible?
- 3 Suggest how the age of a child could be determined by observing bones in the skeleton.

the broken bone pierces the skin. It is a more serious injury because the wound may get infected.

Injuries at **joints** may not be due to broken bones. **Ligaments** and **tendons** are both made of connective tissue and may be damaged. Ligaments allow the bones to move at the joint but they also protect the joint by limiting that movement. Tendons provide a pulling force on a bone when a muscle contracts. Sprains occur when ligaments or tendons are stretched or torn. Sometimes an operation is needed to repair them.

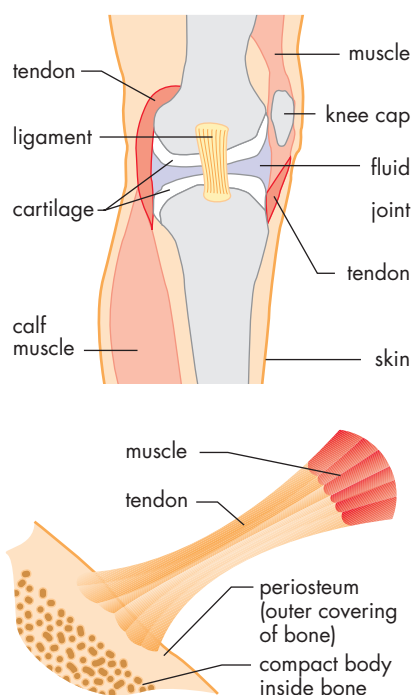


FIGURE 7.12

X-rays are the easiest way to determine if a simple fracture has occurred.

FIGURE 7.13

The structure of a joint. This side view of a knee joint shows one of the ligaments which hold the bones in position and tendons which attach the leg muscles to the bone. The cartilage at the ends of the bone and the lubricating fluid allow smooth movement at the joint.

FIGURE 7.14

A tendon consists of bundles of collagen fibres. It allows a muscle to exert a strong pulling action on a bone.

FACT

Medical researchers have recently isolated a special master cell from adult bone marrow. They used this special 'stem cell' to grow millions of cells in the laboratory. Then they changed these cells into bone, tendon or fat tissue by altering the culture medium.

4 What is the function of bone marrow?

marrow?

5 Write a newspaper article describing new advances in repairing muscles, ligaments and tendons.

Find out

Knee-joint injuries are quite common sporting injuries. What type of injuries can occur to the knee and what treatments are now possible? Summarise your findings in a poster suitable for display in a physiotherapist's waiting room.



7.4

Types of muscles



FIGURE 7.15

(above left)
Highly developed skeletal muscles provide great strength. The action of these muscles is controlled by nerve messages from the brain.

FIGURE 7.16

(above right)
Striated muscle: protein strands give a striped appearance to the muscle fibre.

FACT

FILE

During vigorous exercise, oxygen may not be supplied quickly enough to muscle tissue. The muscle goes into 'oxygen debt'. Another type of respiration takes place which is not as efficient ('anaerobic respiration'). It still supplies some energy to the muscle but it leaves a waste product called lactic acid. The build-up of lactic acid causes muscle pain.

All muscles act by contracting and relaxing. This is controlled by nerve messages, usually sent from the brain. Stimulation of a muscle by a nerve causes the muscle to contract. Lack of stimulation allows it to relax.

The **contraction** of a muscle requires energy which comes from **respiration** within the muscle cells. This process usually uses glucose and oxygen which are supplied to the muscle by a network of blood capillaries. **Glycogen** stored in muscles can also be broken down to supply glucose for respiration.

There are three types of muscles in the human body: **striated**, **smooth** and **cardiac** muscle.

Striated muscles

Striated muscles are also called skeletal muscles. They can make up approximately 50% of body weight. Skeletal muscles support the skeleton and allow movement. They are also known as voluntary muscles because they can be consciously controlled by the brain. For example, you choose to

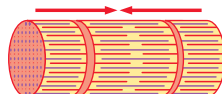
walk by sending nerve messages to the muscles in your legs and feet.

The long thin cells that make up this muscle tissue are called muscle fibres. They can be up to 30 cm long. Each fibre contains thousands of tiny overlapping protein strands. As they slide over each other, they cause the muscle to get shorter (i.e. to contract).

Smooth muscle

Smooth muscle is found in the walls of some blood vessels like arteries and hollow organs such as the stomach, intestine and bladder. The muscle fibres are usually arranged in layers which run in different directions and they are much shorter than striated muscle fibres. They contract slowly and rhythmically and do not tire easily. Smooth muscle is sometimes called involuntary muscle because you cannot consciously decide to make it contract. The contractions are controlled by nerve messages sent from a lower part of the brain.

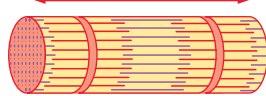
two different types of protein strands slide to overlap when the muscle contracts



bands are closer together

Contracted muscle fibre

protein strands move apart when muscle relaxes

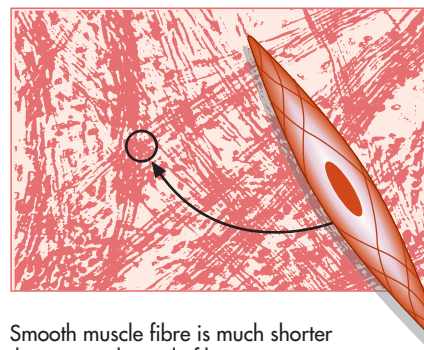


bands move further apart

Relaxed muscle fibre

FIGURE 7.17

There are two different types of protein strands in a muscle fibre. As they slide past each other to overlap, they make the whole muscle fibre shorter. Energy is needed for this process.



Smooth muscle fibre is much shorter than striated muscle fibre

FIGURE 7.18

Overlapping layers of smooth muscle in the wall of the bladder run in different directions. When these muscles contract, urine is emptied from the bladder.

EXPERIMENT 2**MUSCLES IN A CHICKEN WING****AIM**

To observe the structure of a chicken wing.

MATERIALS

- chicken wing
- dissecting scissors
- dissecting dish or newspaper

METHOD

- 1 Cut the skin from the upper part of a chicken wing to expose the bone and muscle.
- 2 Trace the pink muscle to the white tendon where it attaches to the bone.

- 3 Pull on the tendon and note the movement that occurs.

- 4 Draw a labelled diagram of the chicken wing.

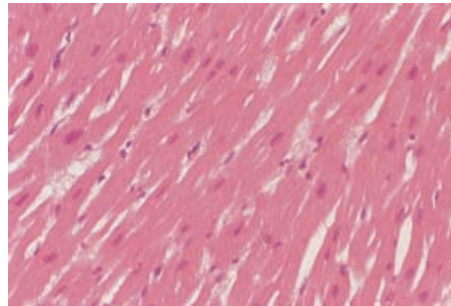
- 5 Cut the meat (muscle) from around the joint to observe the ligaments.

DISCUSSION

- 1 Describe any differences you have observed between muscle, tendon and ligament.
- 2 What type of movement occurs at the joint in the chicken wing.
- 3 Give a summary of the role of tendon, ligament and muscle around the chicken joint.

Cardiac muscle

Cardiac muscle is unique. It makes up most of the inner wall of the heart and consists of Y-shaped muscle fibres. When they contract, they work together to make the wall of the heart shrink. This squeezes the blood inside the ventricle and pumps it out into the arteries. Cardiac muscle is classified as involuntary because we do not control the beating of our heart consciously. However, it is striped like voluntary muscle.

**FIGURE 7.19**

Cardiac muscle fibres join up to form a linked network in the walls of the heart. They are striated like voluntary muscle but our brain does not consciously control the contraction of these fibres.

Questions



- 1 What is another name for a muscle cell? Name three types of muscle cells.
- 2 What process supplies energy to muscles?
- 3 Name the types of muscle not controlled consciously by the brain.
- 4 What type of muscle would you expect to find in each of the following tissues or organs: leg muscle, small intestine, diaphragm, neck muscle, heart muscle, uterus, muscle controlling the iris in the eye, muscle in the wall of an artery.
- 5 Explain why cardiac muscle cells have large numbers of mitochondria.
- 6 What type of muscles do you think tendons connect to? Explain your choice.

Think about



Voluntary muscles contain two types of fibres: fast twitch and slow twitch fibres. Fast twitch fibres contract more quickly but get tired more easily than slow twitch fibres. Different people have different proportions of these fast twitch and slow twitch fibres in their muscles. Which type of fibre would a successful 100 m sprint runner have in higher proportion? Would the answer be different for a marathon runner?

Find out



The largest muscle in your body is the gluteus maximus, which is in your buttocks. The smallest muscle in your body is called the stapedius. It is inside your ear. What is its function?

FACT

FILE

Mitochondria are tiny organelles found in the cytoplasm of all our body cells. They release energy to the cell as part of cell respiration. Cardiac muscle is particularly rich in mitochondria.

Movement

FIGURE 7.20

The skeleton provides sites for muscles to attach. The interaction of muscles about joints in the skeleton allows movement.

Types of joints

One of the important functions of the skeleton is to allow movement. Most joints are designed to make movement possible, but not always. The bones of the skull are not joined together in a newborn baby, but gradually join as the baby grows. These fixed joints or **sutures** can be observed on a model of a human skull.

Four types of joints in the skeleton which do allow movement are shown in Table 7.1.

Muscles and joints

Movement at joints is caused by muscles pulling on bones. There are two types of muscle at the joint, acting in opposite or 'antagonistic' ways. It is the contraction of a muscle that moves a bone. Therefore, a joint may be bent by the contraction of one muscle, but another muscle must contract if the joint is to be straightened.

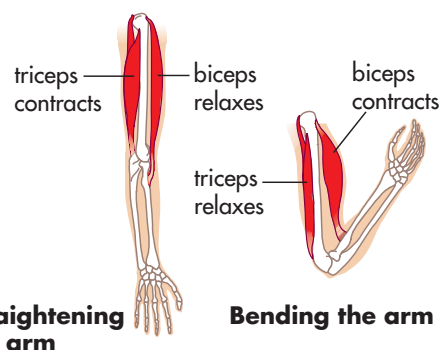
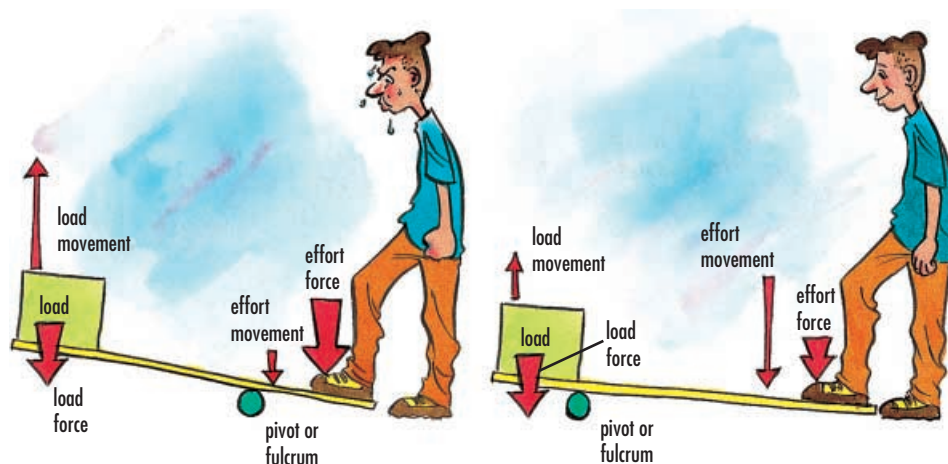


FIGURE 7.21

Bending the arm involves the biceps contracting while the triceps relaxes. When the arm straightens the triceps muscles contracts while the biceps muscle relaxes.

TABLE 7.1 Examples of joints.

Joint	Hinge	Ball and socket	Condyloid	Pivot
Model				
Example	 hinge joint elbow joint	 pelvis socket hip joint	 carpal bones wrist joint	 atlas vertebra axis vertebra below skull at top of backbone
Movement	Back and forth movement in one dimension	Good rotation	Some rotation as well as back and forth movement	Turning from side to side

**FIGURE 7.22**

A plank can be used as a lever which is a force magnifier or distance magnifier, depending on the position of the fulcrum. (a) A man must push down with a large effort over a small distance to lift a load, but the load moves a large distance. (b) He applies a smaller effort over a large distance to lift the load a small distance.

Action of levers

Levers are used in everyday life to help make jobs easier. They are rigid structures which pivot about a point called a **fulcrum**. Sometimes they might be used to change the direction of a force and to reduce the force that is needed to lift something. In this case, they are said to be 'force magnifiers' because a small effort can lift a

large load. At other times, they might convert movement over a small distance to a much larger movement elsewhere. Then they are 'distance magnifiers'.

The bones in our skeletons also act as levers. The joints often provide the pivot point or fulcrum and the muscles provide the effort. The fulcrum is not always between the load and the effort. Sometimes it is at the end of a lever.

USING SCIENCE 3

A SIMPLE LEVER

Design a simple experiment to measure the effort needed to lift a load at different distances from a

fulcrum (like the examples in Figure 7.22). Perform the experiment and write up the experimental report.

Questions

- 1 Name five types of joints. Give an example of each one.
- 2 Bending and straightening of the arm requires antagonistic action of muscles. Explain this statement.

Activities

- 1 Place your fingers on your arm just above the middle of your elbow. Bend the arm to feel the contraction of the biceps muscle. Follow the muscle down towards the elbow to feel the tendon that attaches the biceps muscle to the ulna in the forearm. Now place your fingers at the back of your upper arm to feel the contraction of the triceps muscle as you straighten your arm.

- 2 Design an experiment to test the relative strength of your biceps and triceps muscles. If possible, perform the experiment and compare results with other class members.
- 3 Design and make a model arm, to show the action of biceps and triceps muscles.
- 4 Use the model of the human skeleton to observe different types of joints and the movement they allow. Draw up a table to classify the different joints you observe.
- 5 Look at Figure 7.21. Do the lower arm bones act as a force magnifier or distance magnifier lever when the arm is bent?



Back care

FIGURE 7.23

Muscle tone helps to maintain body posture. Muscles in the neck keep the head upright and muscles attached to vertebrae keep the back straight and upright.

Talk about

If someone is sitting upright and watching television, what changes do you notice in their posture if they start to fall asleep? What accounts for these changes?

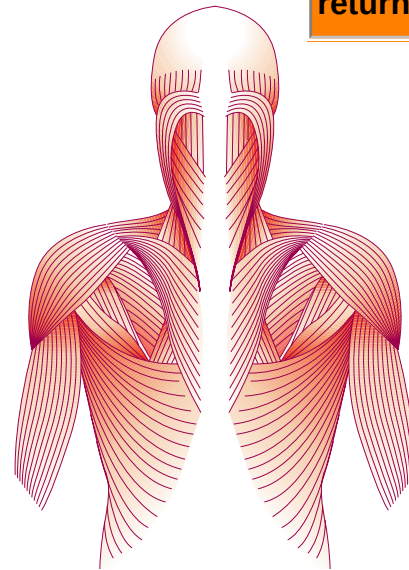
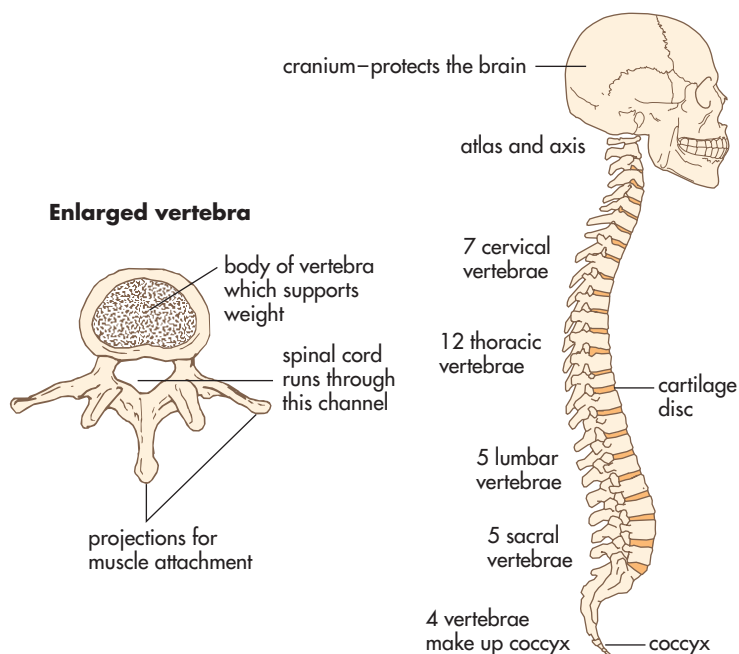
Maintaining posture

Our brain constantly sends messages to our skeletal muscles, making slight adjustments to their state of contraction. The muscle firmness produced is called **muscle tone**. Some relaxing of muscle tone occurs when we fall asleep which is why our body slumps when we fall asleep while sitting upright.

Good muscle tone is important for maintaining posture but our upright posture is dependent on more than this. It

FIGURE 7.24

Structure of the backbone. Use a model skeleton to identify the types of vertebrae labelled. Compare the size of the projections on the thoracic vertebrae with other vertebrae.



requires an interaction between bones, muscles, ligaments, tendons and nerves to keep our body balanced and to support our body weight.

The S shape of our backbone provides balance for walking upright on two legs. Our backbone consists of 33 vertebrae. Each vertebra has projections to which muscles and ligaments attach. The **spinal cord** passes through each vertebra and is protected by it. It allows nerve messages to pass to and from the brain and other parts of the body. The surfaces between each vertebra are flat, forming a type of sliding joint which allows a little flexibility and movement. A cushion of cartilage between each vertebra, called a **disc**, helps protect the joint. Five vertebrae in the lower back are fused to form the **sacrum**, a region which, as part of the pelvic girdle, has to support most of our body weight.

Back pain

Back pain is a common complaint which may have a variety of causes. These include:

- **damage** to muscles or ligaments, e.g. from being very overweight, having poor posture or muscle strain due to lifting heavy loads.
- **osteoarthritis**—wear and tear at joints which is often associated with age. Discs between vertebrae become worn and the spaces between the vertebrae narrow.
- a '**slipped disc**' where cartilage between vertebrae is ruptured or bulges. Nerves become compressed.
- **osteoporosis**—seen more often in older women. There is a loss of bone density

**FIGURE 7.25**

Bending over to lift a heavy load puts stress on muscles in the back. Bending the knees and keeping the load close to the body allows the leg and hip muscles to do most of the work. This protects the back from injury.

due to loss of calcium minerals from the bone. The bones become weaker and are fractured more easily. A very stooped posture (dowager's hump) becomes noticeable in older women suffering from osteoporosis.

Protecting the back

Some forms of back damage and pain can be prevented by keeping in mind the following.

- Maintain **good posture** while awake and asleep. Sleep on your side or your back and use a pillow designed to keep your vertebral column in a straight line.

Questions



- 1 What is muscle tone?
- 2 What is the function of the projections found on the vertebrae?
- 3 Which part of the vertebral column supports most of our body weight? How do the vertebrae in this region differ from the rest of the vertebral column?
- 4 List some of the common causes of back pain.
- 5 What is the difference between osteoarthritis and osteoporosis?
- 6 Design a poster to educate people about back care.

**FIGURE 7.26**

A workout such as this is a weight-bearing activity which has a positive effect on bone density. Such an activity improves muscle tone, but also helps avoid osteoporosis.

- **Exercise** regularly to maintain a healthy body weight and keep muscles well toned. Exercises to strengthen abdominal muscles will help provide extra support for the back. Walking and swimming help strengthen your lower back.
- **Lift** objects **safely**. Always lift from a squatting position, using muscles in hips and legs to do most of the work in lifting.
- Include foods which contain **calcium** in your diet (like milk and cheese) to help maintain healthy bones.

Think about



Smoking and high alcohol intake increase the likelihood of osteoporosis. Weight-bearing exercise improves bone density in older people and thus prevents osteoporosis. What type of exercises would be suitable for older people to help protect against osteoporosis?

Find out



Strong abdominal muscles help support your back. What exercises help develop these muscles? What other exercises are important in strengthening the back?

FACT

FILE

Osteoarthritis can occur at any joint, including joints in the vertebral column. **Cartilage** breaks down at the joint. In severe cases, surgery may be necessary, e.g. for knee and hip joint replacement. Joints between vertebrae cannot be replaced in this way, so other techniques are being investigated. One example is a liquid polymer used to coat the damaged cartilage at the joint.

FACT

FILE

Enzymes found in pawpaw have been used to treat severe cases of slipped disc. The enzyme is injected near the compressed nerve to shrink the problem disc.

CURRENT ISSUES, RESEARCH AND DEVELOPMENT

Cyborgs

Talk about

Science fiction stories and movies often mention robots. Some also include **cyborgs**, characters who are part human and part machine. Brainstorm the variety of robots and cyborgs you know from movies or books.

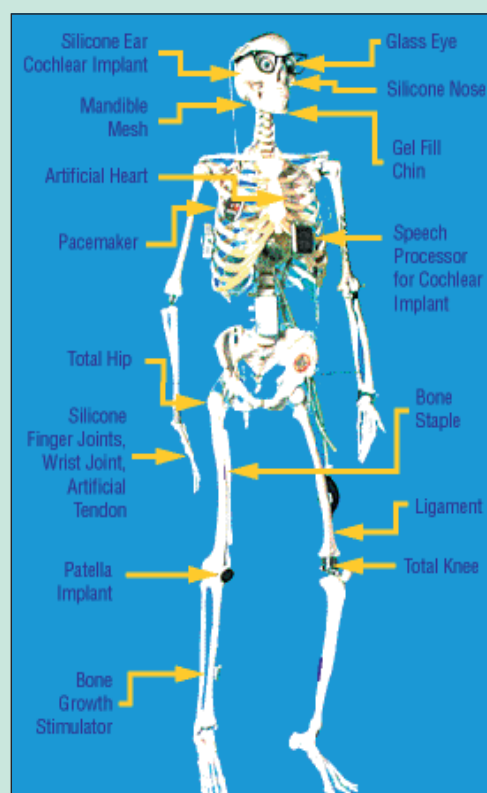
FIGURE 7.27 (right)
The hero of the TV series 'The Six Million Dollar Man', Steve Austin, was claimed to be the first bionic man.



FIGURE 7.28
(far right)
Yorick can be viewed on the Internet. This skeleton illustrates the types of spare body parts which are currently available.
(<http://www.fda.gov/oc/opacom/kids> and choose Meet Yorick from the menu.)

The Bionic Man

In the mid 1970s, a science fiction TV show called 'The Six Million Dollar Man' was produced. It was based on a novel called *Cyborg*, written by Martin Caidin in 1972. The hero of the story, Steve Austin, was the first **bionic** man. He had been an astronaut for NASA and had suffered serious injury when testing an experimental aircraft. Both legs, a left arm and a right eye were replaced with bionic parts. The bionic legs enabled him to run at superhuman speed and make huge leaps. His bionic arm gave him superhuman strength and his bionic eye gave much better vision than any human eye and also allowed him to detect infrared light. This TV series was followed soon after by 'The Bionic Woman'.



Replacement body parts

Today, medical researchers are far less ambitious in designing replacement body parts compared with the writers of 'The Six Million Dollar Man', but the designs are much more sophisticated than 50 years ago. At that time, wooden legs and arms were attached to the body by leather straps.

After World War II, many soldiers returned home with injuries to legs and arms. At the same time, aircraft industries were looking for peace-time applications for some of the new technologies that had been developed such as nylon and lightweight metals. The development of new prosthetic (artificial) limbs began and has continued with the development of materials like copolymers and alloys of titanium metal.

The first improvements in prosthetic limbs made them much lighter and therefore much easier to lift and move. Development of silicone technology then allowed the production of synthetic coverings which matched skin tone. Computer technology and advances in electronics also helped in developing ways of using electrical signals to control movements of these limbs.

In 1999 a young Scottish man, Campbell Aird, was fitted with a bionic arm. His own arm had been amputated 16 years before because he was suffering from a very serious cancer. The bionic arm contains tiny motors and gearboxes. Campbell is able to control the arm, elbow and hand movements by using muscles in his shoulder to send electrical messages to the limb.

Some medical researchers are working on ways to attach bionic arms to bone by using titanium implants. Others have developed artificial limbs which are capable of detecting touch. Electrodes implanted in

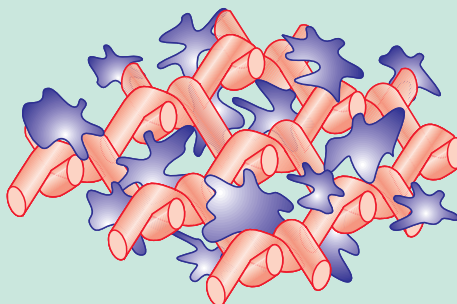


FIGURE 7.29

Polymers with a network of tiny pores are being used to provide a scaffold into which cells can be placed. Given the right nutrients, these cells divide and become organised into living tissue as the scaffold dissolves away. This tissue can then be used in a transplant.

the artificial limb send messages back to electrodes implanted in the skin where the touch is then detected.

In 1999, medical researchers in Australia developed a microprocessor unit which can be attached around the waist of paraplegics. It controls a wide variety of electrical signals to muscles in the lower limbs, causing them to contract in a controlled way, allowing the user to walk.

Future research in replacement body parts

While some medical researchers are developing more sophisticated bionic parts, others have followed totally different research. Transplanting of organs such as heart, lungs and liver have been performed in recent decades. In 1998 surgeons performed the first hand transplant in France. One problem with these transplantation procedures is that the person's immune system recognises the transplanted organ as foreign material and responds by producing chemicals to destroy the tissue. This **rejection process**

FACT

FILE

Some deaf people can now have hearing restored by the use of cochlear implants. This bionic ear has two parts. The external part is made up of a microphone, a speech processor, a transmitting coil and a cable. The internal part is an implant which is placed in the skull and threaded into the cochlea. It produces an electrical signal which stimulates the auditory nerve to send messages to the brain.

USING SCIENCE 4

DESIGNING A MODEL LIMB

Design a model limb, using available materials. Will your design be a simple mechanical model or can

you use computer technology to make a working model?

FACT

FILE

Pacemakers are used to restore healthy heart rhythms. They consist of a pulse generator and a lead which carries the electric pulses to the heart. It is implanted under the skin of the chest and can be adjusted by placing a computerised programming device over the skin above the pacemaker.

must be controlled by the use of special drugs which patients must take for the rest of their lives.

Other researchers are looking for ways to use human tissue to make replacement body parts. Surgeons in the United Kingdom have developed a technique to use a person's own cartilage to repair a damaged knee joint and so avoid knee replacement surgery. They remove some cartilage cells and then mix them with a special enzyme to break down the **collagen** fibres which bind them together. These cells are then grown in the laboratory and reinserted into the knee where they grow, coating the bones with a new layer of cartilage. Similar research is showing ways to grow ligaments, tendons and muscles which can then be used to replace damaged tissue in the body. If these tissues are grown from the patient's own cells, then the problems of tissue rejection are

avoided.

In 1999 medical researchers in the United Kingdom reported growing bone cells to repair the bone of a patient who had lopped off the top of his thumb. Some of his bone cells were removed and injected into a piece of coral shaped like the bone which had been amputated. The bone cells grew through the pores and channels in the coral, breaking it down as they grew. This new bone was then transplanted back to replace the missing part of the thumb.

In the future, it may be possible to manufacture replacement organs like human hearts from just a few simple cells. Because it would take many weeks to grow the heart from the patient's own cells, medical researchers would like to have replacement hearts already prepared, using human heart cells which have been specially engineered to block any immune response in the heart recipient.

Questions



- 1 What is a cyborg? Name one example of a cyborg found in science fiction.**
- 2 What factors influenced the development of better artificial limbs directly after World War II?**
- 3 What types of materials were used in the first artificial limbs? What types are now used?**
- 4 A prosthetic arm may not necessarily be a bionic arm. What needs to be present before the limb can be described as bionic?**
- 5 Describe one example of cell replacement technology being used to provide replacement body parts.**

Think about



All the cell replacement techniques mentioned in this article use examples of adult human tissue. There are special stem cells which can be obtained from human embryos which are much easier to grow in the laboratory to form a whole range of specialised tissue. Should this tissue from human embryos be used to provide the raw materials for cell replacement technology?

Find out



Huge advances are being made in cell replacement research. Use newspaper articles, popular science journals or the Internet to research recent discoveries in this field.

Skin structure and function

7.8

FACT

FILE

Australians have one of the highest rates of skin cancer in the world. This is associated with lifestyles where skin is exposed to sunlight on a regular basis for a long period of time or where severe sunburn has occurred. There are more cases of skin cancer the further north you travel in Australia.

Talk about

What is the largest organ in the body? The liver? Brain? You might be surprised!

FIGURE 7.30

Variation of human skin colour is due to the presence of a pigment called melanin.

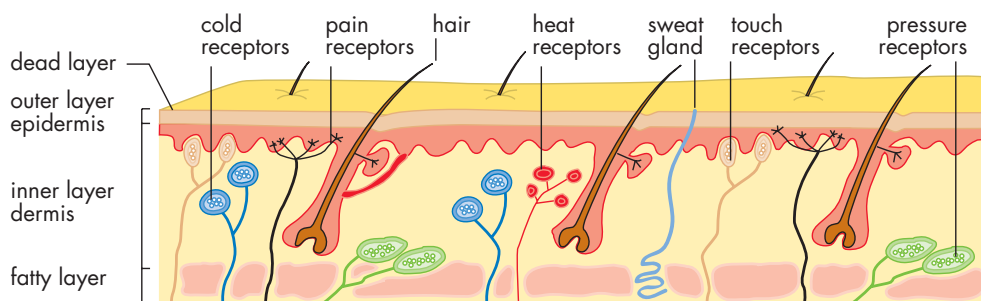


FIGURE 7.31

Skin has three layers: an epidermis from which dead skin cells are shed; a dermis which contains sweat glands and most of the skin receptors; and a layer of fat which provides insulation.

USING SCIENCE 5

TEMPERATURE SENSORS IN SKIN

WHAT YOU NEED

A beaker of ice water, a beaker with water at room temperature and a beaker with very warm water (not more than 40°C).

WHAT YOU DO

Place your hand in the ice water for 5 seconds and then place it in the water at room temperature. Then place your hand in the warm water for 5 seconds and back into the water at room temperature. Does the water at room temperature seem to be at a different temperature depending on whether you put your hand in ice water or warm water first?

Skin is the largest organ in our body. It makes up about 16% of the total body weight of an average adult and has an average area of about 1.8 square metres. Its main functions are:

- acting as a **barrier** to protect the body. It is tough, flexible and waterproof. It contains chemicals which can destroy invading organisms. Some of the bacteria which grow on the skin surface also help to protect it.
- acting as an **excretory organ**. Sweat glands excrete water, urea, mineral salts and some glucose. The evaporation of water from the skin surface requires heat energy and this is the main function of sweating. On a hot day, blood in capillaries close to the skin surface is cooled as sweat evaporates from the skin. This cooling process is important in maintaining a constant body temperature. The other wastes in sweat provide nutrients for colonies of harmless bacteria which grow on the skin.
- acting as a **sense organ**. There are a number of different sense receptors in the skin. Some detect touch or pressure, some detect heat or cold and others detect pain.



FIGURE 7.32
Early detection of skin cancer leads to a cure in most cases.

EXPERIMENT 3

SUNSCREENS: ARE THEY A COVER-UP?



AIM

To see how effective sunscreens are at absorbing UV light.

MATERIALS

- sunscreens with different SPF ratings
- UV-sensitive paper
- piece of glass or beaker

METHOD

- 1 Write the name and SPF rating of each of the sunscreens onto the back of a piece of UV-sensitive paper so that you have one piece of paper for each sunscreen.
- 2 Smear some of each sunscreen onto the shiny side (sensitive side) of the corresponding piece of UV paper. Only cover half the paper with sunscreen.

- 3 Put each piece of paper into the sunlight (making sure they are sunscreen-side up). Put one more piece of UV paper in the sun with a piece of glass or a beaker covering half of it.

- 4 Leave the paper in the sun for about 10 minutes. Observe and record the results.

DISCUSSION

- 1 What happened to the paper? Glue a strip of each piece of paper into your book and write beside it the name of the sunscreen you used on it.
- 2 When UV light hits the sensitive film on the paper, it turns it black. Sunscreen absorbs UV light and stops it reaching the sensitive film. Explain why your pieces of paper look the way they do.
- 3 Did the glass have any effect on the amount of UV light getting through? Could you get sunburnt through glass?

USING SCIENCE 6

THE SENSE OF TOUCH

Design an experiment to determine if the skin on fingers is more sensitive to touch than skin on other parts of the hand or arm. Use two toothpicks. If the two toothpicks touch skin at the same time, but a long distance apart, they can be detected as separate stimuli. If skin is very sensitive, they can be closer

together and still felt as two separate touches. Use this idea to help design your experiment. Be sure to make your experiment a fair test.

Perform your experiment and write up the experimental report.

Hair on the skin has a number of functions. It helps detect movement. For example, if a hair is moved by the wind, it will stimulate pressure receptors near its base. Hairs also stand upright in cold weather, helping to trap a layer of still air close to the skin. The still air helps to provide some insulation to reduce heat loss to the surroundings on a cold day.

Pigmentation

The colour of skin is due to a pigment called **melanin**. Fair-skinned people have very little melanin pigment in their skin, but exposure to sunlight causes more melanin production. Dark-skinned people have far more melanin in their skin. This pigment helps to provide extra protection for skin, so dark-skinned people have a lower incidence of skin cancer.



FIGURE 7.33

Exposure to sunlight not only causes sunburn. It can cause long-term damage to the skin.

FACT

FILE

The Australian Cancer Council's 'Slip, Slop, Slap, Wrap' campaign advises everyone to slip on a shirt, slop on some sunscreen, slap on a hat, and wrap on some sunglasses to protect the skin and eyes from exposure to sunlight. The Council also advises people to stay out of the sun during the hottest part of the day when radiation is more intense.

Questions



- 1 What are the three main functions of the skin?
- 2 Explain how sweating cools the skin.
- 3 Why are fair-skinned people more likely to suffer from skin cancer than dark-skinned people?
- 4 Why do Australians have such a high risk of getting skin cancer?
- 5 Prepare an educational brochure for families to show how sun damage to skin can be avoided.



Think about



Today we live in a multicultural society. Thousands of years ago, when opportunities for travel were very limited, the different races were found in different parts of the world. In what climate were dark-skinned people found? In what climate were fair-skinned people found?

APPLICATIONS AND USES OF SCIENCE

Artificial skin

FIGURE 7.34

Patients with severe burns need expert medical attention.



FACT

Collagen fibres in skin help give it firmness, elasticity and structure (much like cellulose in the cell walls of plants). To make one type of scaffold which can be used for artificial skin, collagen fibres from animal tendons are bonded with special molecules from bone cartilage.

FILE

Burns to the skin can have a variety of causes, such as direct heat, electrical burns, exposure to radioactivity or sunlight, corrosive chemicals and friction. Burns can be classified according to how deep the skin damage is.

- **First-degree burns**—where only the epidermis is damaged. There may be reddening, swelling and pain.
- **Second-degree burns**—where damage reaches into the dermis. Blisters form as plasma fluid seeps into the damaged tissue. The fluid contains antibodies to fight infection at the burn site.
- **Third-degree burns**—where all layers of the skin are damaged. Sometimes fat, nerve muscle, tendon and bone tissue can be affected. There is an open wound which can easily become infected if not protected. Infection is the main cause of death in burn patients.

Treatment of burns

First-degree burns and less severe second-degree burns can be treated by first aid measures at home. The first step is to wash the burned area in water at room temperature for about 20 minutes. This keeps the burn cool and stops the heat damage spreading. The burned skin can then be covered with a sterile dressing to reduce the risk of infection.

Third-degree burns need expert medical attention. The wound must be covered to protect against dehydration and infection. In less severe cases, new skin tissue can grow to repair the damaged area, but for very deep burns, **skin grafts** are necessary.

In a skin graft, surgeons remove healthy skin from another part of the body to cover the wound. The grafted skin is very thin and although it covers the burn, it is not as good as the original skin. There is often

scarring at the edge of the skin graft. The grafted skin contains no oil glands, sweat glands or hair follicles.

Skin substitutes

Skin grafts are not possible for patients who have severe burns over most of the body. In these cases, the burns must be covered by synthetic wound coverings or skin substitutes. In the past, surgeons used skin from dead bodies as temporary cover for a wound, until a patient's own skin could be used for repair. Eventually, the foreign skin tissue was rejected by the body.

Today, synthetic materials are available which do more than cover the wound. They actually provide a matrix or scaffold in which new skin cells can grow. One example of this skin substitute has two layers. The top layer is a sheet of silicone rubber which serves as an **epidermis**. The lower layer is made of **collagen** fibres that form a lattice in which replacement skin cells can start to grow. Eventually the lattice breaks down, leaving skin cells which have collected together to form new skin tissue.

The regeneration of skin cells can also be achieved in the laboratory. Skin cells from a burns victim can be cultured in a collagen lattice matrix to eventually form a layer of skin tissue which can be transplanted back to the patient. This cultured skin tissue can also be used for testing new cosmetics.

Artificial skin has saved the lives of many burns victims. The scarring obtained with ordinary skin grafts does not occur with artificial skin, which is gradually replaced by the patient's own skin cells. The next challenge in tissue engineering of skin is to make it even more like human skin, with pigment cells, sweat and oil glands and hair follicles.

Questions

- 1 List common causes of burns.
- 2 Outline briefly the difference between first-degree, second-degree and third-degree burns.
- 3 What is a skin graft? What type of burns victim needs skin grafts?
- 4 What type of material was used to cover severe burns in the past? What



FIGURE 7.35

A skin graft provides protection for damaged skin tissue.

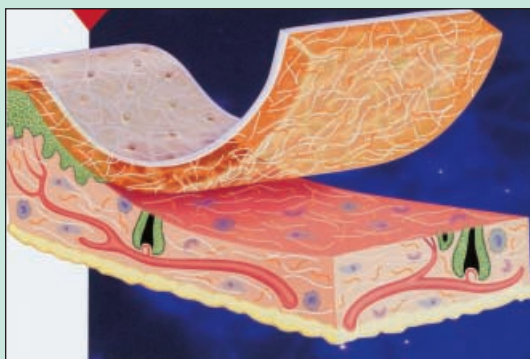


FIGURE 7.36

Artificial skin has saved the lives of many burns victims.

was one problem with using this?

5 What is artificial skin? Describe how it protects the area of a severe burn.

6 Prepare a first aid poster to outline treatment for burns.

Think about

Patients with large skin grafts are advised to avoid strenuous exercise. What problem would they have as their bodies start to get hotter as they exercise?



Find out

Patients with moderate or severe burns may go into shock. Find out the first aid measures you need to take for a person in shock.



Extension worksheet

Skin care

FIGURE 7.37
During a facial, any blocked pores are cleared and the skin is thoroughly cleansed. The face and neck are also massaged to increase blood flow to the face and to relax the client.

Skin care is important for both adults and children. In our Australian climate, it is particularly important to wear sunscreen to help avoid the damaging effects of the sun's ultraviolet rays. In older people, the effect of the sun on skin can be seen by comparing the skin normally exposed to sunlight (e.g. hands, face) with skin that is normally protected from sunlight by clothing. Moisturisers can also help reduce dehydration due to sun exposure or very dry conditions (e.g. an air-conditioned environment).

Special skin problems

Acne

Acne is a common skin condition caused by inflammation of the hair follicles and the sebaceous (oil-producing) glands in the skin. It affects about 80% of adolescents but may also occur in adults. The skin's hair follicles are blocked by sticky cells which trap oil inside the follicle. Bacteria multiply in the oil within the blocked follicle and cause swelling and redness. When white

blood cells die as they attempt to fight the infection, they form pus and a pimple develops.

Treatment of acne

Many treatments for acne involve using creams or washes which increase the rate of skin peeling. This helps reduce the chance of hair follicles becoming blocked. Other treatments concentrate on controlling bacteria on the skin, for example, low doses of antibiotics which may be prescribed by a doctor.

Wrinkles

Wrinkles occur with ageing as connective tissues in the skin (collagen and elastin) weaken or decline in number. The skin becomes thinner and it loses some of its elasticity. Fat deposits under the skin also are reduced, especially on the face. Sun damage to the skin also causes premature ageing. Smoking has also been shown to reduce skin elasticity and cause wrinkles.

Using skin creams to reduce wrinkles

Dermatologists have used skin peeling techniques for many years to treat skin which has been scarred by severe acne. Some noticed that when they were using chemicals to increase skin peeling, for example, retin A, there was also an improvement in very fine skin wrinkles. As the epidermis was peeled away, smoother, younger looking skin was exposed.

One group of chemicals used by dermatologists to peel skin, the alpha hydroxy acids (AHAs), has been available in lower concentrations in cosmetic creams since about 1992. Dermatology studies have shown that there is a slight

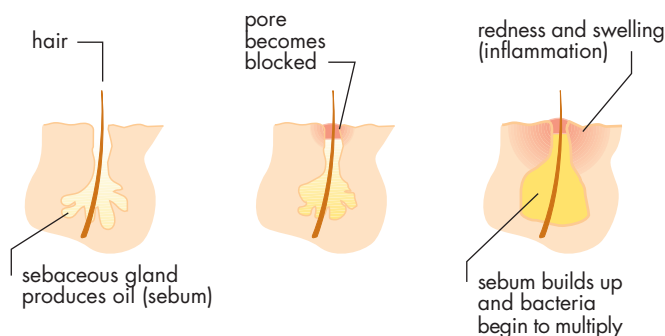


FIGURE 7.38
Acne begins with a hair follicle becoming blocked so that oil (sebum) cannot reach the skin surface.

improvement in the appearance of fine wrinkles when these creams are used. They can also help break down extra pigment which has formed in the skin. One

problem with these creams is that they make the skin more sensitive to sunlight and in some people, irritation of the skin may occur.



FIGURE 7.39
A wide variety of cosmetics is marketed for skin care.

FACT

FILE

Cosmetics have a long history. In Roman times, powdered crocodile intestine and dung were used to remove freckles. In Elizabethan times, women slowly poisoned themselves by using white face powders made of lead oxide compounds and mercury compounds.

USING SCIENCE 7

MARKETING OF COSMETICS

Very sophisticated advertising campaigns are used to sell cosmetics. Many cosmetics have the same basic ingredients, e.g. sorbolene as a moisturiser, but there is a huge difference in cost.

Choose one particular type of cosmetic and obtain a variety of examples of it. Design an experiment to

test whether one product is more appealing to a customer than the other. Is there any relationship between its appeal and the cost of the product? Make sure your experiment is a fair test. Perform the experiment and write it up in an experimental report.

Questions

- 1 Why could sunscreen be considered a vital ingredient in cosmetics?
- 2 Explain why skin peeling can help stop the formation of acne.
- 3 What are AHAs? What do they do?
- 4 What lifestyle habits can a person adopt in order to reduce the risk of premature wrinkles?



Think about

Design a marketing campaign for a cosmetic.



Find out

Vitamins C and E are both claimed to have a beneficial effect on the skin. Find out what role they play in the formation of healthy skin.



**Using scientific language**

- 1 From the following word list, choose a word which matches each of the statements below.
- | | | |
|----------|--------|-----------|
| hinge | suture | cartilage |
| sternum | marrow | tendon |
| ligament | pivot | |
- a holds bones in position at a joint
 b makes red and white blood cells
 c pulls on a bone to cause it to move
 d a type of joint which allows no movement
 e a soft flexible tissue which protects bones at a joint
 f a joint at the base of the skull
 g a joint which allows movement back and forth in one plane

- 2 Unscramble each word in the list below with the help of the hints. Then unscramble the boxed letters to solve the clue.

Scrambled words:

- a tspropu _ _ _ _ _ _ _
 b rneeojitc _ _ _ _ _ _ _ _
 c oioaihrmdcnt _ _ _ _ _ _ _ _
 d tenovmme _ _ _ _ _ _ _
 e caficdl _ _ _ _ _ _ _
 f recpamkea _ _ _ _ _ _ _
 g cgobry _ _ _ _ _
 h sueturs _ _ _ _ _ _

Hints:

- a turgid plant cells provide this for a plant
 b a process which can occur after organ transplants
 c organelles where respiration takes place in a cell
 d one important function of a skeleton
 e describes a plant cell which has lost some of its water content
 f a device which can be used to regulate heartbeat
 g name given to a human who has bionic parts
 h fixed joints found on the skull

Clue:

the type of fracture where the broken bone pierces the skin

Solution:

Check your knowledge

- What structures provide support in plants?
- Why can animals in the ocean grow to be heavier than animals on land?
- Describe the similarities and differences between striated, smooth and cardiac muscle.
- Draw two diagrams to explain how the biceps and triceps muscles work together to allow your arm to bend and straighten.
- Name four possible causes of back pain.
- Describe the three different types of burn.
- Describe how cell replacement technology can be used in medicine.

Apply your skills

- How could a skeleton be used to determine the age of a dead body?
- List joints in the skeleton in order, from the joint which allows most movement to the one which allows least movement.
- Obtain a variety of moisturisers. Use the labels to determine whether these creams contain sunscreens or AHAs.

Challenge yourself

- Write a descriptive report on the latest developments in cell replacement technology.
- Find out about the flight muscles of insects. Draw a diagram to explain how they function.
- Research causes of acne and the treatments which are available. Present your findings on a poster which could be displayed in the waiting room of a dermatologist (a doctor who specialises in skin disorders).
- Find a science fiction novel which features cyborgs. Read the novel and present a review of it to your classmates.
- Design a poster that shows young adolescents how to avoid skin damage and wrinkles.



CHAPTER

8

SURVIVAL IN A HARSH ENVIRONMENT

About 200 million years ago, Australia was joined to Africa, South America, India and Antarctica as part of a giant southern super-continent, known as Gondwana. Around 100 million years ago, these continents started to drift apart. At about the same time, dinosaurs died out very suddenly.

Australia finally separated from the other continents about 50 million years ago. At that time, the continent was mostly covered with dense rainforest. As it moved slowly northwards, the climate gradually became cooler and drier and a wide variety of new plants and animals started to form. Marsupials became a dominant land animal.

Today, there are at least 20 000 species of native flowering plants in Australia. Of these, 85% are endemic (found nowhere else in the world). Of 850 species of native bird, 70% are endemic. Most native Australian mammals are marsupials, another feature that makes Australia unique. Monotremes, like the echidna and platypus, are egg-laying mammals. New Guinea is the only other part of the world which has monotremes.

In this chapter, you will be looking at the Australian environment and why it is so different from that of any other continent. You will learn about adaptations of some plants and animals which help them survive Australia's harsh conditions. You will also learn about some of the threats to habitats and how this could affect us.

CHAPTER OUTCOMES

Successful completion of this chapter will allow you to:

- Describe examples of adaptations of animals for a harsh environment.
- Explain why Australian soils are poorer and contain fewer minerals than soils on other continents.
- Explain why droughts and floods occur regularly in Australia.
- Identify adaptations in Australian plants which help them survive bushfires.
- Give examples of the impact that human activities have had on the environment.
- Discuss how introduced species have affected the Australian environment.
- Understand that water and soil are precious resources and discuss the conflict about their use.

Adaptations for a harsh environment

FIGURE 8.1 (above left) When a frill-necked lizard is approached by a swooping bird, it stands and faces the predator with its frill unfurled and its mouth wide open. This fierce display is usually enough to stop the attack. If not, the lizard becomes more aggressive and lunges at the bird. At other times the frill is folded around its body, helping to camouflage it.

FIGURE 8.2 (above right) Marine mammals like whales and dolphins have blowholes near the back of their heads to allow them to get oxygen without lifting their head completely out of the water.

FIGURE 8.3 The greater bilby (*Macrotis lagotis*) is the largest member of the bandicoot family of marsupials. Its very strong forelimbs and claws are adapted for digging burrows and digging for food such as bulbs and insects. The animal has poor eyesight but a good sense of smell and hearing. Its large, almost hairless, ears also provide a large skin surface which can help keep the animal cool.

The frill of the frill-necked lizard shown above contains supporting spokes which are extensions of a bone at the base of its tongue. The opening and closing of the lizard's jaw controls this display. This unique adaptation protects it from predators.

An **adaptation** is any characteristic of a living thing that helps it survive in its environment. An adaptation may be general (e.g. fish have fins to aid in swimming), or more specialised (e.g. some varieties of angler fish have luminous filaments in their mouths that act as a lure for inquisitive fish). These adaptations are physical characteristics. Sometimes, the way an animal behaves helps it survive. This is a behavioural adaptation. For example, the frillneck lizard suns itself to increase its body temperature.

Talk about

What adaptations would you expect in an animal living in water compared with an animal living on land? Compare how a fish and a lizard get air and how they move.



Marine mammals

Marine mammals like whales, dolphins and seals have special adaptations to allow their survival in the ocean. They have a constant body temperature which is much higher than the surrounding water temperature. To prevent the surrounding water absorbing heat from their bodies, they are insulated by layers of fat or blubber. The blubber also has other uses, such as an energy source at times of food shortage. Because they can survive in water at different temperatures, marine mammals can migrate from the warm tropics to the icy waters of the Arctic or Antarctic.

Like all mammals, marine mammals have lungs. Therefore, they must obtain oxygen directly from air at the water surface. They also have a remarkable ability to dive for long periods of time. This is partly because their muscles store more oxygen than other mammals, using a pigment called **myoglobin**. Only about 13% of oxygen in the human body is stored in muscles, but in whales it is 41%. Also, during a dive, blood flow is concentrated to vital organs like the brain and heart, which means that less oxygen is used.

Desert mammals

Land provides a much harsher environment than water. **Dehydration** (losing water from cells) is a particular problem in a desert environment, especially for mammals, because they have two conflicting

USING SCIENCE 1

WATER BALANCE IN THE KANGAROO RAT

Use the data in Table 8.1 about a desert kangaroo rat (collected when humidity in the burrow was 20%) to answer the questions below.

TABLE 8.1

Water gained (mL/day)		Water lost (mL/day)	
Respiration in cells	54	Evaporation	43.9
Water in food or from dew	6	In urine	13.5
		In faeces	2.6

QUESTIONS

- 1 Is the kangaroo rat in danger of dehydration when in its burrow? Explain your answer.
- 2 Calculate the percentage of total water gained from respiration of food and the percentage of total water lost through evaporation.
- 3 Describe how the kangaroo rat would lose water by evaporation.
- 4 If humidity in the burrow fell to 10%, what would happen to the water balance for the kangaroo rat?

needs. They must conserve water but they must also maintain constant body temperature by allowing some evaporation of water from their skin surface.

Water is lost from an animal's body through **excretion**. To reduce water loss, desert mammals excrete very concentrated urine. Also, their droppings are very hard and dry. To reduce the amount of water lost when they breathe air from their lungs, red kangaroos find a shady spot to rest in the hottest part of the day. Their metabolic rate slows down and their breathing rate is reduced. Other mammals, like the bilby, live in burrows where the temperature remains fairly constant and the humidity is high. This reduces the loss of water vapour from their lungs and skin.

Desert temperatures can range from about 40°C in the middle of the day to below 0°C at night. The forearms of kangaroos contain a dense network of blood vessels close to the skin. On a hot day, kangaroos lick their forearms to add moisture which then evaporates. This helps cool the blood flowing just below the skin surface. Other desert mammals only come out from their burrows in the early morning or early evening to forage for food when the temperatures are mild.

Desert animals cannot rely on a regular supply of water for drinking. They obtain water mostly from **respiration** of food.

food + oxygen → carbon + water + energy
dioxide

FACT

FILE

Outback spinifex ants gather resin from spinifex and mix it with sand to make a cement. They make covered runways to protect them from hot sand, and shelters for scale insects. These insects secrete honeydew, which the ants use for food.

Questions



- 1 Define the term **adaptation**.
- 2 Give two adaptations that a frill-necked lizard has to protect it from predators.
- 3 Describe two adaptations of whales which allow them to dive for lengthy periods.
- 4 Describe the ways that water is lost from the body of a red kangaroo.
- 5 The relationship between spinifex ants and scale insects is an example of mutualism—a close relationship between two species where both benefit. What is the benefit to each organism?
- 6 Copy and complete Table 8.2.

TABLE 8.2 Adaptations of some desert marsupials.

Adaptation	Advantage(s) to the animal
Excretion of concentrated urine	
Resting in shade in middle of day	
Strong forearm with claws	
Living in burrows	
Forearms richly supplied with blood vessels close to the skin	
Large hairless ears	
Obtaining water from respiration of food	



Internet assignment

The sunburnt country

FIGURE 8.4

Forty per cent of the Australian continent is desert. The area surrounding Kings Canyon in Central Australia is typical of much of inland Australia.

In 1791, three years after European settlement of Australia had begun in Sydney, Governor Phillip wrote back to the colonial secretary in London: 'I do not think it probable that so dry a season often occurs. Our crops of corn have suffered greatly from the dry weather'. But Governor Phillip was wrong. In the next few years crops failed and food had to be imported from Cape Town in South Africa.

The British settlers found that drought was a regular occurrence in Australia. It was often followed by flooding. As exploration of the country continued, it became apparent that there were no rich, fertile soils west of the Blue Mountains. Nor was there an inland sea. It was very hard for these new settlers to appreciate that Australia was an arid continent, with very poor, infertile soils.

Why Australia is different

Australia has some of the most ancient landscapes in the world. The rocks have been subjected to continuous weathering and erosion over tens of millions of years without the formation of new rock and soil that has occurred on the other continents. There are two main reasons: very little volcanic activity and the lack of ice sheets during the last Ice Age.

Volcanic activity results in molten material from deep within the Earth's crust being brought to the surface to cool and form **igneous** rocks such as basalt. The weathering of these new rocks produces rich soils. Australia has been a very stable land mass in the past 45 million years, with very little volcanic activity. Some volcanoes formed as the continent moved over hot spots deep within the Earth on its slow journey northward (6–7 cm each year). The remnants of these volcanoes are found down the east coast of Australia in the Great Dividing Range, with the oldest towards the north because this part of Australia passed over the hot spots first. The basalt rock in these areas forms rich soil which can support rainforests or agriculture.

Ice Ages played an important part in producing rich soils. In the most recent Ice Age, which lasted over 2.5 million years, **glaciers** formed and rocks were ground up to form fresh, rich soil as ice sheets moved to cover all the continents of the Northern Hemisphere. This process happened repeatedly, leaving rich, fertile soils on these continents. In Australia, glaciers affected only a small part of the Southern

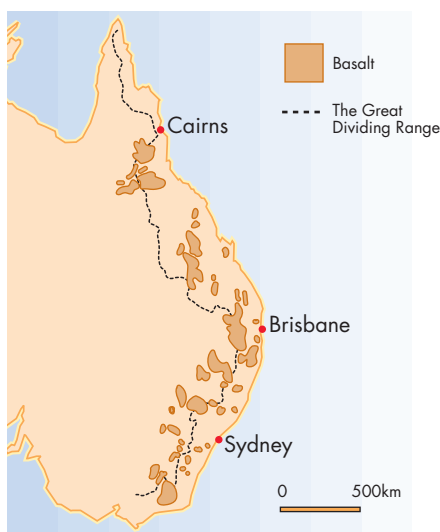


FIGURE 8.5

The basalt outcrops found in eastern Australia range in age from a few thousand years near Melbourne to about 40 million years in the north.

Highlands. The rest of the continent was very dry because so much of the Earth's water was tied up in the huge ice sheets over the rest of the planet. Strong winds caused extensive erosion and there were probably fierce fires. Australia became a desert island, covered with blown dunes and sand for 80% of its area. Some plants

survived in isolated, hospitable pockets across the continent. Apart from Antarctica, Australia is still the driest continent, with 75% of its land surface classified as arid and 40% still dune desert. In most areas, the Australian flora which have survived are adapted to survival in poor soils and they are drought tolerant.

FACT

FILE

Many Australian rivers flow inland. Early European settlers believed that they must drain into an inland sea. Instead they found that they formed salt marshes which dried up in times of drought.



FIGURE 8.6

A glacier grinds up rock as it moves forward and then deposits the broken rock as it retreats. If the process is repeated a number of times, rich soils can build up.

USING SCIENCE 2

WATER PERMEABILITY OF SOILS

Design an experiment to compare the water-holding capacity of two different soils. Think carefully about making your experiment a fair test. Collect a sample of soil that has weathered from a sandstone rock and a second sample from a different area (e.g. from a garden). Perform the experiment and write up your experimental report.

DISCUSSION

- 1 Which soil is more **permeable** to water (i.e. which allows water to pass through more easily)?
- 2 Explain how the water **permeability** of a soil can affect its fertility.

Questions



- 1 What major problem faced British settlers in their first years in Sydney?
- 2 Name a rock which is formed as a result of volcanic activity. What type of soil does it form?
- 3 Why was rainfall very low in Australia during the last Ice Age?
- 4 Explain why the Australian continent does not have the deep rich soils found in continents of the Northern Hemisphere.

Find out



Some commercial crystals and liquids are capable of improving the amount of water which garden soil can absorb and hold. Find out the names of these products and design an experiment to test their effectiveness.

Think about



Over 75% of Australia's population live around the fertile edge of the continent within about 1 hour's drive of the ocean. What special problem does this pose for the environment?

The Australian bush

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FIGURE 8.7
Buds, flowers and fruit of a eucalyptus tree.

The Australian bush is largely dominated by one genus, *Eucalyptus*, except for arid regions where acacias are predominant. **Eucalypts** grow in most parts of Australia, from snow-covered mountains to the dry inland. Tall eucalypts grow close together to form forests. Smaller trees further apart form open woodlands. Other varieties are smaller with many stems. They grow in dry inland areas and are called **mallees**, from an Aboriginal word *mali* meaning water. The roots of some mallees store water, used in times of water shortage by Aboriginal people.

All leaves have pores called stomata at the leaf surface which allow air to move in and out. This process is important for photosynthesis, which uses carbon

dioxide and releases oxygen. Water vapour is also lost from the leaf surface through open stomata. This evaporation of water from a leaf is called **transpiration**.

Eucalypts are well adapted to growing in dry, infertile Australian soils. They have deep roots to absorb water efficiently from the soil. They also have adaptations to help them conserve water. The leaves hang vertically to reduce the surface area exposed to the sun and thus reduce water loss by transpiration. The leaves are hardened with thick waxy cuticles so that they do not wilt in times of water shortage. They share this feature with many other native Australian plants. Plants with such hardened leaves are known as **sclerophylls**.

EXPERIMENT 1

WATER UPTAKE IN PLANTS



AIM

To investigate water uptake by a plant.

MATERIALS

- 4 identical plant stems with leaves
- 5 test-tubes and test-tube rack
- petroleum jelly

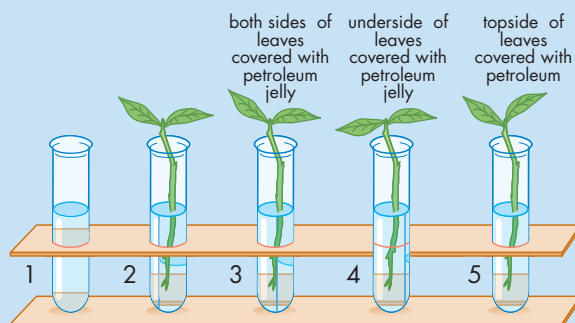


FIGURE 8.8
Experimental set-up to measure water uptake.

METHOD

Three-quarters fill five test-tubes with water. Place the plant stems in four of the test-tubes, some covered with petroleum jelly as shown in Figure 8.8.

RESULTS

Observe any changes that occur in the water level over the next few days. Mark the changes on a diagram of the experimental set-up.

DISCUSSION

- 1 What is the purpose of the petroleum jelly?
- 2 Account for any differences in water uptake between test-tubes 2 and 3.
- 3 What process must be occurring in test-tube 2?
- 4 Account for any differences between test-tubes 4 and 5.
- 5 Why was it important to include test-tube 1 as part of the experiment?

A second genus of plants, *Acacia*, dominates the drier regions of Australia. The roots of acacia plants penetrate deeply into the soil and contain nodules with bacteria that change nitrogen from air in the soil into nitrates. This helps enrich the soil. Some acacia have very fine, narrow leaves. This reduces water loss by transpiration because the leaves have a small surface area. In other acacias, leaves are replaced by flattened stalks called phyllodes which are much harder than leaves and better adapted to arid con-

ditions. To reduce water loss, the phyllodes have thick cuticles, sunken stomata, abundant hairs and high oil content. They hang vertically to reduce the surface area exposed to the sun and to channel rain-water down to the roots of the plant.

FIGURE 8.9

Wattle (acacia) is the national floral emblem of Australia. Wattle seeds also provide good bush tucker. The seeds can be ground to form a flour for cooking. In 1997 the CSIRO began a project where African communities learnt to use acacia seed (from introduced Australian plants) as a food source. The knowledge obtained from Australian Aboriginal communities was a vital part of this project.



EXPERIMENT 2

OBSERVATION OF STOMATA

AIM

To observe and compare stomata from a variety of leaves.

MATERIALS

- variety of leaves, e.g. fern, eucalyptus
- bottle of clear, colourless nail polish
- forceps
- coverslip
- microscope slide
- light microscope

METHOD

- 1 Paint the bottom surface of a leaf with nail polish.
- 2 Allow the nail polish to dry and then use forceps to peel it off.
- 3 Place the clear layer on a microscope slide and place a coverslip over it to flatten it.
- 4 Place the slide on the stage of the microscope and focus it under low power. Observe the appearance of stomata and surrounding guard cells,

together with the waxy cuticle of the leaf surface.

- 5 Draw a labelled diagram.
- 6 Paint a layer of nail polish on the top surface of the same leaf and repeat steps 2–5.
- 7 Repeat steps 1–6 for the other leaves.

RESULTS

Draw labelled diagrams for each slide that was prepared.

DISCUSSION

- 1 Describe any difference between the number of stomates for different leaf types.
- 2 Can you relate any difference in number of stomata to the type of environment to which the plant is adapted? Explain.
- 3 Are there any differences between the number of stomata found on the top surface and bottom surface of the leaves? Can you account for any differences?



Questions

- 1 Define the term transpiration.
- 2 What are stomata? What is their function?
- 3 Name the two plant genera which dominate the Australian bush.
- 4 List the ways that eucalypt and acacia leaves are adapted to reduce water loss by transpiration.

Find out

Visit a local native bushland area. Identify native plants and their

adaptations. What other plants have invaded the area (e.g. privet, lantana)?

Think about

- 1 Some native plants have fine hairs on the leaf surface. Explain how these hairs could help reduce transpiration from the leaf.
- 2 Which animals and plants are on the Australian coat of arms? Design another coat of arms for Australia, featuring different native animals and plants.

FACT

FILE

Water loss from a leaf surface is controlled by the guard cells around each stoma. At times of water shortage they become flaccid, collapse and close the stomata.

Bushfire

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FIGURE 8.10

Many Australian native plants have special adaptations to help them survive bushfire.

FACT

FILE

Early European explorers described Australia as the 'burning continent', because of the fires they saw along the coast. Most of these fires were lit by Aboriginal communities. Some of these fires promoted germination of plants used for food. Since European settlement, fires are much more frequent.

Talk about

What impact has bushfire had on your life? What bushfires have been reported recently on the news? How do local communities respond to these bushfires?

Fire has been a regular part of the Australian environment since the last Ice Age. The oil in the leaves of eucalypts and acacias that helps to reduce water loss by transpiration also makes these plants more flammable. As a result fire spreads quickly through dry Australian bushland.

Many Australian plants have developed special adaptations to help them survive bushfire. Eucalypts have buds beneath the bark on their trunks. These buds start to shoot if fire burns the outer bark. They also have woody roots called **lignotubers** which can regenerate after the main part of the tree has been destroyed by fire. Other native plants need fire to help them reproduce. Acacia seeds need fire to burn the outer seed casing before they can germinate. Other plants like banksias have fruit which opens up after fire to release the seeds.

Before any humans arrived on the Australian continent, bushfires were started by lightning strikes or other natural occurrences. The first human inhabitants

arrived in Australia perhaps 60 000 years ago. These Australian Aborigines learned to use fire for more than just warmth, cooking and sending signals. It was also used to help hunt animals. They learned that new plant growth occurred after bushfire. This would attract more wildlife for hunting. Their patchwork burning of the landscape allowed areas where plants and animals remained unharmed, while the burnt areas had the opportunity to regenerate. There were probably more woodland areas as a result of their bushfires and some fires which went out of control may have changed open forest into grasslands in regions of northern Australia.

The first European settlers had a much greater impact on the environment. They used fire to clear huge areas of land for grazing and agriculture. In the process, habitats for animals and plants were destroyed. In addition, the lack of vegetation allowed top soils to be washed away in heavy rains. They did not foresee the impact of their activities on a fragile environment.

Today, half of Australia's native forests and woodlands have been cleared for farming and urban development. **Soil degradation** is a major problem facing Australian farmers.

Although bushfire can be useful in regenerating areas of sclerophyll forests, very intense fires and very frequent fires cause many problems. For example:

- 1 Acacia seeds will germinate after fire, but the parent plants will be destroyed. If a second fire occurs before the new plants have matured and set seed, there can be no further regeneration of acacias in that area.
- 2 The fire changes the structure of the soil itself and destroys the decomposer

FIGURE 8.11

The woody fruits on a banksia head open up after bushfire to release seeds.



micro-organisms which help break down dead animal and plant material to produce humus. This results in less fertile soil.

- Ground that is exposed after vegetation has been burnt is prone to wind and water erosion. This leads to further degradation of top soil.
- Populations of animals which depend on the vegetation for food and shelter are affected, for example, native birds such

as the noisy scrub bird have decreased in numbers and range because of increased frequency of fires.

Controlled burning is used in some areas to reduce the likelihood of intense bushfires that are a threat to life and property. In this process, local fire brigades burn off selected areas of bushland to form firebreaks and to reduce the amount of plant material available to fuel any future fire.

USING SCIENCE 3

FIRE AND SMALL MAMMAL POPULATIONS

Populations of many native Australian mammals are affected by the frequency of bushfire in their habitat. The data in Table 8.3 gives the results of a survey of animal populations in an area that was burnt by a bushfire 44 years previously.

TABLE 8.3 Number of animals caught in 1000 trap nights.

Years since last fire	Introduced mouse	Bush rat	Yellow-footed marsupial mouse
1	96	24	0
2	5	72	2
5	0	66	2
7	0	83	14
24	0	62	19
44	0	38	125

Draw a line graph to show how the population of each mammal has changed over 44 years since the bushfire. Represent each species by a different line on your graph.

DISCUSSION

- Which animal is likely to die out if the area is burnt each year?
- What period between fires would maintain the largest population of bush rats?
- If you were responsible for deciding how often to control burn this area, what period between burns would you choose in order to maintain healthy populations of the native mammals?

Questions

- What natural occurrence can cause a bushfire?
- Describe the special adaptations that Australian plants have to help them survive bushfire.
- What use did Aboriginal communities make of bushfire?
- What problems can occur when large areas of land are cleared by fire?
- What is control burning? Why is it carried out?

Find out

Many Australian native plants are flammable because of their high oil content. The CSIRO has carried out research to find out which native

plants are less flammable. To reduce the risk of fire, it is recommended that these plants be grown in gardens of homes which are close to bushland areas. Find out which native plants can be grown in your area to reduce the risk of fire.

Think about

- Why would it be important not to do a controlled burn in an area just days before heavy rains are due?
- What other issues must be considered if a controlled burn is to be carried out in a pocket of bushland close to a heavily populated urban area? Write a discussion about the value of controlled burning and how often it should be done.

First drought and then flood

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FIGURE 8.12
Cleaning up after the Sydney hailstorms of April 1999.

In April 1999, parts of Sydney were hit by violent hailstorms. There are estimates that over \$1.5 billion worth of damage was done to roofs and buildings in a very short time. Sydneysiders were dismayed that there had

been no warnings about the hailstorm from the Bureau of Meteorology, but the weather forecasters had expected the storm to move out to sea and miss the city.

Weather forecasting is not a precise science. Satellite technology helps forecasters but there are so many factors influencing weather patterns, that the best they can do is give the most likely weather pattern (e.g. there is a 70% chance of rain today).

A very important advance in our understanding of weather patterns over the last 25 years has been the realisation that drought in Australia is associated with an ocean current, called El Niño. For a long time we have understood that weather

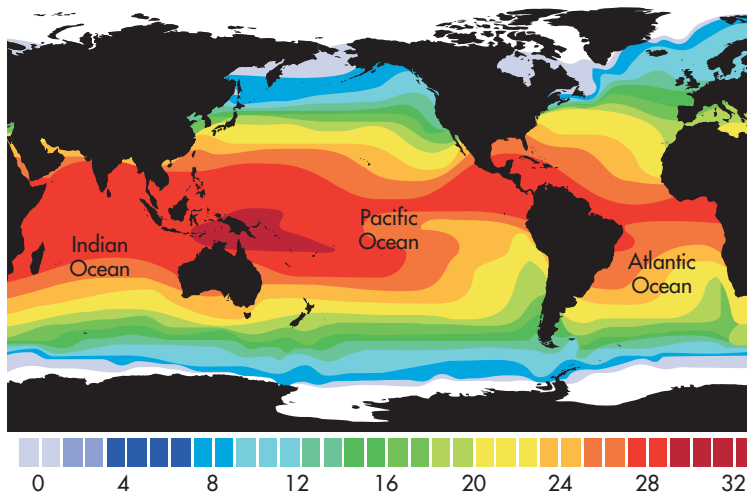


FIGURE 8.13

Average sea surface temperatures around the world. As expected, the oceans are generally warmer in the tropics, but note that the Pacific Ocean is usually cooler in the east.

USING SCIENCE 4

INVESTIGATING CONVECTION



WHAT YOU NEED

Aluminium foil, thread, candle, heatproof lid, coloured ice-cube, beaker of warm water.

PART A

Make a snake by marking and cutting a circle of foil as shown in Figure 8.14. Attach a thread to the snake and hold it over a burning candle. (Take care. Do not let the snake get close enough to the candle to singe or catch fire.)

What happens as the snake is lowered and lifted above the flame? What happens to the air which is heated by the flame?

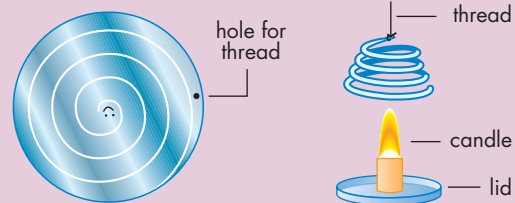


FIGURE 8.14

What happens as the foil snake is placed above the candle?

PART B

Find out what happens when warm water meets cold water by adding a coloured ice-cube to a beaker of warm water. Observe what happens to the coloured water as the ice-cube melts.

What would happen when a warm ocean current meets a cold ocean current?

patterns are affected by the seasons and the rotation of the Earth. Warm air rises in the tropics and sinks in the more temperate zones where it is cooler and denser. This, combined with the rotation of the Earth, means that prevailing trade winds blow around the Earth and are responsible for regular weather patterns.

El Niño is associated with a disruption of the normal pattern of **trade winds** and has been described as the 'world's most powerful weather warper'. It affects weather systems around the world.

Normally trade winds blowing westward across the Pacific Ocean drive warm surface water with them towards Australia and Indonesia. This results in the sea level being a few centimetres higher on the western side of the Pacific. The air above this warm water rises, forming clouds which will result in rainfall in the region.

Under El Niño conditions, a huge mass of warm water builds up on the eastern side of the Pacific, off the coast of South America. The trade winds blowing westwards die down or may even be replaced by winds blowing eastwards. There is a lot of rainfall on the eastern side of the Pacific but on the other side, the Australian continent goes into drought.

Weather satellites can detect build-up of warm water in an ocean region, so weather forecasters can predict the likelihood of drought in Australia.

La Niña is the opposite of El Niño and tends to occur slightly less frequently. Sea surface temperatures are much cooler in the eastern Pacific and regions to the west such as Australia have much warmer sea

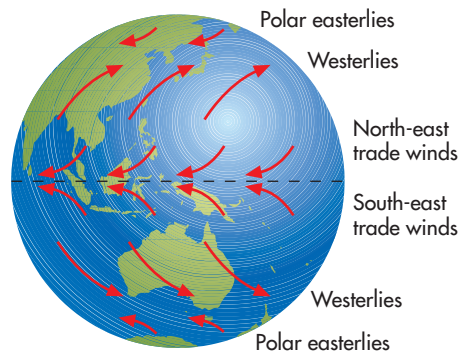


FIGURE 8.15

Trade winds are caused by a combination of convection currents and the rotation of the Earth. Hundreds of years ago, sailors relied on these winds to sail around the world.

surface temperatures. This can result in heavy rains and flooding in Australia. The changes from El Niño to La Niña are responsible for the drought and flooding rain that is part of the Australian climate pattern.

Australia's wildlife has adapted to these changes. The breeding of green turtles and ducks can be affected by an El Niño event. Many plants in arid regions have adapted to drought and flooding rains by having very short life cycles. In the desert after rain, plants like mulga and saltbush flower and produce lots of seeds. Many birds migrate to areas where there are water supplies and their breeding cycles are not dependent on the seasons but, instead, are triggered by heavy rains. Some birds produce multiple broods in a good season but do not reproduce at all in drought. Red kangaroos cease breeding in a prolonged drought and only start to breed again when rains return.

FACT

FILE

The ocean off the coast of Peru normally provides good fishing because there is an upward movement of a cold ocean current. This current is laden with nutrients to support microscopic life which provides food for fish. Peruvian fishermen noticed that they had seasons of poor fishing in some years around Christmas, and that this was associated with a warmer ocean current and more rainfall than usual. They called the current 'El Niño' which means 'the Christ Child'.

Questions

- 1 What are trade winds? Explain why they have a regular pattern of movement across the Earth.
- 2 What is El Niño? Explain what is happening when Australia experiences a drought. What measurements can be made by meteorologists to detect El Niño events?

- 3 What is La Niña? What type of weather occurs in Australia when La Niña is in the western Pacific?
- 4 Give two examples of how wildlife in Australia has adapted to drought and flooding rain.

Think about

How can farmers make use of information about Southern Oscillation?

Human impact

FIGURE 8.16

Chimneys pumping pollution into the air.

FACT

FILE

Each spring and autumn, school students across New South Wales take part in a Water Bug Survey. They collect and identify invertebrates in local waterways to determine the stream pollution index. Some invertebrates are very sensitive to pollution (e.g. stonefly and mayfly nymphs), so if they are present, it indicates that the waters are very healthy. You could find out about these water bug surveys (<http://www.streamwatch.org.au>) and take part in them.

Bushfires, floods and droughts have had an impact on the environment for millions of years. However, greater impacts are being made now as a result of human activity.

Talk about

What do you think is meant by the term pollution? What types of things would be regarded as pollutants?

When a smog alert day is declared it is not because more pollution is being created but because there is no wind to blow it away. Sydney beaches are sometimes closed, not because of increased waste being pumped out into the ocean but mainly because the wastes fail to spread out and end up back in close to the shore. Chimneys from factories are tall so that the waste gases mix with air higher in the sky and don't cause a pollution problem around the factory.

This solution to pollution is dilution: to get rid of something you just spread it out or put it somewhere else. This has been and still is the most common way we deal with pollution.

Pollution is the spoiling or contaminating of the environment. A **pollutant** is anything which does this.

Water pollution

There are many ways in which pollutants enter our rivers and oceans.

- Factories often discharge their polluted water directly into rivers or the sea.
- The storm water system which is designed to carry away rainwater from

roads and houses often picks up pollutants before it drains into a river or sea.

- Excess fertilisers or pesticides are often washed from farms into rivers.
- Mud from building sites or unmade roads can make rivers muddy, and this can affect plant growth in rivers.

All these forms of pollution will affect plants and animals which live in or around rivers, creeks, lakes and oceans. Not only can human activity add pollutants to the river, it can also alter such things as the temperature and the amount of oxygen. This may also affect the organisms that live in the water.

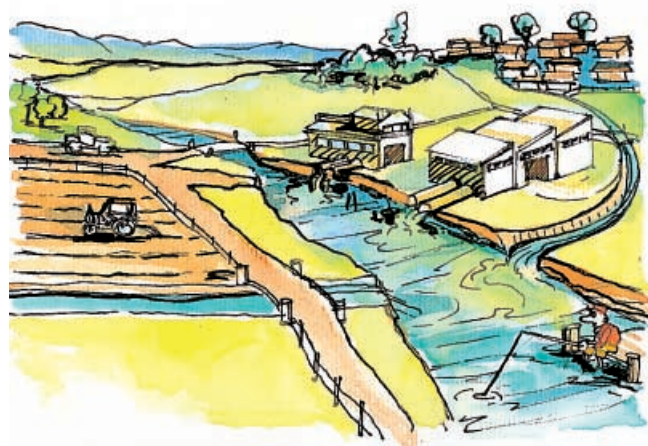


FIGURE 8.17

Pollution can cause considerable damage when it enters a water system.

EXPERIMENT 3

WATER QUALITY OF A LOCAL STREAM, RIVER OR LAKE

AIM

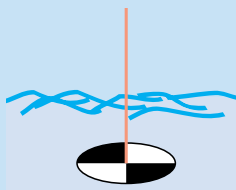
To conduct some basic tests of water quality.

MATERIALS

- turbidity disk (black and white quartered disk on a string)
- thermometer
- collecting jar
- microscope or magnifying glass
- Bunsen burner
- evaporating dish

FIGURE 8.18

A turbidity disk is lowered into the water until the markings can no longer be seen.



METHOD

- Make a sketch of your site, indicating where north is and other main features.
- Lower the **turbidity** disk into the water until it cannot be seen, measure the depth at which this happens by measuring how far the string goes into

the water.

3 Measure the temperature of the water with the thermometer. If the creek has storm water drains from roads these can pour warm water off the road into the creek during a summer storm. This may affect organisms living in the water.

4 Describe the colour and smell of the water. Does it have oily rainbows?

5 Collect a sample of water from the river and shake it to see if bubbles form. Then test to see if it contains dissolved particles by evaporating the water in an evaporating dish over a Bunsen burner.

6 Examine a sample of water under a microscope or magnifying glass for signs of living organisms. You may wish to sample the water at other locations and make comparisons. Other activities which could be carried are:

- a visual assessment of pollution.
- a survey of drains leading into the river.
- measurement of acidity or alkalinity.
- speed of water flow—this may affect how quickly any pollution is moved away.



Questions



- What is meant by the term **pollution**? What is a **pollutant**?
- Could some substances be **beneficial** in some situations but be a **pollutant** in others? Explain your answer.
- During what sort of weather conditions do you normally have **bad air pollution** in cities? Explain.
- When mud was washed into a river it was found that many of the water plants in the river died. The mud was not known to be **poisonous** to the plants. Explain why the plants may have died.
- As the temperature of water increases it has less **oxygen** dissolved in it. If a factory dumps thousands of litres of hot water into a river, what effect might this have on the **organisms** that live in the river?

Think about



A water sample was taken at the fishing jetty in Figure 8.17 which showed that a lot of oil was being released into the river. Further samples were taken at various points along the river. The results are shown below.

TABLE 8.4

Site number	Site	Oil (%)
1	Upstream from factory	1
2	Downstream from fishing jetty	3
3	Next to factory outlet	1.5
4	Next to storm water drain	5

- Where is the oil pollution most likely to have come from?
- Explain why the result from site 2 is lower than from site 4.
- What might have been the original source of the oil?

FACT

FILE

1998 was 'International Year of the Oceans' but it also was the year that saw the largest deaths of coral in modern times. In 1998 ocean temperatures were the warmest on record. Many scientists believe that the two events are directly related.

Developing or destroying?

FIGURE 8.19

How many different types of animals and plants can be seen in this photograph? Would your answer be different if the photo was taken before the land was cleared and the pasture was planted?



About 5% of Australia's land and some areas of ocean have been set aside for National Parks and nature reserves.

FIGURE 8.20

Are the basic habitat requirements for kookaburras being met here?



Impact of humans on ecosystems

A **habitat** is the place where an organism lives. Most organisms have special needs which their habitat can provide. If the habitat is changed in any way, some of these requirements may no longer be met. Some requirements are:

- a dependable year-round supply of suitable food
- a supply of water
- places to nest or shelter
- mating partners.

A number of different human activities have caused changes to habitats.

Agriculture: Humans have converted millions of hectares from its natural state for producing food either by growing crops or grazing animals. In Australia two-thirds of the total land area is used for agriculture and most of this has resulted in an almost complete change in habitat. Most of the

native vegetation has been removed and replaced by crops or new types of pasture.

Logging: Forests provide a habitat for many different types of animals and plants. Many of the trees that once occupied the areas now used for farming were sawn up and used as building timber or firewood. The demand for timber, paper and firewood has increased but not enough new trees are planted to keep up with the demand. New areas are constantly being sought out for logging. Many of the new timber plantations are of pine trees which provide a suitable habitat for very few of our native wildlife.



FIGURE 8.21

A pine plantation. Why wouldn't this provide a suitable habitat for our native animals?

Mining: The effects of mining on the environment are not as widespread as those of agriculture. However, damage to wildlife habitat still occurs. The main damage is often caused to rivers and creeks

where toxic chemicals can seep into the water, or soil washed into the river makes it muddy, which can stop plant growth.

Urbanisation: The growth of cities has gobbled up quite a bit of natural habitat. With new housing areas come new roads and traffic, new gardens with a variety of new plants, new sources of pollution, domestic pets who might view the existing wildlife as sport, and a lot of noisy people. The overall result is that very few of our native animals or plants can cope with this change and either move out or in most cases die out.

Questions



- 1 If farmers provided nesting sites such as nesting boxes on their farms, would this be enough to encourage wildlife back into these areas?
- 2 Describe the habitat a pine forest provides and how this might be different from normal bushland.
- 3 Examine Figure 8.20. If the bushland in the background was cleared, how might this affect the kookaburras?

Think about



The following statements are the answers a student gave in a test. What might the questions have been? Write down some possible questions to match these answers.

- 1 Places to nest or shelter and mating partners.
- 2 Not enough new trees are being planted to keep up with the demand.
- 3 Wheat crops, oat crops, rice crops.

Activity

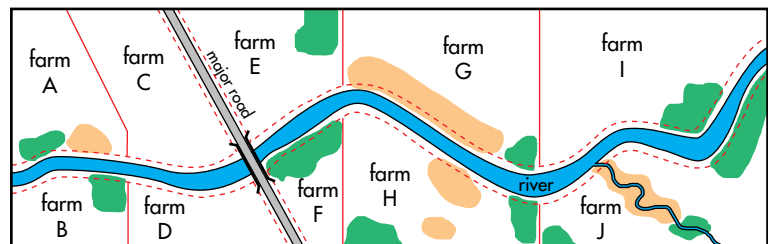


A wildlife corridor is a strip of native vegetation which links patches of bushland. Many of our native animals are having a hard time because of decreasing habitat. In many cases these isolated patches are

not big enough to provide all of the habitat requirements. By linking the patches of bushland together many animals can reach food sources, shelter and breeding partners. Examine the map of an area of farmland (Figure 8.23). There are several isolated patches of bushland. Redesign this area and include some wildlife corridors so that animals can move between the various areas. You may need to consider how to form links across roads.



FIGURE 8.22
Damage caused by mining.



- farm boundaries
- present bushland
- swampy areas – unproductive farmland

- Conditions
- Farmers don't own land along the river. It is public land 15 m either side of the river's edge
 - Farmers have the right to use water for stock
 - Farm A, G and C unlikely to cooperate in the wildlife corridor project

FIGURE 8.23
Re-draw this map in your workbook and link the various areas with wildlife corridors.

FACT

FILE

Planting belts of native trees on cleared land has many advantages for agriculture. The trees provide a windbreak to reduce wind erosion and damage for some crops. They also help stabilise the soil to lessen erosion during rains or floods and they reduce soil salinity. The trees provide a habitat for birds, which can eat some farm pests. Their presence reduces lambing losses and increases the weight of beef cattle.

Introduced animals and plants

FIGURE 8.24

The fox was introduced from Europe in the 1860s to provide sport for shooters. It adapted very well to its new home and spread widely throughout all parts of the mainland. It has caused considerable problems for sheep and poultry farmers and will eat a wide variety of native animals, such as lizards, snakes, frogs, fish, snails and spiders, as well as birds and mammals.

Introduced animals and plants

Australia's greatest explorers, the rabbit and the fox, were just some of the huge range of foreigners brought into Australia by the early settlers. The Victorian Zoological Society was set up in 1857 for the purpose of importing animals and plants into Australia to be set free into the wild.

Talk about

Why do you think this was done? Do you think people would behave the same way if settling a new area today?

The list of **introduced** animals and plants is enormous. Most of the familiar four-legged animals are introduced, as are many of the plants in our gardens. Our major sources of food—fruit, vegetables, meat, eggs and dairy foods—are produced from introduced species. Wool, cotton and leather are also from foreign species. Most of the introduced species have been of benefit to humans, but many have had an undesirable effect on the environment. Added to this are the ones that got away, many of which have caused considerable damage to our native wildlife.

Questions



- 1 Why were the rabbit and fox released into Australia?
- 2 Why is the cane toad a serious threat to Australian wildlife?
- 3 Name five imported animal species and five imported plant species that are useful to humans.



FIGURE 8.25

Rabbits arrived in Australia with the First Fleet in 1788. They caused few problems in those early times. However, following deliberate releases in many areas over the country their numbers increased dramatically. Between 1880 and 1950 their population was estimated at around 800 million. Rabbits cause damage by eating vast areas of grass right down to the roots and therefore leaving the ground exposed to erosion. Their eating habits have caused major concern for farmers and also for native animals such as wallabies, kangaroos and birds.

FACT

FILE

Two hundred years ago, there were 131 native species of mammal in New South Wales. Since then 27 species have become extinct and 50 are on the endangered list. Contributing factors have been habitat destruction, competition for food by rabbits and predation by foxes and feral cats.

**FIGURE 8.26**

The cane toad, a native of South America, was introduced to control cane beetles which had accidentally arrived with the sugar cane plant. The cane toads were a failure. Not only did they not control the cane beetle, but a cane toad will eat just about anything it can swallow, including insects, mice, small snakes and lizards. Cane toads can weigh up to 1.25 kg, live up to 15 years, and produce 40 000 eggs per year. Unfortunately many potential predators find the toad poisonous.

4 Imagine you are a scientist investigating the possibility of importing a new type of predator to control rabbits. How could you be sure that it was not going to harm any of our native species or be a pest to humans?

5 Blackberries have invaded the banks of many creeks in Victoria.

a How might the blackberry bush have affected native plants?

b Rabbits often dig burrows in the centre of blackberry bushes. What is the advantage for rabbits to do this?

Think about

Examine Figure 8.28.

1 What is the general trend in the populations of native and introduced animals shown in the graph?

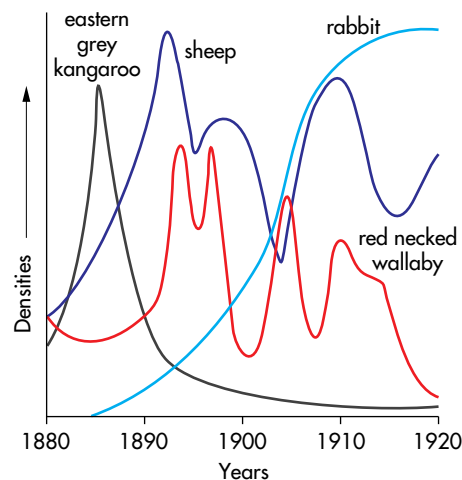
2 What may have caused the drop in numbers of grey kangaroos around 1885?

Activity

Collect a map of the world from your teacher. Using the list of introduced species draw arrows from their countries of origin to Australia.

**FIGURE 8.27**

Boneseed was initially a garden plant introduced from South Africa, but was also used in some areas to stabilise sand dunes. It spreads very rapidly as its fruits are eaten by birds which then deposit the seeds in other areas. Boneseed has the ability to take over bushland areas as it can re-establish itself more quickly than the native vegetation after a bushfire.

**FIGURE 8.28**

Changes in numbers of native and introduced animals in the New England tablelands (New South Wales) 1880–1920.

Include the name of the plant or animal and the date of introduction to Australia. You may like to do some research and add some extra species to your map.

Find out

Two viruses have been introduced into Australia to control rabbit populations. The myxoma virus was introduced in 1950. The calici virus was released in 1996. Find out about each of these viruses and how effective they have been in controlling rabbit populations.

FACT

European carp are widespread in Australian rivers. The problems they pose to our waterways can be compared with the effect of rabbits on land. These introduced fish disturb sediments and create turbidity which destroys aquatic plants because they don't get enough light. They also release nutrients which contribute to the formation of algal blooms. These blooms release toxins which poison rivers.

IMPLICATIONS FOR SOCIETY AND THE ENVIRONMENT

Water—the precious resource

FIGURE 8.29

Mound springs are found where artesian water surfaces. This one is called The Bubbler. Some mound springs have dried up because water from the Great Artesian Basin has been drawn off elsewhere.

Early European settlers in Australia faced the problem of finding reliable water supplies. In the 1880s an apparently limitless water source was found in inland areas. Farmers dug bores and tapped into underground water which was under pressure. They had discovered the water-bearing layers (**aquifers**) of the Great Artesian Basin which lies under one-fifth of the continent. At first, a lot of the water was allowed to run freely and was wasted, but

now farmers are aware that the water supply is limited and must be conserved.

Changes in soil

Irrigation of areas like the Murray River and Murrumbidgee River valleys has been vital to Australia's agriculture. Production of many crops and irrigated pastures are important to our economy. However, irrigation has also contributed to a problem that already existed naturally in many soils—a high salt level (or **salinity**).

Salt levels in soils rise because of low rainfall and high evaporation rates. Higher rainfalls can dissolve salt and carry it off in rivers to the ocean. High evaporation rates mean that traces of salt in groundwater are left as residue when the water evaporates. In addition, many rocks in Australia were laid down when seas covered areas of the continent and they already contain a lot more salt (sodium chloride) than rocks laid down in freshwater conditions. When they weather, they form a salty soil. Artesian water also contains a lot of dissolved salt so when it reaches the ground level and evaporates, it contributes to salt residues in soil. Wind can contribute to **soil salinity** by carrying salt onto the land in sea breezes.

Human activities have increased soil salinity. Clearing of land has contributed for two reasons: top soil has been washed away, exposing subsoil layers which contain more salt, and **water tables** have risen, bringing dissolved salts closer to the surface. Irrigation also causes the water

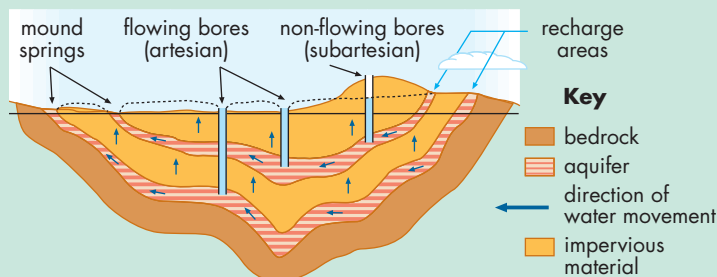
FIGURE 8.30

The Great Artesian Basin. Rainfall in the Great Dividing Range replenishes water at the edge of the basin and it moves under gravity at the rate of about 1 m each year.



FIGURE 8.31

An aquifer. In the Great Artesian Basin rain enters permeable rocks in the Great Dividing Range. This water can take up to 2 million years to move through to springs on the other side of the basin.



USING SCIENCE 5

SALINITY AND SEED GERMINATION



Prepare a series of salt solutions ranging in concentration from 1 to 5%. Design an experiment to test the effect of these solutions on the germination of seeds and the subsequent growth of the plant. Take

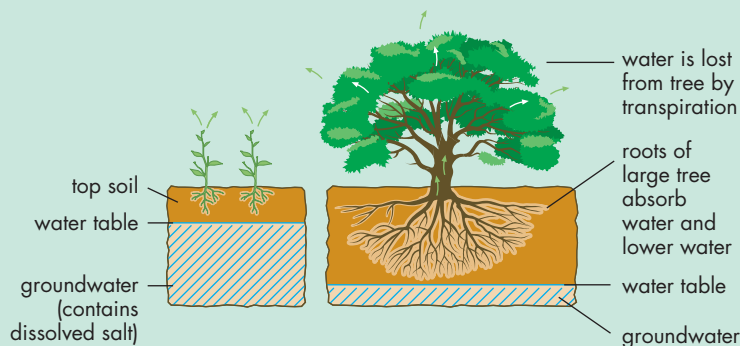
care to make your experiment a fair test. Perform the experiment and write up your report for presentation to the class.

table to rise as extra water is absorbed into the groundwater. If the soils are very sandy and porous, water penetrates deeply.

Murray–Darling River system

The Murray–Darling river system is the largest river system in Australia. About 70% of its water is used for irrigation and less than 20% reaches the ocean. Australian wildlife has adapted to seasonal changes in river volume but the amount of water taken from the river for storage in dams or irrigation means that drought conditions along the river occur about six in every 10 years instead of about once every 20 years. Floods which flush out the river system now occur only once every 14 years compared with once every 3 years before waters were diverted for other purposes.

In 1991 and 1992, the Darling River became the longest stretch of poisoned river ever found, with 1000 km of its length choked by poisonous blue-green algae. Pollution of the river by sewage run-off and fertilisers contributed to this algal growth. Stock losses occurred immediately but it is also believed the toxins produced by the algae will have long-term effects on sur-



viving sheep and cattle. The levels of blue-green algae only dropped when sufficient rainfall flushed out the river. Blue-green algal blooms continue to break out along the river system.

Increasing salt levels, caused by run-off from soils with increased salinity, also pose a problem for the river's health even though salt is currently being pumped away at the rate of about 270 000 tonnes per year. The CSIRO has estimated that 12 billion trees need to be planted in the Murray–Darling Basin to keep salinity at a moderate level.

In 1996, huge bank collapses occurred along a 2000 km stretch of the Darling River. This happened at a time when a record amount of water was being extracted from the river for agricultural purposes.

FIGURE 8.32

The term 'water table' refers to the top surface of groundwater. If the water table rises, salty groundwater affects the top soil. Large trees will absorb a lot of water from soil, lowering the water table. This keeps the salty groundwater away from the top soil.

FACT

FILE

Salinisation of soils has been a problem for thousands of years. Flood plains of the Tigris and Euphrates were being irrigated for growing wheat in 4000 bc. By 3500 bc, half the land could not be used because of higher salt levels. People moved away as the soil became less fertile.

Questions



- 1 What is the Great Artesian Basin?
- 2 Where does the water in the Great Artesian Basin originate?
- 3 What natural causes contribute to high salt levels in Australian soils?
- 4 Which human activities contribute to soil salinity?
- 5 What probably caused the blue-green algal bloom in the Darling River in 1991?

Think about



The Murray–Darling River Basin is sometimes called Australia's bread basket because it contributes over \$10 billion to the national economy. Most of its waters are used for irrigation. Do you think this level of irrigation can be maintained in the future? Write an exposition to argue your point of view.

IMPLICATIONS FOR SOCIETY AND THE ENVIRONMENT

How many people for Australia?

FIGURE 8.33 (above)
New York City is a graphic example of the increasing urbanisation of the planet. It has a population of 17.8 million people, about the same as the total population of Australia.

FIGURE 8.34 (above right)
What will life be like in the future?

It took more than 3.5 million years for the human population to reach 2.5 billion. To add another 2.5 billion people has taken just 30 years. At the moment the world's population increases every year by about 95 million people. That is more than five times Australia's population every year.

The areas of most rapid population growth are Asia, Africa and South America. As the population continues to increase, the effects of human activities such as clearing of natural habitats, production and release of pollutants, industrialisation and urban-

isation will also increase.

If the population continues to grow at its current rate the impact on the environment could be disastrous, with the worst possible outcome being our own extinction.

Huge increases in the population of a species are not uncommon in nature but usually they are eventually controlled by factors such as food supply, space shelter, predators, competitors and disease.

Will the human population stabilise under these controls or will we control it before disaster occurs?

USING SCIENCE 6

OUR INCREASING POPULATION

Table 8.5 gives an estimate of the world's population from 1650 to 1990. Graph this on graph paper leaving room for figures up to the year 3000.

TABLE 8.5

Year	Population in billions
1650	500
1700	575
1750	675
1800	825
1850	1050
1900	1525
1950	2500
1990	5500

1 Why are these said to be estimated rather than actual world population figures?

2 By how much did the population increase from 1650 to 1950?

3 By how much did it grow from 1950 to 1990?

4 Use your graph to estimate the world's population in the year:

a 2050

b 3000

5 What do you think life will be like in the year 3000? What problems might the people of this time face? What solutions might be available?

6 Some countries such as China have laws restricting the number of children permitted to a couple. Is this fair? Should all countries do this? (People in rich countries have more impact on the environment.) Should religious beliefs interfere?

7 What else do you think could be done to cope with the world's increasing population? Discuss your ideas with others in the class.

Talk about

Why do most Australians live in cities? Think about the disadvantages and disadvantages of city life.

Some limits to growth

Improved technology has brought about a 'Green Revolution' in recent decades. Developments in plant and animal breeding and the use of fertilisers and irrigation have helped to increase food production for a growing world population. However, there is a price to pay for these advances.

Water and soil are two essential resources for human survival and they are both finite. Only 3.5% of water on Earth occurs as fresh water and most of this is in the form of ice in polar regions. The demand for fresh water is increasing twice as fast as human population growth. At the same time water supplies are becoming polluted with fertilisers, sewage and pesticides. Finding reliable water supplies has always been a problem in Australia because it is such a dry continent, and water pollution adds to this problem.

Land degradation is a major global problem. Arid regions such as Australia are especially fragile. Human activities such as clearing of land, irrigation and increasing frequency of fires have contributed to loss of top soil and increasing soil salinity. It has been estimated that the survival of 250 million people around the world is currently threatened by land degradation. More of the Australian continent is being turned into desert each year as a result of both natural and human activities.

Questions



- 1 List the factors that can control population numbers in natural populations.
- 2 What two natural resources are under particular threat?
- 3 Why is Australia not capable of supporting the large populations found in other parts of the world?
- 4 Design a poster entitled 'Threat to survival'.



Some possible solutions

Large-scale clearing of land for agriculture can reduce rainfall in that area. Large-scale planting of native trees may help to modify rainfall patterns as well as helping to prevent soil erosion and reduce soil salinity.

Farmers should not overstock their lands and they need to review their use of irrigation. Mulching of ground can be a helpful step in regenerating degraded land.

Conservation is the protection and careful use of natural resources. There are many ways that we can contribute to conservation. Examples include planting trees, re-using and recycling materials such as glass and paper, composting food wastes, and buying products that are less harmful to the environment.

Find out



In the past our natural resources (e.g. from mining) provided a large proportion of Australia's export income. What are the main sources of income now?

Think about



- 1 Air pollution from burning of fossil fuels (coal and petroleum) is a major environmental concern. How can you help conserve these fuels?
- 2 What else can you do to help conserve the Australian environment?

FIGURE 8.35

Contours seen in this Darling Downs landscape have been designed to reduce soil loss by water erosion. The fertile soils in this area have weathered from basalt rock.

FACT

FILE

The Darling Downs is a rich agricultural area where farmers have taken special care to develop good management practices. In spite of this, it has been estimated that the Darling Downs is losing 8 million tonnes of top soil each year. At that rate, top soil will be completely gone from the area in the next 30–50 years.



Using scientific language

Write a descriptive report about what has happened to the Australian environment in the past 200 years. Include the following paragraphs: changes to habitat, pollution, introduced species.

Also include the following terms as part of your description: agriculture, logging, mining, urbanisation, extinction, salinity, conservation.



Check your knowledge

- 1 Describe the adaptations of desert mammals that help them conserve water.
- 2 Explain why the Australian continent does not have the rich soils found on other continents.
- 3 What is Southern Oscillation? How does it explain drought in Australia?
- 4 What special adaptations do Australian plants have to help them survive bushfire?
- 5 What is the connection between rubbish thrown into the street and rubbish in rivers and the sea?
- 6 What should scientists have tested before introducing the cane toad from South America?
- 7 Tree planting can help conserve native wildlife by providing suitable habitats. Are there any other benefits associated with tree planting?
- 8 What are the main reasons for establishing national parks? Can you name three national parks in Australia?



Apply your skills

Choose the best answer. Refer to Figure 8.36, the causes of eucalypt dieback.

- 1 The most direct cause of the eucalypt destruction by sheep is:
 - A greater use of imported pasture species.
 - B greater use of fertiliser.
 - C damage to young trees by sheep grazing.
 - D increased soil nutrients due to sheep droppings.
- 2 Which of the following does not have a direct connection to 'increased local population of tree-feeding insects'?
 - A Application of fertiliser.
 - B Destruction of habitat for predators.
 - C Increase in insect reproduction rate.
 - D Increased soil nutrients.

- 3 What is the major cause of damage to young trees?
 - A Fertiliser application.
 - B The larvae of leaf-eating insects.
 - C Increased soil nutrients.
 - D Grazing of sheep.

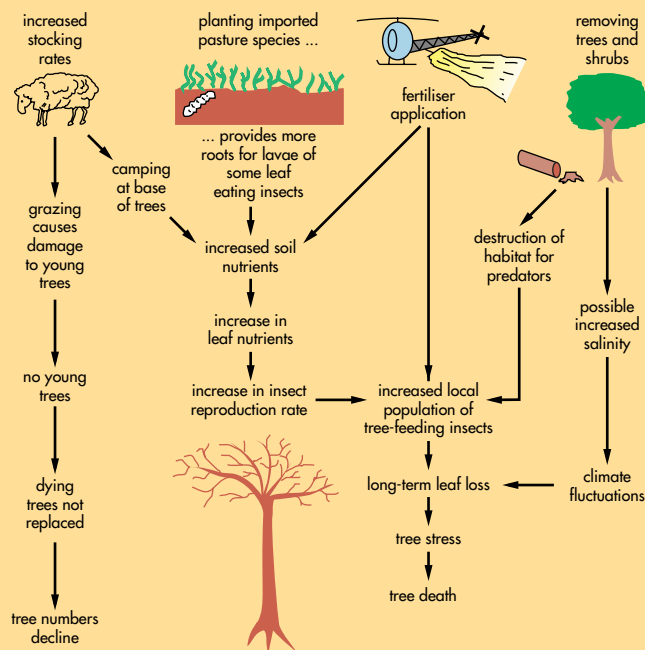


FIGURE 8.36

Challenge yourself

- 1 Monitor your energy use. Most electricity and gas bills have a small graph indicating energy consumption over the previous months. See if you can lower your power consumption by reducing the number of lights left on, and limiting the use of heavy electricity consumers such as electric heaters, hairdryers, clothes dryers and washing machines.
- 2 Research other ways in which you can help your environment by doing simple things at home or at school. You could present these ideas on a poster.
- 3 Many Australian desert animals are in danger of extinction or may be extinct. Choose one of the following: desert rat kangaroo, stick-nest rat, long-tailed dunnart, the lesser bilby or yallara. Write a descriptive report about the animal, including its appearance and adaptations, its behaviour and its current known distribution or previous distribution. 'Environment Australia Online' (<http://www.environment.gov.au>) is a useful starter reference.



Web page



Tester



CHAPTER

9

NATURAL RESOURCES

CHAPTER OUTCOMES

Successful completion of this chapter will allow you to:

- Describe examples of the use of natural resources.
- Distinguish between natural and manmade resources.
- Give examples of resources obtained from the Earth, the ocean and the air.
- Understand the importance of collecting and classifying materials.
- Identify possible careers in geology.
- Identify some common minerals and their properties.
- Describe materials manufactured from sand.
- Give examples of resources obtained from plants.
- Understand the importance of biodiversity and maintaining rainforests.
- Understand the importance of recycling at home and at school.

Since ancient times, human communities have used resources around them for more than just food and shelter. They have developed new technologies and continued to find new uses for materials in nature.

There are more than 210 species of frogs in Australia. One of these, the green tree frog, has a number of different types of glands at its skin surface. One of the glands secretes a type of glue which probably protects the frog by gluing the mouth parts of any insect which bites it. This glue is currently being tested for use in repairing knee cartilage. It may have another application in gluing titanium metal in facial surgery and is being tested as an anticancer agent.

A granular gland in the same frog produces chemicals which repel mosquitoes, and a third gland produces a wide range of substances with antibiotic properties.

These discoveries come at a time when there is worldwide concern for the decline in frog populations. It is a timely reminder about the importance of biodiversity—the conservation of the wide variety of living things which exist on Earth today.

It is not only living things which can be an unexpected source of new materials. NASA scientists have recently developed the use of zeolite minerals as a growth medium for plants. These naturally occurring minerals have a crystal structure that forms a honeycomb shape. The tunnels which they provide can be used to store mineral nutrients for use by plants. The plant can absorb these nutrients very efficiently, resulting in less wastage of fertiliser and little loss of fertiliser into surrounding groundwater to cause pollution.

In this chapter, you will learn about some of the resources used in our society and the importance of managing them carefully for the sake of future generations.

Resources we depend on

FIGURE 9.1

Red ochre is a natural pigment made from crushed rock rich in iron oxide. It was used by Aboriginal people as a pigment in cave paintings and as a body paint for sacred ceremonies and burials. The red colour was associated with blood and death.

Like the ochre used in cave paintings and sacred ceremonies by Aboriginal people for tens of thousands of years, other Earth materials were used every day by ancient humans. Tools, weapons, rock shelters and ceremonial objects, all made from rock materials, were as essential to them as food and water.

A **resource** can be defined as anything that is useful to humans. **Natural resources** include soil, water, wind and materials from mining. The ways in which we use these resources have changed, and we have become more demanding of our planet as our society has developed.

Once water was used only for drinking and washing. Now we use it to generate electricity.

The wind once pushed huge sailing ships around the world. Now it also generates electricity, pushes windsurfers, pumps water and lifts hang-gliders.

Mining provides metals for cutlery, railway tracks and electrical wiring. Coal, oil and gas from deep under the ground power our cars, heat our homes and make most of our plastic products.

This book you are reading now could not have been produced without these resources. Computers were used when the writers looked up information in a library

and on the Internet. They were used for word processing and when the typesetter and designer laid out the final pages. Computers are made from metals, plastics, glass and silicon microchips, all of which are manufactured from materials found in the Earth. They also need electricity generated by coal, gas or water. Then there is the paper from trees grown in soil, and finally other resources from the Earth are used to make the inks, plastics and other chemicals needed to produce this book.

Talk about

Some resources are said to be 'renewable', and others are classified as 'non-renewable'. What do you think these two terms mean? Discuss which of the following would be 'renewable' and which would not: oil, iron ore, wind, water, soil, sunlight, timber. Does everyone agree?

USING SCIENCE 1

WHERE DOES IT COME FROM?

In this activity you will investigate one of these products and the natural resources used to make it: bronze, gunpowder, glass, plastic, steel, microchips, synthetic fibres, polystyrene, petrol.

YOU WILL NEED

Access to a library and a large copy of the data grid shown in Figure 9.2.

Questions Sources	When was this product first made?	Who first made the product?	What Earth resources are needed to make this product?

FIGURE 9.2

Data grid for collecting information in note form.

WHAT YOU DO

Use your data grid to collect information in note form. The boxes across the top should contain the questions you would like to ask about your topic (three have been put in for you). In the boxes down the left-hand side you should record the details of the sources you have used (such as author or volume, title, date of publication, page numbers).

Use your notes to make a large flow chart which tells the story of the development and use of your chosen product. The chart should be colourful and readable from 1 metre away. Glue your data grid to the back of the chart.

Share your information with others in the class, using your chart as a starting point. Whose product was developed most recently? Why were some products made much earlier in human history than others?



FIGURE 9.3

All of these plastic items are made from oil which comes from underground reservoirs.

Questions



- 1 What is a resource?
- 2 Who might have been the first people to use the wind as a resource? How might this have happened?
- 3 Name three objects made, or dependent on, natural resources which you have used today. Include something you ate, something you put on your body and something you operated with a switch. State how the resources were used in each case.



Think about

Imagine your world without metals or plastics. List all of the activities you are involved in for one day. Cross out all of those that involve the use of something metal or something plastic. What do you have left? Compare your remaining list with others in the class.



FIGURE 9.4

Water stored behind purpose-built dams can be channelled through huge turbines to generate electricity. How else do modern humans use water?

The salt of the Earth



FIGURE 9.5

There are over 3000 known minerals, most of which form natural crystals in some locations. These crystals (left to right) are ruby, emerald, pyrite and quartz.

FACT

FILE

Rubies and emeralds (Figure 9.5) are used in jewellery because they are hard and have striking colours. Their colours are due to the presence of the element chromium in the crystal structure. Because the atoms surrounding the chromium in the ruby are different from those around the chromium in the emerald, different coloured light is reflected. Thus the ruby appears red and the emerald appears green.

‘You are the salt of the Earth. But if the salt loses its saltiness, how can it be made salty again?’

Matthew 5:13

This quote from the Bible highlights the importance to early human communities of one of the simplest natural substances—common salt. ‘The salt of the Earth’ is an expression we use to describe a person who, though ‘ordinary’ in many ways, has qualities we admire, and so it is with salt.

Salt is probably one of the first minerals for which humans found a use. The term **mineral** applies to a naturally occurring solid substance which has a definite chemical composition. Every particle of common salt in the world is made up of the chemical **elements** sodium and chlorine,

combined to form the compound sodium chloride. If you look at table salt under a microscope you will find that most of the particles have a cube shape. We call this regular shape a crystal. In nature cubic salt **crystals** can grow to be quite large.

Other minerals have different crystal shapes, due to different chemical elements and different arrangements of the atoms or ions of these elements within the mineral structure. Some minerals occur as single elements in nature. These **native elements** include diamond (made of pure carbon) and gold. Most other minerals are naturally occurring **compounds**. These minerals include quartz (silicon dioxide) and rutile (titanium dioxide).

Salt has long been prized for its effectiveness as a food flavouring, and as a food preservative before refrigeration was available. More recently salt has been used in large amounts in the manufacture of industrial chemicals, paper products and plastic.

Australia has plentiful supplies of many other minerals. The ones that provide us with the materials we need for everyday life, and are thus worth mining, are called **ore minerals**. Such mineral ores include bauxite from which we extract aluminium, haematite, which is an iron oxide, and deposits of gold in its pure elemental form.



FIGURE 9.6

In nature common salt (sodium chloride) crystals often encrust plant roots in very salty environments.

EXPERIMENT 1

GROWING SALT CRYSTALS

**AIM**

To show how minerals can crystallise from solution, and that all salt (sodium chloride) crystals have a similar form.

MATERIALS

- table salt
- spatula
- 100 mL beaker
- magnifying glass or stereomicroscope
- clean dry watch-glass
- stirring rod
- hot water

METHOD

- 1 Add 50 mL of hot water to the beaker.
- 2 Use the spatula to add salt, small amounts at a time, and stir with the glass rod. Do this until you are sure no more salt will dissolve (there should be some undissolved salt on the bottom of the beaker).

- 3 Allow excess salt to settle, then pour off a little of the salty water into the watch-glass.

- 4 Put the watch-glass with the solution in it where it will not be disturbed for a few days.

- 5 Examine the contents of the watch-glass with the microscope or magnifying glass.

RESULTS

What has formed in the watchglass? Describe and sketch what you see.

EXTENSION

- 1 Try adding food dye to the salt solution to see if it colours the resulting crystals.
- 2 Other household minerals that can be dissolved and grown into crystals again include sugar, washing soda, borax (an insect killer) and Epsom salts (used to clean the washing machine, and as bath salts). Have a try!

**FIGURE 9.7**

In Australia we produce all our own salt by evaporation of sea water or salt lakes. We also export large quantities to Japan.

**FIGURE 9.8**

Everyday objects such as steak knives and drink cans are manufactured from minerals. From iron ore we can make steel for the knife and from bauxite we can extract aluminium for the can.

Questions

- 1 Name a mineral which occurs naturally as an element and one that occurs naturally as a compound.
- 2 Name the mineral from which aluminium can be extracted and the mineral from which iron can be extracted.
- 3 Give two properties of ruby and emerald which show that they have atoms arranged in different ways.

Think about

Rubies and emeralds are minerals which are also called

precious stones. Salt is very useful but is not classified as precious. What properties would you expect a mineral to have to be considered 'precious'?

Find out

Useful resources can also be obtained from air. The air must first be cooled until the gases condense into liquids. They can then be separated by a process called fractional distillation because they boil at different temperatures. Find out the boiling point of oxygen and nitrogen. Give examples of the use of these gases in medicine and industry.

FACT**FILE**

Rutile is a major source of titanium metal. Titanium is a light, strong metal which resists corrosion. It is used in medicine (e.g. for joint replacements), and in industry (e.g. in aircraft).

CURRENT ISSUES, RESEARCH AND DEVELOPMENT

Collections and careers



FIGURE 9.9 (above)
Do you have a collection?

FIGURE 9.10 (above
right)
Most garden rocks are made up
of many different mineral
grains. Some specimens might
consist of one mineral only, like
the white quartz pebble in the
front here.

Talk about

Are you a collector? Do you have a drawer full of stamps or coins? Do you swap basketball cards?

Have you ever thought of starting a collection of **rocks** and minerals? Perhaps you could start with that piece of raw opal you brought back from Coober Pedy last holiday, or maybe that beautiful milky-white quartz pebble you found in a creek could form the basis of a great collection. The rocks in your garden might even be a good place to start.

Humans have collected rocks and minerals from the beginning of their time. They ground minerals for paints, fashioned them into body ornaments, and shaped rocks into tools and weapons. Sam Peyton, a middle school student from Victoria, is a keen collector of ancient **stone artifacts** and is most interested in the cultures which surrounded their use.

'I guess I have always had a fascination for rocks. Today I am mostly interested in stone tools and the waste pieces of rock (called flakes) which give us clues as to how prehistoric people lived. Working with the encouragement and permission of the Victorian Archaeological Society I have collected rock flakes in my school grounds which have been left behind by Aboriginal tool makers who lived in the area long before European settlement. I made a map of the locations of my finds and registered the chips of **silcrete**—a hard, creamy-white rock—with the Survey.

'My fascination with these ancient artifacts started when I was six. While I was on holiday in New Zealand a collector gave me a Maori adze, a type of hand axe. Since then I have added to my collection an Egyptian scraping tool and some very old stone arrow heads from Northern Europe. I even do some identification work for other collectors at times.'

USING SCIENCE 2

ROCKS AND MINERALS



Collect some small rocks or pebbles from a garden or roadside. Wear sunglasses or safety glasses for the next step to avoid eye damage. Use a hammer to break each rock so that you can see a fresh surface (the outer surface will probably be very weathered). Try to decide if the rock contains more than one mineral. (A stereomicroscope may help if it is available.)

If it seems to be one uniform colour, even under the microscope, it may be a mineral specimen. It is more likely it will consist of several different colours which usually means that it is a rock. **Rock** is the geologist's term for a solid mixture of minerals. Return the broken fragments to the garden.

Questions



- 1 What is the main difference between a rock and a mineral?
- 2 How would ancient people have been dependent on rocks and minerals in their everyday lives? You may wish to do a series of illustrations to answer this question.
- 3 Have you ever collected anything? Write a letter to a friend about your own experiences of 'collecting', or the collections of someone you know well.

Think about



Why do you think humans like to keep collections?

Do some collections serve different purposes from others? Are there any animals which collect things? Why do these animals collect these objects? Make a list of all the things people collect!

Activity



Many jobs involve the study of rocks, minerals, ancient stone artifacts and the structure of the solid parts of the Earth. Some of the people who have these jobs started out just like Sam, with an interest in the Earth and the rock and mineral materials we use from it.

Go through the employment section of a newspaper. Try to find some of these jobs: geologist, mining engineer, geochemist, geophysicist, hydrogeologist, soil scientist, environmental geologist, mineralogist, palaeontologist, archaeologist. Most of them involve some experience in the study of geology (the Earth, its structure, rocks and minerals).

Choose one of these and photocopy or cut out the advertisement. Visit the school library or your careers teacher



FIGURE 9.11

Sam Peyton, a middle school student, is a collector of rocks which have been modified by ancient humans to make tools.

and find out more about the job. Make a poster about it, showing what qualifications you need, what type of environment you would work in, what type of work you would be doing, who you would be working for, and other relevant details.

FIGURE 9.12

Earth scientists at work: an archaeologist (top) and geologists (bottom).



Which mineral is which?



FIGURE 9.13

These three mineral specimens have a similar colour but different hardnesses. The one on the left can be scratched by glass but not by an iron nail. The one in the middle cannot be easily scratched by glass. The one on the right can be scratched by a copper coin.

Collecting rock and mineral specimens is one thing. Identifying them is another. Imagine you have found the beautiful mineral specimens in Figure 9.13.

The first observation you would make is that they are all pink.



FIGURE 9.14

Minerals with different crystal shapes.

FIGURE 9.15 (below)

In 1822 Friedrich Mohs devised this scale for the hardness of minerals. Some mineralogists carry a hardness kit in the field to do quick hardness tests on unknown minerals. What do you think a hardness of $3\frac{1}{2}$ means? Why is glass listed as 6–7?

Talc	1
Gypsum	2
Calcite	3
Fluorite	4
Apatite	5
Orthoclase	6
Quartz	7
Topaz	8
Corundum	9
Diamond	10

Field hardness kit

Fingernail (on finger!)	$2\frac{1}{2}$
Copper coin	$3\frac{1}{2}$
Pen knife or iron nail	$5\frac{1}{2}$
Small piece of glass	6–7
Small piece of quartz	7

The colour of minerals

Colour is an important property, but it can be misleading. Many minerals, though different in chemical composition, can have similar colours. To make things more complicated, specimens of the same mineral can vary greatly in colour. For example, quartz, a very common mineral, can be pink, purple, black, yellow, white, brown or clear.

Crystal shape

A **mineralogist** (a **geologist** who specialises in the study of minerals) also observes crystal shape. Not all mineral specimens are perfect crystals but if they can find even a tiny specimen with regularly arranged faces, classification is much easier.

The hardness test

The next step is to establish how hard the mineral is. For this we use a special scale called **Mohs' scale** of mineral hardness.

The softest mineral on the scale is talc and is given a value of 1. Diamond, the hardest natural substance, is given a value of 10. The way mineralogists do this test is to try to scratch the mystery mineral with known specimens from the scale. For

USING SCIENCE 3

MINERAL IDENTIFICATION

Use a mineral kit provided by your teacher to practise identifying minerals. You will need to check which of the tests described here you are allowed to do. Record information about each specimen in a table similar to this one.

Mineral name	Colour	Crystal shape (sketch)	Hardness	Streak	Density (g/cm ³)

example, if the mineral can be scratched by **quartz**, but not by **apatite**, then the unknown mineral must be between hardnesses 5 and 7. If specimens of some of the minerals on Mohs' scale are not handy, some everyday objects of known hardness can be used to scratch the unknown specimens.

The streak test

The **streak** is the coloured mark left on the unglazed side of a white ceramic tile when the mineral is scraped across it.

Density of minerals

Another easy test is to measure the **density** of the mineral. If two mineral specimens are the same size, but one is heavier, the heavier one has a higher density. Calculating density is done by measuring the mass of the specimen (how much matter is in the mineral), and dividing this number by the volume (how much space the specimen takes up). Figure 9.17 shows how this can be done in the laboratory.

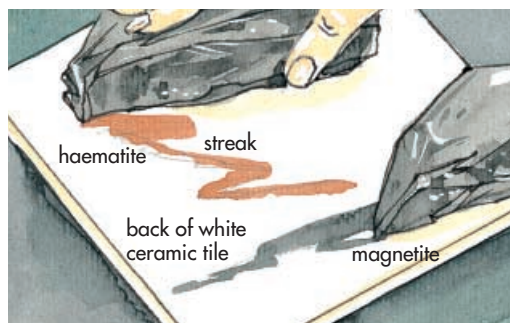
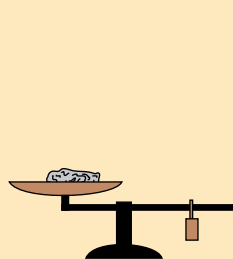


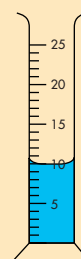
FIGURE 9.16

Often two mineral specimens with a similar colour have different streak colours. Haematite produces a rusty red streak while magnetite produces a black streak. Both minerals are grey-black when whole.

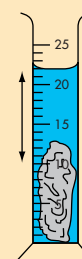
calculating density of a mineral specimen



1. Measure mass on a beam balance or an electronic balance.



2. Note volume of water in a measuring cylinder.



3. Put mineral specimen in water. Rise in volume of water equals volume of mineral specimen.

3 Which of the tests described on these pages could be classed as 'destructive' tests? Why?

4 Use Mohs' scale (Figure 9.15) to identify the three minerals in Figure 9.13. One is calcite, one is quartz, and one is orthoclase.

Think about

Discuss with your classmates and teacher the possibility of starting a gemstone club for students interested in collecting and swapping rock and mineral specimens. Invite a guest collector to talk to the class, or organise a trip to a gem show.

FIGURE 9.17

Measuring density: $D = \frac{m}{V}$

Questions

1 A geologist working for a mining company found two mineral specimens that he believed to be ore minerals. They were both dark grey and had a metallic shine. Describe two quick tests he could do to find out if they were specimens of the same mineral, or were different.

2 Why is the colour test not sufficient on its own to identify most minerals? Give two reasons.

Using sand

FIGURE 9.18

To make stained-glass windows different minerals are added to the glass to produce the variety of colours.

Sand is a basic ingredient in two useful **synthetic materials**: **glass** and **concrete**. Sand (silica or silicon dioxide) is a plentiful resource so both glass and concrete are fairly inexpensive materials. They are used extensively in modern society.

Talk about

Work in pairs to brainstorm uses of glass and concrete. Compare your results with those of other class members and prepare a mind map on the uses of each material.

Glass

Glass is a man-made material but forms of glass do occur in nature from the action of intense heat on sand. **Obsidian** is a volcanic rock formed when the heat of a volcano causes sand to melt. **Tektites** are dumbbell-shaped glassy objects formed from molten rock thrown into the air after a meteorite hits the Earth. Another example is a **fulgurite**, a brittle, glassy tube that can form when lightning strikes sand.

In the making of glass, sand is mixed with lime and soda and heated to about 2000°C until it is molten. Unlike most other materials, it does not form crystals as it cools. Instead, the long-chain glass molecules arrange in random order much like the arrangement of particles in a liquid. For this reason, glass is sometimes

called a **supercooled liquid**. As the glass cools, the molecules become fixed in their position and resist any change in their arrangement. The molten glass can be manipulated into any shape and will retain that shape as it cools.

One important property of glass is its **transparency**, or its ability to transmit light. In 1676, it was discovered that the addition of lead oxide to the glass mixture resulted in glass which sparkled but was also softer and easier to cut. Lead crystal can contain up to 33% lead oxide. The lead oxide makes the molten glass more difficult to work, but the final product is more appealing.

Glass is very stable. Most chemicals, even acids, can be stored in it safely because it does not react with them or retain odours. It does not conduct electricity. As it can be moulded into any shape while molten it is very useful for making containers.

Glass is also **recyclable**. Broken pieces of glass called **cullet** are added to accelerate the melting process in the manufacture of more glass.

Concrete

Today concrete is the most frequently used material in construction works, but the use of cementing material has a long history. Natural cement binds together particles in sedimentary rocks. Some insects such as ants and termites have learned to use plant resins as cementing materials to make shelters. Early human cultures used different binding materials in construction of shelters. The Egyptians used mortars made from gypsum and lime in building the pyramids.

Concrete is a synthetic rock, made by mixing cement with other rock material like



FIGURE 9.19

Glassmaking has been part of human culture for about 5000 years. Today most glassblowing is done by machines in a factory.



Internet assignment

FIGURE 9.20

The addition of lead oxide to glass produces crystal which is very heavy but has more sparkle. It can be fashioned into beautiful glassware.



sand, pebbles or gravel ('aggregate') and then adding water. The water causes the mixture to harden as it reacts with chemicals in the cement in a process called **hydration**. This hydration produces crystals that bind the aggregate particles together.

The first **cement** used in making concrete was probably lime. The Romans used this from about 500 BC. The process to make Portland cement was invented in 1824.

Ground chalk and finely divided clay were heated in a kiln to drive off carbon dioxide. The resulting material was ground up to make cement.

One advantage of concrete is that it can be poured into moulds before it sets so that it can take almost any shape. It has very strong resistance to **compression** but it needs to be reinforced with steel if it is to withstand stretching (**tensile**) forces.

FACT

FILE

Plasticisers are added to cement to make concrete more workable with less water.

EXPERIMENT 2

CURING CONCRETE



AIM

To investigate the curing of concrete at different temperatures.

MATERIALS

- 20 g cement
- 30 g sand
- container for water bath
- 3 disposable petri dishes and lids
- oven or incubator
- 10 mL water
- plastic gloves

METHOD

- 1 Weigh each of the empty petri dishes with lids and record the weight.
- 2 Prepare a batch of fresh concrete from the water, cement and sand.
- 3 Divide this mixture evenly into the three petri dishes and replace the lids.
- 4 Weigh each of the petri dishes and lids again.
- 5 Leave one dish in the refrigerator, the second at room temperature and the third in a water bath in an

incubator set at 40°C overnight.

- 6 Next day, place the petri dishes without lids into the incubator set at 60°C and leave for at least 1 day.

- 7 Weigh the petri dishes with lids again.

CALCULATIONS

Determine the initial mass of concrete in each petri dish and the loss in weight after it has been heated in the oven. Express this as percentage of original concrete mass for each sample.

DISCUSSION

- 1 Which sample has used the most water for hydration?
- 2 Is there a pattern for hydration and curing temperature?

EXTENSION

Different substances can change the curing rate of concrete. Investigate the effect of adding calcium chloride or sugar to the original concrete mix.

Questions



- 1 Describe the types of glass that can be found in nature.
- 2 List the advantages of using glass for containers.
- 3 Is glassmaking a physical or chemical change? Give a reason for your answer.
- 4 Is making concrete a physical or chemical change? Give a reason for your answer.
- 5 What is the difference between concrete and cement?

Think about



Name some technical instruments that rely on using high-quality glass.

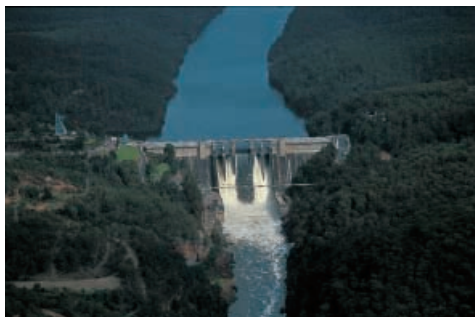


FIGURE 9.21

The reinforced walls of Warragamba Dam hold back waters which are used to supply 80% of Sydney's water.

APPLICATIONS AND USES OF SCIENCE

Resources from plants

FIGURE 9.22

Tasmania has Australia's largest area of cool temperate rainforest. Trees like Huon pine, blackwood, sassafras and myrtle provide timbers for craftwork and cabinet making.

Talk about

What do a rubber eraser, a chocolate bar, a woven basket, a cup of coffee, a golf ball and chewing gum have in common? They all are made from materials obtained from rainforest plants. What parts of a plant can be a useful resource for people?

Plants have been used as a resource for people from ancient times. Australian Aboriginal people used plants to make shelters, weapons, canoes and water containers. They used the bark of trees to make string, fishing line and rope. The string could be woven to make dillybags, which were often dyed with materials obtained from plant roots. They also used cane and reeds to make bags and baskets.



FIGURE 9.23

Basket weaving from plants is a traditional skill of Aboriginal women.

Aborigines knew which plants were a source of bush food and which were poisonous. They knew which plants stored water. Plants were also used for their medicinal properties. The leaves of some eucalyptus plants were pounded into a paste and mixed with animal fat to make an ointment. This was then rubbed directly onto sores, burns or wounds. The young leaves of stringybark eucalypts could be chewed to cure colds or the bark and leaves could be soaked in water to make a mixture to combat diarrhoea as well as aches and pains.

Today, we are still dependent on plants for many of our resources. Tropical rainforests have been a particularly rich source. Eighty per cent of the developed world's diet originated in rainforests: fruit like avocados, oranges, lemons, bananas and tomatoes; vegetables like corn and potatoes; rice; spices like pepper, chocolate, cinnamon and ginger; tea and coffee; and nuts like cashews and Brazil nuts all originated in rainforests.

Plants are still a source of medicines. One-quarter of Western medicines come from rainforest plants. Also, one-quarter of the active ingredients in cancer-fighting drugs come from them. Vincristine is one of the most powerful anticancer drugs. It comes from a rainforest tree called periwinkle and is used to treat childhood leukaemia. There is still potential for many more medicines to be discovered. Only about 1% of rainforest plants have been

EXPERIMENT 3

DISTILLATION OF OILS



Many Australian plants are a source of oils which have medicinal value. Eucalyptus oil and tea tree oil both have antibacterial properties. They are extracted from leaves by distillation.

AIM

To extract oil from the leaves of a eucalypt.

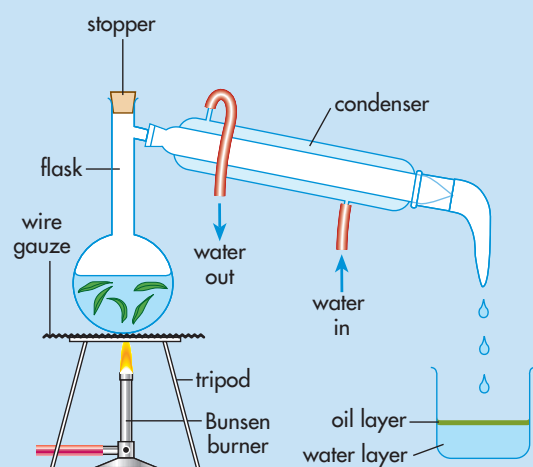


FIGURE 9.24

MATERIALS

- eucalyptus leaves
- distillation apparatus
- Bunsen burner
- water
- tripod
- gauze mat

METHOD

- Set up the distillation apparatus as shown in Figure 9.24.
- Add about 100 mL water to the flask.
- Crush the eucalyptus leaves and add them to the flask, replacing the stopper.
- Turn on the water for the condenser.
- Light the Bunsen burner and boil the water in the flask.
- Collect the water and any oil it contains in a small beaker.

DISCUSSION

- Does the oil have a strong aroma? Does it feel oily?
- How could the oil be separated from the water?

EXTENSION

Compare the oil from different types of eucalypts or tea trees. Observe the leaves under the microscope and look for clear areas which are the site of oil glands.

tested so far and these have yielded 2000 chemicals which have been shown to have some effect on cancer cells.

Gums and resins from rainforest plants provide us with rubber, chicle (for chewing gum) and other materials which can be used to make golf balls, paint, varnish and perfumes. The leaves of these plants can provide us with many essential oils used for perfumes, shampoos, cosmetics and their healing properties. They are called 'essential' because they have an essence or aroma.

Rainforest timbers like teak, mahogany and rosewood, provide fine materials for cabinetwork. Timber is valuable as a building material because unlike steel and aluminium, it is a **renewable** resource. There is also less energy consumed in its production, as shown in Table 9.1. The fossil fuel energy used to make each material can be compared by weight (per

TABLE 9.1 Energy used in the production of building materials.

Material	Fossil fuel energy	
	(MJ/kg)	(MJ/m ³)
Rough sawn timber	1.5	750
Steel	35	266 000
Concrete	2	4 800
Aluminium	435	1 100 000

kilogram) and by volume (per cubic metre). The energy used is measured in megajoules (MJ).

Wood can be broken down into smaller particles and recombined to form paper, cardboard and various types of building board like particle board. It is also burned in homes to provide heat.

Forest management is an important issue for all Australians. Existing state forests must be managed carefully so that

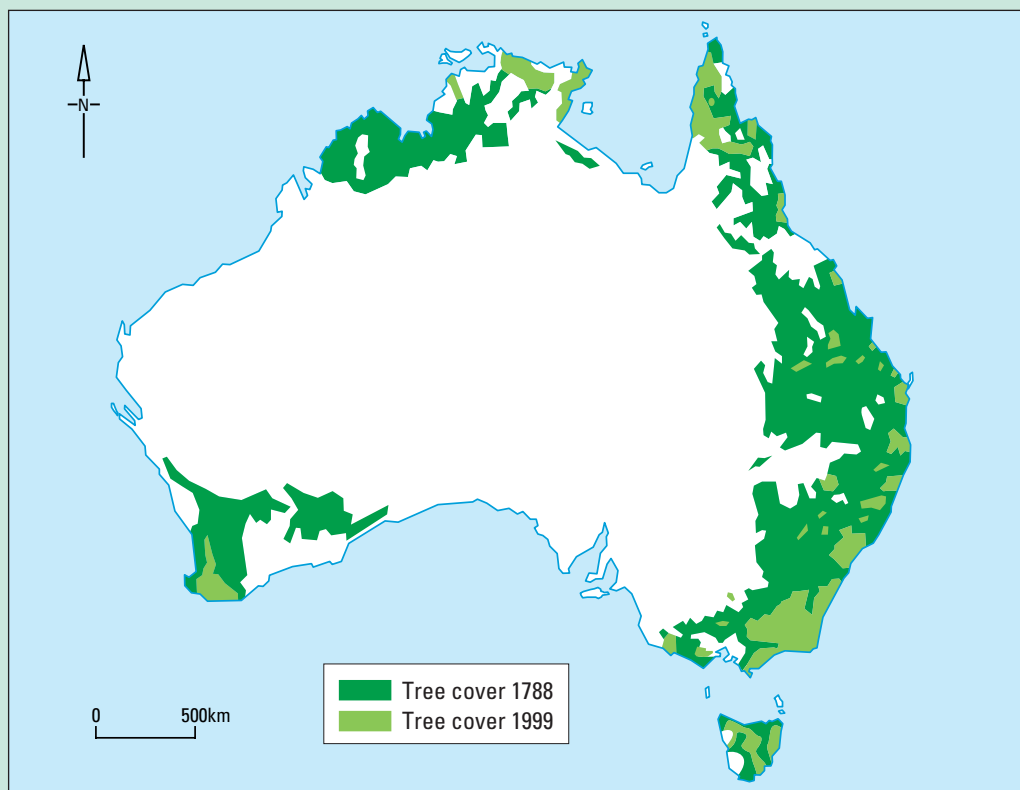


FIGURE 9.25

The extent of tree clearing in Australia since 1788.

they are a renewable resource, with new growth replacing timber which has been cut down. Plantation forests, mostly radiata pine, are also planted to provide a sustainable source of timber.

Sustainable development

Sustainable development refers to development that leaves resources for future use and does no long-term harm to the environment. Over half of all the species of animals and plants on Earth come from rainforests. Two hundred years ago, 14% of the Earth's land surface was occupied by rainforest. Today, it is only 6%. In Australia, 75% of rainforests have been cleared. Rainforests are such a valuable resource that many scientists are trying to convince developers that there is more economic value in leaving them intact and harvesting products from them than clearing the land for grazing cattle or for providing timber.



FIGURE 9.26

Timber is an excellent building material as it is a renewable resource. It requires less energy to produce than other alternatives.

Questions



- 1 Name some Australian plants which have medicinal properties.
- 2 Name six examples of food which grew in rainforests originally.
- 3 What types of valuable materials, apart from food, can be obtained from rainforest plants?
- 4 What is sustainable development? Why can state forests be considered a sustainable resource?

Think about



Write an exposition entitled 'Save the Rainforest' which argues for all existing rainforests to be protected.

Find out



- 1 The terms 'softwood' and 'hardwood' have nothing to do with

the hardness of the wood itself. Australian native trees are mostly hardwood. Radiata pine is a softwood, but the wood itself is quite hard. What do softwood and hardwood mean?

- 2 A process called steam distillation is used to extract oils like eucalyptus, tea tree and lavender from leaves. It gives a greater percentage yield of oil than the process you used in experiment 3 on page 189. What equipment is needed for this experiment? Draw a diagram of the experimental set-up.

- 3 Investigate the variety of Australian bush tucker foods which are being served in Australian restaurants.

FACT

FILE

In recent years there has been an interest in using bush tucker to develop restaurant food which is unique to Australia. Macadamia nuts have already been exported. Trees being grown in northern New South Wales include the Queensland Davidson plum, the riberry and the black apple. They produce fruit which can be used for making jams, or for flavouring sauces and icecream.



FIGURE 9.27

Native plants used by Australian aborigines as bush tucker are now being used in some restaurants.

Dyes from plants



FIGURE 9.28

Plants are the source of many commercial dyes used to colour fabrics and food. The way that dyes react with different substances depends, to some extent, on the acids and bases they contain.

Dyes from plants

Most plants produce colour. The use of plant materials like bark and roots to make dyes has been a common practice for thousands of years. The first synthetic (man-made) dye was invented by accident in 1856 by an 18-year-old chemistry student, who was trying to extract an antimalarial drug, quinine, from a plant. Instead, he made a very deep purple dye.

Although there are many synthetic dyes available today, natural dyes are still in use and are present in plants which we see every day. Weeds often produce strong dyes. Garden flowers, such as dahlias, give

good dyes and iris leaves give strong olive green dyes. Onion skins produce yellow-brown colours and fennel dyes range from yellow to green. Pine cones, nut hulls and barks from eucalypts and *Banksia serrata* produce brown shades. Wattle flowers and stems produce good dyes, especially the Cootamundra, and the leaves will produce a range of colours from bronze to green, and tan to orange. Eucalypt leaves produce colours ranging from beige to orange to red, especially the silver dollar gum and *Eucalyptus nicholii*. The leaves and bark of the native oak produce fawn colours. Lichen and some seaweed also produce dyes.

TABLE 9.2

Mordant	Colour notes	Special dyeing and safety notes
Alum (potassium aluminium sulfate), or potash	white powder	Too much may cause wool to be sticky.
Copper sulfate	blue powder	Warning: Copper sulfate is caustic . Use an eyewash bath for eye contact or running water for skin contact.
Cream of tartar	white powder	Can be used in addition to alum.
Vinegar	colourless liquid	Often useful to acidify a dye bath solution or to neutralise alkaline water.
Tannin (tannic acid)	pale brown powder	Found naturally in eucalypts, nut hulls and tree barks.
Baking soda (bicarbonate of soda)	white powder	Can change purple dyes to a bluish colour. Add it to the dye bath and let the bubbles subside before immersing fabric.
Common salt (sodium chloride); Calgon; Glauber's salts (sodium sulfate)	white crystals or powders	Distribute colour more evenly through fabric fibres.
Ammonia (ammonium hydroxide)	clear, non-sudsy, non-detergent ammonia is the best type to use	Can extract colour from lichen. Intensifies yellow and green dyes.

EXPERIMENT 4

DYING TO GET STARTED

**AIM**

To investigate the effects on wool of using different natural plant dyes or different mordants.

MATERIALS

- pieces of white wool fabric, and 2 m lengths of white wool. Before dyeing, fabrics must be washed to remove dirt and grease.
- plant materials. Leaves and stems should be shredded and soaked in water for a few days. Bark should be shredded and soaked for 2 weeks. Each member of the class should choose one plant.
- other fabrics for comparison

METHOD

- Simmer the plant material for between 20 minutes and 3 hours, until you get a strong colour.
- Strain out the solid matter, filtering if necessary, and add required mordants to the dye bath.
- Put fabric or wool into the dye. Top up with more water if necessary.
- Simmer for 30–40 minutes, stirring occasionally. Do not allow wool to remain at the bottom of the container.
- Strain the hot material or wool in a colander and leave to cool.

- Rinse in cool water. Hang to dry.

Safety notes:

- use rubber gloves
- wear safety goggles when heating dye solutions

- Investigate how effectively different materials are dyed by the natural plant dye.

- Investigate how the plant dye is affected by combining it with different mordants.

- Devise recipes using your plant dye and different mordants that produce different colours. Present your findings as a reference chart for others to use.

RESULTS AND DISCUSSION

Write a report which summarises your experimental findings. For your plant, what was the effect of adding different:

- mordants?
- combinations of mordants?
- amounts of mordants?
- fabrics?

EXTENSION

Keep your fabric and wool pieces, then as a group or class, decide on an environmental message you would like to promote. Combine your pieces to produce a mural which illustrates your message.

Mordants

If materials are dyed directly with a natural plant dye, the new colour may wash out or fade. For this reason, nearly all plants need a **mordant** to fix the colour in the fabric. Alum and copper sulfate are the most commonly used mordants. More than one mordant can be used in the same dye bath, but only a small amount is needed.

Questions



- Outline the safety precautions that you should take when producing dye solutions.

- Suggest why each of these safety precautions should be followed.

- Use Table 9.3 to answer the questions below.

- Suggest reasons why James didn't add a mordant to his dye mixture.

TABLE 9.3

Student	Colour	Dye mixture
Susie	Blue	Lichen, non-cloudy ammonia and washing soda
Kwang	Yellow	Onion skins boiled with alum
Robin	Green	Wattle flowers and leaves with copper sulfate
Maria	Red	Lichen, non-cloudy ammonia and alum
James	Brown	Stringybark soaked in water

- Explain how the same plant could be used to produce different dye colours.

- Outline the method used by Robin to produce his dye mixture.

- What difficulties could Kwang experience during the preparation of his dye solution? How could they be minimised?

FACT

FILE

Most natural dyes come from plant material. An exception is the red dye, cochineal. It comes from the female cochineal insect which lives on prickly pear cactus in Mexico and Central and South America.

Wasted resources



FIGURE 9.29
The accidental oil spillage from the *Exxon Valdez* oil-tanker on 23 March 1989 caused widespread environmental damage.

FIGURE 9.30
Many birds which landed on the oil-covered water died. The oil coated their wings so that they could not fly.



The *Exxon Valdez* disaster

Our home, planet Earth is finite; all life shares its resources and the energy from the Sun and therefore has limits to growth. For the first time we have touched those limits.

... When we compromise the air, the water, the soil and the variety of life, we steal from the endless future to serve the fleeting present.

Excerpts from the *Certificate of Interdependence*,
David Suzuki Foundation



FIGURE 9.31
Car bodies and other household waste materials are often seen in our waterways.

Close to midnight on a winter's night, one of the world's worst environmental disasters occurred off the coast of Alaska. The *Exxon Valdez*, a supertanker with modern navigation equipment, ran aground on a reef. Millions of litres of crude oil spewed out into the crystal clear waters of Prince William Sound, causing widespread damage and death to many species of wildlife.

The oil floated on the surface at first, at the mercy of the wind and water currents. Water birds landing on it became coated with oil and many were drowned. Fish were unable to breathe in the oxygen-starved water. Large scavenging birds such as rare bald eagles ate the oil-covered smaller birds and fish; the eagles' stomachs and intestines became coated with oil and they could no longer absorb nutrients or water. They died of starvation and dehydration, or were poisoned by the chemicals in oil.

Deposits in our backyard

Conservation of resources is a worldwide problem. There is not a lot that we can do about oil spills but we can act locally in other areas.

The materials we throw away or accidentally release into the environment are picked up by wind, water and even ice. They are transported just as natural materials are, and are often deposited in places where they can do a lot of damage to plants and animals, including humans.

There is plenty we can do on a personal basis to reduce these and many other problems which arise from humans' deposits. Campaigns like 'Clean Up Australia Day' encourage people to reduce the amount of litter released into our water-

EXPERIMENT 5

CLEAN UP YOUR SCHOOL

AIM

To involve your local community in reducing litter in your surroundings.

MATERIALS

- the involvement of the whole class
- butcher's paper and felt-tip pens
- regular meeting times
- other materials as the need arises

METHOD

As a class, you are to plan a campaign to reduce litter within your school grounds. Once you have a sound plan of action, carry it out and evaluate the results. Here are some suggestions.

- 1 Start with a map of the school grounds and conduct a survey to find out the trouble spots.

Note where bins are at present.

- 2 Determine whether wind or water might be concentrating litter in certain areas.
- 3 Organise and advertise a clean-up day that involves the whole school.
- 4 Find out if your school has any recycling system going. If so, what is recycled? Can this be improved? Find out from your local council if there is any way this scheme can be expanded to include other items.
- 5 Invite the press to do a story on your campaign.

DISCUSSION

It is most important that you reflect on what you have done at regular intervals during the campaign. Which things are working? Do we need to change our strategies? Do we need help?

ways. Education and recycling programs are the major tools we have in the fight to minimise the impact of unwanted human deposits in our fragile environment.

Questions

- 1 Look at Figure 9.30.
 - a What parts of the car could pollute the water?
 - b What other problems could the abandoned car present?
- 2a Construct a bar graph or sector graph for the data in Table 9.4.

TABLE 9.4 Contents of the typical household rubbish bin.

Food scraps	45%
Bottles and other glass	20%
Paper	19%
Metal (e.g. cans, foil)	10%
Plastic (e.g. containers, film wrap)	4%
Other	2%

- b If this represents a typical garbage bin that was accidentally spilled into the gutter, imagine what would happen to each type of waste.
 - i Which ones would rot away?
 - ii Which types might be transported by the wind? Why?

iii Which might be transported by water? Why?

iv Which types would cause the most damage? Explain.

c Construct a similar graph for the contents of your own rubbish bin.

Think about

Consider the wildlife affected by the oil spill from the Exxon Valdez. Imagine you are one of the wild animals and describe your experience as you are affected by this unknown substance.

Find out

Find out about recycling services in your suburb. Does your family take part? If not, design a recycling strategy for your family, and try to involve them in the planning and execution of your program. Work out how to monitor the effectiveness of your program. You should consider ways of creating less waste in the first place, as well as how to dispose of waste appropriately.



Word process



Using scientific language

- Complete these sentences.
 - A resource is something which ____
 - Three examples of natural resources are ____
 - Minerals which are mined are called ____ minerals.
 - A mixture of two or more metals is called an ____.
- Explain how the two terms in each pair are related, then explain how they are different.
 - mineral and crystal
 - streak and hardness
 - density and mass
 - non-renewable resource and renewable resource
 - cement and concrete
 - dye and mordant
 - conservation and sustainable development



Check your knowledge

- You have just started a rock and mineral collection. Describe how you would divide up your collection into two categories—rocks and minerals. Now explain how to divide the rocks up into igneous and sedimentary rocks (assume you have a microscope). Finally, you will divide your minerals up by using the following tests: colour, crystal shape, hardness and streak. Explain what needs to be done for each of these tests. Now draw a poster-sized picture of your imaginary rock and mineral collection in a big box with a compartment for each specimen. Draw in the specimens in a way that shows how you arranged them in the categories you have just made. Label the sections of your collection box.
- Look at the following list of resources: concrete, glass, bauxite, haematite, eucalyptus oil, radiata pine.
 - For which of these resources must an extraction process be used?
 - Which ones must be manufactured?
 - Which resources are non-renewable?
- Write a descriptive report on resources which are available from plants. Include in your report paragraphs about resources obtained from leaves, those obtained from stems and those obtained from reproductive structures (e.g. seeds and fruit).

Apply your skills



- Consider this list of resources which are obtained from mining: coal, oil, aluminium, iron, salt, gold, sand. How would each of these types of people use these resources?
 - a baby
 - an astronaut
 - a poet
 - a rollerblade enthusiast
 - a teacher
- Silicates, the biggest group of minerals, all contain silicon (Si) and oxygen (O) together. Look at the chemical formulae for the minerals on the Mohs' hardness scale shown, and decide which ones are silicates.

1	Talc	$\text{Mg}_3\text{Si}_4\text{O}_{10}(\text{OH})_2$
2	Gypsum	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
3	Calcite	CaCO_3
4	Fluorite	CaF_2
5	Apatite	$\text{Ca}_5(\text{PO}_4)_3(\text{F},\text{OH})$
6	Orthoclase	KAlSi_3O_8
7	Quartz	SiO_2
8	Topaz	$\text{Al}_2\text{SiO}_4\text{F}_2$
9	Corundum	Al_2O_3
10	Diamond	C

- Which other chemical symbols appear frequently in this table? Find out what chemical elements these symbols represent, and how abundant they are in the crust.
- Which of these minerals are elements?
- Which ones are compounds?

Challenge yourself



- With a group of friends, design and make a game which allows players to reach and win an environmental award. You could use chance cards, dice and counters. Include good resource-management practices as part of your game and also include environmental hazards to be used as stumbling blocks along the way.
- The developed countries of the world, including Australia, are contributing more than their fair share of greenhouse gases into the environment. However, greenhouse gas emission is a global problem which already seems to be affecting climate and wildlife (e.g. coral reefs). Find out what plans are being made on a worldwide basis to reduce greenhouse gas emissions. What is Australia contributing to these efforts? Draft a letter to the editor of a local paper to argue for individuals making their own contributions to reduced greenhouse gas emission (i.e. think globally, but act locally).



CHAPTER

10

EXPLORING SPACE

Humans have been exploring space for thousands of years. When people began to chart the night sky as early as 4000 BC they were exploring with their eyes. Detailed celestial maps and calendars were developed by many ancient civilisations in China, the Middle East and Central America.

This was the only type of space exploration possible until the invention of the telescope in 1608. Galileo used one of the first telescopes to observe the mountains and craters of the Moon, the phases of Venus, the moons of Jupiter, sunspots and millions of stars in the Milky Way. Over time larger and more powerful telescopes were built. These allowed an enormous number of objects to be discovered in deep space. During the 20th century exploration of space took giant steps forward with the use of radio, infrared and other types of telescopes.

The development of rockets resulted in the first artificial satellite being placed in orbit around the Earth. Sputnik 1, launched in 1957, marked the beginning of the space age. Remarkable progress has been made since then. Telescopes orbit the Earth above the atmosphere and view the most distant objects in the Universe. Men have visited the Moon, and unmanned probes have visited all the planets except Pluto. Probes have even passed through a comet's tail. Space stations have also been put into orbit, allowing people to spend long periods of time in space.

The urge to explore further and unravel the mysteries of the Universe will continue. During your life-time you may witness a permanent base on the Moon, astronauts visiting Mars and probes exploring space beyond the solar system.

In this chapter you will learn about the Universe and how people have made advances in the exploration of space.

CHAPTER OUTCOMES

Successful completion of this chapter will allow you to:

- Describe how different cultures developed ideas to explain the Universe.
- Locate and describe features of the night sky.
- Make a model of the local group of galaxies.
- Use scales to describe differences in the sizes of, and distances between, objects in the solar system.
- Explain how a satellite remains in orbit.
- Discuss technological developments that have extended our ability to explore space, including the use of telescopes, satellites and space probes.
- Describe crewed space flight both in orbit and to the Moon.
- Give reasons for and against space exploration.

HISTORY OF SCIENCE

Ideas about the Universe



FIGURE 10.2

Thales, an early Greek astronomer, thought the Earth was floating in a sea. Many previous cultures believed this also.



FIGURE 10.3

People on Easter Island thought the world hatched from an egg like the one the figure is holding.



FIGURE 10.4

Ancient Hindu belief.

FIGURE 10.1

This COBE (Cosmic Background Explorer) satellite photograph shows changes in the cosmic microwave background. Many astronomers believe this is evidence for the Big Bang.

Space is everywhere. It is between your eyes and this page. It is between you and everything else on Earth, and beyond Earth. It is so big we use words like vast, immense, immeasurable, infinite and continuous to try to describe it. Scientists often use the word space to mean outer space, the area outside Earth. The **Universe** is all space and everything it contains.

People have always looked into space and asked questions.

- How did the Universe begin?
- What size is it?
- What shape is it?
- Will it ever end?
- How will it end?
- Why am I in it?

Talk about

Try to answer the questions posed above. Discuss these with other class members and see if their answers agree with your ideas.

Early ideas

Early **astronomers** used their eyes to look at the sky. Today, astronomers look into space with powerful equipment. Their ideas about how the Universe began and what it's like change as new pictures are sent back from space to Earth by **satellite**.

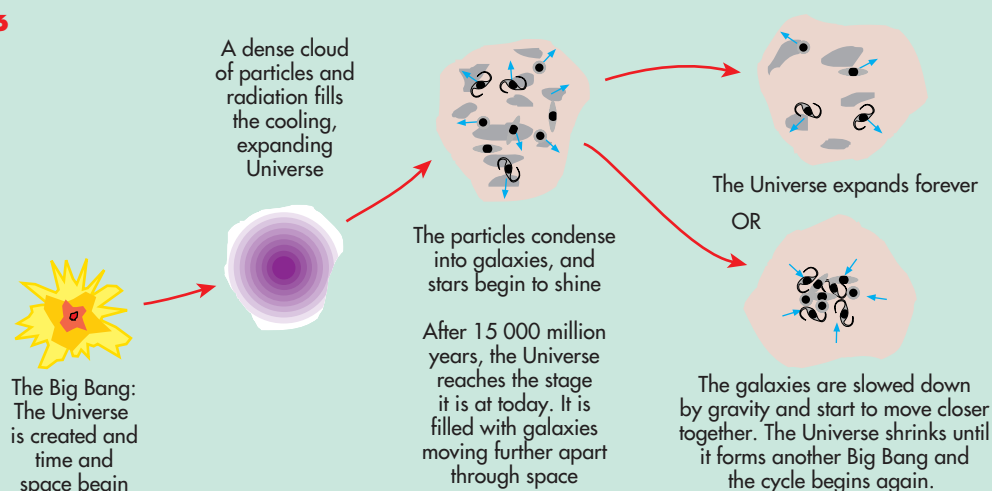
FIGURE 10.5

The ancient Egyptians thought the goddess Nut gave birth to the Universe.



FIGURE 10.6

The Big Bang theory.



USING SCIENCE 1

NIGHT ACTIVITIES

PART A

On a cloudless night, take a torch covered with red cellophane (to reduce glare) and a groundsheet outside. How many stars are there? Are some brighter than others? Do they seem to twinkle? Where are these 'twinkling' stars? Can you see different colours? Draw and describe what you can see.

PART B

Fasten a cardboard tube to a solid base so that when you look through the tube, you can line it up so you can see a bright star in the middle. Watch the star for 10 minutes. Does it move? If so, in which direction? Write down a list of your observations and try to explain them.

Modern ideas

Today, most astronomers agree that the Universe began around 12 billion years ago with a gigantic explosion of gas and matter. This is called the **Big Bang theory**.

By measuring the colour of the light from other stars, astronomers have found evidence that the galaxies are expanding. Galaxies further away are moving fastest. There is also lots of **dark matter**, which is invisible but which affects the way stars move. Astronomers are trying to find out about this, as well as working out new ways to discover and measure the different sorts of **background radiation** left over from the Big Bang. Different astronomers interpret things in different ways. There will always be controversy about the Universe!

Questions

- 1 What is an astronomer?
- 2 What is the Universe? How old is it thought to be?
- 3 Look at Figures 10.2–10.5. Try to



explain each belief.

4 What is the Big Bang theory?

What evidence supports this theory?

Think about

Choose one early belief about the Universe and write a short story, poem or legend which describes how the Universe was created.

Find out

- 1 How do we know the galaxies are expanding? Use the library to find out about the Hubble constant.
- 2 What happens between the galaxies when they move apart? Does more matter come in to fill up the spaces? If so, what matter? Do new galaxies form of their own accord? What does this suggest about the beginning of the Universe?



FACT

FILE

Many of the words we use about space today come from the Greeks because they were among the first people to write down their observations and thoughts about the Universe. Aster is a Greek word for star. Cosmos means Universe.

What is in space?

FIGURE 10.7

The Andromeda galaxy is a spiral shaped galaxy 2 million light-years away. The Milky Way is also a spiral galaxy, and the Earth and our Sun are in one of the spiral arms. The furthest known galaxies are 5 billion light-years away.

FACT

Light travels at the fastest speed we know: the speed of light is 300 000 km/s. Astronomers use light-years to help us understand the huge distances between objects in space. Knowing the speed of light, you can calculate that light travels about 10 million million kilometres in a year. This distance is called a light-year.

FILE

When we look at the night sky, we see mainly stars. There are billions (1 billion = 1000 million) of stars in the Universe, grouped together in great spinning star clusters called **galaxies**. Most of the stars we see belong to our home galaxy, the **Milky Way**.

The Milky Way is one of perhaps 100 billion galaxies in the Universe. Each galaxy contains billions of stars, as well as gases, dust clouds and planets. There are about 20 galaxies near us. The Milky Way and the Andromeda galaxy are the two biggest members of a small cluster of galaxies in our neighbourhood in space, called the **Local Group**.

Earth is located in the Milky Way galaxy. Almost every point of light you can see around you at night is also in the Milky Way galaxy, except for a few other nearby galaxies. People in the Southern Hemisphere can see two galaxies, the **Large** and **Small Magellanic Clouds**. These appear as small smudges of light, like faint clouds. People in the Northern Hemisphere can see the Andromeda galaxy.

Constellations

Small star groups which form patterns, called **constellations**, are easily seen in the sky. If you have seen the 'saucer' in the sky, then you have seen part of the constellation Orion. Over 90 different constellations have been named by astronomers.

Constellations are important in a study of space since they help to form a map of the sky. Any location in the sky can be described in relation to the position of a known constellation.

Twelve special constellations are known as the constellations of the **Zodiac**. The Sun appears to occupy a different constellation each month.

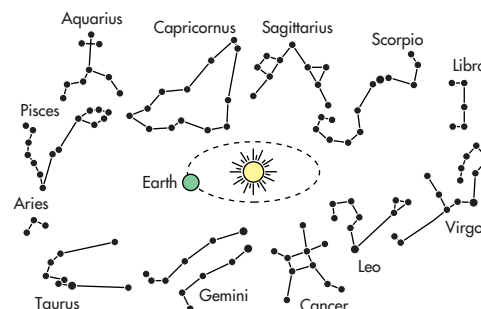


FIGURE 10.8

Constellations of the Zodiac. Suggest how your astrological sign relates to the constellations of the Zodiac.

What else is out there?

The Universe also contains clouds of gas and dust called **nebulae**, **black holes**, objects so dense that not even light can escape their gravitational pull, **quasars**, puzzling sources of intense radiation, and **pulsars**.

THE LOCAL GROUP OF GALAXIES

WHAT YOU NEED

(Per group of four or more students):

scissors, white card, paints (including black, yellow and blue), paintbrush, ruler, cotton wool, glue, wooden satay sticks, polystyrene foam sheet, compass, pen.

WHAT YOU DO

- 1 Cut card spirals with diameters of 5 cm, 4 cm and 3 cm to represent the galaxies Andromeda, the Milky Way and M33 respectively. Paint the galaxy centres yellow and the arms blue.
- 2 Cut 26 wooden sticks to the lengths shown in

Table 10.1. Paint them black. Attach each spiral to its correct stick, and cotton wool balls to the others. Paint the cotton wool 'galaxies'.

- 3** Cut the polystyrene to measure 40 cm by 36 cm. Make a grid (4 cm intervals) of 9 cm by 10 cm.
- 4** Use Figure 10.9 to mark the position of each galaxy and place each stick in its correct position. This is how our Local Group looks from the outside.

DISCUSSION

Compare the spread of galaxies, and size. Why do you think some galaxies are so close together?

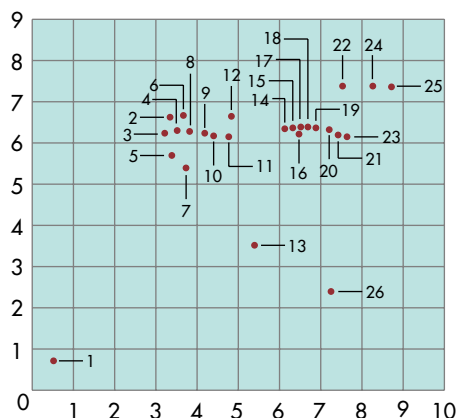


FIGURE 10.9
Position of each galaxy in our Local Group.

Questions

- 1 Distinguish between a star, a constellation and a galaxy.
- 2 How many kilometres is the Andromeda galaxy from Earth?
- 3 Proxima Centauri, the closest star to Earth apart from the Sun, is 4.2 light-years away. How many kilometres is this?

Think about

If we travelled by car, it would take us 48 million years to reach Proxima Centauri. It would take 80 000 years for the fastest probes to reach this star. If light from a star takes millions of years to reach us, what are we looking at when we look into space? How is looking at the stars like looking back in time?



FIGURE 10.10
The Pleiades star constellation in the cluster Taurus.

Night activity

To the ancient Greeks, the Pleiades (Figure 10.10) were seven sisters who were pursued by Orion, the hunter. Zeus, the chief god, took pity on them and changed them into stars. To some Aboriginal people, these seven stars were kangaroos.

- a** Locate the Pleiades in the constellation Taurus in the northern sky from November to April. (With a telescope, more than seven stars can be seen in the cluster Pleiades.)
- b** Find out the Aboriginal legend relating to the Pleiades.
- c** The Southern Cross may be found in the Southern Hemisphere at any time of the year if you look south. Locate the Southern Cross in the sky. Make up a legend about the Southern Cross.

TABLE 10.1 Stick heights (in millimetres).

1. GR8 = 210
2. Leo I = 110
3. Leo II = 105
4. Sextans = 95
5. Ursa Minor = 120
6. Carina = 80
7. Draco = 125
8. Milky Way = 95
9. S.M.C. = 80
10. L.M.C. = 90
11. Sculptor = 90
12. Fornax = 80
13. NGC 6822 = 80
14. NGC 147 = 200
15. M32 = 190
16. NGC 185 = 200
17. NGC 205 = 170
18. Andromeda = 180
19. A-II = 165
20. A-I = 175
21. A-III = 170
22. M33 = 175
23. WLM = 70
24. IC 1613 = 95
25. Pisces = 155
26. DDO 210 = 70
These are all the names of galaxies in the Local Group.

The vastness of space



FIGURE 10.11

It is very hard to appreciate the size of the solar system, never mind that of our galaxy or the Universe! We are used to very small distances compared to those needed to describe our place in space.

When astronomers discover a new object in space they determine its size and distance from Earth. Within the **solar system** distances such as those from the Sun to the planets are usually measured in millions of kilometres. However, once beyond our solar system, the immense **interstellar** and intergalactic distances are measured in **light-years**. (Although modern astronomers prefer to use the parsec, which is equal to 3.25 light-years.)

The closest star to our own is Proxima Centauri which is 4.2 light-years away. This means that light from this star takes 4.2 years to reach us. Our own galaxy is so large it measures 100 000 light-years from edge to edge. The large Cloud of Magellan, one of the brightest galaxies near our own, is 169 000 light-years away. The furthest galaxy seen with a light telescope is about 5000 million light-years away. Radio tele-

scopes have measured galaxies at twice this distance.

These distances are so vast that they are impossible to imagine, totally beyond our daily experiences. Instead, we will concentrate on the sizes of the planets and their distances from the Sun. A scale model of the solar system can be constructed to aid our understanding of these distances.

Questions



1 If interstellar distance measures the distance between stars, what does intergalactic distance measure?

From your study of the scale model of the solar system:

2 Which planet is the largest?

3 The Moon is about 250 000 km

USING SCIENCE 3

THE VASTNESS OF THE SOLAR SYSTEM

WHAT YOU NEED

Card, scissors, sticky tape, textas, rulers, compasses, trundle wheel, 30 metre tape.

WHAT YOU DO

In groups of two or three, you are going to make a scale model of the solar system based on a scale where the Earth is represented by a disc just 1 mm in diameter. Table 10.2 lists the planets with their diameters and their distances from the Sun.

TABLE 10.2

Planet	Diameter (km)	Distance from Sun (km)
Mercury	5 000	58 000 000
Venus	12 400	110 000 000
Earth	12 800	150 000 000
Mars	6 800	230 000 000
Jupiter	140 000	780 000 000
Saturn	120 000	1 400 000 000
Uranus	52 000	2 900 000 000
Neptune	50 000	4 500 000 000
Pluto	5 000	6 000 000 000

Scaling down these distances can get difficult so Table 10.3 has done most of the work for you. The table tells you that the Sun, to the same scale, will be 109 mm in diameter and some 12 m away from the Earth.

TABLE 10.3

Object	Diameter	Distance from Sun
Sun	109 mm	
Mercury	0.33 mm	5 m
Venus	1 mm	8.5 m
Earth	1 mm	12 m
Mars	0.5 mm	18 m
Jupiter	11 mm	62 m
Saturn	9 mm	114 m
Uranus	2.5 mm	230 m
Neptune	4 mm	360 m
Pluto	0.25 mm	480 m
Alpha Centauri	120 mm	3200 km!
Sirius (brightest star)	20 mm	6500 km!

Cut out circles to represent the size of the stars and planets from the card provided or use dots on a piece of paper. When you have done this, find a suitable place to construct your model. It might be a long corridor or even outside the school building. Stick labels near each planet.

Cut out two signposts to indicate where the two stars Alpha Centauri and Sirius might be found. To the scale you are using, where on the Earth's surface might they be?

from the Earth. Using the same scale, how far from the Earth should the Moon be placed?

Think about

- 1 Within the constellation Orion lies the star Rigel, 900 light-years from Earth. In which Earth year did the light reaching us now leave Rigel?
- 2 In the constellation of Canis Major (next to Orion) is Sirius, only 8.6 light-years away. If Sirius was destroyed today in which year would astronomers on Earth first know about it?

Find out

How do astronomers measure the distance to stars?

Activity

A Boeing 747 can cruise at about 1000 km per hour and a rocket could achieve about 40 000 km per hour.

- a Using Table 10.2, construct a timetable for an inter-planetary shuttle service. Each group member could take a different trip. Check values from one person to another to see if the results make sense.
- b Which planet would be the furthest a human astronaut might reach while travelling in a rocket at this sort of speed?



FACT

FILE

The most distant objects so far detected are called quasars. These lie 10 billion light-years or more from the Earth and are moving further apart as the Universe expands.

Exploring with satellites

FACT

The Hubble space telescope was placed in orbit around the Earth, by astronauts aboard the space shuttle, in 1990. There were some problems with the telescope's main mirror which prevented it from being focused properly. These problems were overcome by 1994. Since then, the Hubble telescope has taken some of the sharpest images of celestial objects ever seen.

FILE

Satellites are objects which travel around or **orbit** another object. The Earth is a natural satellite of the Sun and the Moon is a natural satellite of the Earth. Artificial satellites may be unmanned spacecraft or manned vehicles such as the space shuttle



FIGURE 10.12
The first artificial satellite to orbit the Earth—Sputnik 1.

and space stations launched into orbit by powerful rockets.

But what keeps a satellite in orbit? Surprisingly, it is the Earth's gravitational pull on the satellite! However, satellites don't crash into the Earth because of their tremendous speed. Although they are pulled towards the Earth by gravity, their speed enables them to keep falling 'over the horizon'. Thus, if they are travelling with exactly the correct speed they never

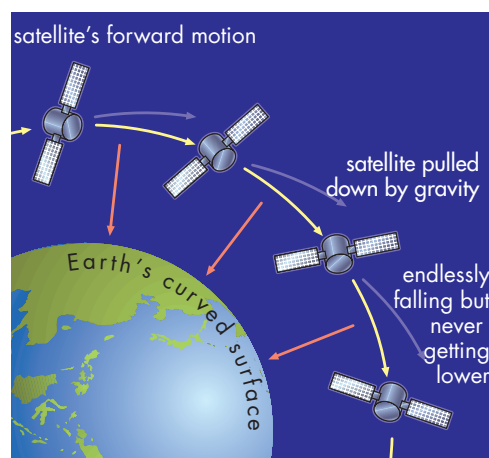


FIGURE 10.13
Satellites travel in straight lines, unless forces act on them. Earth's gravity pulls satellites down. But satellites are also moving sideways so they follow the curve of the Earth's surface, endlessly falling as they orbit the planet.

USING SCIENCE 4

FORCES THAT KEEP A SATELLITE IN ORBIT

This activity is best done outside. Attach a tennis ball or squash ball to a 1 m length of cotton or thread by piercing the surface. Carefully, whirl the ball around your head.

1 What direction is the force applied to the thread? What direction is the force on the ball?

2 When nobody is close, let go of the thread and observe exactly what direction the ball moves.

3 Does the ball move away from or towards your hand?

4 Does the ball still keep moving in a circular path?

5 How does your hand provide a force to keep the ball moving in a circular path?

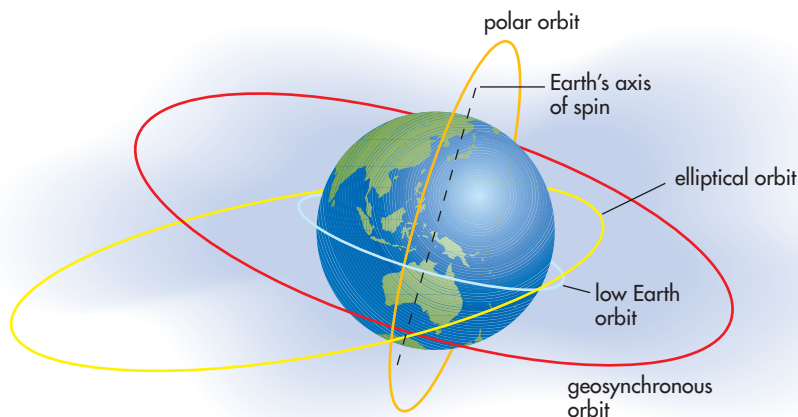


FIGURE 10.14

Satellites can orbit the Earth in a number of positions. Low Earth orbit occurs at 150–500 km. These are roughly circular orbits used by the space shuttle and space stations. Many communication, navigation and weather satellites orbit at around 37 000 km above the Earth's equator. These are called **geosynchronous orbits** because the satellites travel at the same speed as the Earth turns. Other orbits are elliptical, **polar** and **Sun-synchronous**.

actually get any closer to the Earth's surface. Isaac Newton reasoned that the Moon orbited the Earth in exactly this manner over 300 years ago!

Satellites have many uses. Some are used for communication by relaying telephone and television signals around the globe. Other satellites collect data to enable **meteorologists** to make accurate short- and long-term weather forecasts.

Highly detailed images of the Earth's surface can tell us about land management, agriculture, forestry, fisheries, mineral exploration, environmental management, urban planning and tourism. During the devastating Sydney bushfires of January 1994, satellite images provided useful information to firefighters and emergency services. Landsat 5 takes pictures of the

entire globe every 16 days and its images can reveal items as small as a highway or football oval—and all from 705 km above the Earth's surface!

Some satellites are designed to explore deep space. Instead of looking down at planet Earth they look at the Moon, Sun, planets, stars and galaxies. Research satellites carry devices to detect all sorts of electromagnetic radiation such as microwaves, radio waves, infrared rays, ultraviolet rays, X-rays and visible light emitted from objects in space. The information collected is used to construct images which are much clearer than those produced by ground-based instruments. Above the Earth's atmosphere these devices avoid the dust and the scattering effects of the air.

Questions

1 What keeps satellites in orbit?

2 Make a list of as many different uses of satellites as possible.

3 What is remote sensing?

Think about

Satellites can now be used to take clear photographs of people on Earth. Do you think this is desirable? Explain.

Find out

1 Satellites can be placed into polar orbit. What is a polar orbit, and what type of satellites are placed into a polar orbit?

2 The news of Sputnik's launch caused great dismay in the US. Why was the US government keen to better the achievements of the USSR?

3 What discoveries have been made by the Hubble space telescope?

FACT

FILE

The world's first artificial satellite was launched by the then Soviet Union on 4 October 1957. It was called **Sputnik**, which means 'fellow traveller' in Russian. Its mass was only 84 kg and its diameter only 53 cm. Its orbit ranged from 227 to 941 km from the Earth and it carried apparatus to measure the temperature and density of the upper atmosphere. A small battery-powered radio transmitter beamed results down to receiving stations on Earth.

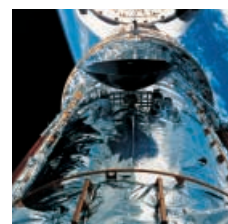


FIGURE 10.15
The Hubble space telescope.



Glossary

Leaving Earth



FIGURE 10.16

All objects (including the crew) 'float' in an orbiting space station.

FACT

FILE

NASA has a 145 m deep drop-tower facility that begins on the surface and continues into a mine shaft. The 'fall area' has most of the air removed to reduce drag. The bottom is filled with polystyrene balls to break the fall. Weightlessness under these conditions only lasts for 5.2 seconds! A mine shaft in Japan 490 m deep allows free-fall conditions for 11.7 seconds.

The USSR's Sputnik 1, launched in October 1957, was the first artificial satellite, but Sputnik 2, launched only days later on 3 November 1957, paved the way for crewed flight since it contained a dog named Laika. This was followed by many satellites launched by the Americans and Soviets containing animals of one type or another. Medical experiments on these animals confirmed that it seemed possible for humans to survive the enormous forces during launch and re-entry and for space

vehicles to provide an environment that could support life safely and reliably.

Yuri Gagarin became the first man in space when he completed one orbit of the Earth in a Vostok craft launched on 12 April 1961. On 5 May 1961, Alan B. Shepard Jr became the first American in space, although, unlike Gagarin, he did not perform a complete orbit but simply left the Earth's atmosphere. John H. Glenn Jr became the first American to orbit the Earth, on 20 February 1962. On 16 June 1963, Valentina Tereshkova, on board Vostok 8, became the first woman in space, completing 45 orbits of the Earth.

A major problem faced by early space scientists was predicting how the human body would react to **weightlessness**. Throughout its evolution, the human body has been used to a **gravitational force** or weight of 9.8 N for every kilogram of body mass. This means that a person with a mass of 50 kg feels a gravitational force or weight of 490 N. This normal weight is often referred to as '**1g force**' or simply '**1g**'.

Since a satellite or space station in orbit is actually falling freely under the influence of gravity, all the objects inside it (including the crew) are also falling freely and appear



FIGURE 10.17

Yuri Gagarin—the first human in space, 12 April 1961.

EXPERIMENT 1**A DEMONSTRATION OF WEIGHTLESSNESS**

Do this experiment outside!

AIM

To demonstrate the effect of water in free fall.

MATERIALS

- polystyrene drink cup
- pencil
- water

METHOD

- 1 Using a pencil, carefully make a hole near the bottom of the polystyrene cup.
- 2 Cover the hole with your finger and fill the cup with water.
- 3 Take your finger off the hole and record what happens to the water.

4 Place your finger back over the hole and re-fill the cup. Lift the cup as high as possible. Remove your finger and release the cup at the same time. Carefully watch what happens to the water before the cup hits the ground.

5 If you have access to a video camera, photograph the cup during its fall. Look at the resulting video by using the pause button or freeze-frame control.

DISCUSSION

- 1 Why does the water escape through the hole while the cup is being held?
- 2 Why does the water inside the cup behave differently while they are both falling?
- 3 Why is the 'free fall' condition similar to astronauts in an orbiting space craft?

to have no weight. Both the astronauts and objects inside a spacecraft in orbit would float relative to each other.

A tradition of making a 'weightlessness indicator' has grown up among American astronauts. They design and make a small device which will tell them when they have achieved Earth orbit.

Questions

- 1 What was the first living thing into space? Why was this significant?
- 2 Explain what is meant when a fighter pilot experiences a force of '3g'.
- 3 You are in a lift at the top of a very tall office building. The lift cable suddenly breaks! Describe your final few seconds.

Think about

Imagine you have taken a pair of bathroom scales into a lift in a tall city office block. You stand on the scales and watch the readings as the lift moves up and down. Write down

what you think might happen to the indication of your weight. Discuss your results with other students and your teacher. What would have to happen for the scales to read zero?

Find out

Research the names of the first ten humans to go into space. Draw a time line of when their missions occurred and mark on the chart how long each astronaut was in space, which country was responsible for the launch, and the purpose of the mission.

Activity

Imagine that you have been chosen to go on a space mission. Design a small device that will indicate that you are in a truly weightless condition. (It will have to fit in your pocket.) You should be able to test your device by dropping it and videotaping the results.



The exploration of the Moon

FIGURE 10.18

Lunar rocks were brought back to Earth in this pod.

FACT

FILE

NASA's Lunar Prospector was launched in January 1998. This spacecraft orbited the Moon, surveying the gravity and magnetic fields on the Moon's surface as well as its composition. Analysis indicated that water ice may be located in deep craters around the poles. The Lunar Prospector was deliberately crashed into the crater Mawson, near the Moon's south pole in July 1999. Scientists observed the impact, hoping for more evidence of water.

On 20 July 1969 (21 July in Australia), Neil A. Armstrong, an American, became the first human to set foot on the Moon, uttering the famous lines 'That's one small step for (a) man; one giant leap for mankind'.

Five more successful landings were undertaken by the Americans. They returned a total of 382 kg of **lunar rocks** and travelled somewhere near 100 km on the lunar surface.

The USSR brought a 100 g sample of lunar soil back to the Earth during September 1970. They also landed two lunar roving vehicles but did not ever land a person on the Moon.

The rocks brought back from the Moon gave scientists a valuable insight into the Moon's (and also the Earth's) geological history. The lunar highlands have been

dated at about 4.3 billion years old. Up to about 4 billion years ago it seems that the Moon was bombarded by huge **asteroids**, some hundreds of kilometres in diameter. These caused huge **craters** to be formed and left the surface covered in shattered rocks and dust. Internal heating caused **lava** to flow out onto the surface to form large dark areas we call the 'seas' or **maria**. It seems that very little major activity has occurred in the last 3 billion years. With no atmosphere, the surface only changes when hit by something arriving from space. Footprints left by the 12 humans who walked on the Moon will in all probability remain sharp and clear for thousands of years.

This knowledge of the Moon helps us understand the forces that have shaped the Earth. The Earth too must have been bombarded with similar asteroids, and this may account for features we can recognise even now on its surface. However, because Earth has an atmosphere, weathering by wind and water soon smooths over Earth's craters.

How the Moon originally formed is a question that has been posed for many centuries. The chemical difference in the rocks from the Moon and the Earth mean it is unlikely that it was part of the Earth that somehow broke free. Two plausible alternatives are that it formed separately somewhere near the Earth at about the same time as the Earth, or that it was captured by the Earth's gravity at some stage many billions of years ago. Perhaps further lunar exploration will give us the answer.



FIGURE 10.19

Rock samples returned from the Moon.

EXPERIMENT 2**IMPACT CRATERS ON THE MOON****AIM**

To demonstrate the formation and weathering of impact craters.

MATERIALS

- dry sand
- spray bottle
- water
- a variety of spherical objects—ball bearings, golf ball, tennis ball etc.
- tray
- flour

METHOD

- 1 Fill the tray with sand to a depth of about 5 cm and smooth the surface.
- 2 Put some flour in a strainer and shake it over the sand until the sand is covered in a thin layer of flour.
- 3 Drop the lightest sphere onto the tray of sand from a height of about 2 m.
- 4 Copy Table 10.4 and record the diameter and appearance of the crater.
- 5 Repeat steps 1–4 with the heavier spheres.
- 6 Choose one of the spheres and drop it into the sand from three different heights. Copy Table 10.5 and record your observations.
- 7 Using an empty spray bottle, blow air, gently at first, across the rim of the craters you have made. Copy Table 10.6 to record your results.
- 8 Put a small amount of water in the spray bottle and repeat step 7.

RESULTS**TABLE 10.4** Dropping from 2 m height.

Ball	Diameter of crater	Maximum depth of crater	Appearance
lightest			
medium			
heaviest			

TABLE 10.5 Single sphere from different heights.

Heights	Diameter of crater	Maximum depth of crater	Appearance

TABLE 10.6 Effect of air and water.

Air blown across the crater	Water sprayed across the crater
Before	Before
After	After

DISCUSSION

- 1 Is there any relationship between crater size and the weight of the sphere that was dropped?
- 2 Does the height from which the sphere is dropped affect the crater size or shape?
- 3 What different types of weather conditions found on Earth are simulated in this experiment?

Questions

- 1 How old were the rocks brought back from the lunar highlands?
- 2 Footprints left in sand on the Earth's surface last a very short time. Why should footprints on the Moon last for thousands of years?

Think about

Like our Moon, some planets show many impact sites. Others do not. The Earth is likely to have

suffered many impacts, so why are so few seen on the Earth's surface?

Find out

- 1 Why do you think that the Soviets did not land anyone on the Moon? Why was President Kennedy so determined that the US should land humans on the lunar surface by the end of the 1960s?
- 2 Research the composition of Moon rocks and find how they compare to common rocks found on Earth.



FIGURE 10.20
A US astronaut walks on the surface of the Moon.

Space stations

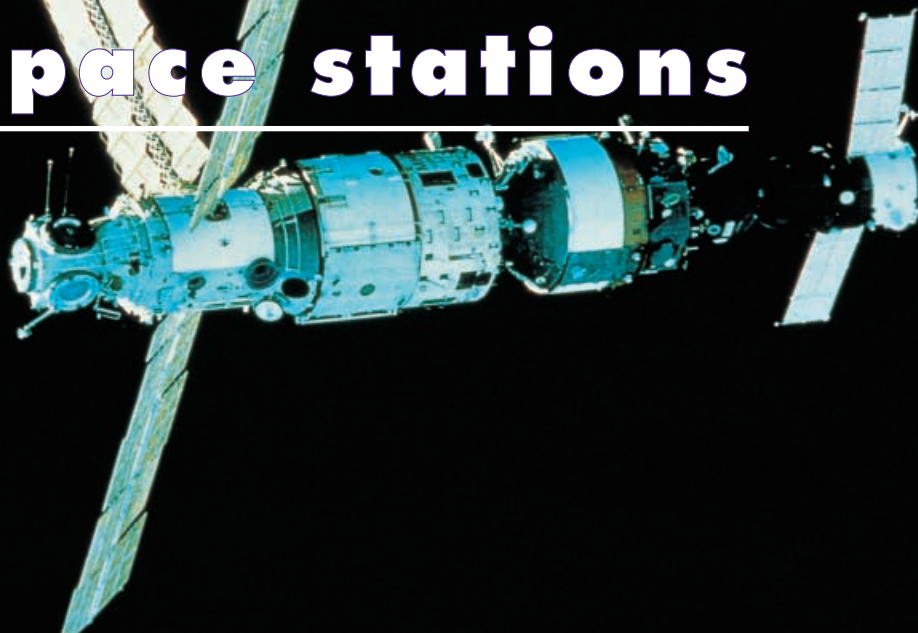


FIGURE 10.21
The Mir space station.



Animation

FACT

FILE

Cosmonaut Valeri Polykov established the world record for the number of days spent in space. He arrived on Mir in January 1994 and returned to Earth in March 1995, a total of 438 days.

Since the early days of space exploration, the advantages of operating a **space station** in Earth orbit have been apparent. Because an orbiting space craft is continually 'falling' towards the surface of the Earth, any objects inside it appear to be weightless. (This does not mean that no gravitational force is acting; the Earth's gravitational field acting on an orbiting satellite is only a little weaker than the gravitational field on the surface of the Earth, especially for close-orbiting satellites.) If humans are to explore the solar system and beyond, lengthy periods in near-weightless conditions would cause physiological changes in the body. An orbiting space station provides the opportunity to research changes in body function and the problems associated with maintaining life for the years necessary for exploring other planets.

The military objective of being able to monitor not only your own country but also less friendly states was an important motive behind establishing both the Soviet Salyut series of space stations and the US station Skylab. The first Salyut station carried high-resolution spy cameras which even ejected film which fell to Earth for developing!

The Soviets launched seven Salyut space stations over the period 1971 to 1982. They were fairly small, with a mass

of about 20 tonnes each. Soyuz spacecraft ferried crew back and forth to conduct military and scientific research. Salyut 7 is still in Earth orbit but unused. All the others gradually moved closer to the Earth and disintegrated as they re-entered the Earth's atmosphere.

The USA launched Skylab in 1973. This was much larger than Salyut, with a mass approaching 90 tonnes. During its 6-year life, scientists researched biological problems associated with weightlessness and the growing of pure crystals for industry. Forty-six thousand photographs of the Earth's surface were taken. Skylab eventually fell to Earth on 11 July 1979, scattering debris over parts of Western Australia.

The Mir space station was launched by the Soviets in 1986 and was continuously crewed until 1999. Gradually, different modules were launched and successfully docked with the original unit. During its flight, the on-board systems were added to and updated. The normal crew numbered three, but when changeovers occurred there were up to six or seven people on board. Trips from the Earth to Mir became routine. Astronauts from many countries went on board and the US space shuttle docked with the complex in 1995. The present complex has a mass of 100 tonnes. At times school students were able to speak directly to the



cosmonauts by radio about their conditions and work.

The construction of another permanently crewed space station is now under way. The International Space Station Alpha, ISSA for short, is designed to house experiments and a permanent crew for conducting research and science experiments in a **microgravity** environment. ISSA is currently being assembled in orbit from sections or modules built by contributing nations. The first component called Zarya, a control module, was put in orbit by a Russian rocket in 1999. A Russian-built service module and a connecting module from the USA will follow. Additional components from Europe, Canada and Japan will complete the structure. ISSA's final volume will be equal to two jumbo jets and its mass is expected to be over 450 000 kg. In all, 43 flights will be required to launch all the pieces into orbit.

Questions

- 1 What are the advantages of a space station in Earth orbit?
- 2 Are there any disadvantages to maintaining a space station in permanent orbit?
- 3 What products are manufactured in orbiting space stations?

Think about

You have been given the opportunity to plan and design a new space station to be launched in the early years of the 21st century. It is to have a crew of between 10 and 20 engaged on a variety of manufacturing and research projects. As much food as possible is to be produced on board. In groups of three or four, prepare your plans. Each member of the group should be responsible for one aspect of the design. Present your ideas to the rest of the class, perhaps in the form of a poster.

Find out

- 1 Find out the names and nationality of any astronauts (or cosmonauts) presently in orbit. How long have they been in orbit and what is the main purpose of their stay?
- 2 What are some of the biological effects of lengthy periods of weightlessness?

FIGURE 10.23

Andrew Thomas, NASA astronaut.

FIGURE 10.22
The planned International Space Station Alpha.

FACT

Australian-born astronaut Andrew Thomas served as flight engineer aboard the Mir space station for 141 days in space. He made 2250 orbits of the Earth.



Exploring the planets

FIGURE 10.24

Viking Lander 1's view of the surface of Mars.

FACT

NASA intends to extend its exploration of the solar system over the next few years with a number of new missions. The Europa orbiter is planned for launch in 2003. The Pluto-Kuiper express should depart Earth in 2004 and a solar probe is planned for 2007.

FILE

Beyond the Moon, unmanned spacecraft have landed on Mars and Venus and have flown close to Mercury, Jupiter, Saturn, Uranus and Neptune. Space probes have even flown through the tail of Halley's Comet.

Mars

Mars is the planet that is most similar to the Earth, and it seemed the most likely planet to have forms of life. The US Mariner 9 orbited Mars for almost a year from November 1971 to October 1972, taking enough photographs to map almost the entire surface of the planet. In 1976, Vikings 1 and 2 soft-landed on the surface. Each had colour TV cameras, equipment able to detect the presence of bacteria, and weather-recording instruments. Both functioned well for several years and sent back photographs of the Martian landscape throughout the seasons.

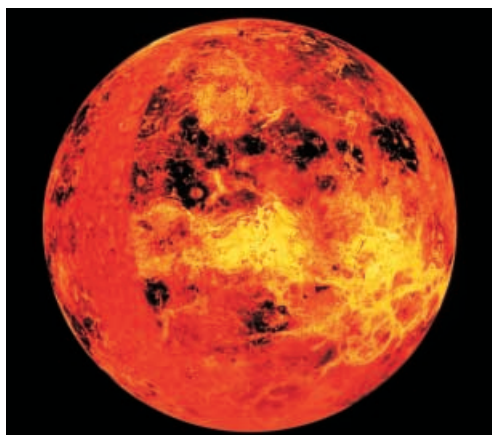


FIGURE 10.25

A map of the surface of Venus constructed from satellite images.

In 1997 the Pathfinder mission was even more successful with the small rover Sojourner exploring the Martian surface. However, at the end of 1999 the US lost two probes: the Mars Climate Orbiter and the Mars Polar Lander. These missions cost NASA \$500 million.

Venus

The Soviet Union had success probing the hostile environment of Venus. The surface temperature was found to be 480°C and the atmosphere exerted a pressure 88 times greater than that on the Earth. Nevertheless, the Venera probes survived long enough to take photographs of the surface and analyse the soil. In 1983, Veneras 15 and 16 entered Venus's orbit and returned radar images to the Earth. The dense clouds of carbon dioxide and sulfuric acid hide the surface from normal visible-light cameras. The US Pioneer Venus 1 and 2 crafts were launched in 1978 and transmitted detailed radar images of the planet to Earth. This enabled continents to be mapped and the influence of the Sun on Venus's atmosphere to be monitored.

The US Magellan probe used radar to produce detailed maps of the surface of Venus between 1990 and 1993.

Close to the Sun

The US sent the Mariner 10 probe to orbit the Sun, at times coming very close to Mercury, the innermost planet. Its first encounter in 1974 sent back images of a

deeply cratered surface not unlike the Moon's. On its second encounter, Mariner 10 detected a magnetic field around the planet.

Jupiter and beyond

The US launched two deep space probes, Pioneers 10 and 11, in 1972 and 1973. They successfully negotiated the asteroid belt on the far side of Mars's orbit and flew close to Jupiter. Pioneer 10 continued on its way out of our solar system to become the first human-made object to enter **interstellar space**. Pioneer 11 flew past Saturn in 1979.

The launch of Voyagers 1 and 2 in 1977 saw the start of a spectacular journey through the outer parts of the solar system. Voyager 2, in particular, was responsible for unexpected discoveries. After flying past Saturn, it was directed to Uranus, passing within 80 000 km of the planet in January 1986, 13 years after its launch. Here it discovered four additional rings around the planet as well as ten new moons. The craft came close enough to one of Uranus's moons, Miranda, to give photographs of that icy body. Voyager 2 then headed for Neptune, coming within 5000 km of the surface and discovering six additional moons before finally leaving the solar system.

The Galileo spacecraft arrived at Jupiter on 7 December 1995, firing its main engine for 49 minutes to go into orbit around Jupiter. On the same day it released a probe into Jupiter's atmosphere which relayed data about atmospheric temperature, pressure and chemical composition before vaporising. Since then the Galileo spacecraft has been busy studying Jupiter's four largest moons. Europa is the most interesting as it is covered with ice and its atmosphere contains oxygen.

The US launched a probe to Saturn in 1997. Called Cassini, it is expected to reach the ringed planet in 2004.

Questions

1 What advantages does sending space probes have over relying on observations from the Earth?



FIGURE 10.26

This photomosaic of Mercury's surface shows the many craters on its surface.

2 Draw a diagram of the solar system. By the side of each planet, write the date that a probe visited and new information about the planet that was obtained.

Think about



'We expect to learn about the conditions that gave rise to the appearance and successful evolution of life in the solar system.' NASA Advisory Council

To what extent can spending large amounts of money on space probes be justified when the world is faced with the inability to feed its growing population?

How might an understanding of how life started benefit the present population of the Earth?

Activity

Use the Internet and connect to NASA's home page to find the latest news on current and future missions to space.

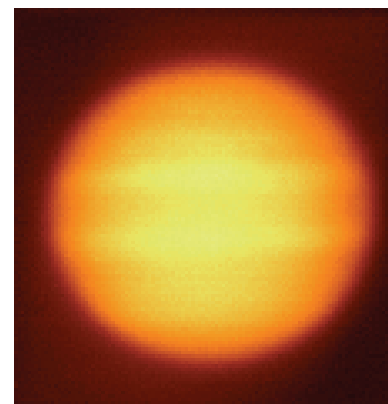


FIGURE 10.27

A recent image of Jupiter from Galileo.

APPLICATIONS AND USES OF SCIENCE



Space programs: are they worth it?

FIGURE 10.28 (right)
How much money would be
needed to eradicate poverty in
the third world?

FIGURE 10.29
(far right)
How much money do space
programs cost? How much money
would be needed to colonise
Mars?

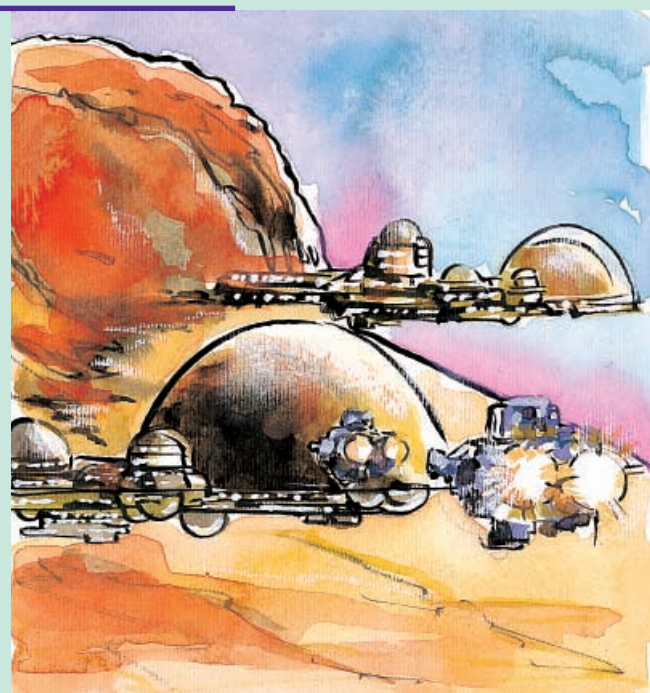
FACT

Some estimates suggest that it would cost \$150 billion each year to feed, clothe, house and educate the world's entire population. The Apollo space program cost close to \$50 billion and about \$5 billion is spent every day on defence worldwide.

FILE



Building telescopes to study space is costly enough, but sending spacecraft into space, with or without astronauts, is extremely expensive. Now that manned flights have been sent to the Moon, many astronomers want to send astronauts to Mars. Flights to Mars would be much more involved because it would take over a year just to get there. (Getting to the Moon only took 3 days.) There is also the expense: it would cost over a million million dollars. Many



people think that if that much money is to be spent, it should go towards feeding starving people and protecting the environment here on Earth. But it is more complicated than that. Building spacecraft is profitable; feeding starving people isn't. There are also the untold benefits of research into space travel.

Some people argue that if we are unable to save the Earth from over-crowding and pollution, perhaps we could move to Mars and start again. Other argue that if we can't save the Earth, we won't last long on Mars.

Activity

Hold a class debate about whether space travel is worthwhile.



Making Mars green again

NASA'S plan to terraform Mars—turn the red planet into Earth II—stands as the most challenging engineering project humans have ever considered.

The temperature on Mars must be lifted by 39°C, the lethal gases that surround the planet must be removed, a shield must cover the planet to spare people from solar radiation and a dead planet must be transformed to offer life.

NASA says it can be done in six steps.

STEP 1 (2015–2030)

People arrive on Mars. A prefabricated colony will be set up for 12–14 astronauts. They will grow experimental crops; study the atmosphere, the dust storms and solar radiation; and search for signs of life. By 2030, they will know if Mars can be humanity's new home.

Cost: Several hundred billion dollars.

STEP 2 (2030–2080)

Mars's temperature must rise from its average –45°C. NASA has suggested the greenhouse effect: have small nuclear reactors pour greenhouse gases into the atmosphere to cover Mars with a thick blanket that will retain heat. As the atmosphere thickens, the planet will give off more carbon dioxide, nitrogen and water vapour—the basic elements to sustain life.

This will require up to 10 000 scientists, engineers, physicists, skilled mechanics and agronomists.

Cost: \$20 billion a year.

STEP 3 (2080–2115)

Mars is now working to sustain life, not destroy it. Clouds will form, canyons will turn into lakes and the planet will warm, but not enough to let plants grow and give off oxygen.

Mars will now be home to 50 000 humans who will throw away their space suits but still need special breathing devices.

Cost: \$60 billion a year.

STEP 4 (2115–2130)

Mars will now have an average temperature of 0°C. Ponds, lakes, streams, small rivers and small oceans will form, and evergreen trees will emerge in the cold climate.

By 2130, Mars will be home to 250 000 people.

Cost: Several billion dollars a year.

STEP 5 (2130–2150)

Mars now has an average temperature of 5°C, still colder than Earth's 15°C average. Great oceans will have formed, millions of trees will be planted and wheat will grow on huge farms, but the planet will still not have enough oxygen to allow humans and animals out of the biospheres without breathing apparatus.

Cost: Mars will now be paying for itself.

STEP 6 (2150–2170)

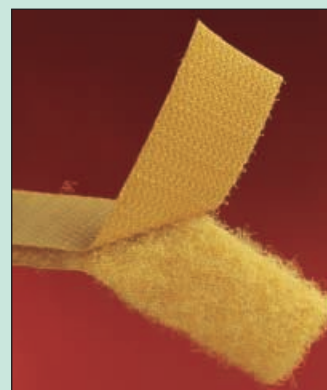
Terraforming of Mars is now complete.

Adapted from an article in *Sunday Mail Magazine*, 21 July 1991.



FIGURE 10.30

Velcro fastenings, Teflon coatings, microwave ovens, advanced computer systems and satellites are just some of the benefits that have been developed through space research.





Using scientific language

- Match the words in the first column with the statements in the second column.

satellite	the study of the Earth from orbiting satellites
polar orbit	the first artificial satellite placed in Earth orbit
Sputnik	distance light travels in one year
geosynchronous	name of our galaxy
Sun-synchronous	a satellite which orbits the Earth once in 24 hours
light-year	a theory about the formation of the Universe
remote sensing	a satellite that crosses the same points on the Earth's surface at approximately the same time each day
Milky Way	the Moon is a natural one
crater	the path of a satellite which crosses the north and south pole
Big Bang	a deep pit caused by meteorite impact
- Choose five of the words from the first list. Use each one in a sentence that clearly shows its meaning.



Check your knowledge

- Explain, with reasons, whether each of the following statements is True or False.
 - An orbiting space station has escaped from the Earth's gravity.
 - For a satellite to be in a geosynchronous orbit, it must be about 3700 km above the Earth's surface.
 - A satellite in polar orbit will fly directly over every part of the Earth in 24 hours.
- Now answer the following questions.
 - What conclusions can be drawn from the fact that there are chemical similarities and differences between Moon and Earth rocks?
 - Why can very pure crystals be grown in orbiting space stations?
 - How many times greater is the surface pressure on Venus than that on the Earth?
 - What was the first artificially produced object to leave the solar system?
- How do constellations differ from galaxies.



Word process

Apply your skills



- Jet planes can achieve weightless conditions for about 30 seconds by flying along a parabolic flight path. During the start and finish of the flight path, the crew experience greater g forces than normal. At the top of the flight path, the crew experience less than normal g forces. Figure 10.31 shows the flight path.

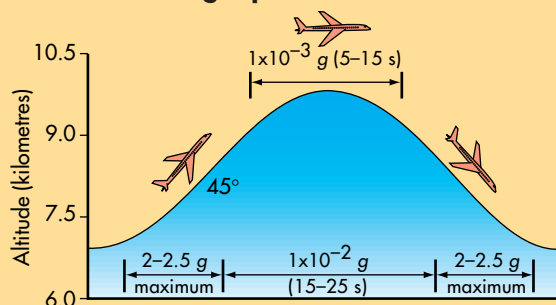


FIGURE 10.31

- What is the maximum g force experienced?
 - For how long is the gravitational force less than $1/100$ and less than $1/1000$ of normal?
 - At what height above the Earth's surface does the special flight path start?
 - What is the maximum height of the flight path?
The unfortunate effects of these flights on the crew's stomachs earned the special jet the nickname 'vomit comet'!
- Prepare a travel brochure for a 'space trip of a lifetime'. Give some idea of how long each stage of the trip will take, the sort of experiences the interplanetary tourist is likely to encounter, the changes in gravity, the food likely to be provided, etc. Illustrate your brochure with some of the sights the intrepid traveller might see.
 - The signals sent back by the spaceprobe Voyager from near Uranus took 2 hours 40 minutes to reach Earth, travelling at the speed of light. How many light-years away is Uranus?

Challenge yourself



Although, from Earth, the stars in a constellation may appear close together, they may be actually quite far apart. Rigel and Betelgeuse, for example, are in the constellation of Orion, but they are 600 light-years away from each other. Draw a diagram or make a simple model to show how this may be possible.



Tester

GLOSSARY

- acacia** the most dominant genus of plant found in arid areas of Australia; includes wattle 161
- acid** a substance with a sour taste that reacts with metals to form hydrogen gas. Acids turn blue litmus red 16
- adaptation** a characteristic possessed by an organism that helps it to survive in its natural habitat or assists with its lifestyle 156
- aerodynamic** shape that reduces air resistance 56
- aether (or ether)** medium which it was suggested must be present if light waves were to travel through space 103
- aggregate** pebbles or gravel to be used as raw material for concrete 187
- air resistance** friction that operates on an object moving through the air 56
- alchemy** early form of chemistry practised from ancient times to the Middle Ages, based on the belief that ordinary metals could be changed into gold using a hypothetical substance called the philosopher's stone 4
- alkali** a base that is soluble in water 17
- alloy** general term for any substance formed by mixing two or more metals; e.g. bronze (copper + tin), brass (copper + zinc) 18
- alveoli** (singular alveolus) tiny sac-like structures found in lungs; where gas exchange occurs 123, 124
- ammeter** device for measuring electric current 67
- amperes (A)** unit of electric current 67
- angle of incidence** angle at which a light ray hits a surface, measured relative to a normal drawn at the point of impact of the ray with the surface 88
- angle of reflection** angle at which a ray of light bounces off a surface, measured relative to a normal drawn at the point of impact of the ray with the surface 88
- anorexia nervosa** a condition where a person refuses to eat healthy amounts of food; results in severe weight loss and health problems 120
- anus** site of elimination of faeces (body wastes) 118
- aquifer** a layer of rock which contains groundwater 172
- artery** blood vessel with thick muscular walls; carries blood away from the heart 126
- asteroid** relatively small rocky objects orbiting the Sun; most are found between Mars and Jupiter 208
- astronomer** scientist who studies the Universe 198
- atoms** small particles that make up all matter 2
- atrium** (plural atria) upper chamber of each side of the heart; left atrium takes in blood from the lungs, right atrium takes in blood from rest of the body 127
- average speed** the distance an object travels in a certain amount of time; usually measured in metres per second or kilometres per hour 45
- background radiation** radiation (left over from the Big Bang) 199
- ball and socket joint** a joint between bones which allows a rotation movement; e.g. hip joint 140
- base** a substance with a soapy feel that neutralises acids. Bases turn red litmus blue 16
- Big Bang theory** theory that the Universe began in a gigantic explosion 199
- bile** detergent-like substance released by the liver into the small intestine; helps make an emulsion of lipids to increase the surface area exposed to the enzymes which digest lipids 118
- bionic parts** artificial body parts which include use of computer technology 145
- black hole** formed from the collapse of a giant star; so dense not even light can escape its gravity 200
- bloodstream** blood flowing through the arteries, veins and capillaries 118
- bone** a living tissue found in vertebrate skeletons 136
- bone marrow** tissue found in bone which divides to form red blood cells, white blood cells and platelets 136
- breathing** taking air in and out of the lungs to ensure sufficient gas exchange 123
- buoyancy** ability to float; buoyancy occurs when an upward force is provided by water to balance the weight of the object (it must be less dense than water for this to occur) 133
- capillaries** small blood vessels which connect arteries to veins; site of exchange of materials with the cells 118, 124, 126
- carbohydrates** substances, such as sugar and starch, found in food; used as a source of energy for a living organism 108, 112
- cardiac muscle** the very specialised muscle found in the heart 138, 139
- cartilage** flexible tissue found as part of the vertebrate skeleton 135, 136
- catalyst** substance that speeds up the rate of a chemical reaction but is itself unaffected 33
- cell (electric)** single device that supplies the energy to make an electric current flow 62
- cellulose** complex carbohydrate found in cell walls of plants; cannot be digested by most animals 117, 132
- cement** a substance which can be used to bind materials together 187
- chemical change** a change in which a new substance or substances are formed (e.g. paper burning) 22
- chemical digestion** chemical breakdown of food into smaller molecules by the action of enzymes 116

- chemical energy** potential energy stored in the chemical bonds of a substance 65
- chemical equation** method of showing reactants involved and products formed in a chemical reaction 10
- chemical formula** group of atomic symbols that shows the kinds and number (or ratio) of atoms in a substance 10
- chemical reaction** reaction in which a chemical change occurs 22
- circuit** a continuous path along which an electric current can flow 65
- circuit diagram** diagram using universally accepted symbols to represent an electric circuit 69
- circulatory system** transport system composed of heart, blood and blood vessels 126
- collagen** a protein fibre which gives flexibility to animal tissue; also helps provide support and structure 136, 146, 151
- combustion** burning or the rapid combination of a substance with oxygen; heat and light energy are usually produced 32
- components** parts of circuit; e.g. globe, switch 65
- compound** a substance composed of two or more different types of atoms; can be broken down into simpler substances 2, 180
- compound fracture** a bone injury where the broken bone pierces the skin 137
- compression (forces)** pushing forces which squeeze an object 187
- compression wave** a type of wave; sound is an example 102
- concrete** a synthetic rock made by mixing cement with other materials 186
- conductivity** ability to conduct an electric current 8
- conductor** a substance that will allow an electric current to flow through it 66
- condyloid joint** a joint between bones which allows some rotation as well as back and forth movement 140
- constellation** small group of stars which forms a pattern 200
- contact force** force that is exerted when two objects are in contact; e.g. friction, buoyancy, tension 42
- controlled burning** burning off areas of bushland during non-dangerous times when the extent of the burn can be limited; reduces the danger of other bushfires 163
- convection** transfer of heat energy in a liquid due to less dense warm matter rising and denser cool matter falling 164
- cornea** thick curved transparent material at the front of the eye; most of the focusing of light onto the retina occurs here 99
- corrosive** will burn through solid substances 16
- crater** scar left by the impact of a meteorite or asteroid 208
- crystal** mineral in which the atoms are arranged in such a way as to produce definite shapes 180
- cullet** broken pieces of glass; useful in recycling of glass 186
- cyborg** term used in science fiction to describe characters that are part human and part machine 144
- dark matter** matter in the Universe which does not emit light 199
- decibel** a unit to measure loudness of sound (symbol dB) 104
- dehydration** loss of water from living cells 120, 156, 194
- density** mass per unit volume of a substance 185
- dermis** the main layer of skin; contains glands and nerve receptors 147
- detergent** anything that combines with water to get things clean 36
- diaphragm** muscular structure at base of rib cage which can change the volume of the chest cavity; contraction and relaxation of the diaphragm allows air to move in and out of the lungs 125
- diffusion** spreading out of the particles in liquids and gases due to their random movement; results in substances moving from areas of high concentration to lower concentration 119
- digestion** breakdown of food into simple soluble substances which can enter cells 116
- disc** a cushion of cartilage between each vertebra in the backbone 142
- ductility** able to be drawn out into wire 8
- echo** reflected sound heard a second time 90
- El Niño** a warm ocean current which flows from the western Pacific eastwards, with cooler waters rising close to Australia; drought conditions occur because the air above the cool water is dense and does not rise to form clouds 165
- elastic** describes a material that returns to its original shape after it has been stretched or compressed 48
- electric current** a flow of charged particles 65
- electrical energy** energy that causes charged particles to move 65
- electricity** the energy that results from the movement of charged particles 62
- electrolysis** causing a chemical reaction in a liquid by passing an electric current through it 79
- electromagnetic spectrum** a group of energy waves including visible light, ultraviolet and infrared rays, microwaves, X-rays, gamma rays and radio waves 94
- electromagnetic wave** a type of energy wave; light is an example 103
- electrons** negatively charged particles which surround the nucleus of atoms 65, 67

- electroplating** depositing a layer of metal on another using electrolysis 79
- element** a substance composed of only one type of atom; it cannot be broken down into simpler substances by chemical reactions 2, 180
- emulsion** a colloid with tiny droplets of one liquid spread through another liquid 113
- endoskeleton** internal skeleton of vertebrates 133
- engine** machine which converts stored energy (in fuel) into useful work or movement 46
- enzymes** chemicals which speed up reactions in living organisms e.g. the digestion of food 116, 118
- epidermis** the outer cell layer of skin or other structures such as leaves 147, 151
- eucalypts** the most dominant genus of Australian plants 160
- evaporate** change of state from a liquid to a gas 27
- excretion** removal of waste products from cell metabolism; e.g. breathing out carbon dioxide from the lungs, excreting urine from kidneys 157
- excretory organ** an organ which helps remove waste products produced by cell activities in the body; e.g. kidney excretes urea 148
- exoskeleton** external skeleton of arthropods such as insects 122, 133
- fat** lipid which is a solid at room temperature 129
- fermentation** conversion of sugar to alcohol and carbon dioxide by microorganisms such as bacteria and yeasts; utilised in bread and wine making 31
- field force** (or non-contact force) force that is exerted on an object from a distance; e.g. gravity, electrostatic and magnetic forces 42
- filaments** finger-like projections which make up gills in aquatic animals 122
- flaccid** describes plant cells which have lost water so that the cell wall has collapsed slightly inwards as the cell membrane shrinks away from it 132
- force** a pull or push; forces can change the way things are moving 42
- friction** force that acts against a moving object 42
- fuel** something that burns to release heat energy 32
- fulcrum** a pivot point; the point about which a lever moves 141
- fulgurite** a glassy tube formed where lightning has struck sand 186
- fuse** a safety device consisting of a wire or metal strip designed to melt when excessive current passes through it 73
- g force** 1g force is the force experienced by someone accelerating at 9.8 m/s² 206
- galaxy** collection of millions of stars 50, 200
- gall bladder** storage organ for bile which helps digest lipids 118
- gas** state of matter with no fixed shape or volume 6
- gas exchange** process in living things where one gas is taken up and another is released; e.g. in animals, oxygen is taken up and carbon dioxide released 122
- gastric juice** highly acidic liquid secreted by the stomach; contains hydrochloric acid, mucus and enzymes which digest protein 118
- geologist** scientist who studies the Earth, its origin, history, structure and composition 184
- geosynchronous (geostationary) satellite** satellite which orbits the Earth at about 37 000 km; orbit is synchronised with rotation of the Earth so it appears stationary from Earth 205
- gills** respiratory surfaces found in fish and other aquatic animals 122
- gizzard** muscular structure found in the digestive system of birds and simpler organisms; contains grit and small stones to grind food in mechanical digestion 116
- glacier** river of ice 158
- glass** a solid classified as a supercooled liquid; made from sand and other materials 186
- glycogen** energy-rich carbohydrate stored by mammals in the liver and muscles 128, 138
- gravitational force** see weight 206
- gravity** the force of attraction between any two objects, e.g. the Earth and a person 42, 50
- habitat** the place where an organism lives 168
- heat** type of energy that can raise the temperature of things 26
- hertz** a unit for measuring frequency; number of cycles per second (symbol Hz) 94
- hinge joint** a joint between bones which allows back and forth movement like the hinge of a door 140
- hologram** an image produced using a laser; the image has a three-dimensional appearance 103
- hydration** a chemical reaction which uses up water molecules 187
- hydrochloric acid** acid found in the stomach; helps break down protein and destroys swallowed microorganisms 118
- igneous rock** rock formed when molten magma or lava solidifies; e.g. granite, basalt 158
- indicator** a substance that indicates a change, often by changing colour itself, that would otherwise be difficult to observe directly 24
- inelastic** describes a material which does not return to its original shape after it has been stretched or compressed 48

- inorganic compounds** simple non-organic substances such as water, minerals, carbon dioxide 16
- insulator** a substance that will not allow an electric current to flow easily through it 66
- intensity** how much of something there is; the brightness of light or the loudness of a sound 95, 96
- interstellar space** outside the solar system; between the stars 202, 213
- introduced species** non-native species 170
- iris** coloured muscular tissue surrounding the pupil of the eye; contracts and relaxes to control the amount of light entering the eye 99
- irrigation** controlled flooding of land with water to allow crops to be grown 172
- joint** junction of two or more bones in a skeleton 137, 140
- kilowatt hour** also known as unit; unit of measurement of the electrical power used in an hour (kW) 81
- kinetic energy** energy possessed by a moving object 65
- La Niña** a warm ocean current which flows from east to west across the Pacific Ocean, bringing warm air close to Australia; associated with heavy rain and flooding 165
- lanthanides** also called the rare earths. Group of metallic elements with atomic numbers 57 to 71 18
- Large and Small Magellanic Clouds** two galaxies visible from the southern hemisphere 200
- large intestine** final organ in the digestive system; site of water reabsorption 118
- larynx** the voice box; contains vocal cords which vibrate to produce sound as air passes over them 125
- laser** optical device which produces intense light of a single wavelength 95
- lava** molten rock that flows on to the surface of the Earth or other body 208
- lever** a simple machine which can act as a force magnifier or a distance magnifier 141
- ligament** connective tissue which holds bones in place at a joint but still allows some movement at the joint 137
- light-year** distance travelled by a beam of light in a vacuum in one year, approximately 9.46 trillion kilometres 202
- lignin** material found in the walls of xylem tissue in woody plants; gives support and provides the wood tissue 132
- lignotuber** a swollen underground root which stores water 162
- lipids** substances, such as fat and oil, found in food; used as a source of energy 108, 112
- liquid** state of matter that changes its shape but not its volume 6
- liver** large important organ found in the abdomen; functions include production of bile, food storage and breakdown of toxic substances in the body 118
- Local Group** small cluster of galaxies relatively close to our own 200
- logging** cutting down and transporting timber 168
- lunar rocks** rocks obtained from the surface of the Moon 208
- lungs** respiratory organ found in amphibians, reptiles, birds and mammals; allows gas exchange 123, 124, 156, 157
- lustre** amount of shine on the surface of a substance; metals are often said to have a metallic lustre 8
- magnetic field** a region in which a magnetic object will experience a force 76
- magnetic force** force experienced by a magnet or magnetic object in a magnetic field 76
- magnetise** cause a magnetic object to become a magnet 76
- malleable** able to be hammered into thin sheets; able to be bent and shaped 8
- mallees** varieties of eucalypts adapted to grow in arid areas 160
- maria** 'seas' of the Moon; formed by ancient lava flows 208
- mass** the amount of material in an object; measure in grams (g) or kilograms (kg) 51
- matter** anything that takes up space and can be weighed 2
- mechanical digestion** physical breakdown of food into smaller particles e.g. by chewing 116
- melanin** a pigment responsible for the dark colour of skin and freckles 149
- metalloids** elements displaying some properties of metals and some properties of non-metals; e.g. arsenic, silicon 8
- metals** elements which tend to be shiny, ductile and malleable, conduct heat and electricity, and form positive ions; e.g. copper, iron, gold 8
- meteorologist** scientist who observes the atmosphere so that weather can be forecast 205
- microgravity** condition where there seems to be little gravitational pull 211
- Milky Way** spiral galaxy containing our solar system 200
- mineralogist** geologist who specialises in the identification of minerals 184
- minerals** naturally occurring chemical compounds; some types of minerals are required in the diet to form part of healthy tissue, e.g. calcium is needed for healthy bones and teeth 108, 180, 184
- mining** excavation of minerals from the ground 168
- mitochondria** (singular mitochondrion) organelle where respiration takes place in cells 139

- mixture** substance made up of a number of different elements or compounds mixed together but not chemically joined 2
- model** simple representation of a complex thing or idea 13
- Mohs' scale** comparative scale of hardness of minerals devised by Friedrich Mohs 184
- molecule** group of two or more atoms bound together 2,12
- monomers** small molecules that are the building blocks of polymers 37
- mordant** a substance used to fix a dye in a fabric, i.e. to stop the dye from washing out when the fabric is wet 192,193
- mouth** first part of the digestive system; mechanical and some chemical digestion occurs here 118
- mucus** slippery substance produced along the digestive tract; helps the movement of food and protects the inside lining of the digestive tract 118
- muscle tone** the healthy condition of muscle, maintained by regular involuntary contractions 142
- myoglobin** a pigment which can store oxygen in muscles; found in higher proportions in marine mammals 156
- native elements** minerals found as elements in nature, e.g. gold, diamond 180
- natural resources** anything useful to humans which can be obtained from the natural environment 178
- nebula** huge mass of dust and gas remaining from a supernova 200
- neutralisation** reaction of an acid with a base to form a salt and water 16
- newton** unit of force (N) 51
- non-metals** diverse group of elements which do not display the characteristics of metals; tend to be poor conductors of heat and electricity; e.g. sulfur, oxygen, carbon 8
- normal** line drawn at right angles to a surface 88
- nucleus** found at the core of an atom; contains protons and usually neutrons 67
- nutrients** substances required by a living thing to provide energy, materials for growth, repair and normal functioning 108
- obsidian** a naturally occurring rock formed from molten sand from a volcano; also called volcanic glass 186
- oesophagus** tube leading from the mouth to the stomach 118
- optic nerve** nerve which takes information from the retina to the brain 99
- orbit** path followed by a planet or satellite around another body such as the Sun; normally elliptical 204
- ore minerals** minerals of sufficient worth to mine 180
- organic compounds** complex carbon-containing compounds; usually also contain hydrogen and often oxygen 16
- osteoarthritis** inflammation at bone joint caused by wear and tear 142
- osteoporosis** a weakening of bones associated with low bone density due to loss of calcium minerals from the bone 142
- oxygen** colourless, odourless gas found in air; needed by most living things to survive; also necessary for burning 32
- pancreas** releases digestive juices containing enzymes into the first part of the small intestine; the enzymes help digest lipid, carbohydrate and protein 118
- parallel circuit** an electric circuit in which current divides and some passes through each component 71
- pepsin** protein-digesting enzyme found in gastric juices in the stomach 118
- peristalsis** wave-like contraction of muscles in the walls of the digestive tract; moves food through the digestive tract 118
- permeability** the tendency or ability of a substance to allow fluids to penetrate through it, e.g. permeability of soil to water 159
- phonon** a bundle or packet of sound energy 103
- photon** a bundle or packet of light energy 103
- physical change** a change in which no new substance is formed (e.g. ice melting) 22
- pitch** how high or low a sound is 96
- pivot joint** a joint which allows turning from side to side 140
- plasma** liquid non-cellular component of blood; carries digested food wastes and hormones 126
- platelets** components of blood, help in the clotting of blood 126
- polar orbiting satellite** satellite that orbits the Earth by flying over the North and South Poles 205
- pollutant** substance or physical factor which contaminates or spoils the environment 166
- pollution** contamination of the environment 166
- polymer** very long molecules made up of repeating units called monomers; plastics are polymers 37
- potential difference (voltage)** electrical energy lost across a component or gained across a cell; measured in volt (V); unit of potential difference 65
- power** how much energy something uses; measured in watts (W) or kilowatts (kW) 80
- precipitate** a solid formed when two solutions are mixed 24
- preservative** food additive that prevents or slows down food spoilage 114
- product** new substance produced in a chemical reaction 10, 23
- proteins** substances found in food; used mostly for growth and repair 108, 112
- pulsar** rapidly spinning neutron star 200

- pupil** hole in the iris of the eye where light enters 99
- pure substance** a substance made up of only one type of element or compound 2
- quasar** astronomical source of intense radiation 200
- rare earth metals** see lanthanides 18
- ray** tiny beam of light that travels in a straight line 88
- reactants** substances that react to form new substances in a chemical reaction. Reactants are written on the left-hand side and products are written on the right-hand side 10, 22
- rectum** final part of the large intestine where wastes (faeces) are stored prior to elimination 118
- recyclable** describes a substance which can be collected and processed again for future use 186
- red blood cells** oxygen-carrying cells found in blood; contain the pigment haemoglobin; mature human red blood cells lack a nucleus 126
- reflection** rays or beams of light bouncing off a smooth surface; sound will also bounce off surfaces 88, 90
- refraction** change in direction of a wave as it passes from one medium to another e.g. a light ray bends as it passes from air to glass or water 99
- rejection process** destruction of transplanted material by antibodies; the transplant is detected by the body's immune system as foreign material invading the body 146
- renewable** term applied to a resource which can be replaced, e.g. timber 189
- resistance (R)** opposition to the flow of electric current through a conductor 67
- resource** any material or substance which humans can make use of 178
- respiration** chemical reaction in living things that releases energy from food for life processes; requires oxygen 32, 110, 122
- respiratory system** system which allows gas exchange to take place; e.g. in mammals, consists of trachea, lungs, alveoli etc. 123, 124
- retina** light-sensitive layer on the interior surface of the eye 99
- reverberation** effect created by reflection of sound by walls in an enclosed space 90
- saliva** watery substance secreted by the salivary glands; contains water, mucus and amylase, a starch-digesting enzyme 118
- salivary glands** three pairs of glands which secrete saliva into the mouth 118
- satellite** any object or body that orbits a planet or other large body 198, 204
- sclerophylls** plants which have hardened leaves as an adaptation to surviving dry conditions 160
- semiconductor** crystalline material with an electrical conductivity between that of metals and insulators, e.g. silicon 68
- series circuit** an electric circuit in which current passes through one component and then another 71
- short circuit** direct connection between two points in an electrical circuit with a low resistance which allows a large current to flow through it. This bypasses the rest of the circuit and can result in the circuit overheating 73
- simple fracture** a broken bone 137
- skin graft** replacement of damaged skin by healthy skin from another part of the body 150
- sliding friction** the force that exists between two surfaces in contact which are moving 54
- small intestine** main digestive organ of the body; digestion of carbohydrates, lipids and protein is completed here. Also the major site of absorption of digested nutrients into the blood 118
- smooth muscle** the type of muscle found in the walls of body organs and tissues; also called involuntary muscle because it is not consciously controlled by the brain 138
- soap** detergent made from animal or plant oil 36
- soil degradation** the breakdown of the structure of soil; includes washing away of topsoil and change in the composition of the soil 162
- soil salinity** the amount of salt in soil 172
- solar system** consists of the Sun and all the things that orbit it including the nine planets 202
- solid** state of matter which has a fixed shape and volume 6
- sonar** sound navigation and ranging; ultrasonic waves used to locate things under water 92
- sonic boom** loud sound caused by an object travelling faster than the speed of sound 105
- space station** spacecraft in a permanent orbit capable of housing a crew 210
- speed** the rate at which distance changes over time 45
- spinal cord** together with the brain, forms part of the central nervous system; carries nerve messages between the brain and other parts of the body; protected by vertebrae in the backbone 142
- starch** energy-rich carbohydrate stored by plants 110, 113, 128
- static electricity** build-up of charge in an object that remains at rest 62
- static friction** the force which exists between two surfaces in contact before they move 54
- stem cells** unspecialised cell tissue which can differentiate into different types of cells 136

- stomach** muscular sac-like organ of the digestive system; helps in mechanical digestion of food and chemical breakdown of protein 118
- stomata** (singular stoma) pores in a leaf that control the entry and exit of gases, including water vapour 160
- streak (test)** mineral identification test where unknown specimen is scraped across the unglazed side of a white ceramic tile so that powder from the specimen can be observed 185
- striated muscle** striped muscle which controls voluntary movement of bones in the skeleton 138
- Sun-synchronous orbiting satellite** close-orbiting satellite that passes over the same part of the Earth at the same time each day 205
- superconductor** a substance which greatly increases its electrical conductivity when cooled to extremely low temperatures, so that charge flows through it without resistance 19, 68
- supercooled liquid** a solid which has no clearly defined crystal form; the molecules are arranged in random order like the molecules in a liquid; glass is an example 186
- supersonic** faster than the speed of sound 105
- sustainable development** development which can continue because resources are used wisely and not completely consumed 190
- suture** a fixed joint between bones 140
- synthetic** describes a substance which has been made by combining two or more other substances in a chemical reaction (called a synthesis reaction) 186, 192
- synthetic detergent** detergent made from oil (not plant or animal oil) 36
- tektite** a glassy object formed from the cooling of molten rock which has been thrown into the air after a meteorite has crashed into the Earth 186
- tendon** tissue which connects a muscle to a bone; pulls against the bone when the muscle contracts 137
- tensile forces** stretching forces which pull an object apart 187
- terminal velocity** point at which a falling body ceases to accelerate due to weight being balanced by air resistance 57
- terminals** connection points on a component of an electric circuit 65
- theory** an idea which is used to explain a wide range of observations 62
- threshold of hearing** softest sound able to be heard 104
- trachea** the windpipe; carries air from the nose down to the bronchi 125
- trade winds** regular pattern of winds which blow because of convection currents formed by temperature differences between tropical, temperate and polar regions; the direction they blow is modified by the rotation of the Earth 165
- transparency** the ability to transmit light, so that the material can be seen through clearly, e.g. glass 186
- transpiration** loss of water vapour by evaporation from a plant 160
- turbidity** the muddiness of water 167
- turgid** describes plant cells which have ample water so that the cell wall is fully extended by the cell membrane pushing against it 132
- Universe** all space and everything in it, including stars and galaxies 198
- urbanisation** development of cities and towns displacing countryside 169
- valve** structure which allows a fluid to flow in one direction only, e.g. in the heart or in veins 126
- vein** blood vessel with thin, non-elastic walls; carries blood back to the heart 126
- ventricle** lower chamber of each side of the heart; left ventricle pumps blood to the body; right ventricle pumps blood to the lungs 127
- vertebrae** (singular vertebra) small bones that protect the spinal cord and make up the backbone 134, 142
- villi** (singular villus) finger-like projections of the small intestine which increase the surface area for the absorption of digested nutrients 118
- vitamins** diverse group of substances such as vitamin C and vitamin E which are required in small amounts for normal body functioning 108
- voltage** see potential difference 65
- vulcanisation** technique for hardening rubber by heating and chemically combining it with sulfur. The rubber becomes stronger and more elastic 34
- water table** the upper surface or boundary of rock which contains groundwater 172, 173
- watt** unit of power (W) 80
- weight** the force of gravity pulling on an object; measured in newtons (N) 51, 206
- weightlessness** condition in which a person seems to have no weight and floats inside a structure such as a spaceship 206
- white blood cells** cells found in the blood which help fight disease, e.g. by engulfing invading micro-organisms 126
- xylem** type of dead cell tissue found in some plants; transports water and minerals to the leaves 132

ACKNOWLEDGMENTS

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